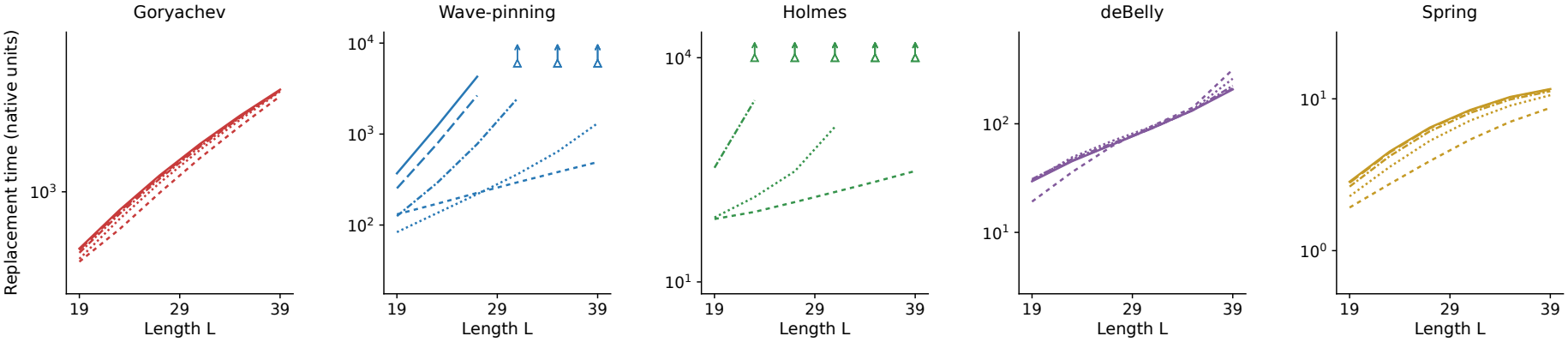
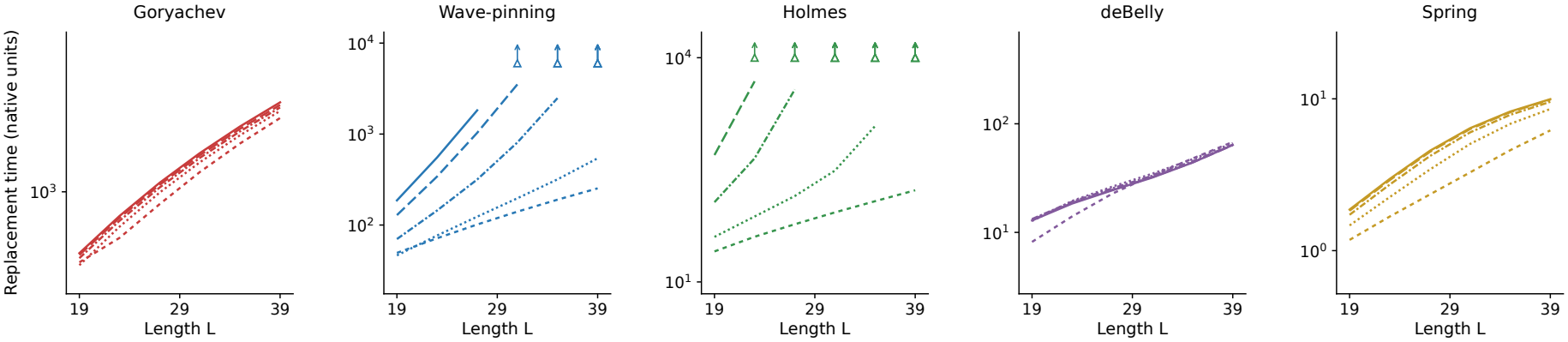
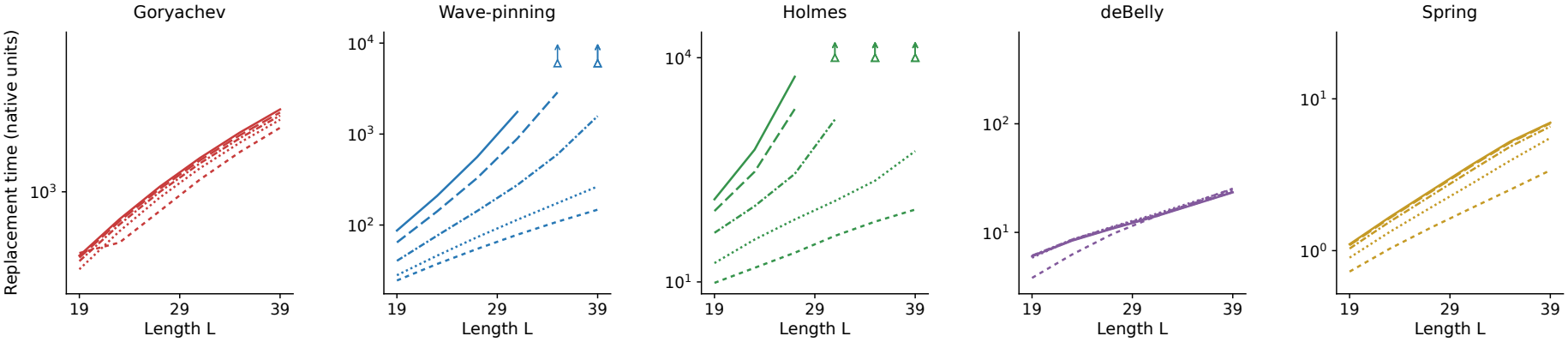

Equal-area width series: area 0.5×; all finite times and lower bounds
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Replacement time (native units)
10^3
10^4
10^2
10^1
10^0
19
29
39
Length L
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2


Equal-area width series: area 1×; all finite times and lower bounds
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Replacement time (native units)
Length L
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2


Equal-area width series: area 2×; all finite times and lower bounds
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Replacement time (native units)
Length L
19
29
39
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2

deBelly L19, gap16: exact native endpoint audit

The first gap does not fully retain visible activity.

Pulse-1 end share: 0.921382.

Gap-1 raw share: 0.102965; raw ratio: 0.111751.

Gap contrast: 0.032750, below the existing 0.05 gate.

Eligible gap share is unavailable because activity is below that visibility gate.

The main D places this known below-gate observation at x = 0 with an open circle.

This is a visibility code, not an assertion of mathematically zero raw activity.

Pulse-2 end share: 0.988950.

Pulse-2 headroom gain: 0.859447.

Independent checks recompute all three endpoints from phase-native active fields.

The native archives and audited endpoint table agree; no endpoint is forward-filled.

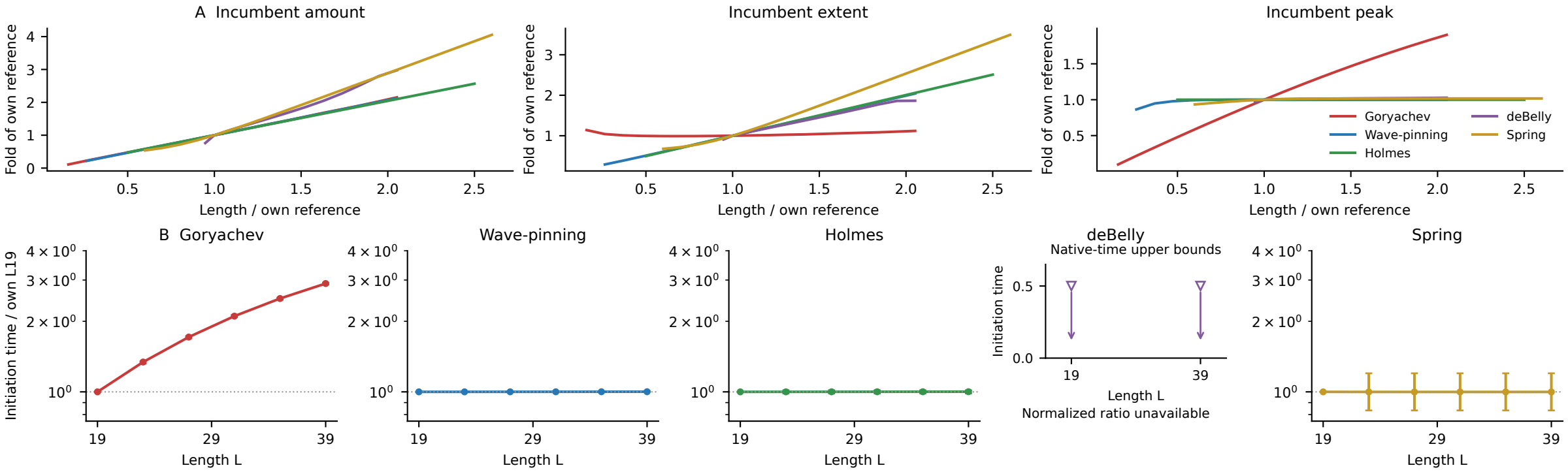
D uses the first gap only. Later gaps must not substitute for it.

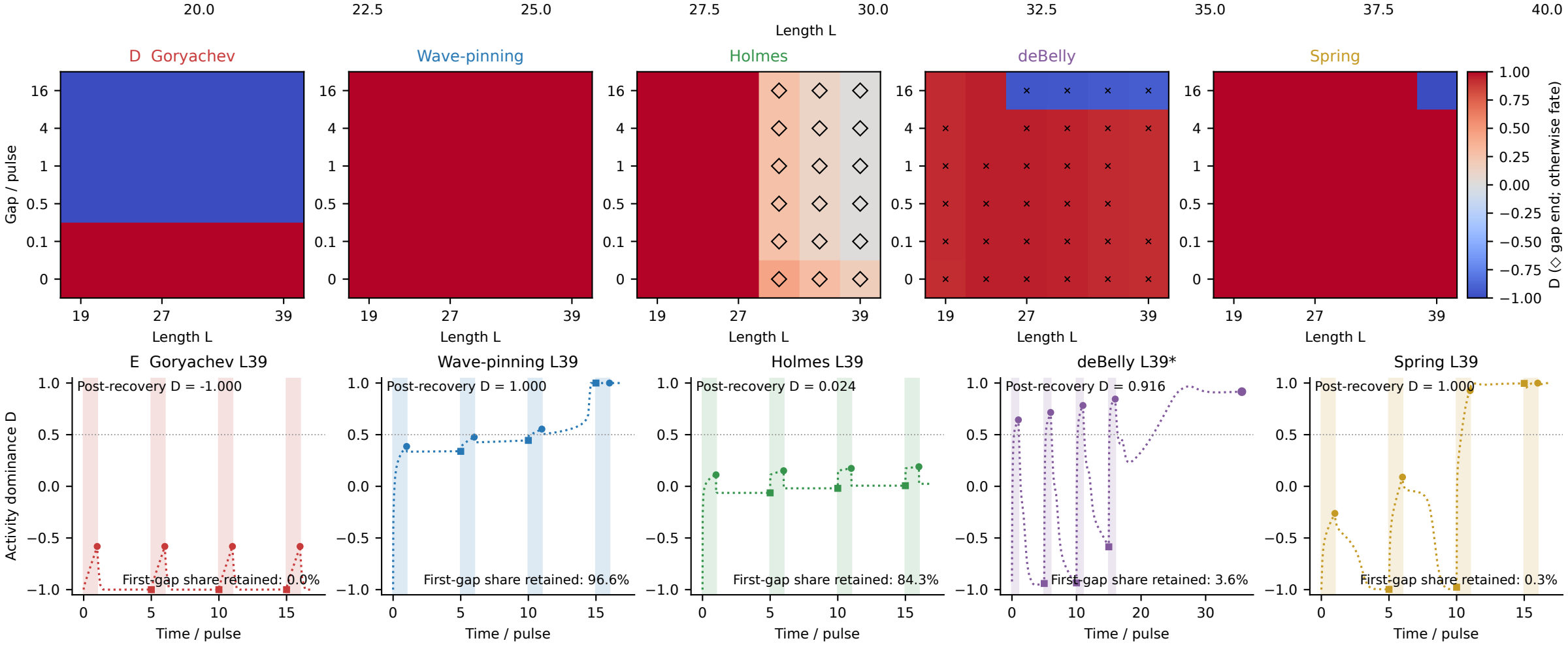
All 180 D coordinates, raw shares, visibility and field-tail flags are in the CSV.

Holmes L31–39 uses the existing L27-referenced train, with no target-length calibration.

deBelly timing and endpoint-field qualifications remain in the retained supplement.

Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures





F Shape and timing signatures distinguish mechanism classes

Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Experimental measurement
Initiation-limited Goryachev	Yes (2.90×)	Modest (1.12×)	Resets	No	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Mass-conserved, saturating Wave-pinning, Holmes	No resolved slowing	Large; grows over resolved L range	Share retained (ratchet)	—	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Slow non-activity state deBelly*, Spring	Unresolved* / no	Modest (~1×)	Collapses early	Yes	Beam width at fixed photon dose; pulse trains; gap-end biosensor share

◇ Holmes L31–L39: last inter-pulse gap-end D, with no target-length calibration. × failed field tail; ? uninformative ratio.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).

Holmes large-L retention: completed frozen design and qualifications

No continuous commitment by 100,000 is an observed finite-horizon finding.

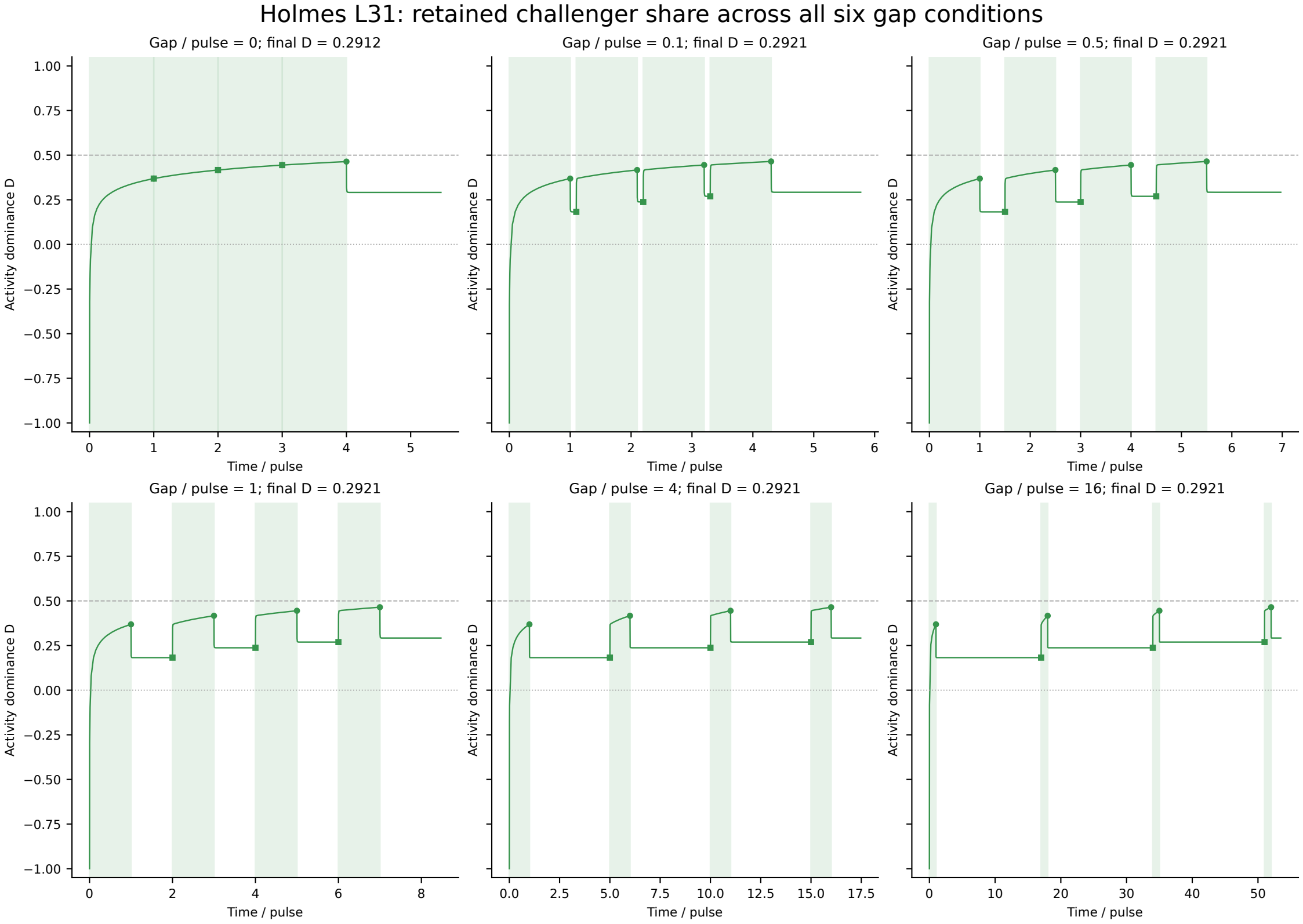
L27 upper committing duration = 18,125; four pulses each 6,796.875.
Target lengths: 31, 35, 39; gaps: 0, 0.1, 0.5, 1, 4, 16 × pulse.
Primary: D and challenger share at every pulse and gap endpoint. Fate is secondary.

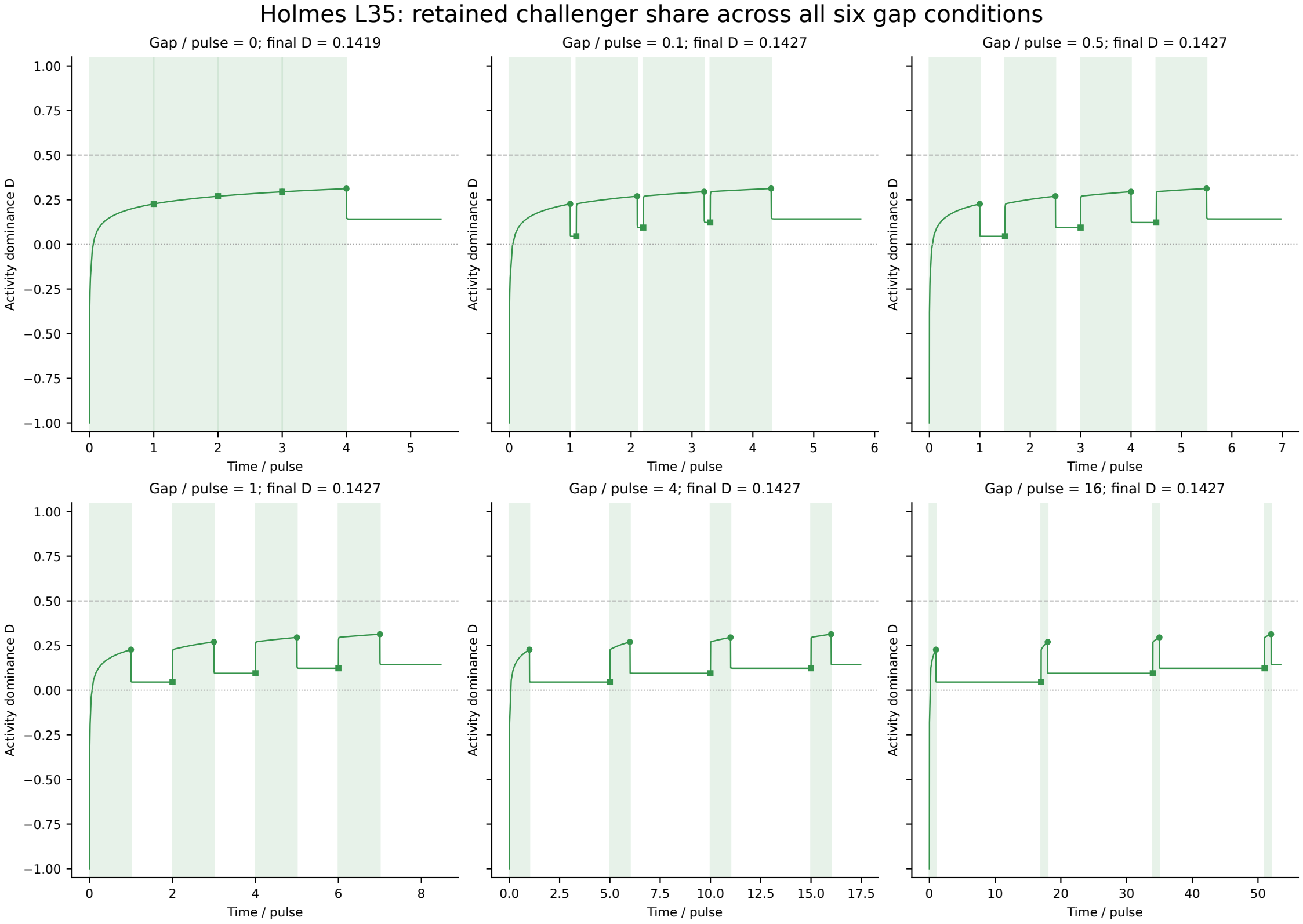
L39 bounded-noise check: gap16, mean 2% native peak, maximum 4%;
12 independent seeds plus matched-mean and zero controls.
S3-matched nonperiodic spatial OU input: ell 1.9, tau 10, knot 2.5.

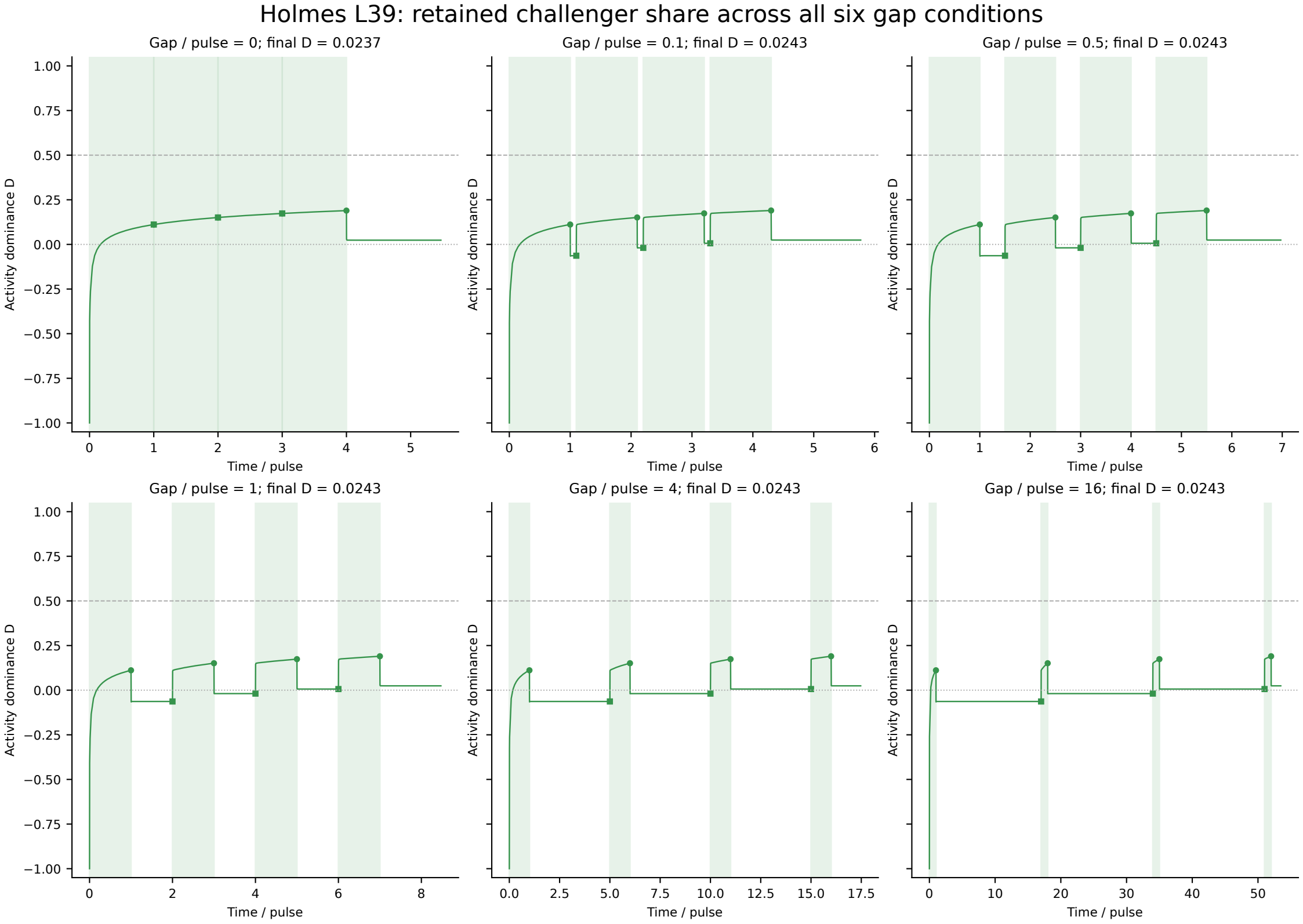
All 32 native SLURM trajectories and independent numeric/phase/restart/tail
audits completed. No kinetics or calibration thresholds were retuned.
Clean L39 gap4 retains >84% raw share even with slightly negative gap-end D.
The secondary S3 criterion requires both D ≥ 0 and raw retention ≥80%.

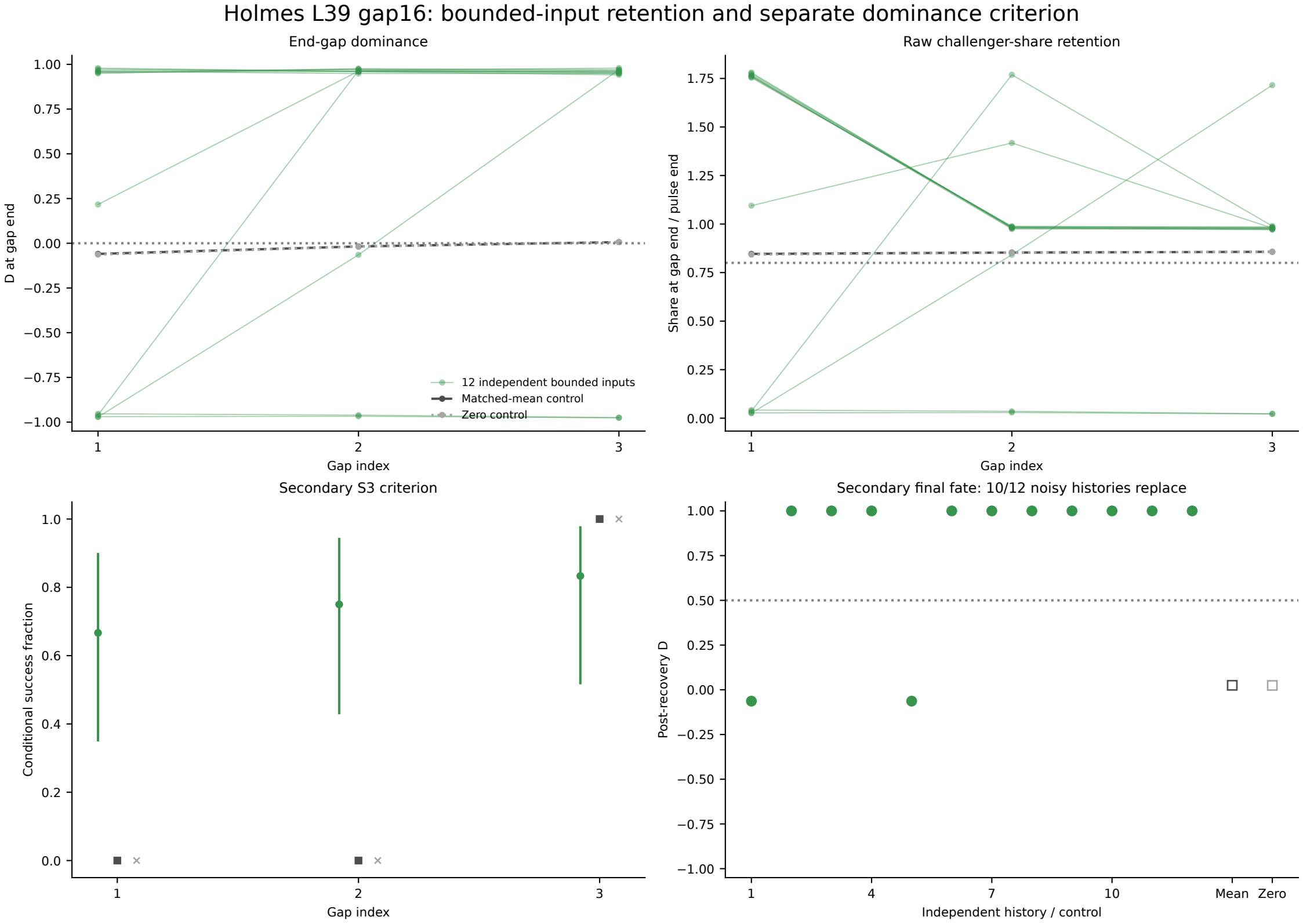
Class membership is unchanged. deBelly initiation remains interval-unresolved;
Holmes width-penalty growth is established only over measured moderate sizes.

The theoretical mean gap-integrated commanded input is 2.46 × ON exposure;
this long-gap assay is weak instantaneously, not a matched-total-dose test.
Heterogeneous input yields variable gap retention and 10/12 final replacements;
the matched-mean and zero controls remain below replacement.


Holmes L31: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.2912
Gap / pulse = 0.1; final D = 0.2921
Gap / pulse = 0.5; final D = 0.2921
Gap / pulse = 1; final D = 0.2921
Gap / pulse = 4; final D = 0.2921
Gap / pulse = 16; final D = 0.2921
Activity dominance D
Time / pulse


Holmes L35: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.1419
Gap / pulse = 0.1; final D = 0.1427
Gap / pulse = 0.5; final D = 0.1427
Gap / pulse = 1; final D = 0.1427
Gap / pulse = 4; final D = 0.1427
Gap / pulse = 16; final D = 0.1427
Activity dominance D
Time / pulse


Holmes L39: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.0237
Gap / pulse = 0.1; final D = 0.0243
Gap / pulse = 0.5; final D = 0.0243
Gap / pulse = 1; final D = 0.0243
Gap / pulse = 4; final D = 0.0243
Gap / pulse = 16; final D = 0.0243
Activity dominance D
Time / pulse


Holmes L39 gap16: bounded-input retention and separate dominance criterion
End-gap dominance
D at gap end
Gap index
12 independent bounded inputs
Matched-mean control
Zero control
Raw challenger-share retention
Share at gap end / pulse end
Gap index
Secondary S3 criterion
Conditional success fraction
Gap index
Secondary final fate: 10/12 noisy histories replace
Post-recovery D
Independent history / control
Mean
Zero

Holmes L31: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.3686983	0.3686983	0.4167275	0.4167275	0.4443979	0.4443979	0.4639236	0.2912072
0.1	0.3686983	0.1823237	0.4169269	0.2374473	0.4448783	0.2694141	0.4647032	0.2920993
0.5	0.3686983	0.1823238	0.4169269	0.2374478	0.4448786	0.2694155	0.4647042	0.2921004
1	0.3686983	0.1823239	0.4169270	0.2374485	0.4448790	0.2694172	0.4647053	0.2921017
4	0.3686983	0.1823247	0.4169273	0.2374523	0.4448812	0.2694278	0.4647122	0.2921096
16	0.3686983	0.1823278	0.4169287	0.2374675	0.4448900	0.2694698	0.4647399	0.2921412

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Holmes L35: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.2268125	0.2268125	0.2701031	0.2701031	0.2949934	0.2949934	0.3125254	0.1419210
0.1	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
0.5	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
1	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
4	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426673
16	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426673

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Holmes L39: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.1113753	0.1113753	0.1508245	0.1508245	0.1734726	0.1734726	0.1894153	0.0236737
0.1	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
0.5	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
1	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
4	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
16	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Class-membership evidence: Goryachev

initiation slows with L
Value: 2.898986378675809; bounds: [2.871794871794872, 2.9220779220779223]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: 1.124834416847597; bounds: [1.124484748557296, 1.1262376237623761]
L19 {'status': 'bounded', 'lower': 1.1700680272108843, 'upper': 1.1849315068493151, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.1798811734705885, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 1.124484748557296, 'upper': 1.1262376237623761, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.124834416847597, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; modest non-growing penalty is distinct from the large saturating-model penalty

share held through gaps
Value: 2.4681882341887874e-21; bounds: [None, None]
L39 gap4 first-gap D -0.58178128 to -1; share 0.20910936 to 5.1612126e-22; all raw retention ratios [2.4681882341887874e-21, 2.468188239315979e-21, 2.4681882814510725e-21]

commits after in-gap collapse
Value: -1.0; bounds: [None, None]
L39 gap4 gap-end D [-1.0, -1.0, -1.0]

Class-membership evidence: Wave-pinning

initiation slows with L
Value: 1.001354777811466; bounds: [1.0, 1.0070175438596491]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: None; bounds: [11.059907834101383, None]
L19 {'status': 'bounded', 'lower': 3.9870689655172415, 'upper': 4.049676025917927, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 4.023952792808644, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'lower_bound_only', 'lower': 11.059907834101383, 'upper': None, 'lower_strict': True, 'upper_strict': False, 'estimate_from_interpolation': None, 'narrow_censored': True, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}

share held through gaps
Value: 0.9656458108363115; bounds: [None, None]
L39 gap4 first-gap D 0.38625115 to 0.33862762; share 0.69312558 to 0.66931381; all raw retention ratios [0.9656458108363115, 0.9783668083224305, 1.2867515519314443]

commits after in-gap collapse
Value: 0.9999999753672253; bounds: [None, None]
L39 gap4 gap-end D [0.3386276200076397, 0.444706256945651, 0.9999999753670956]

Class-membership evidence: Holmes

initiation slows with L
Value: 1.001466041861598; bounds: [0.9848484848484849, 1.0153846153846153]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: None; bounds: [133.15579227696406, None]
Deciding comparison is L23 versus L19; extreme L39 remains uninformative. L19 {'status': 'bounded', 'lower': 28.99002493765586, 'upper': 29.125, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 29.034710721675975, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'lower_bound_only', 'lower': 0.0, 'upper': None, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': None, 'narrow_censored': True, 'broad_censored': True, 'informative': False, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; Holmes growth verified L19 ~29.03 to L23 >133.16; growth through L39 unresolved, not an assigned infinite ratio

share held through gaps
Value: 0.842868741871212; bounds: [None, None]
L39 gap4 first-gap D 0.11137532 to -0.063256484; share 0.55568766 to 0.46837176; all raw retention ratios [0.842868741871212, 0.8521812245120128, 0.8572795615775773]; first two gap D<0 fail the secondary S3 D>=0 criterion but retain >84% raw share

commits after in-gap collapse
Value: 0.024287736083136613; bounds: [None, None]
L39 gap4 gap-end D [-0.06325648449328491, -0.019191515710938245, 0.006269998360605156]

Class-membership evidence: deBelly

initiation slows with L
Value: None; bounds: [0.0, None]
deBelly first-update intervals do not resolve a normalized size effect

narrow-cue penalty and growth
Value: 0.963187372120211; bounds: [0.9548872180451128, 0.9696969696969697]
L19 {'status': 'bounded', 'lower': 0.9615384615384616, 'upper': 1.04, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.011603783514335, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 0.9548872180451128, 'upper': 0.9696969696969697, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 0.963187372120211, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}

share held through gaps
Value: 0.03626279229899842; bounds: [None, None]
L39 gap4 first-gap D 0.64353609 to -0.94040079; share 0.82176804 to 0.029799604; all raw retention ratios [0.03626279229899842, 0.038068019303912876, 0.23249058800910286]

commits after in-gap collapse
Value: 0.9164591847342345; bounds: [None, None]
L39 gap4 gap-end D [-0.9404007923089395, -0.9347088176848134, -0.5855576180817079]; deBelly first two collapse and final replacement agree with quarter-step final D 0.9208534696017405; later-gap timing, third-gap bias sign and endpoint fields remain sensitive

Class-membership evidence: Spring

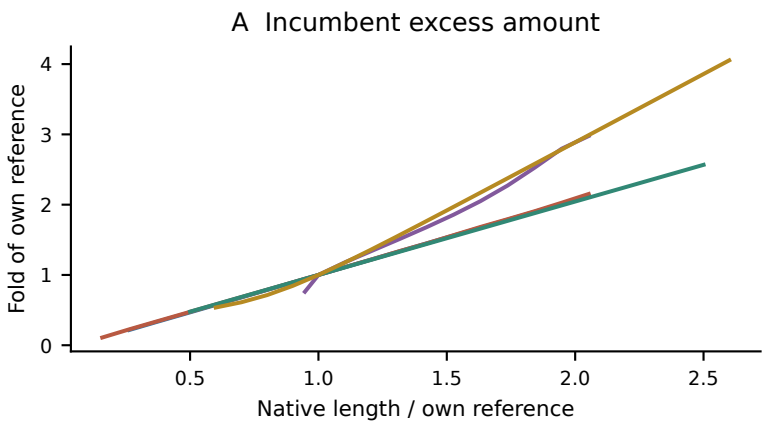
initiation slows with L
Value: 0.9991692546590409; bounds: [0.8333333333333334, 1.2]
L39/L19, central area, fixed 10% L19 reporter

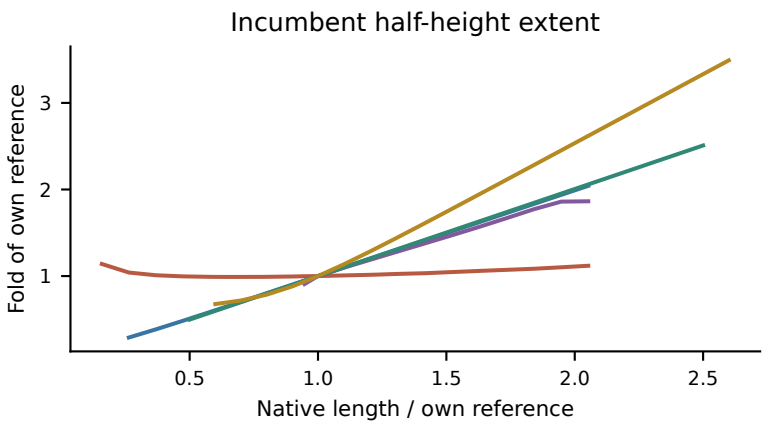
narrow-cue penalty and growth
Value: 1.1610062875918385; bounds: [1.1511627906976745, 1.1764705882352942]
L19 {'status': 'bounded', 'lower': 1.2, 'upper': 1.3571428571428572, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.2728575892292848, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 1.1511627906976745, 'upper': 1.1764705882352942, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.1610062875918385, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; modest non-growing penalty is distinct from the large saturating-model penalty

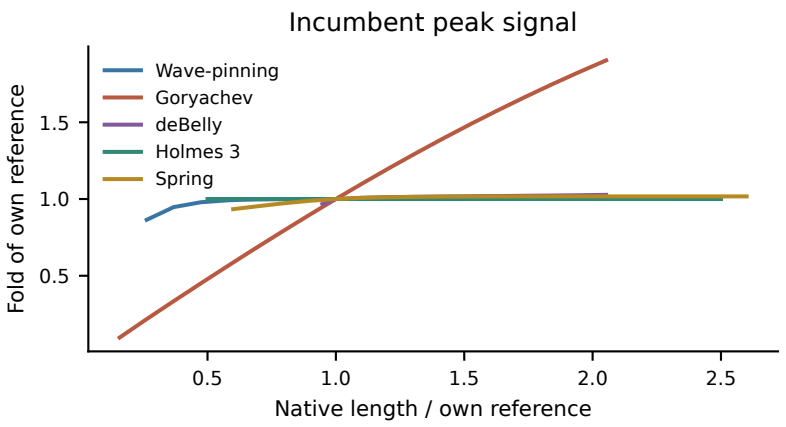
share held through gaps
Value: 0.003315145224882711; bounds: [None, None]
L39 gap4 first-gap D -0.26175354 to -0.99755261; share 0.36912323 to 0.0012236971; all raw retention ratios [0.003315145224882711, 0.021681739713766957, 1.0364261376049886]

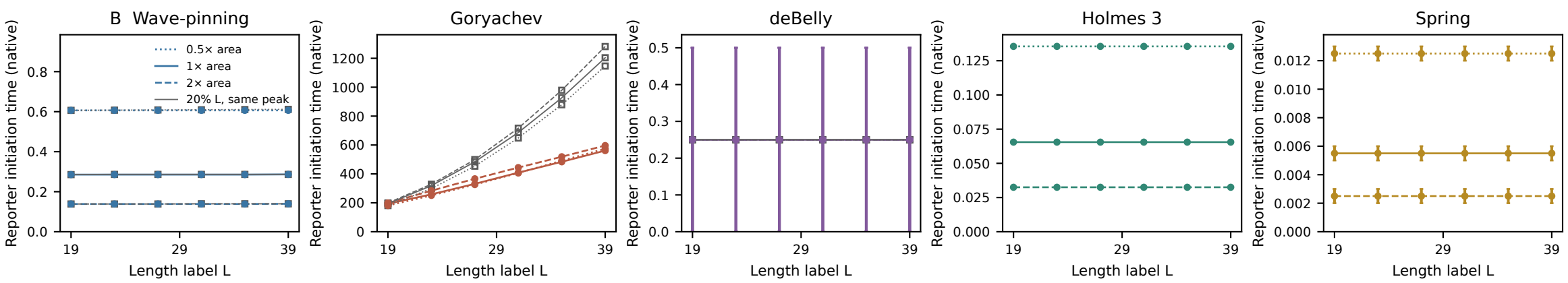
commits after in-gap collapse
Value: 0.9996954239534162; bounds: [None, None]
L39 gap4 gap-end D [-0.9975526057568276, -0.976390262546511, 0.9966415640033679]

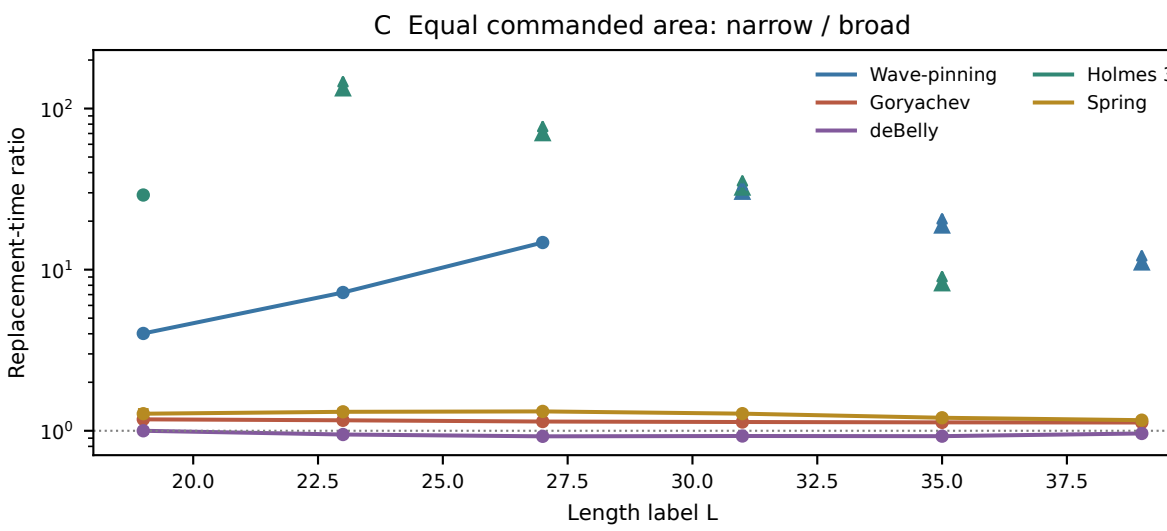
Figure 2. Shape and timing distinguish replacement responses
AUDITED FINAL

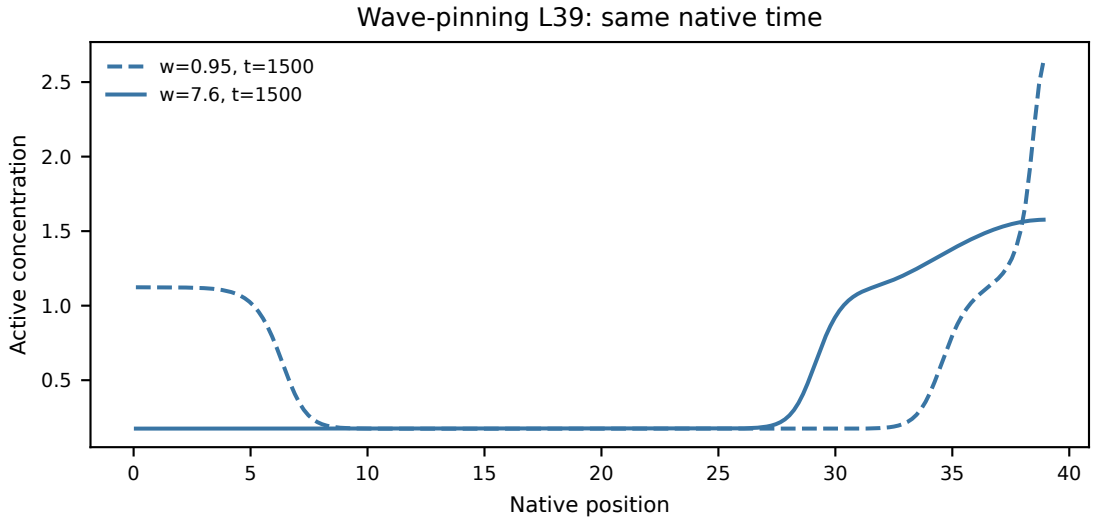


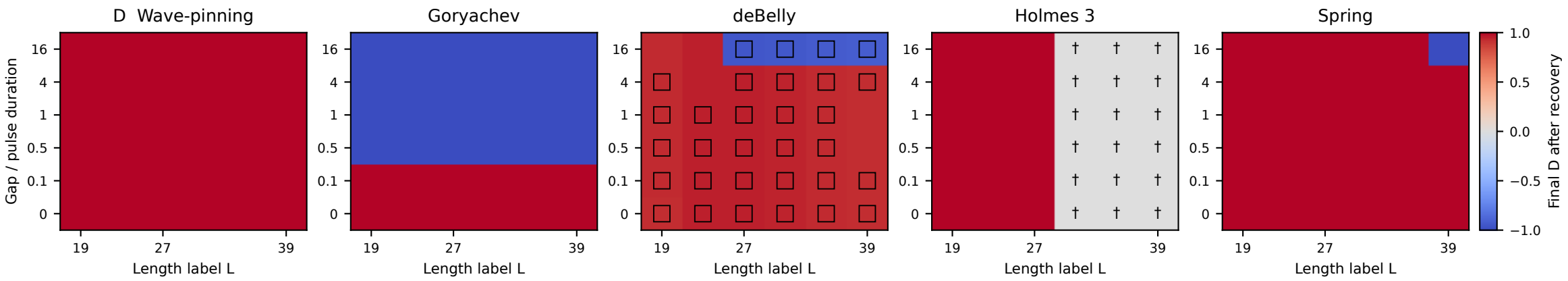


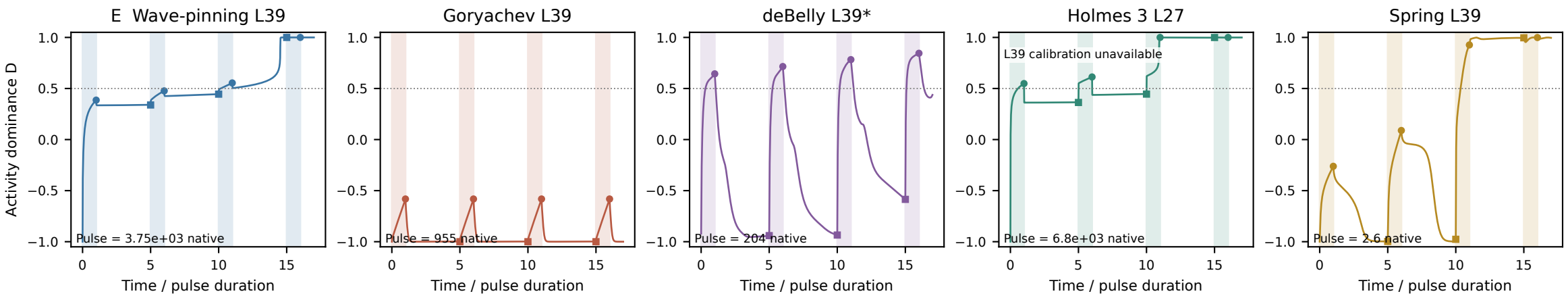












F Signatures tied to measurements; causal interpretation requires native-field support

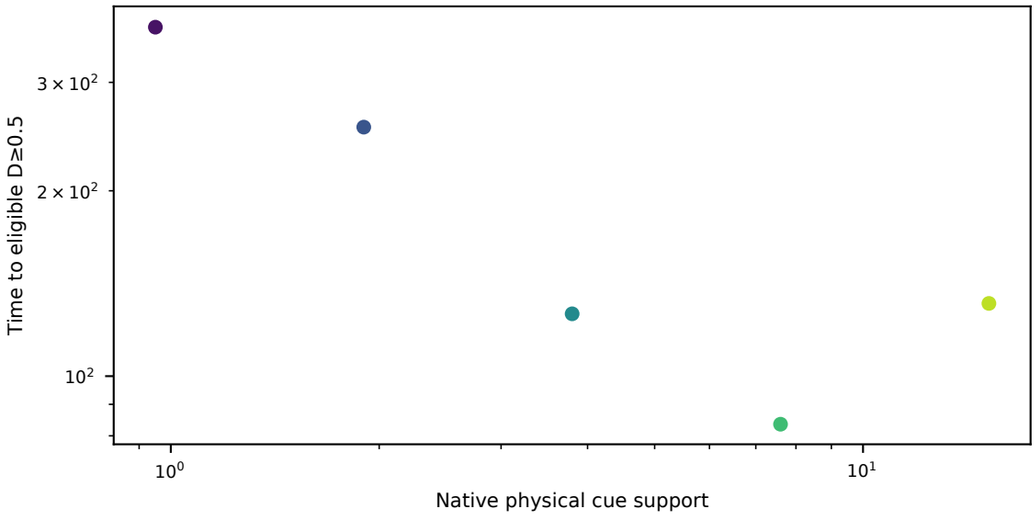
Model	Trace L	Initiation L39/L19	Narrow / broad, L19 → L39	Pulse 1 → gap 1: D	Final D
Wave-pinning	L39	1–1.01	3.99–4.05 → >11.1	0.39 → 0.34	1.00
Goryachev	L39	2.87–2.92	1.17–1.18 → 1.12–1.13	-0.58 → -1.00	-1.00
deBelly*	L39	both in (0, 0.5]	0.962–1.04 → 0.955–0.97	0.64 → -0.94	0.92
Holmes 3	L27	0.985–1.02	29–29.1 → uninformative	0.55 → 0.36	1.00
Spring	L39	0.833–1.2	1.2–1.36 → 1.15–1.18	-0.26 → -1.00	1.00

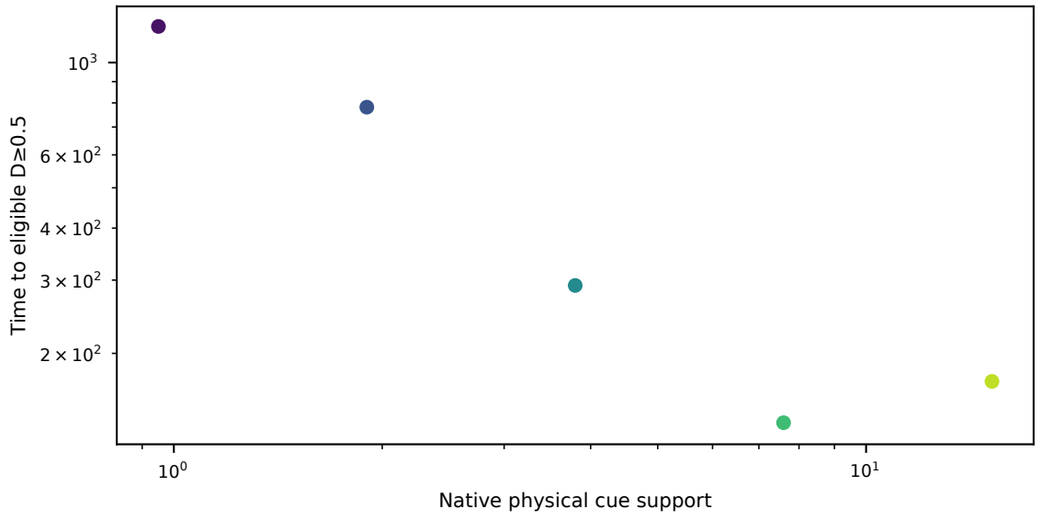
Shape: beam width at fixed input area • Timing: equal total ON pulse trains • Reporter: activity share after each gap

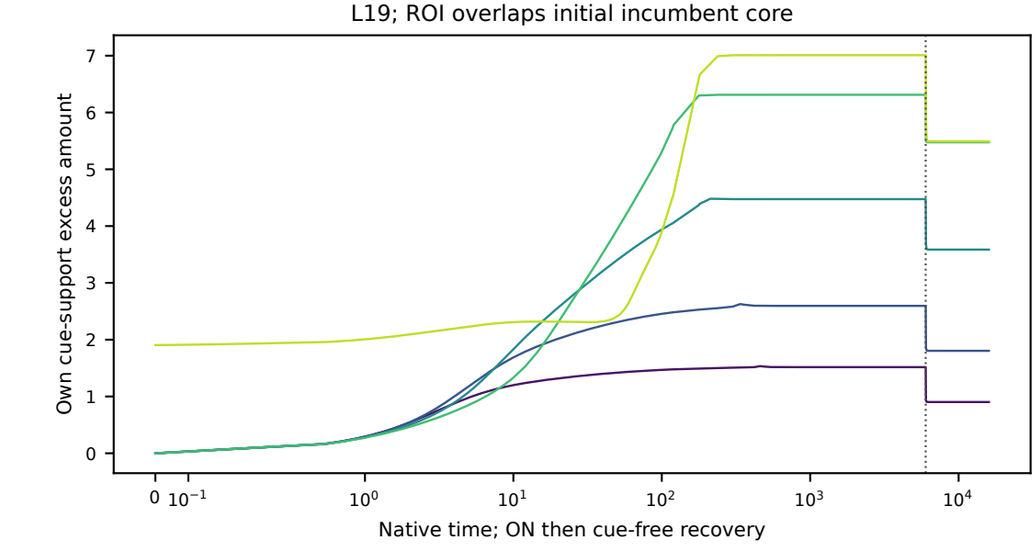
↑ triangles: finite-cap ratio lower bounds; bounds cannot be ranked. †: no calibration; ×: low activity; □: failed field tail.

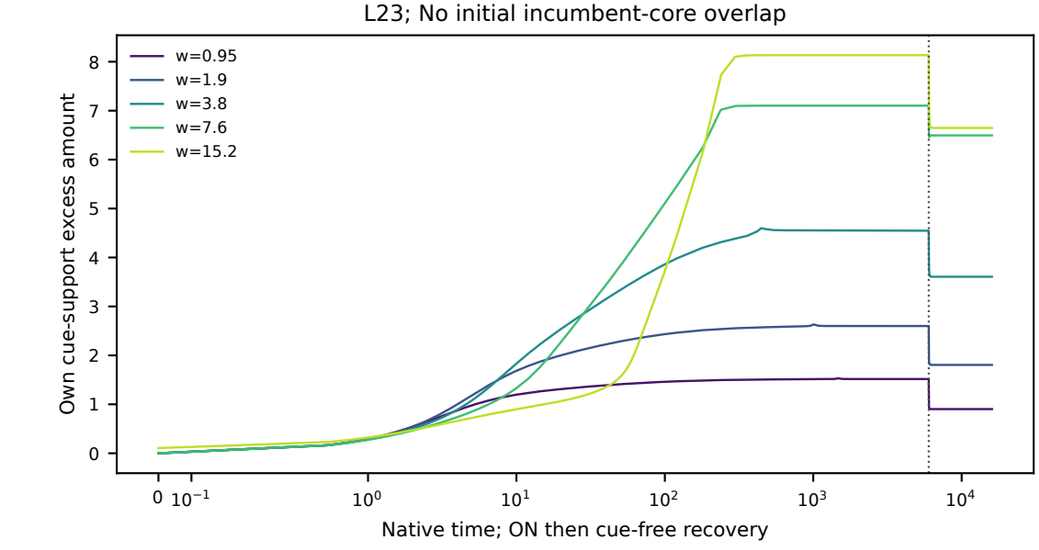
* deBelly later-gap timing and endpoint fields have numerical qualifications (S5). Native clocks differ; optical dose is uncalibrated.

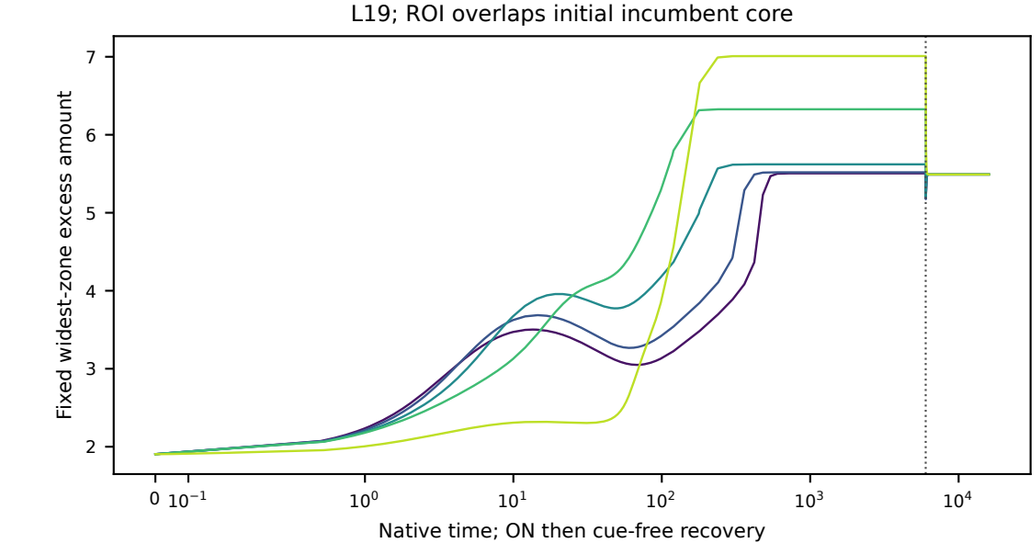
S1 Wave-pinning • commanded area 0.5× • native kinetics

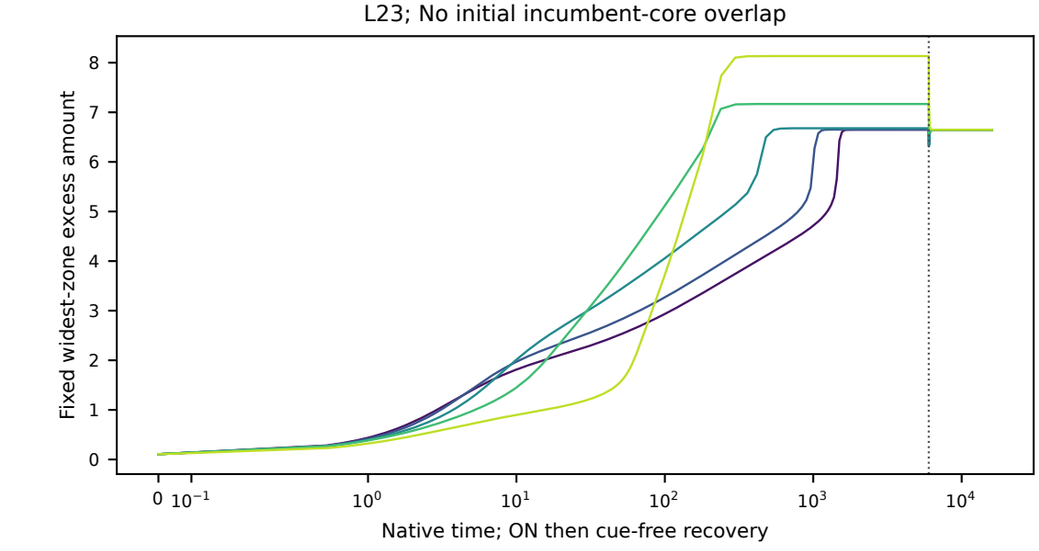






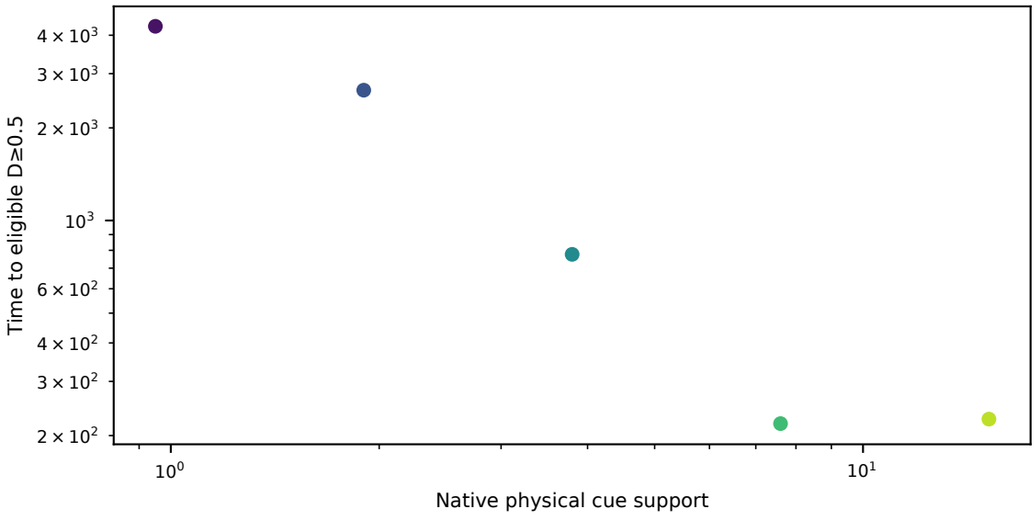


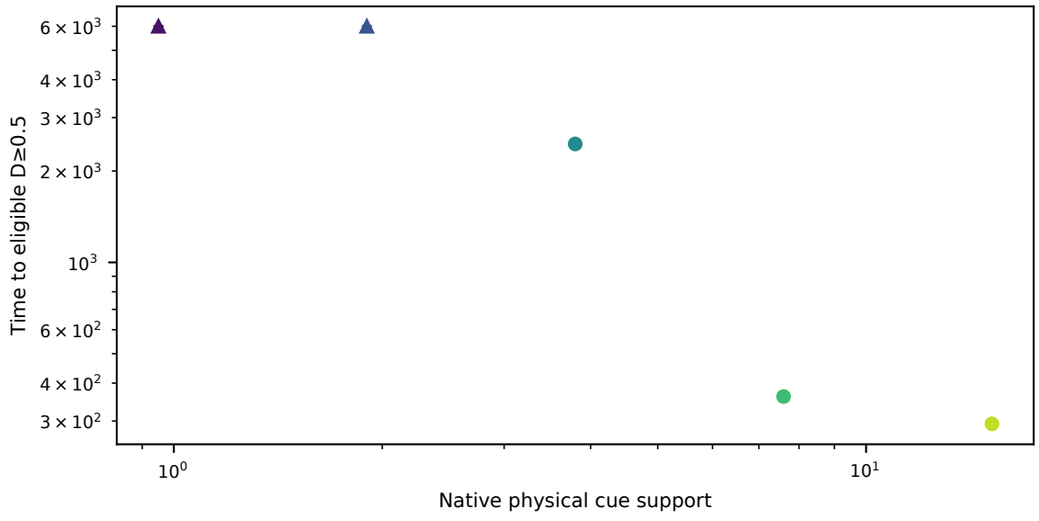


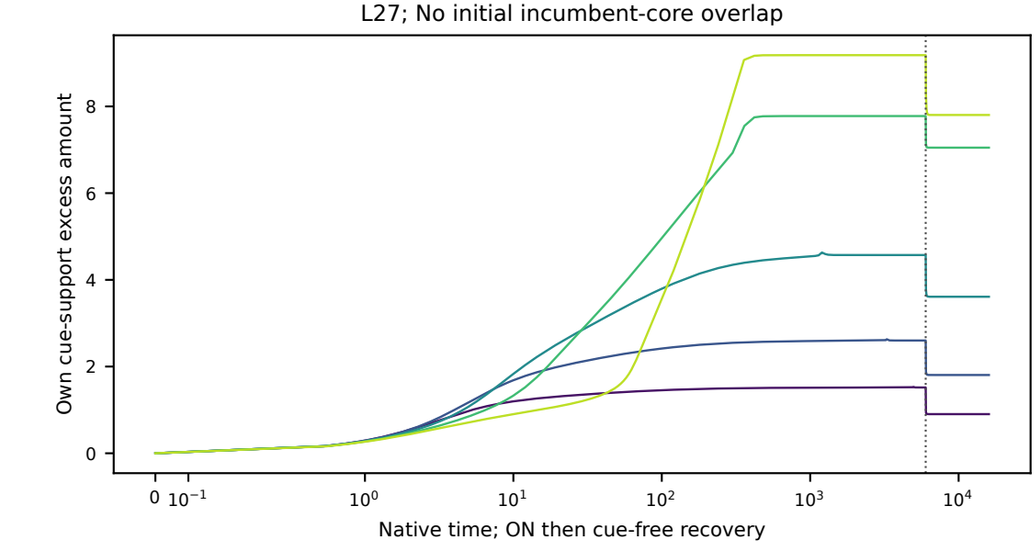


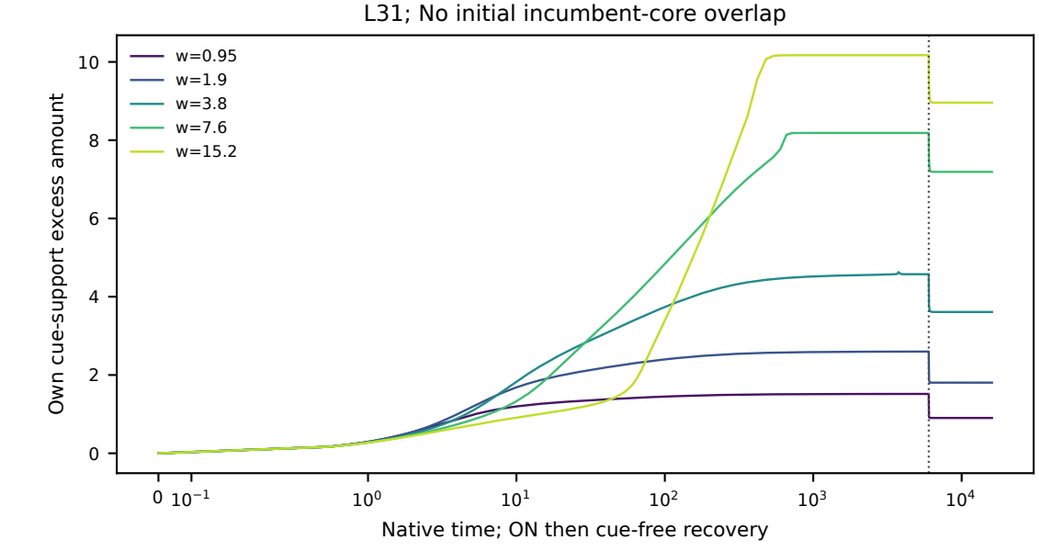
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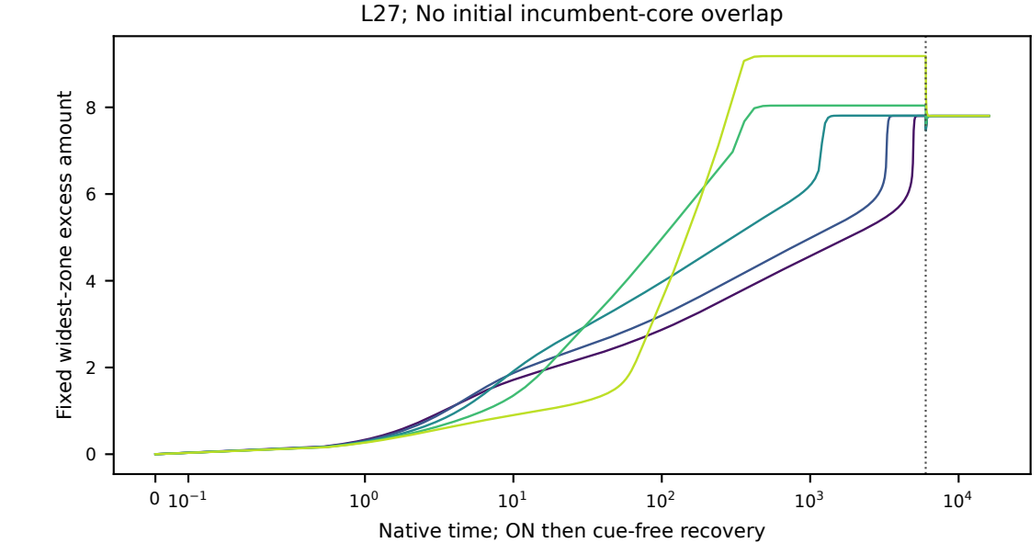
S1 Wave-pinning • commanded area 0.5× • native kinetics

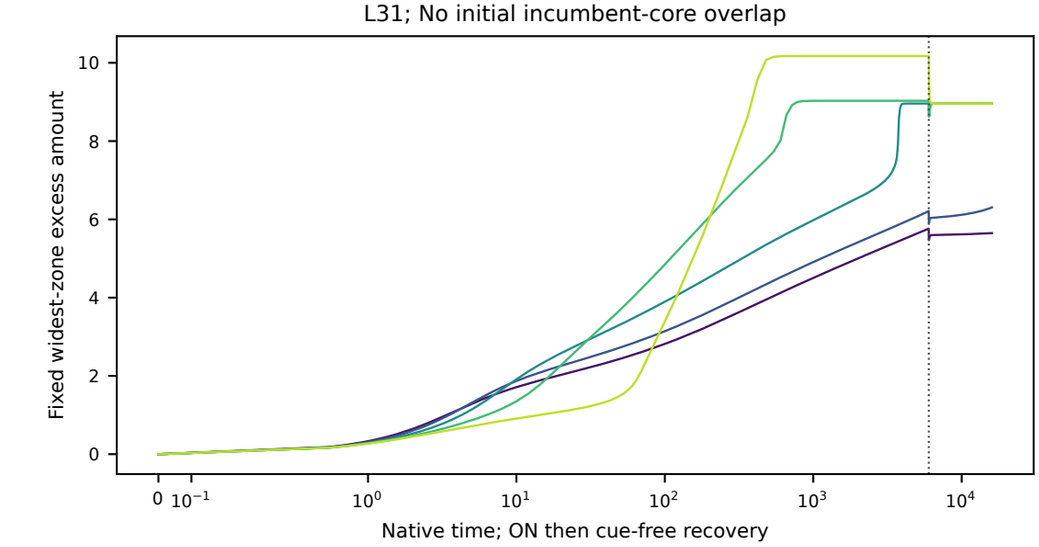






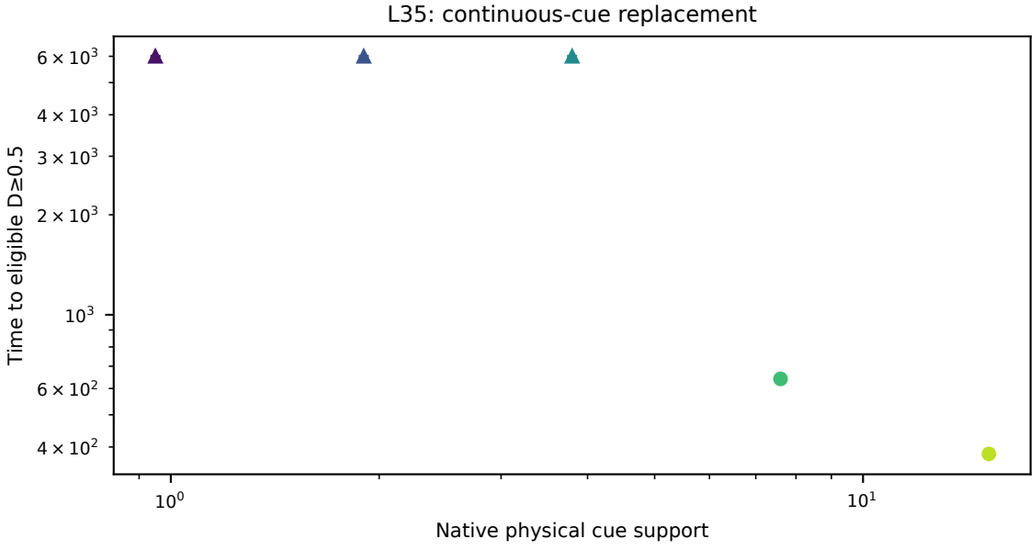


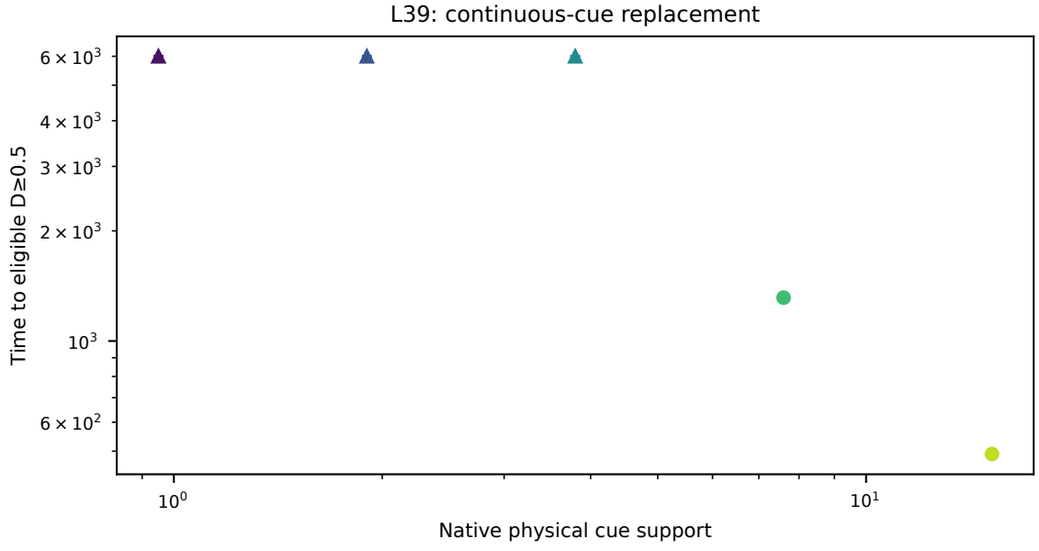


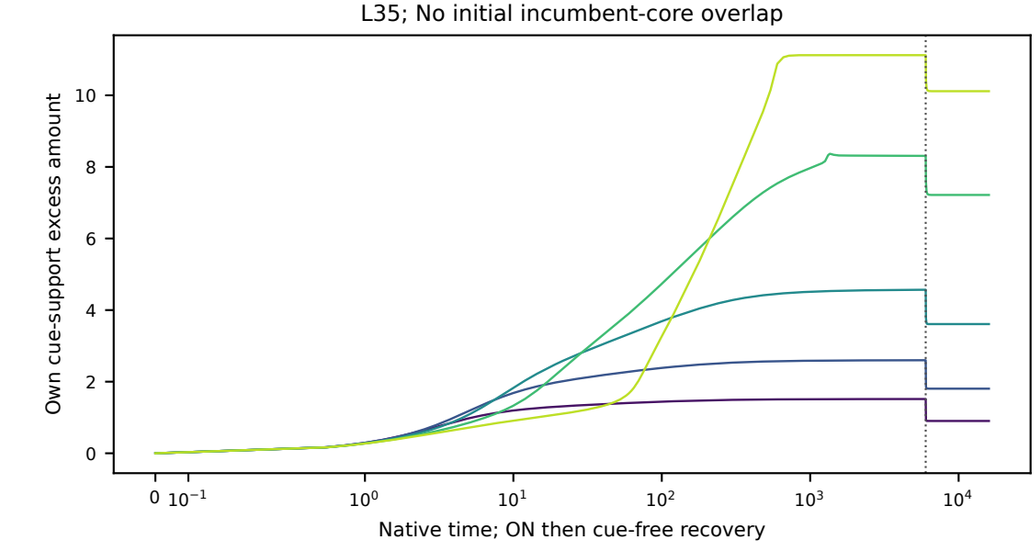


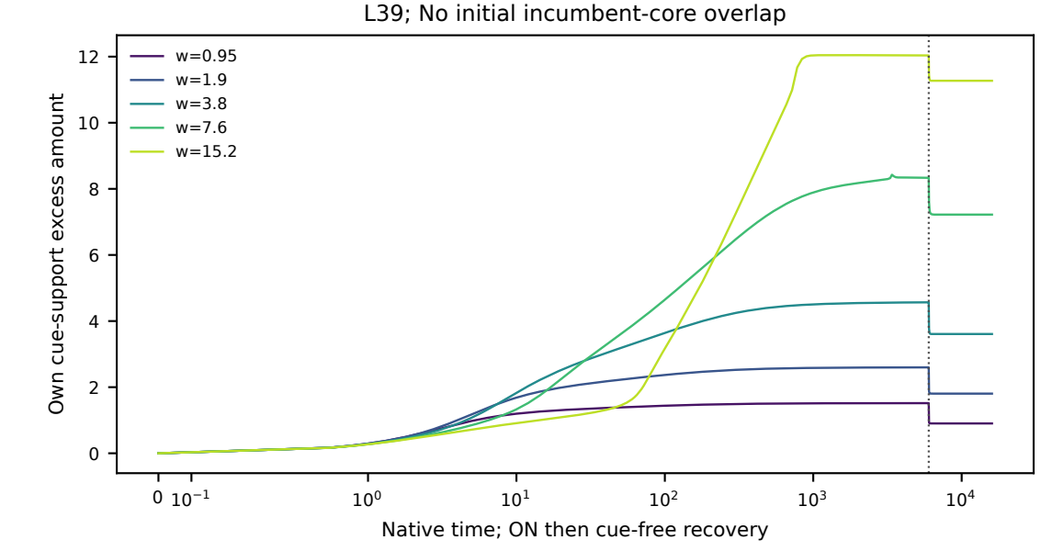
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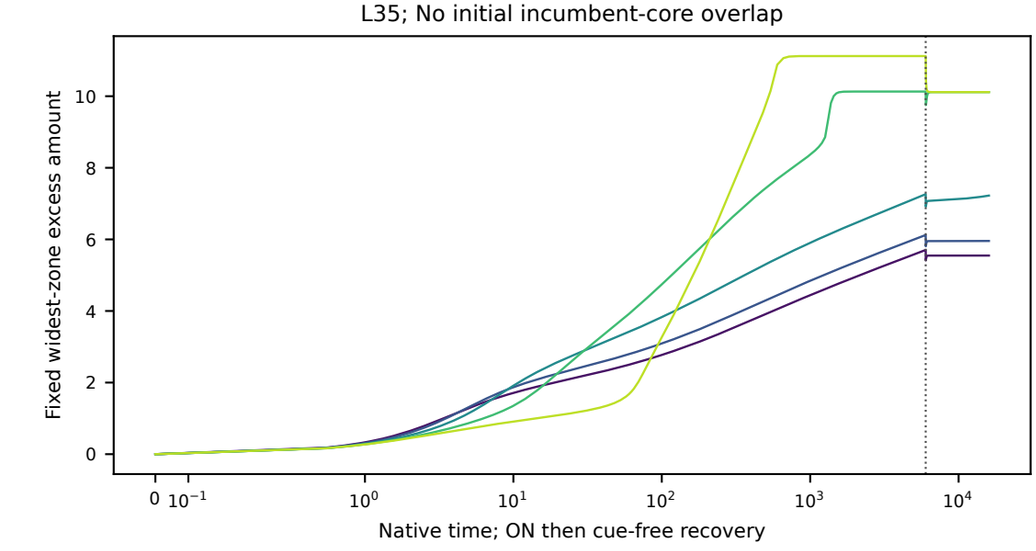
S1 Wave-pinning • commanded area 0.5× • native kinetics

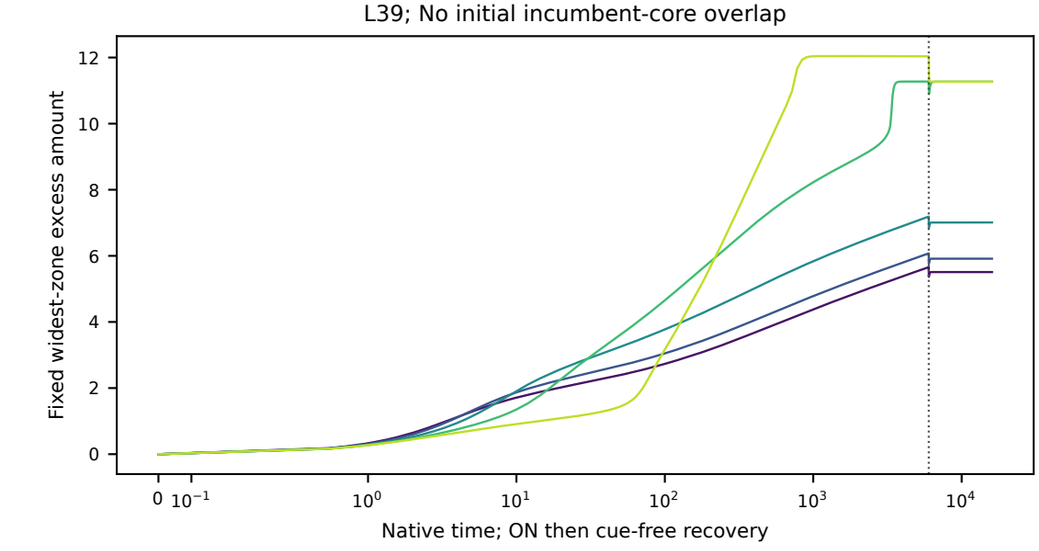






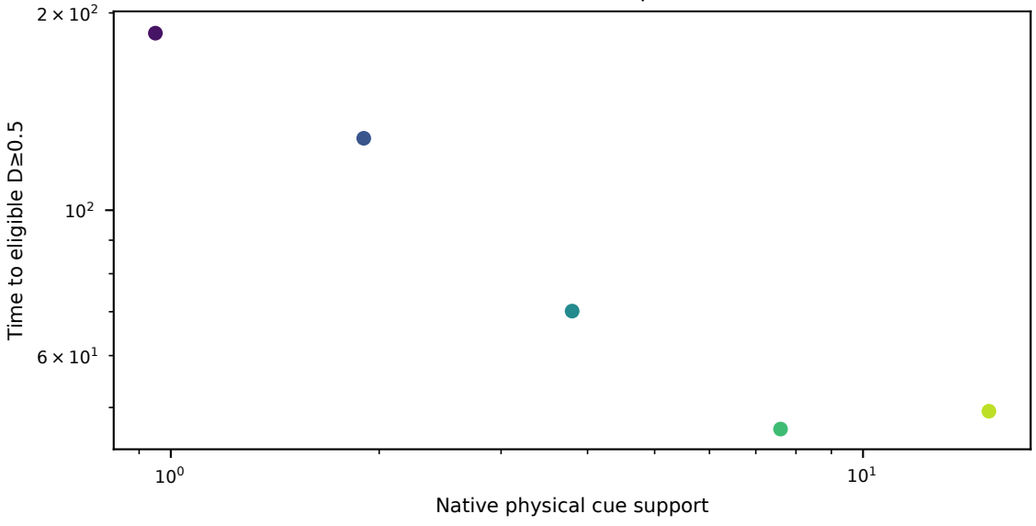


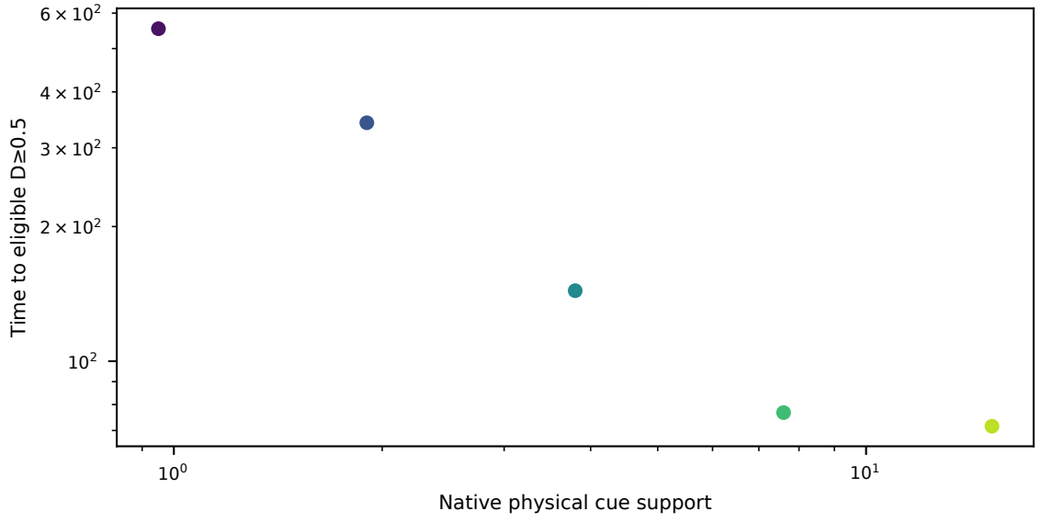


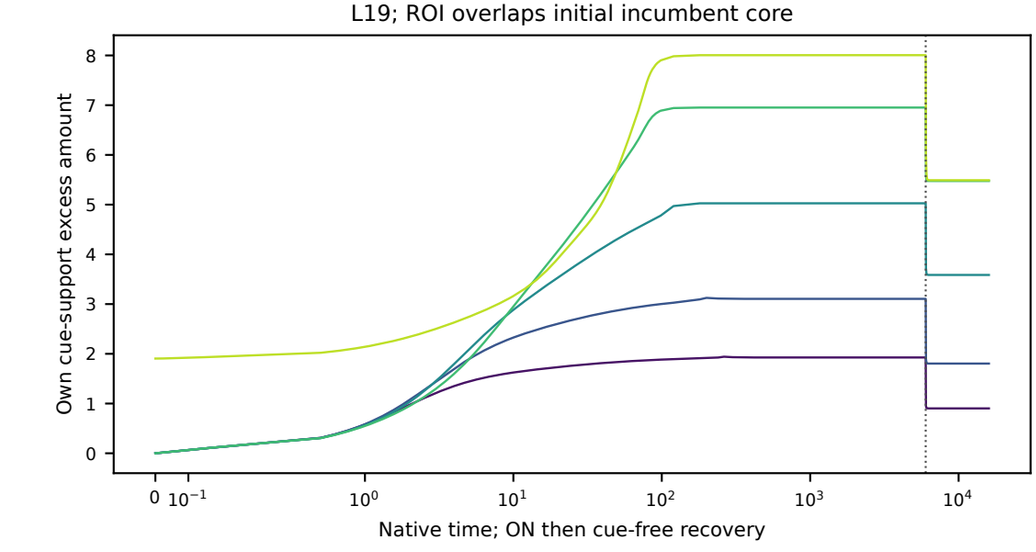


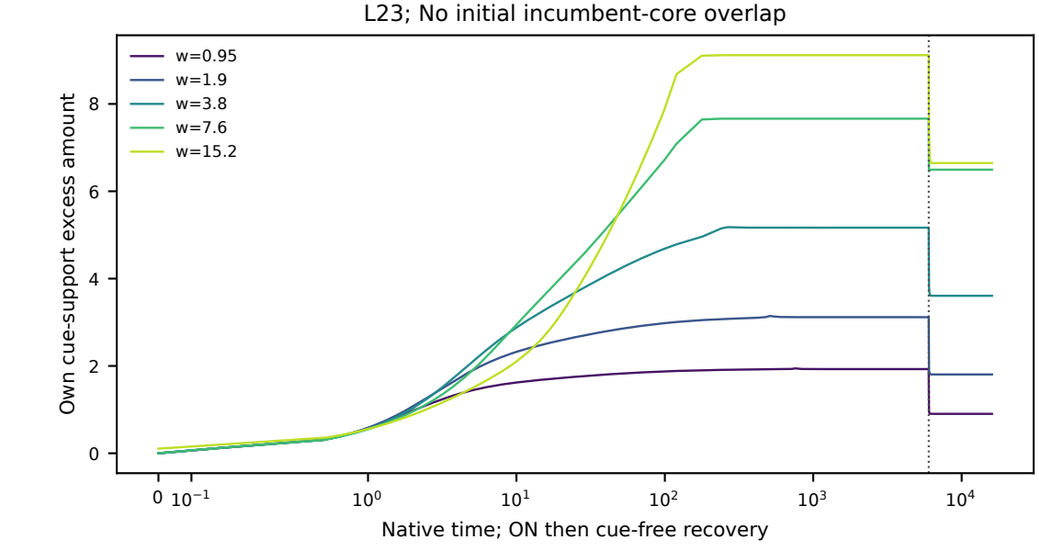
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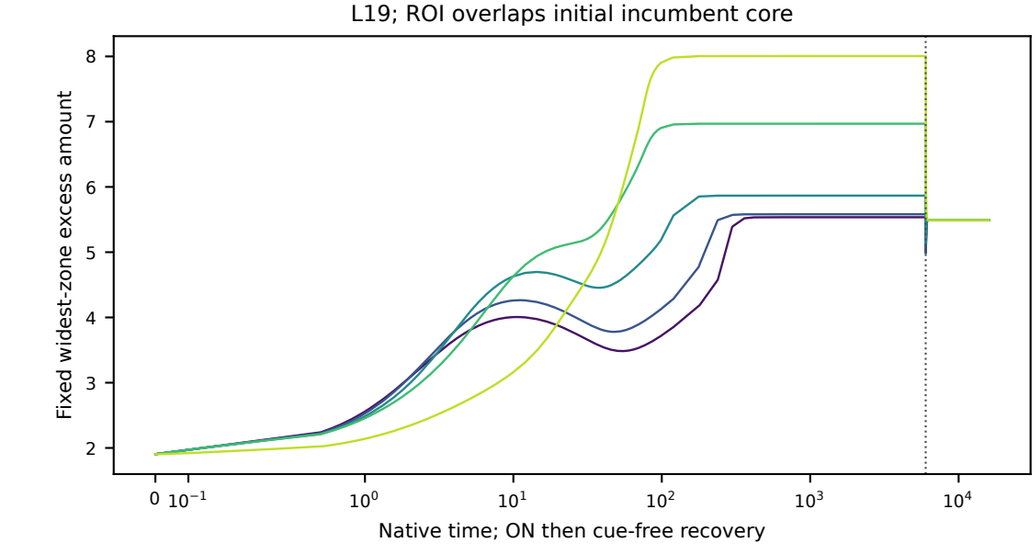
S1 Wave-pinning • commanded area 1× • native kinetics

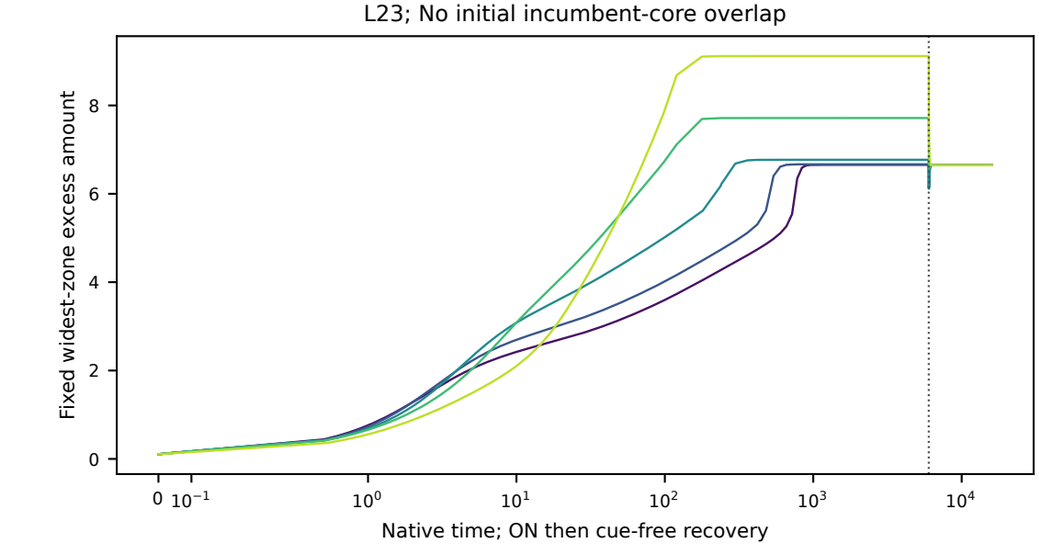






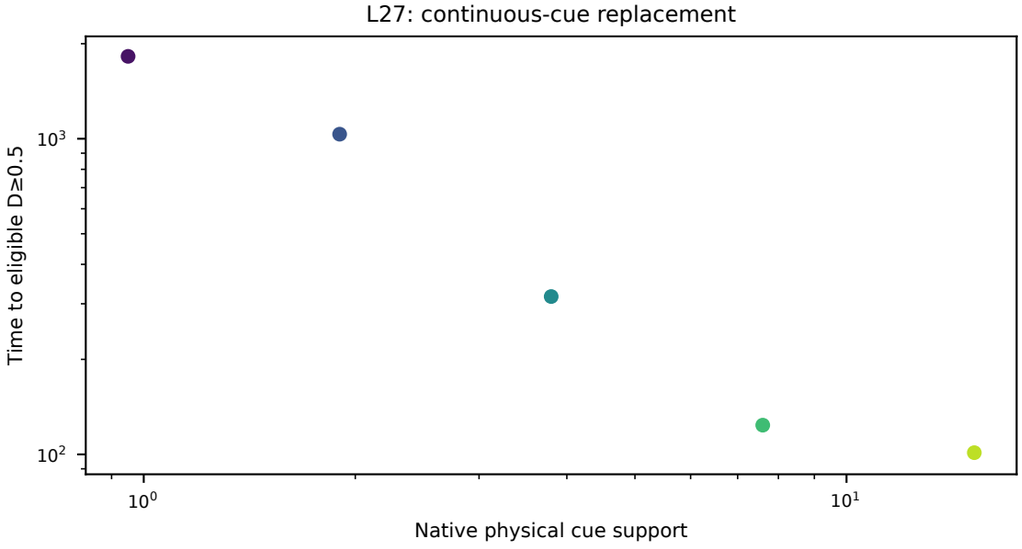


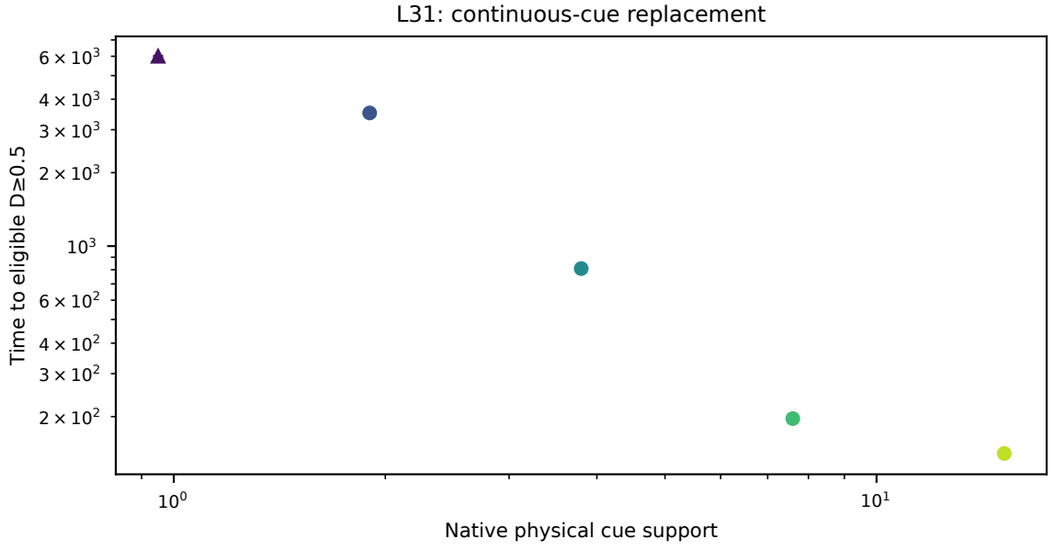


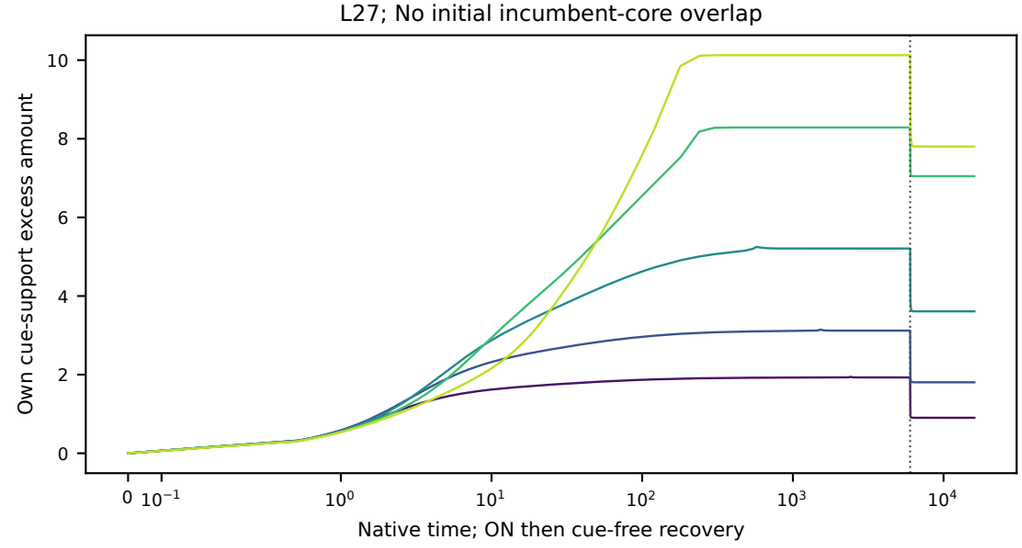


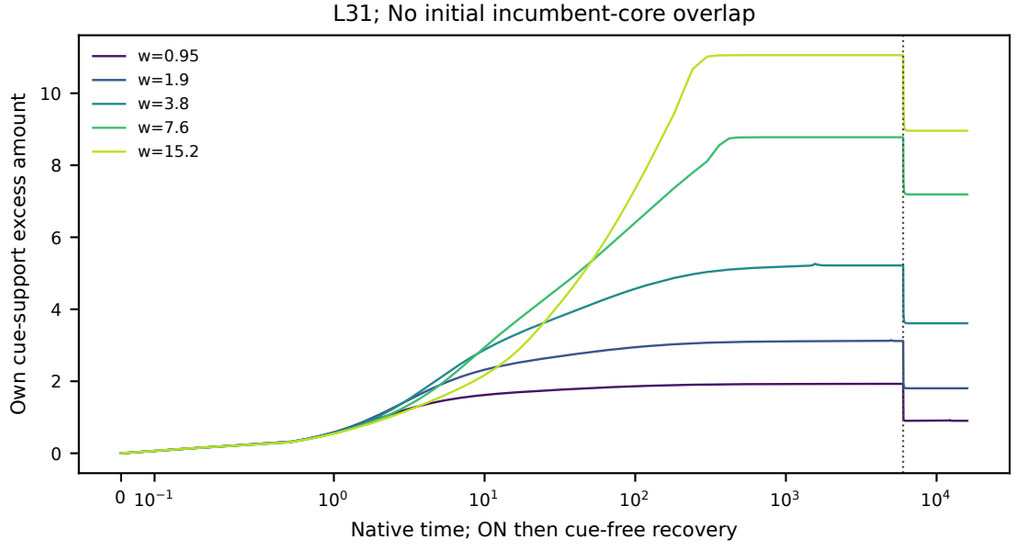
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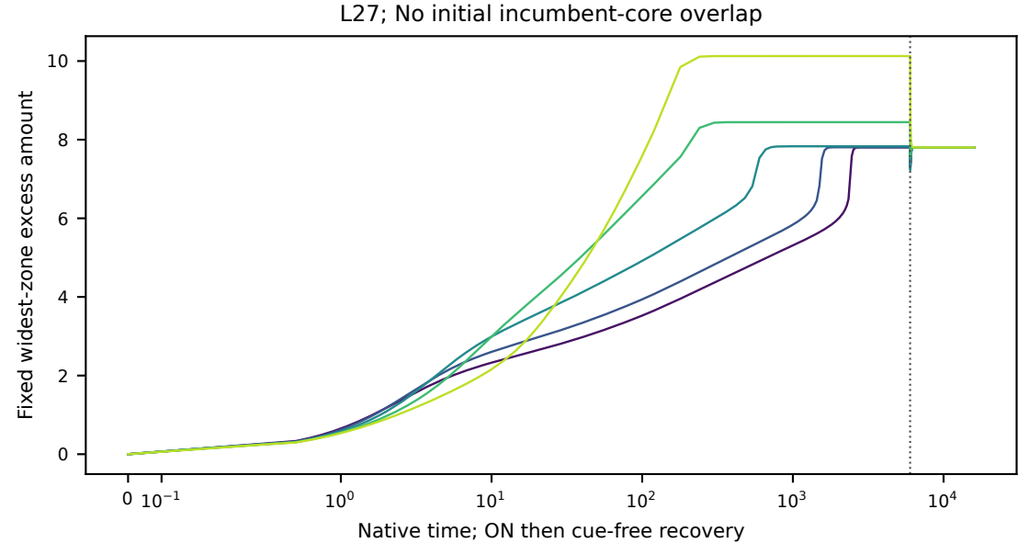
S1 Wave-pinning • commanded area 1× • native kinetics

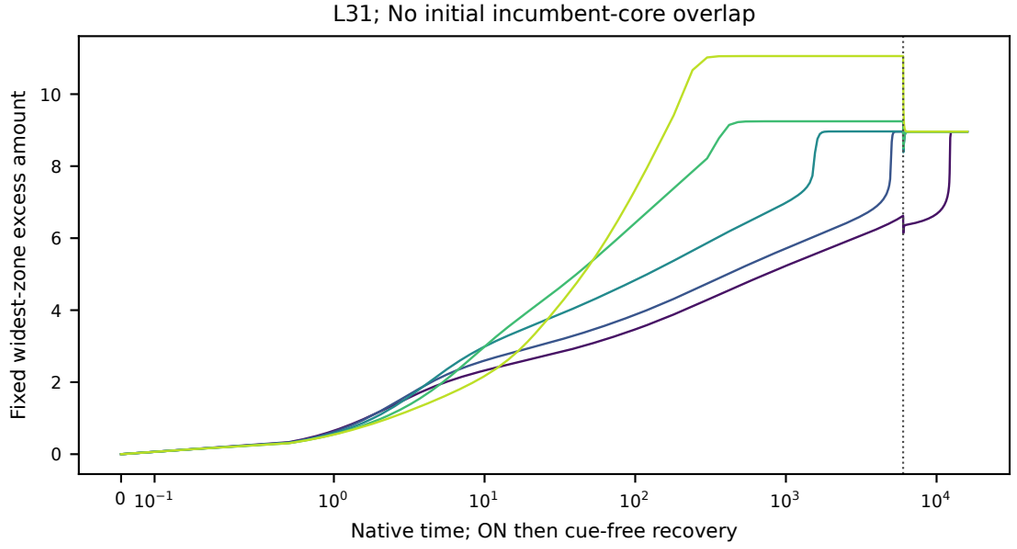






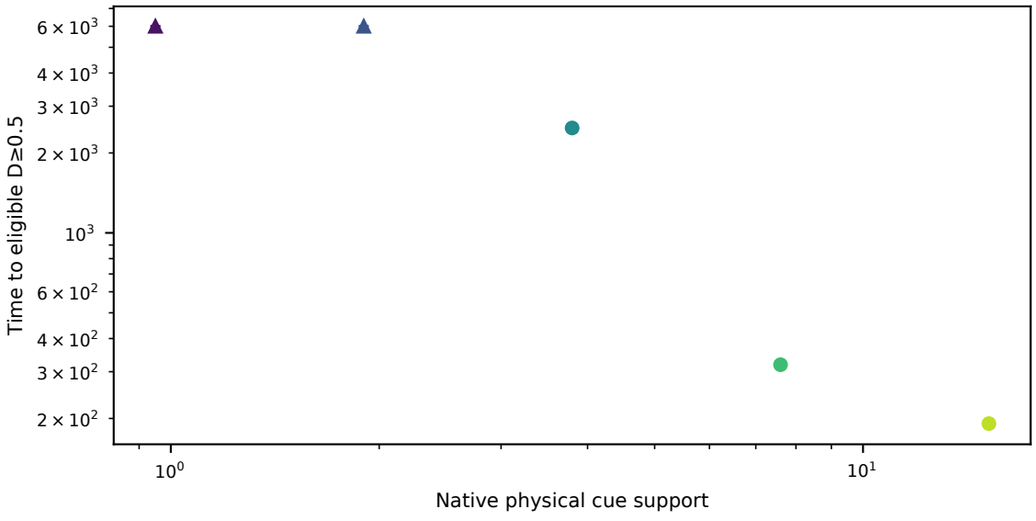


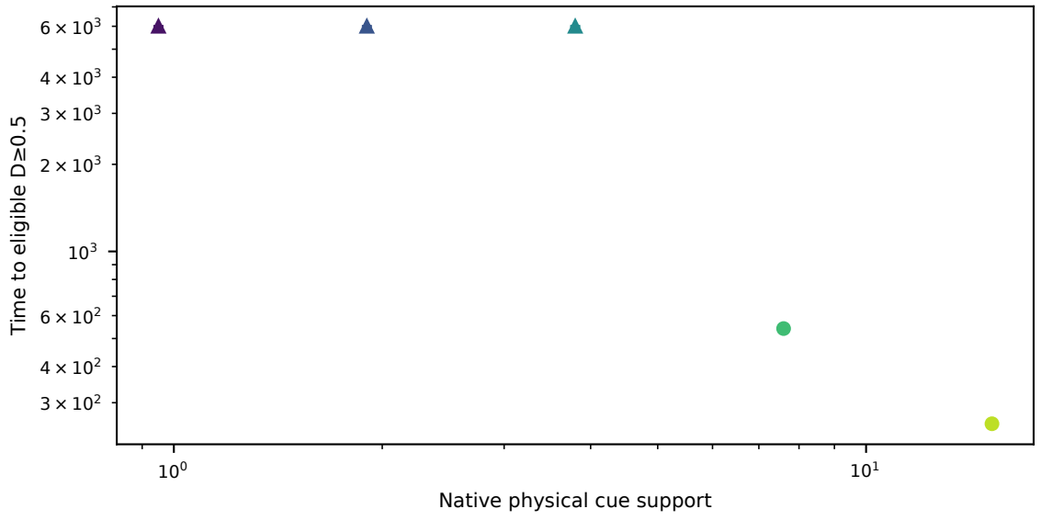


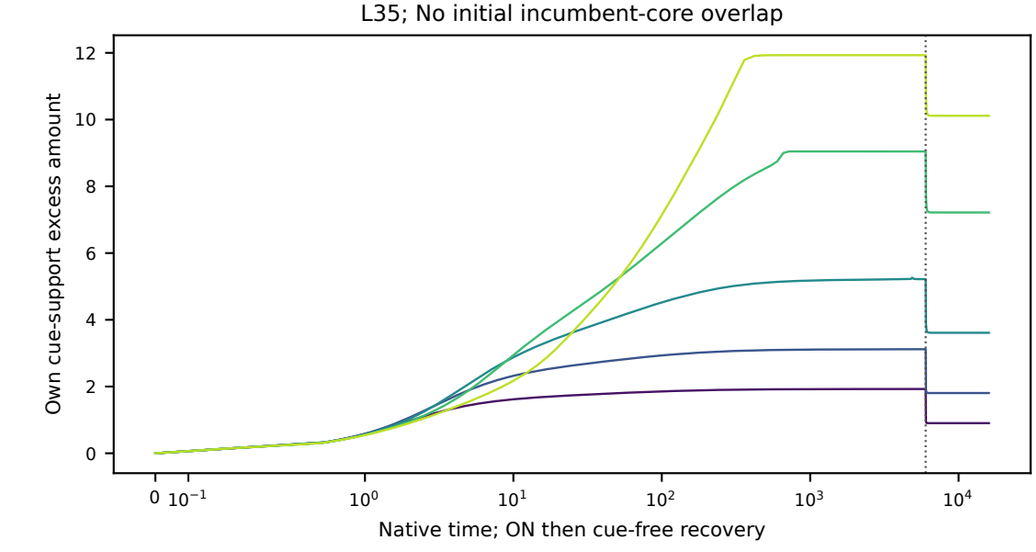


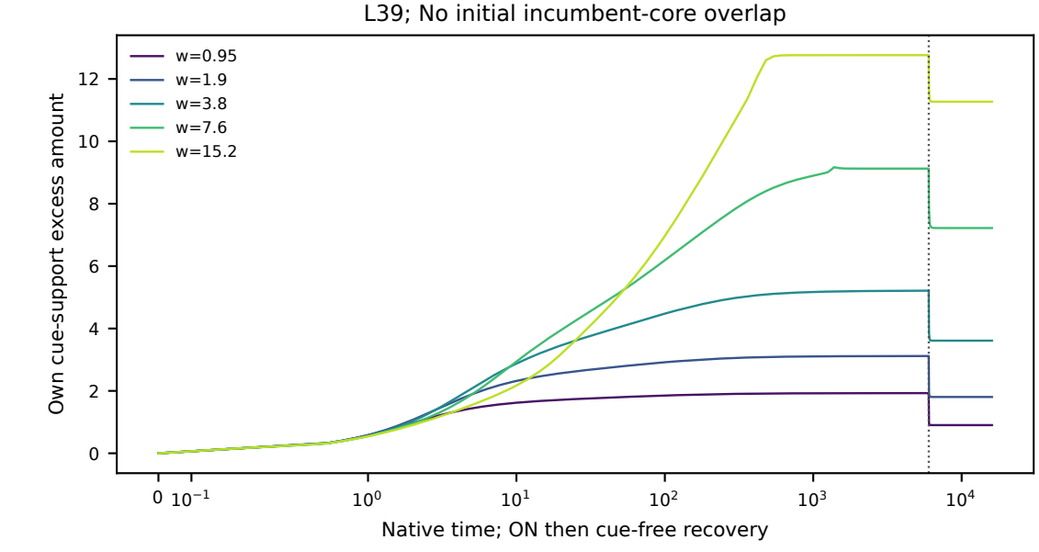
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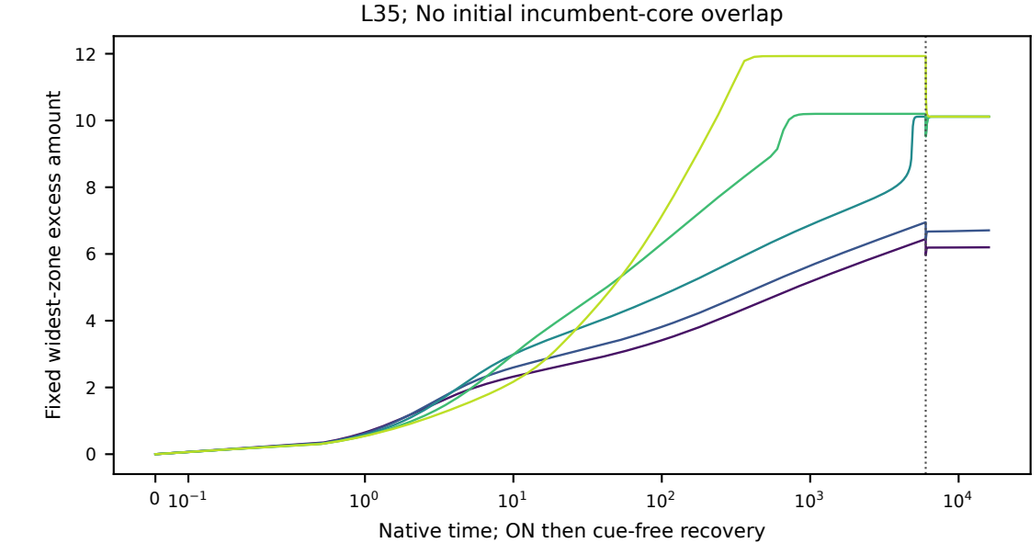
S1 Wave-pinning • commanded area 1× • native kinetics

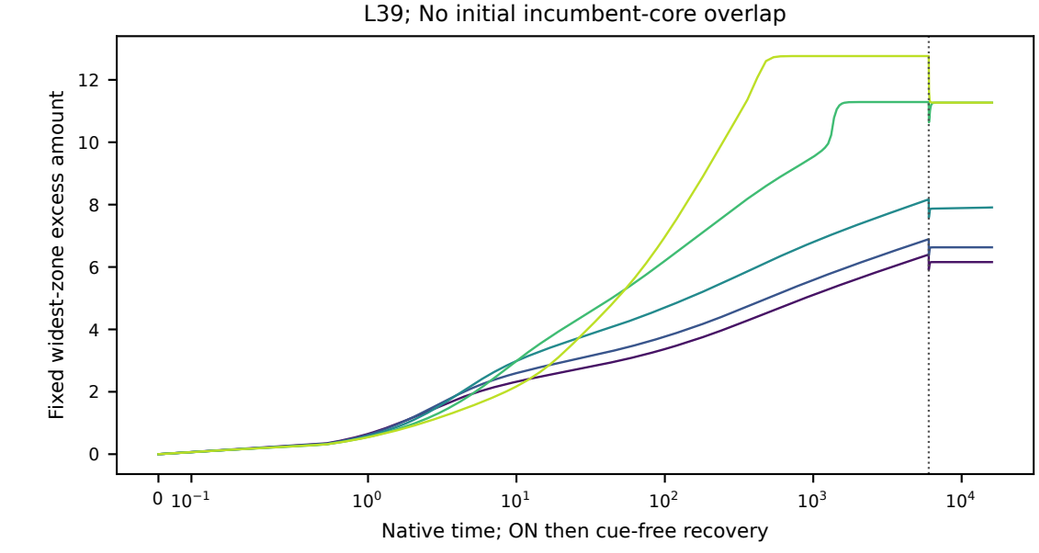






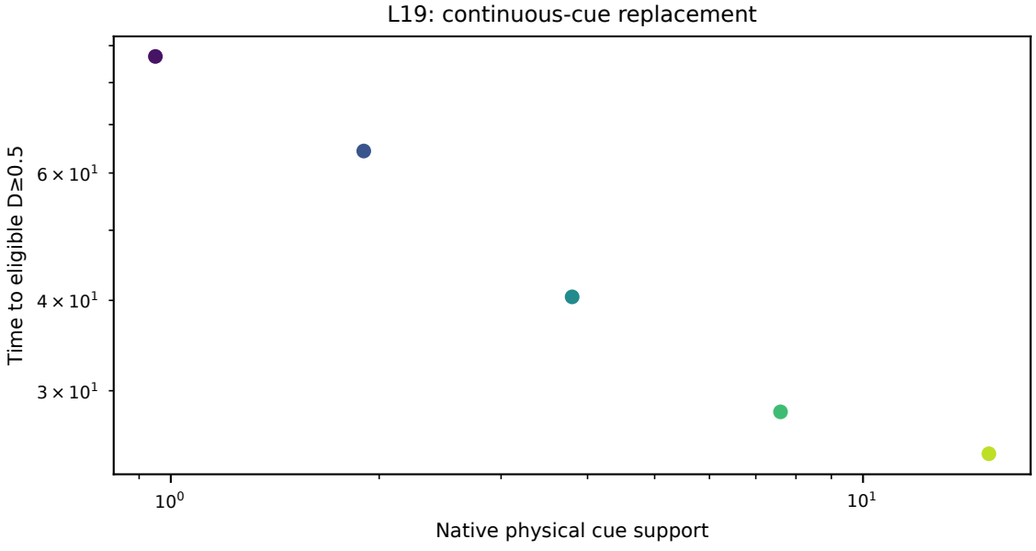


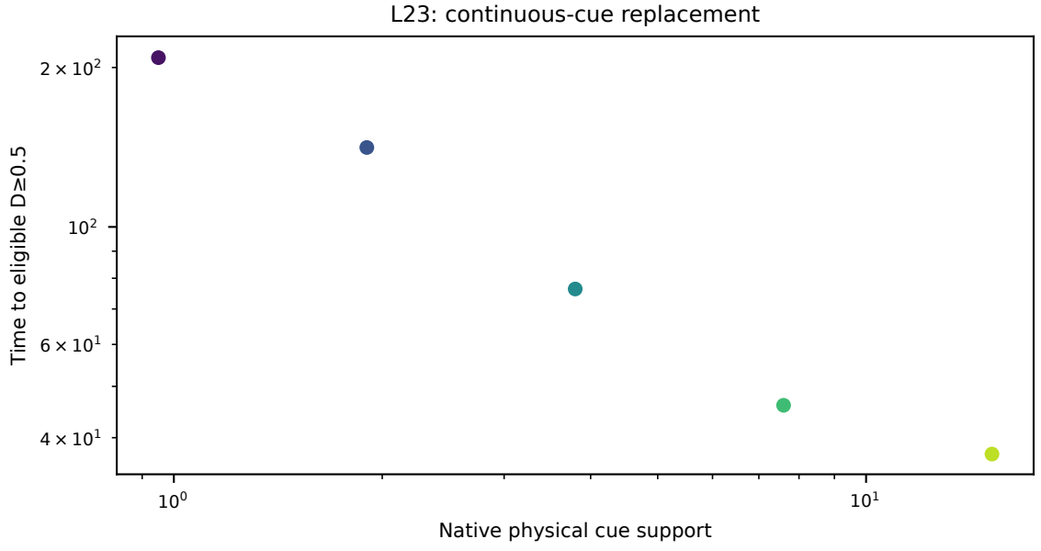


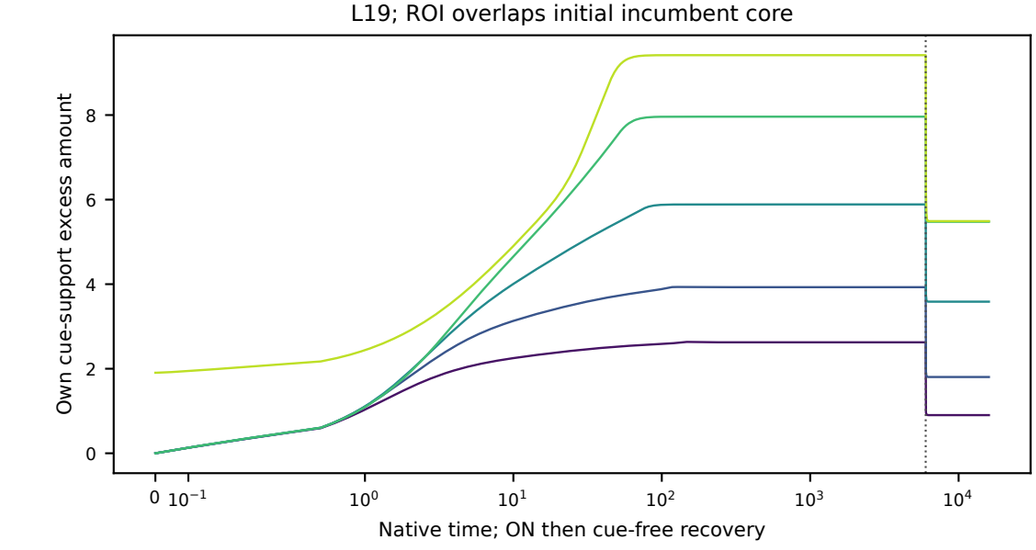


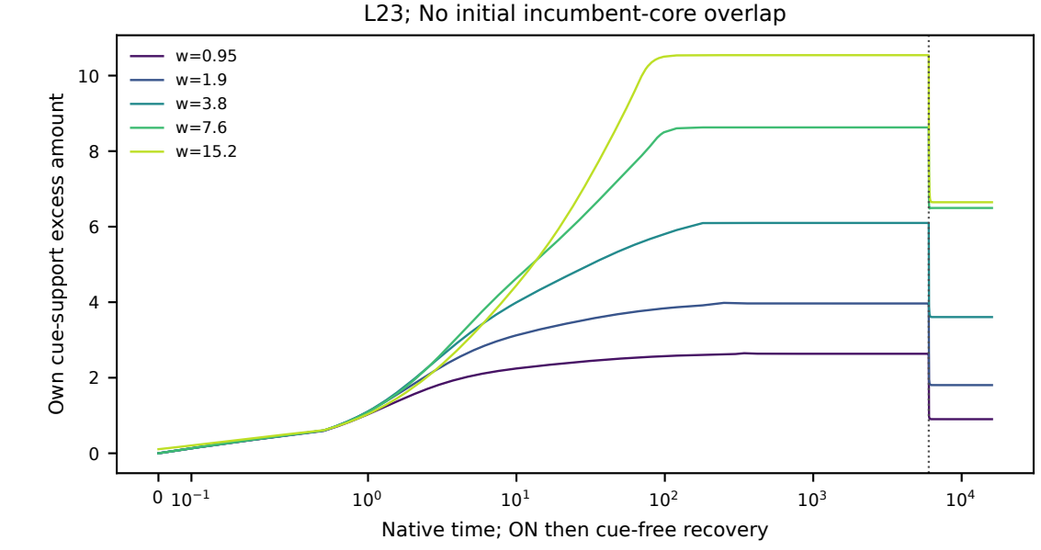
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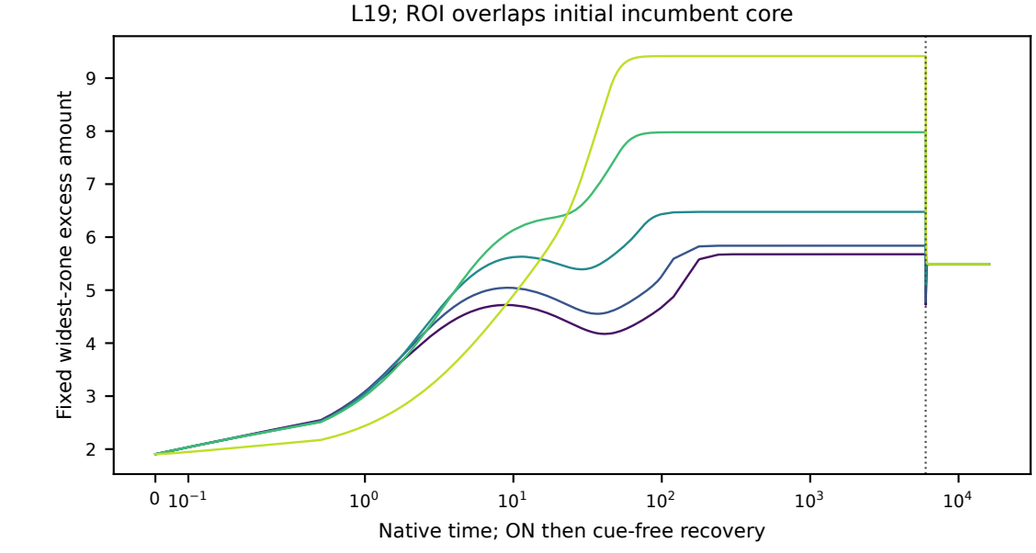
S1 Wave-pinning • commanded area 2× • native kinetics

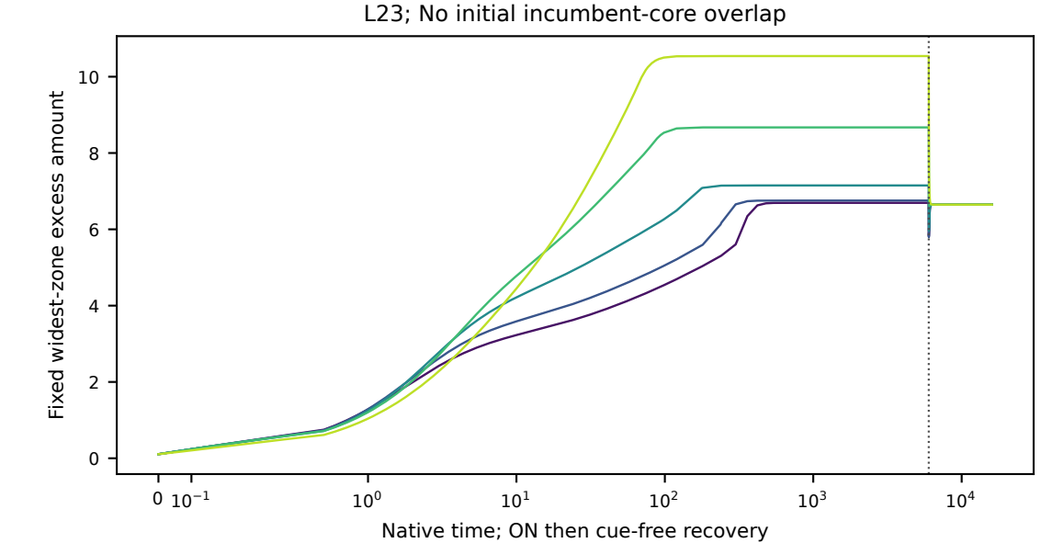






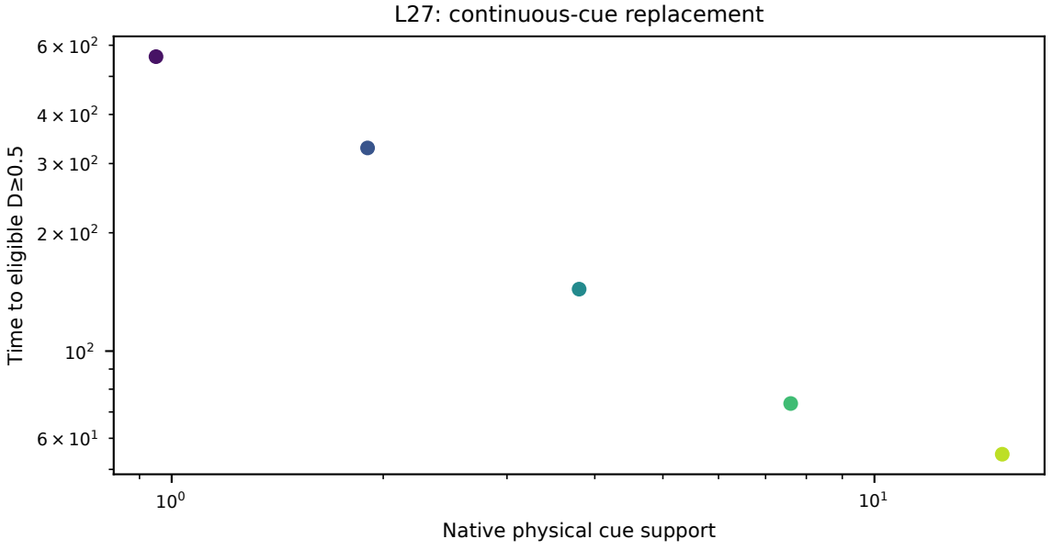


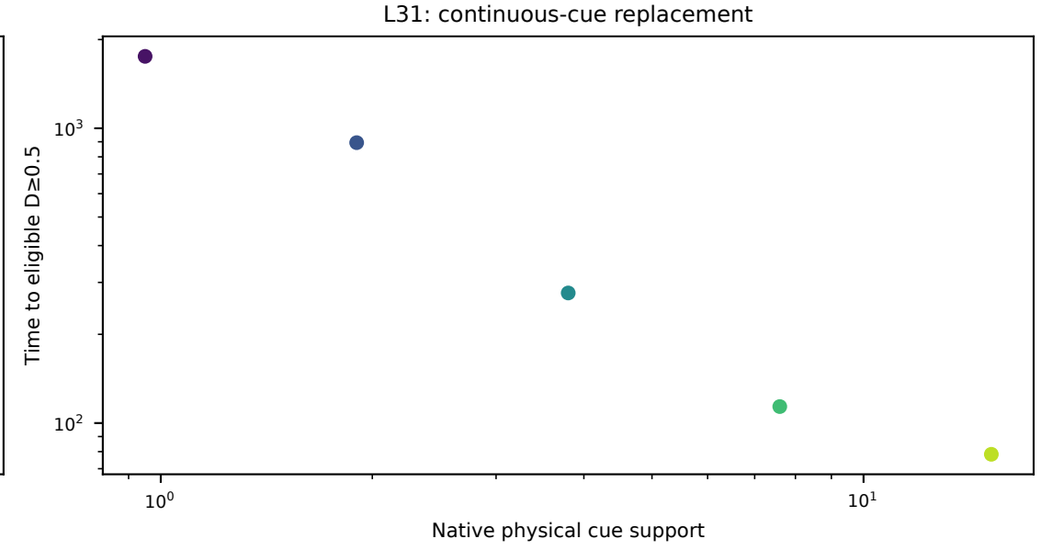


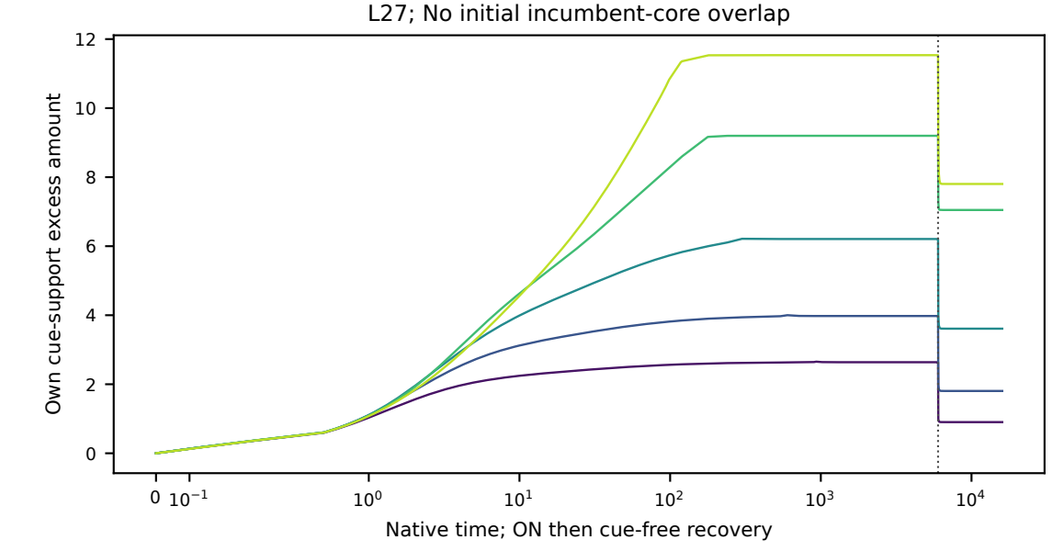


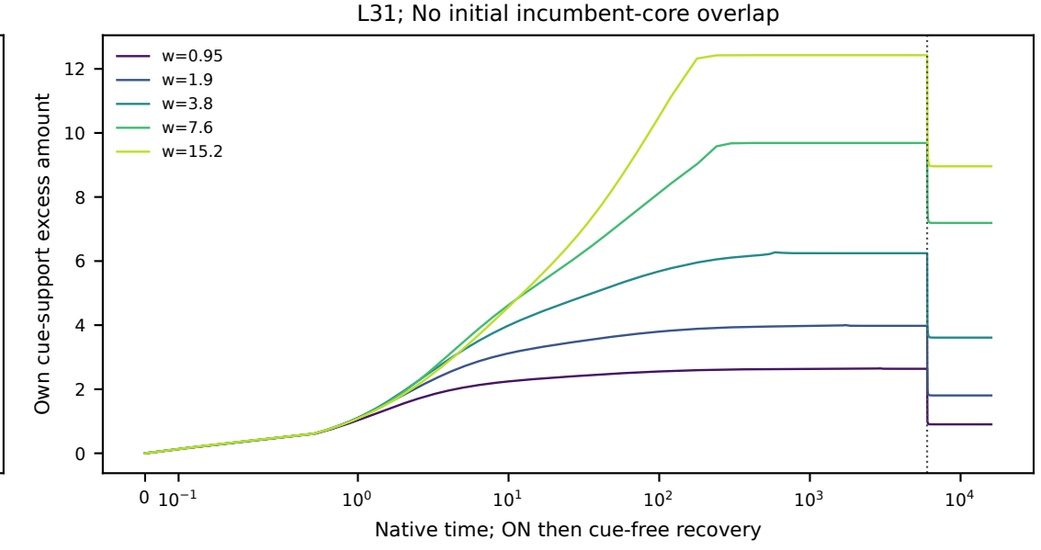
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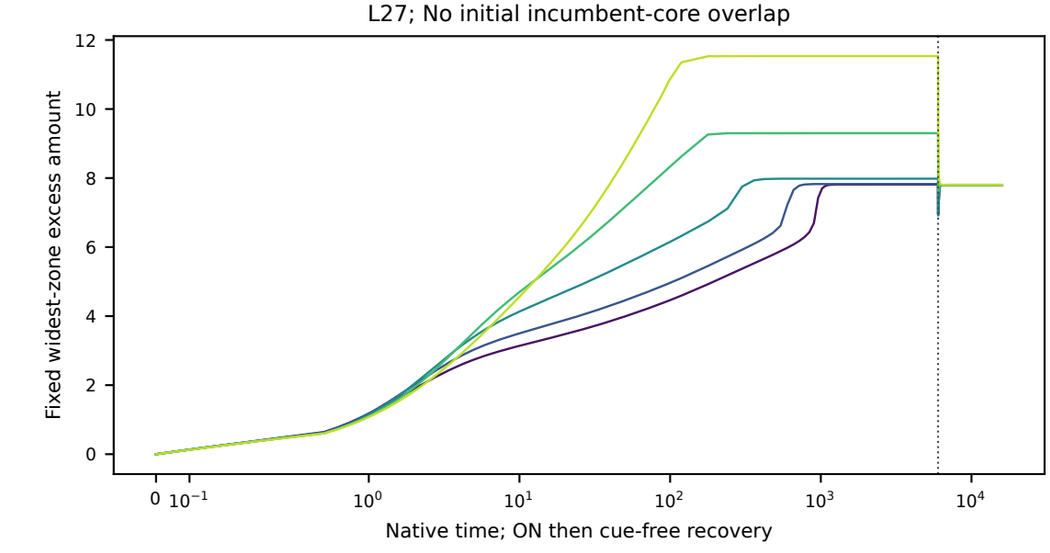
S1 Wave-pinning • commanded area 2× • native kinetics

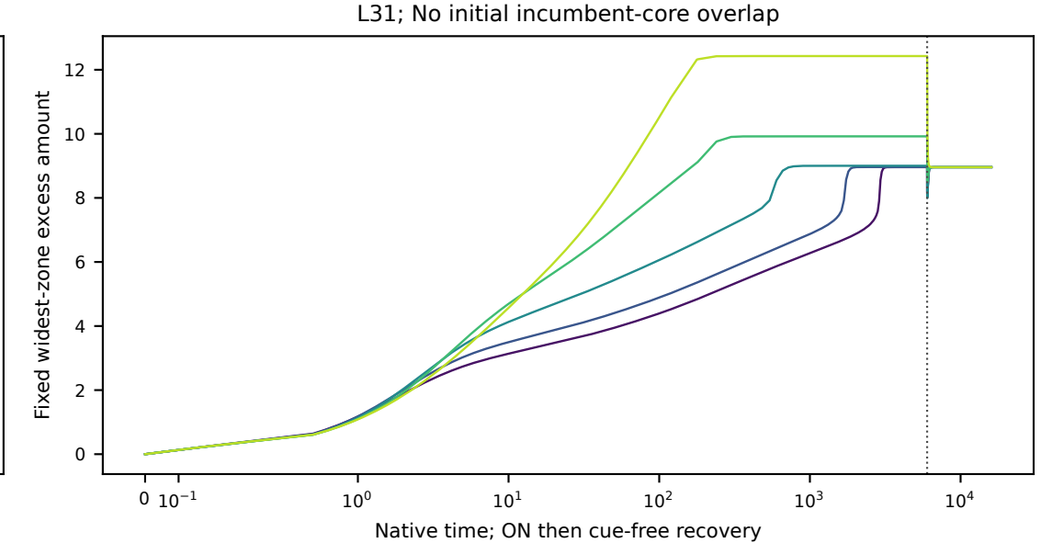






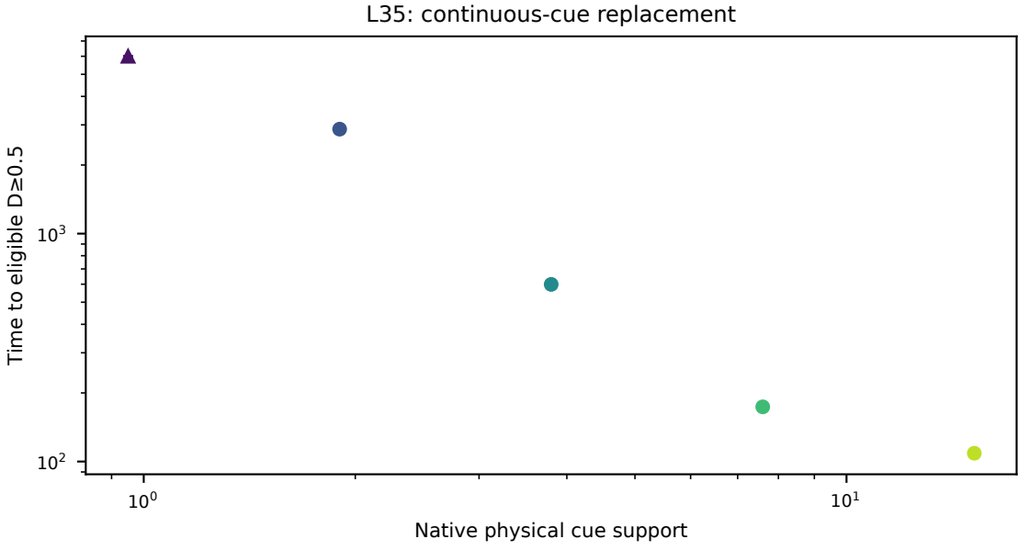


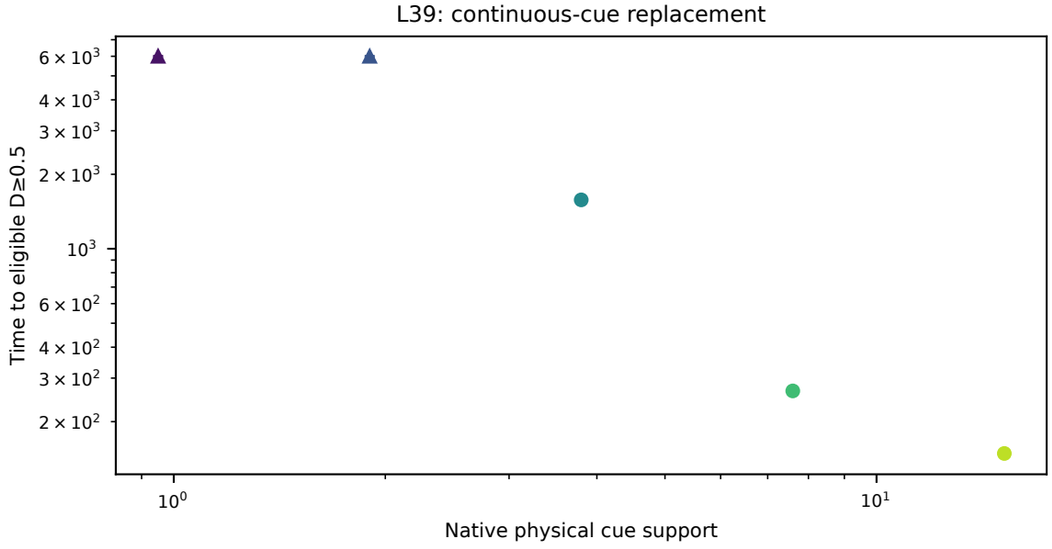


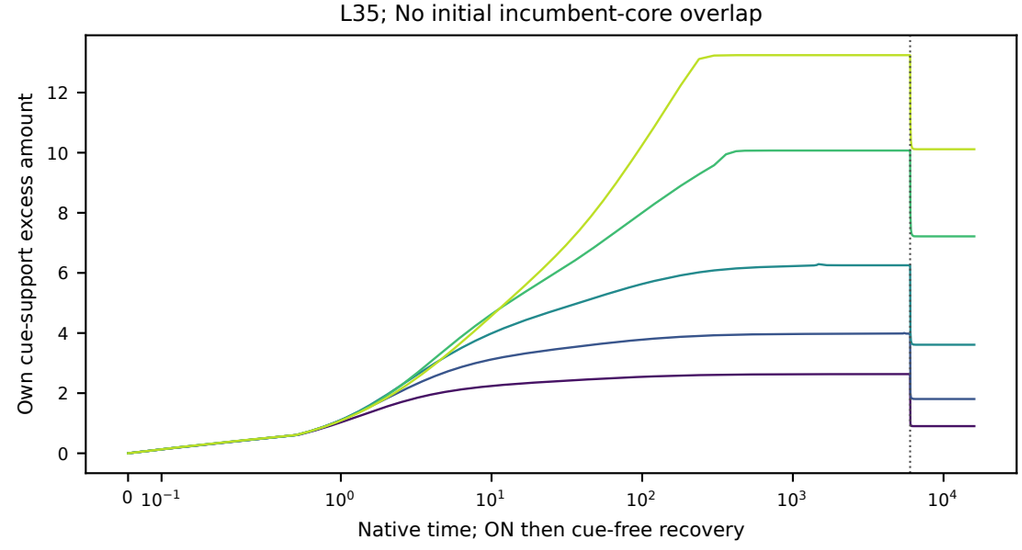


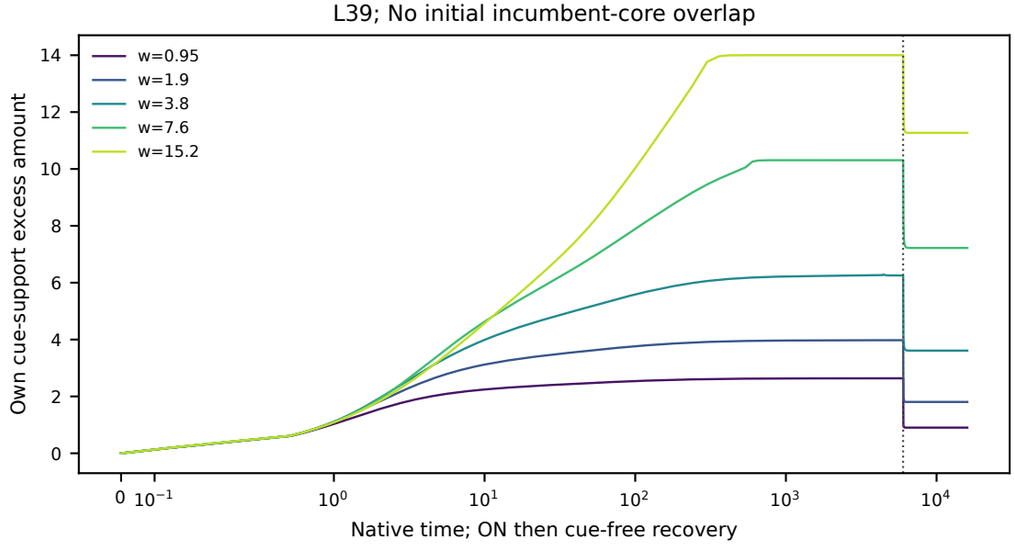
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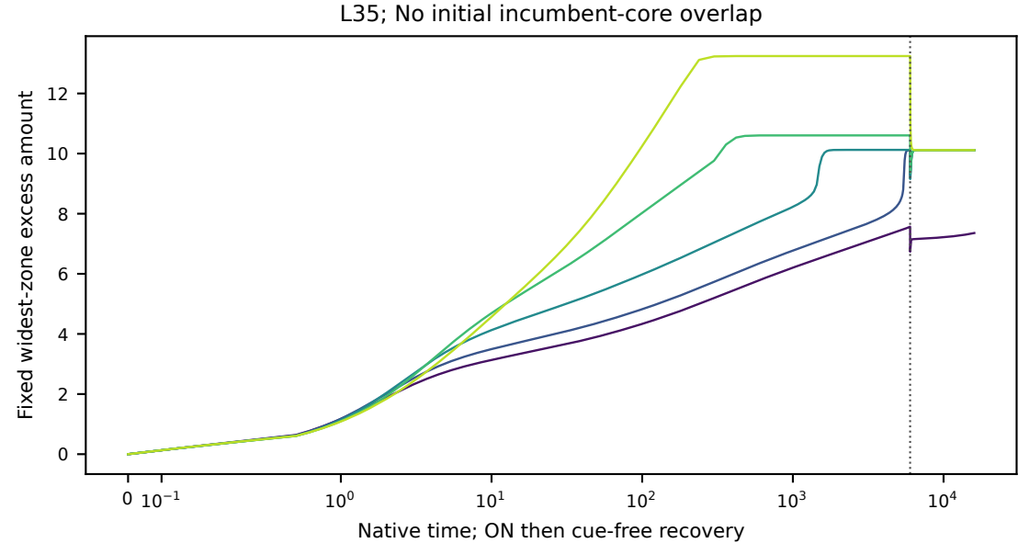
S1 Wave-pinning • commanded area 2× • native kinetics

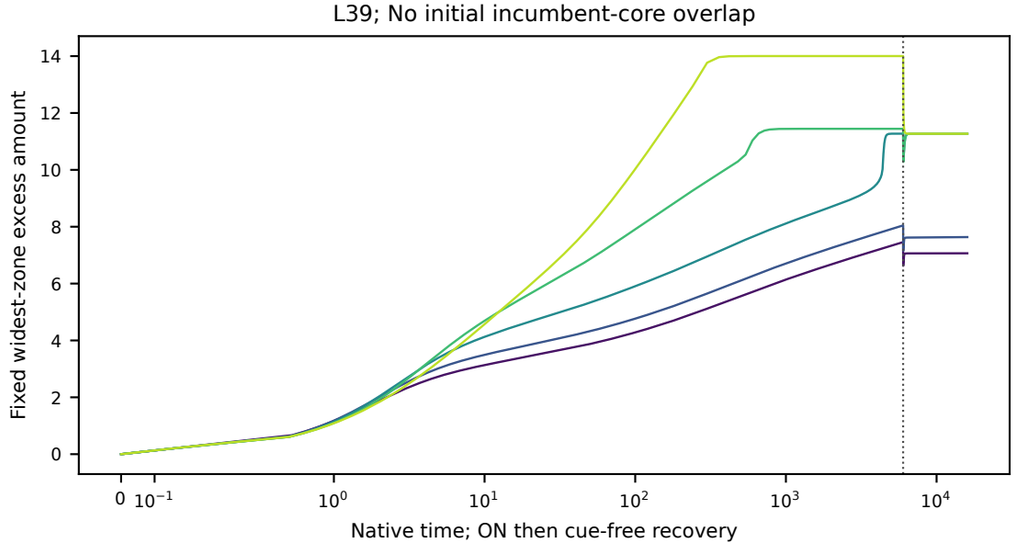






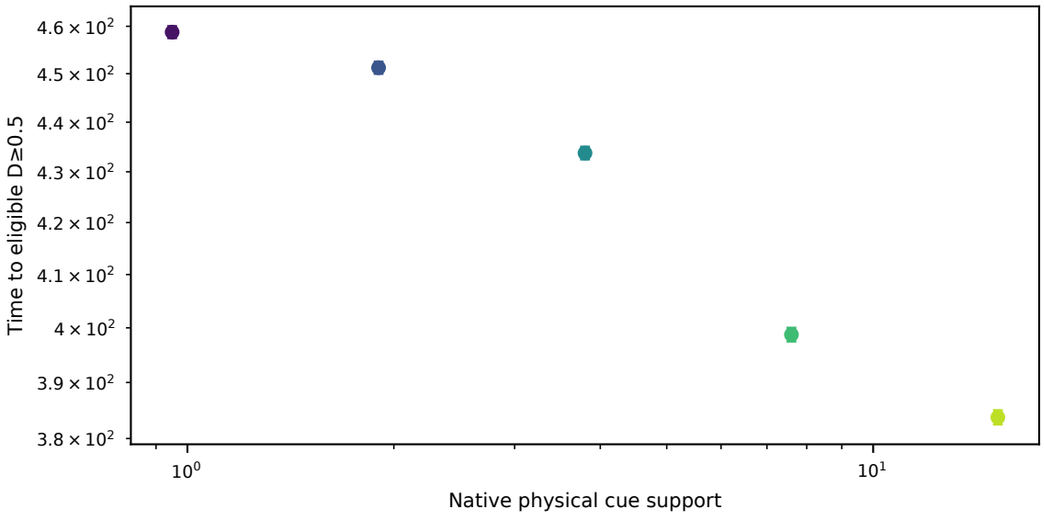


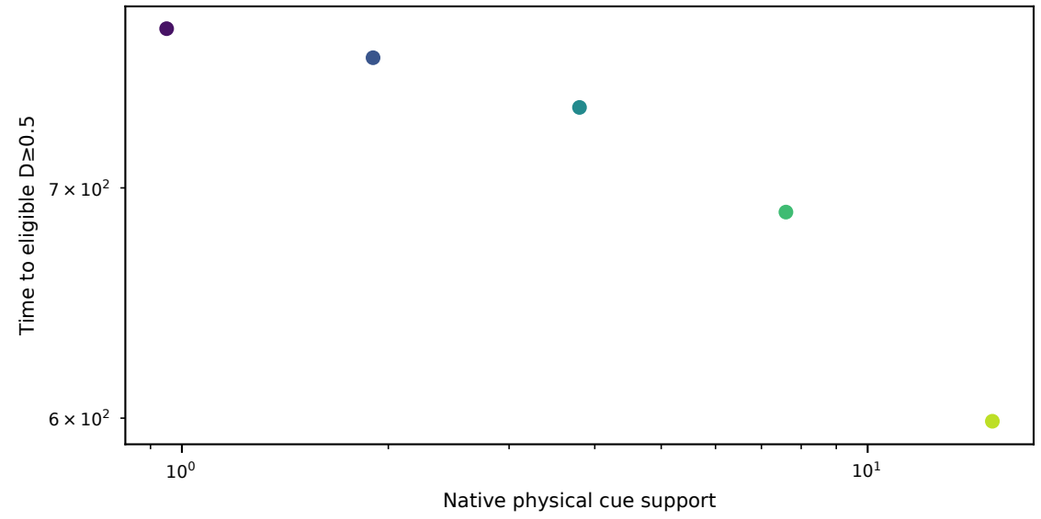


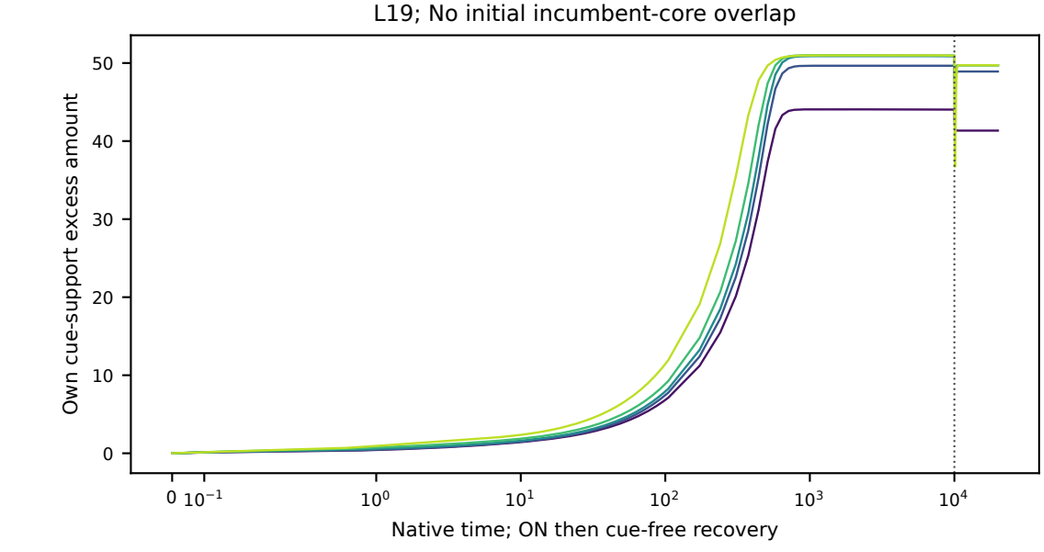


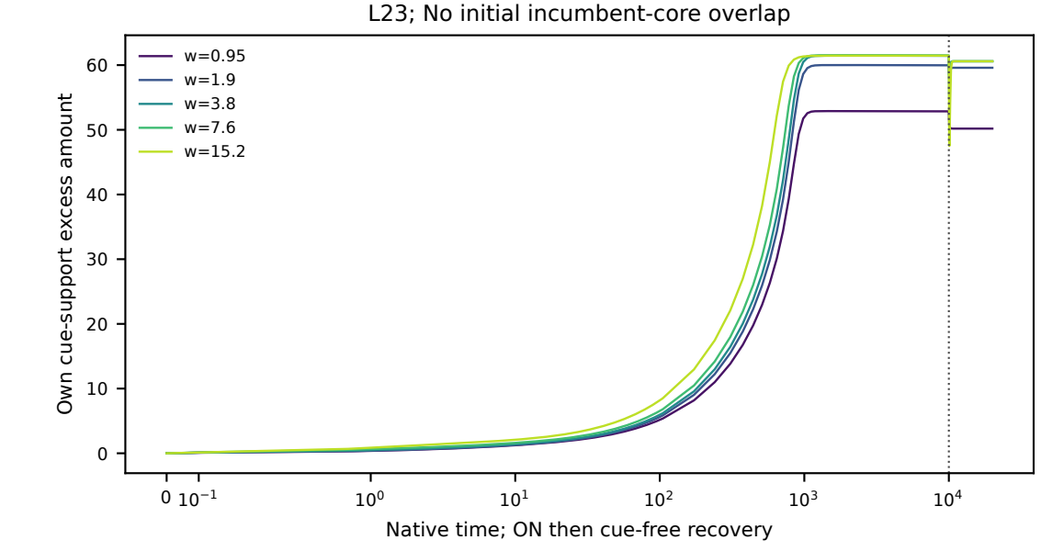
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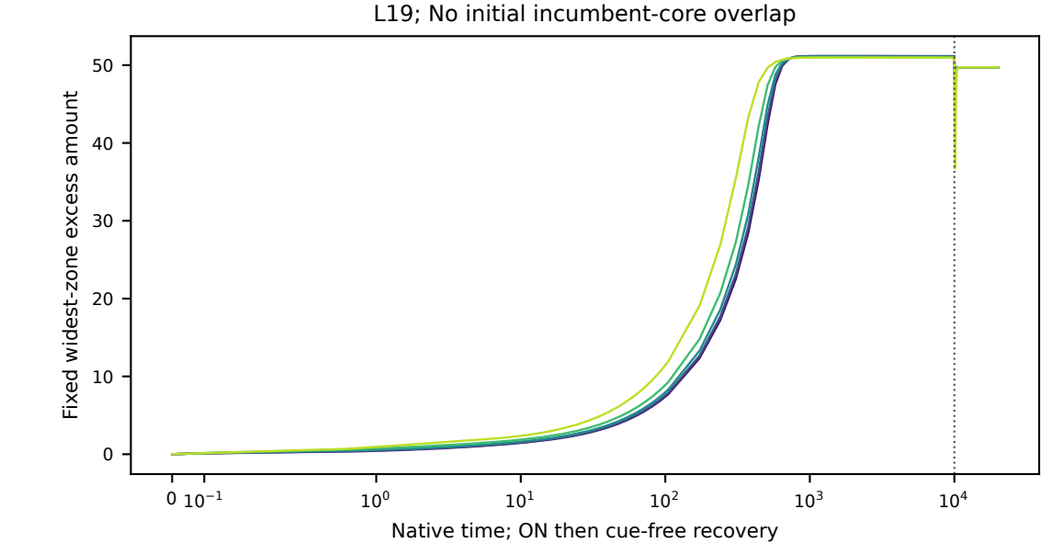
S1 Goryachev • commanded area 0.5× • native kinetics

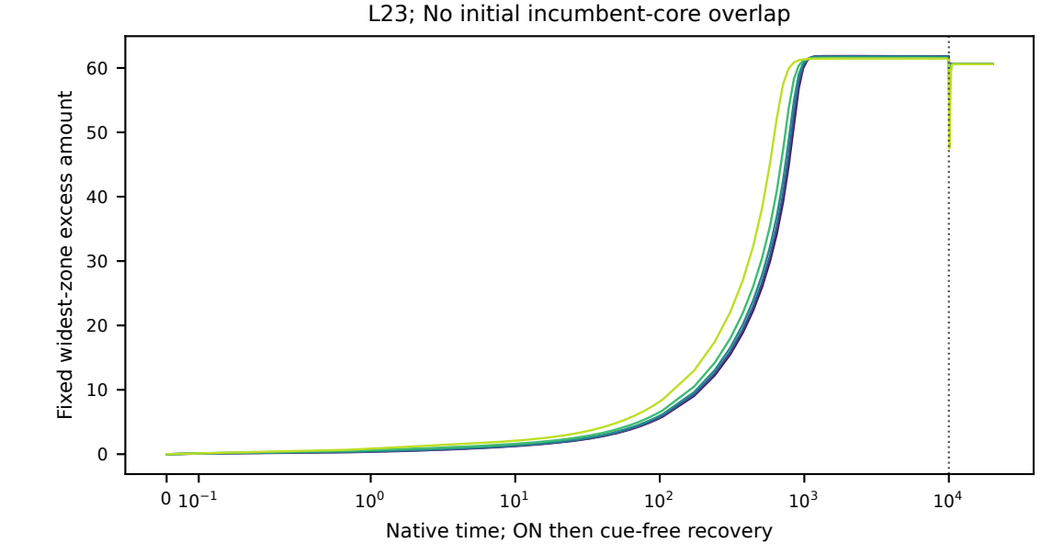






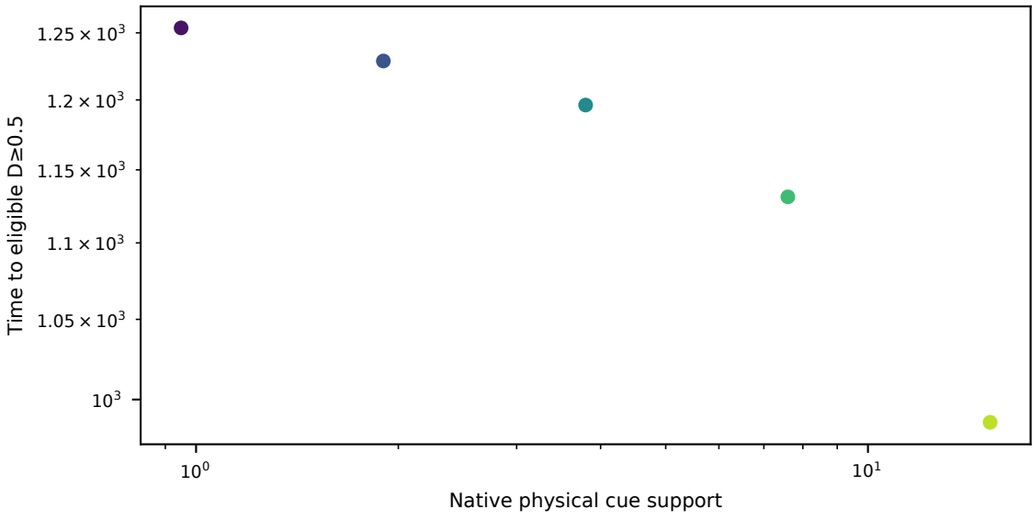


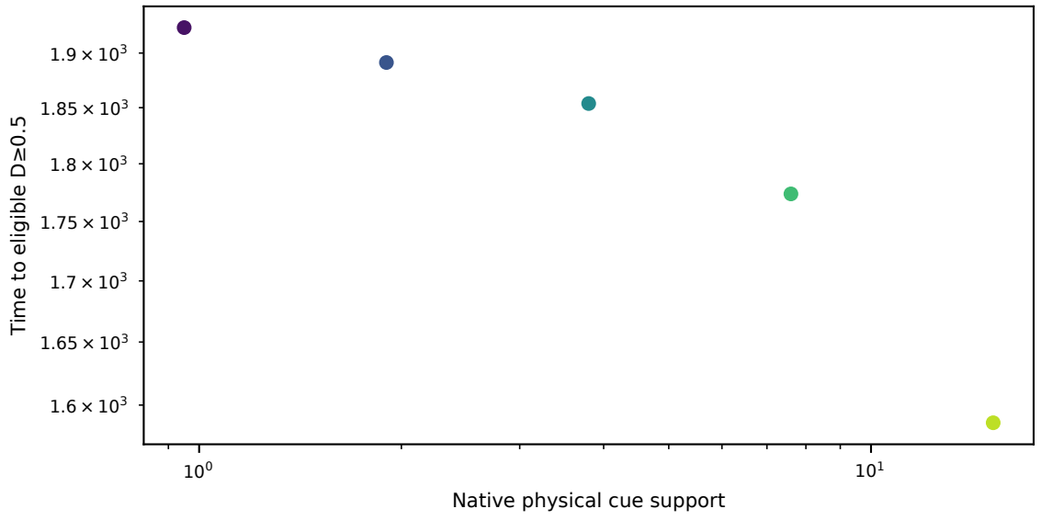


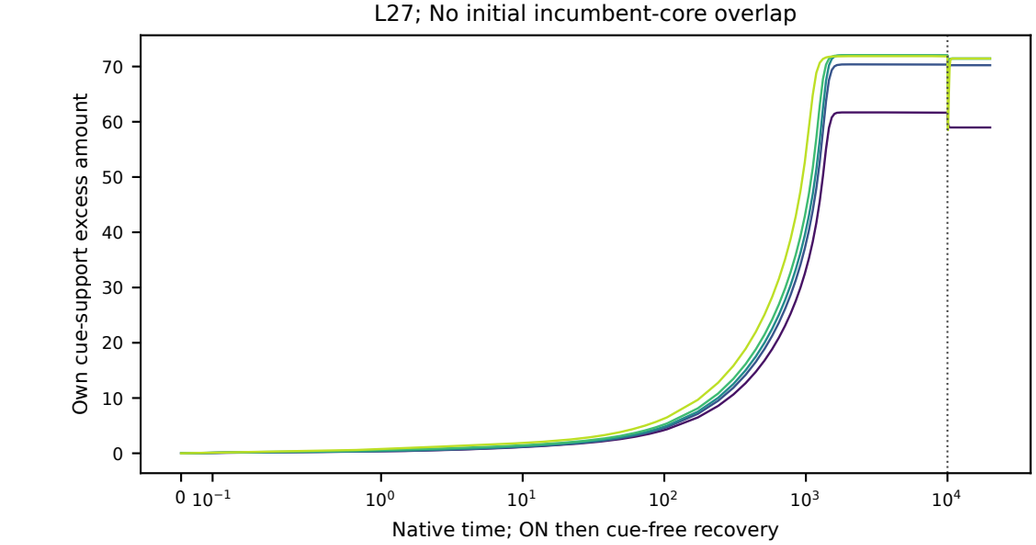


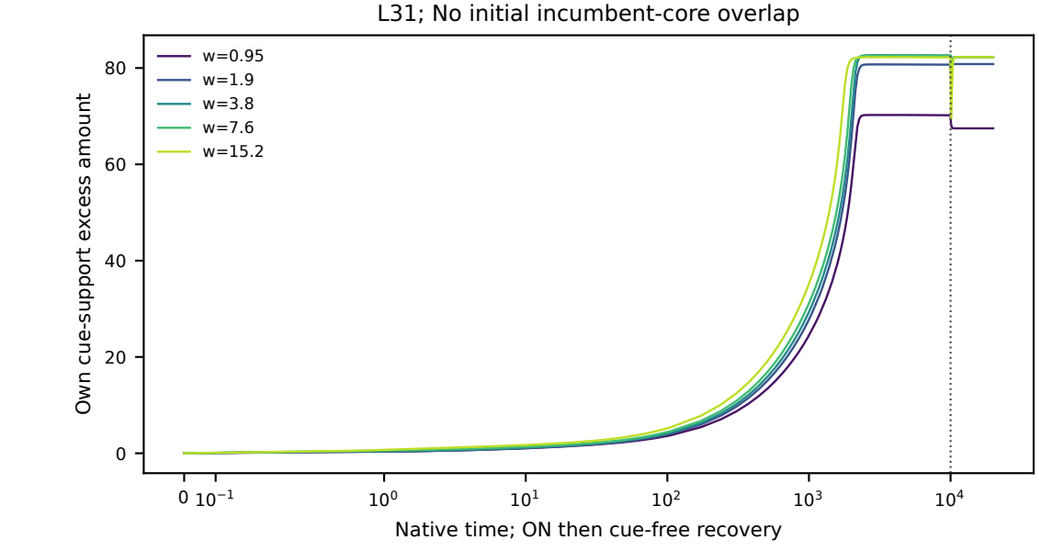
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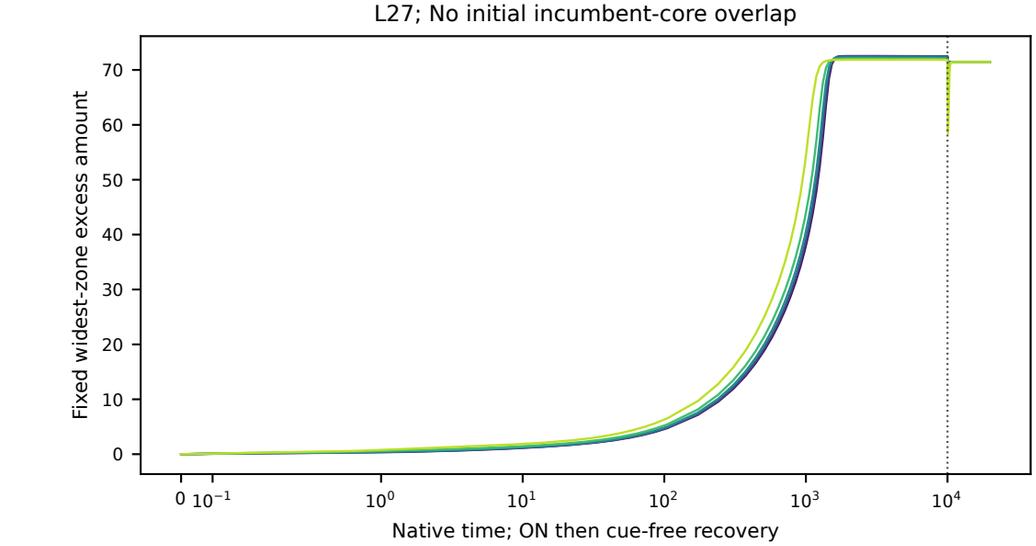
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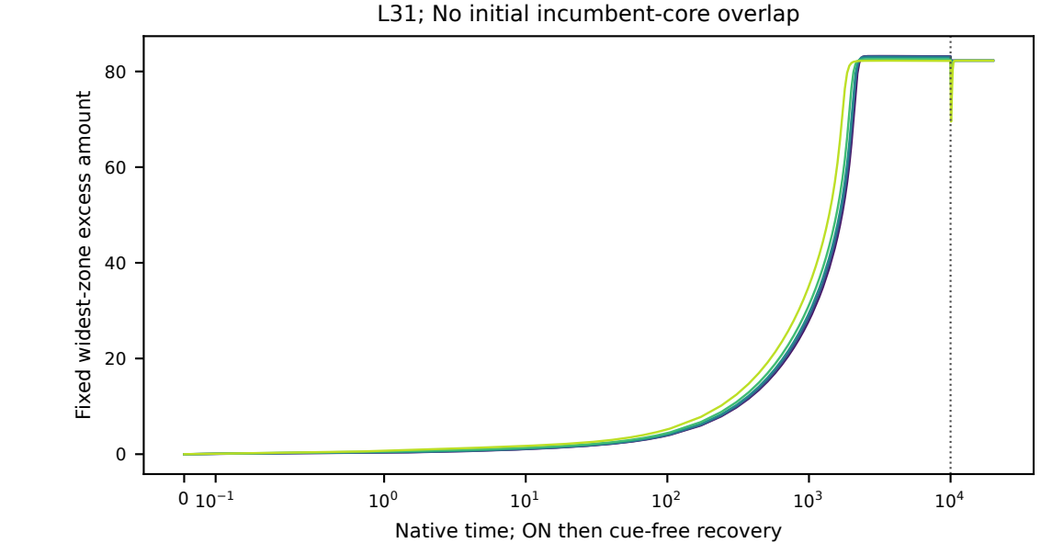






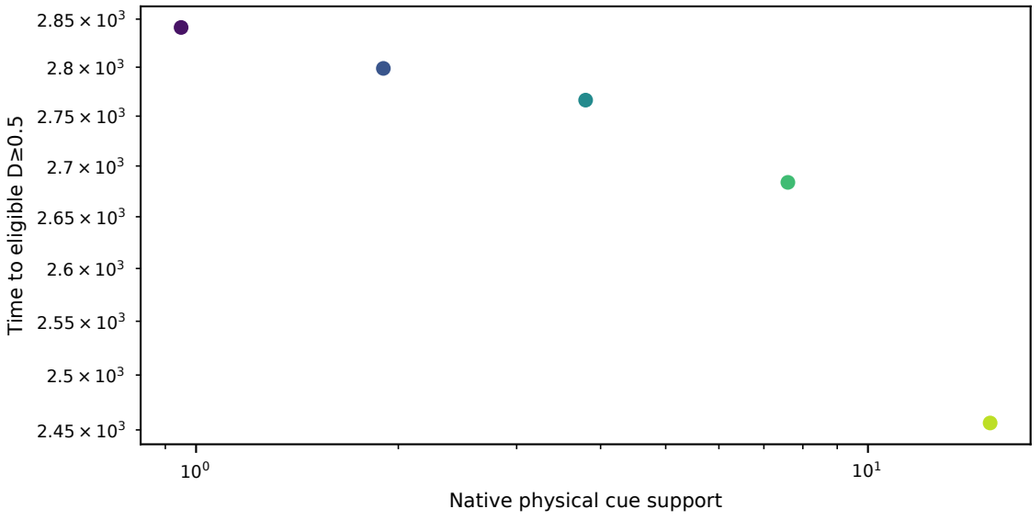


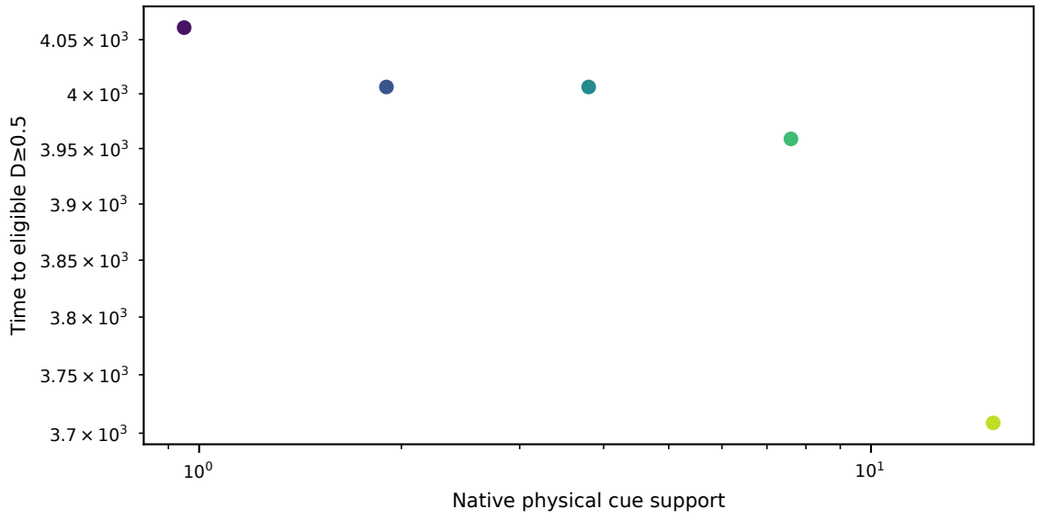


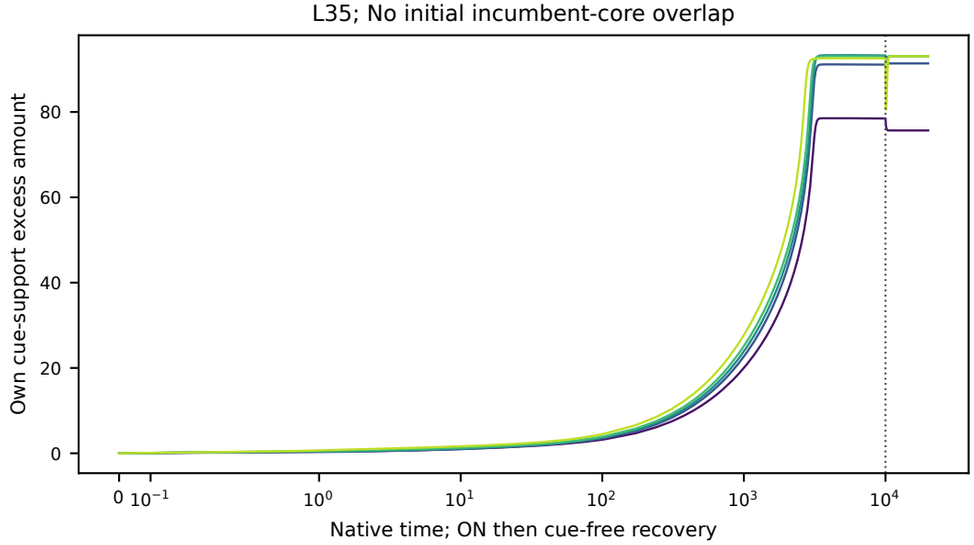


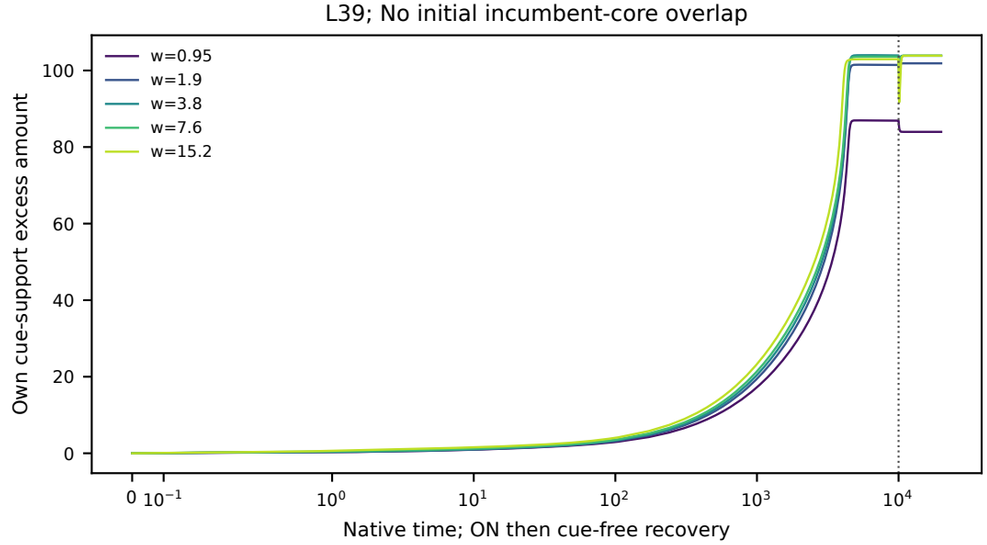
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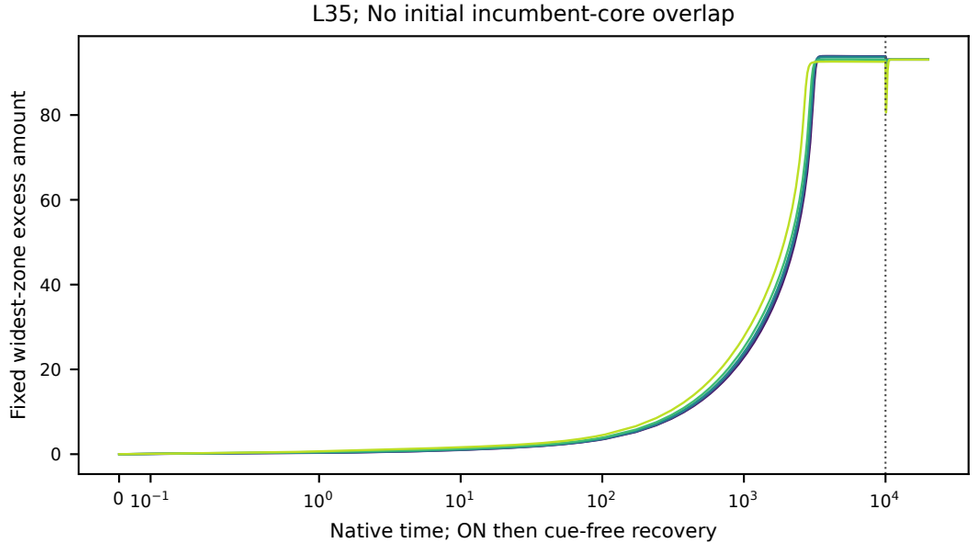
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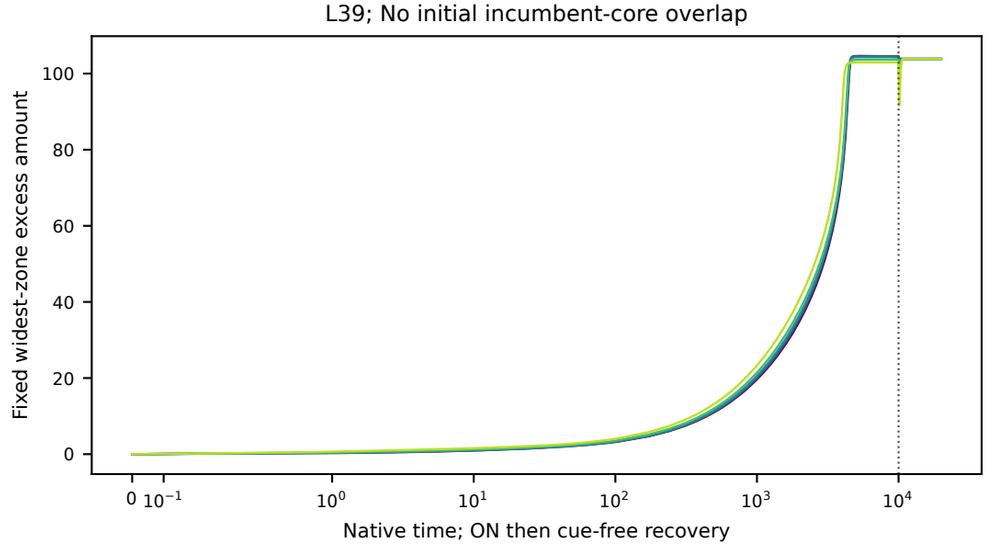






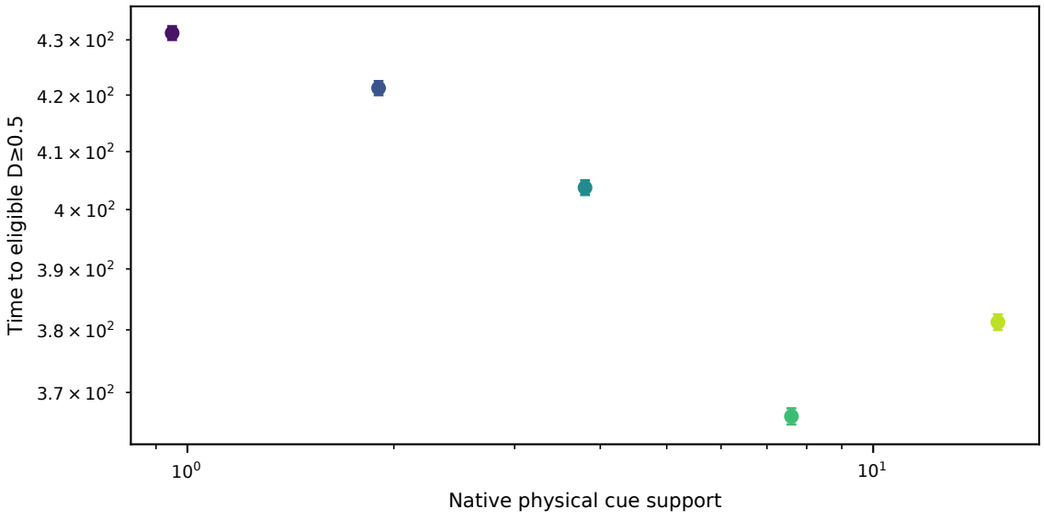


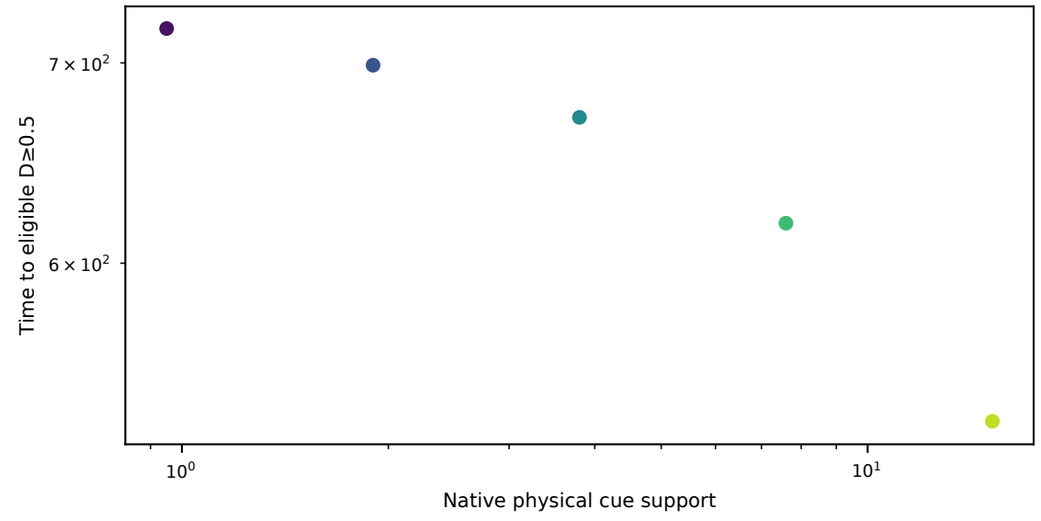


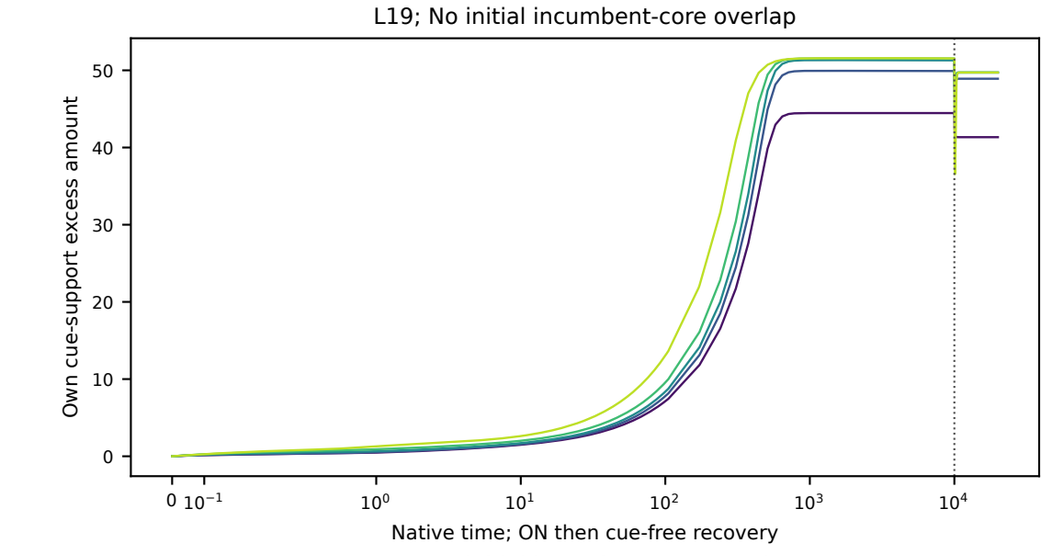


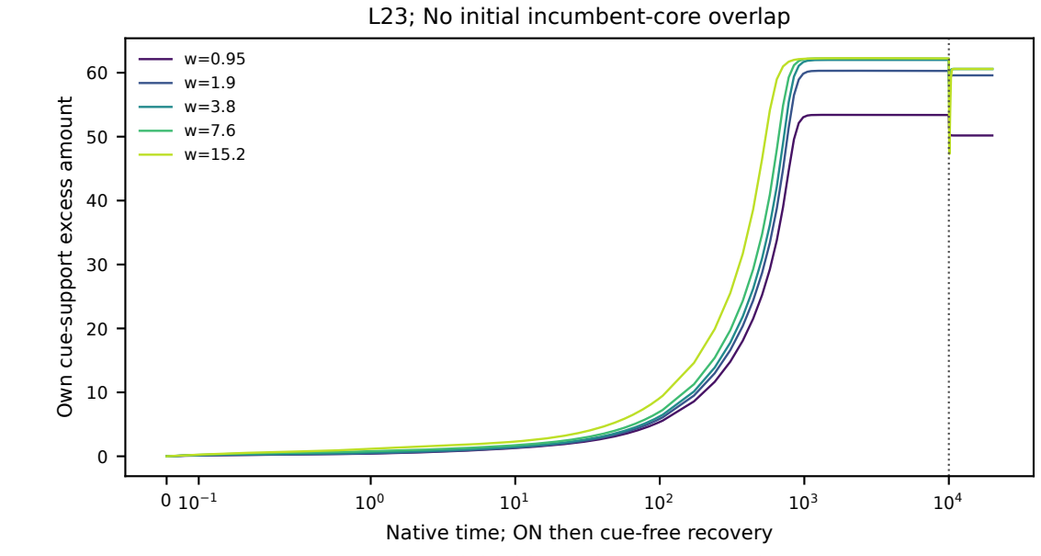
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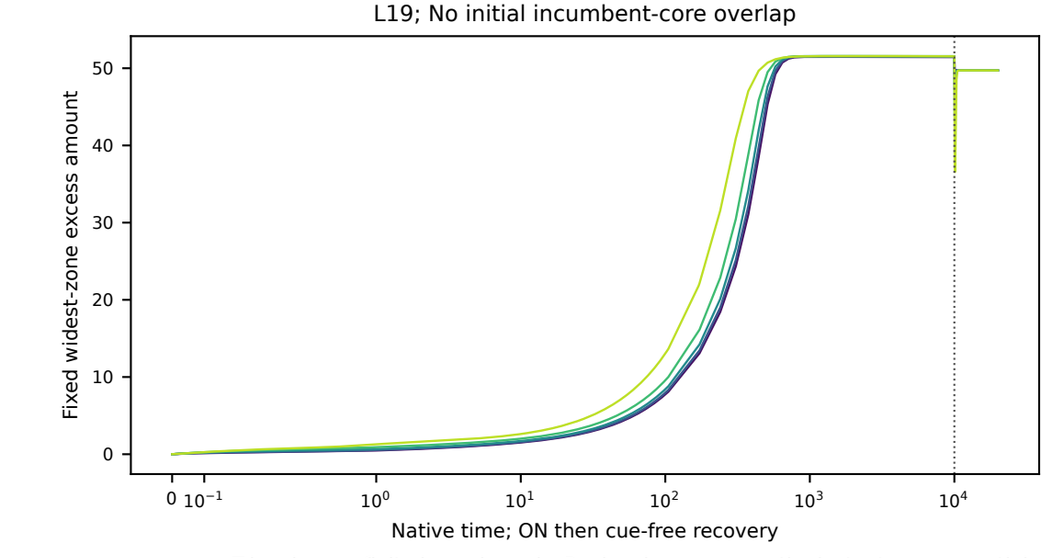
S1 Goryachev • commanded area 1× • native kinetics

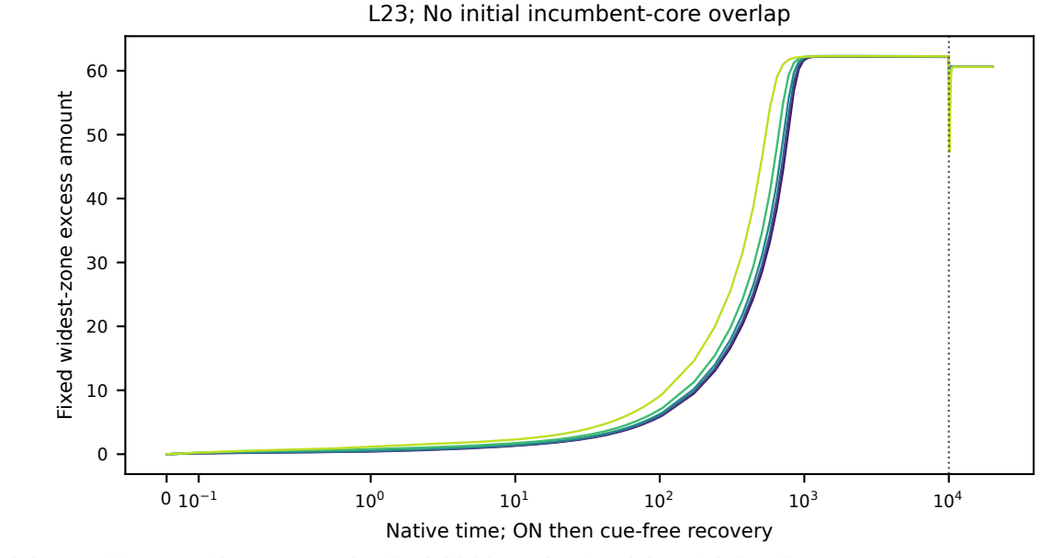






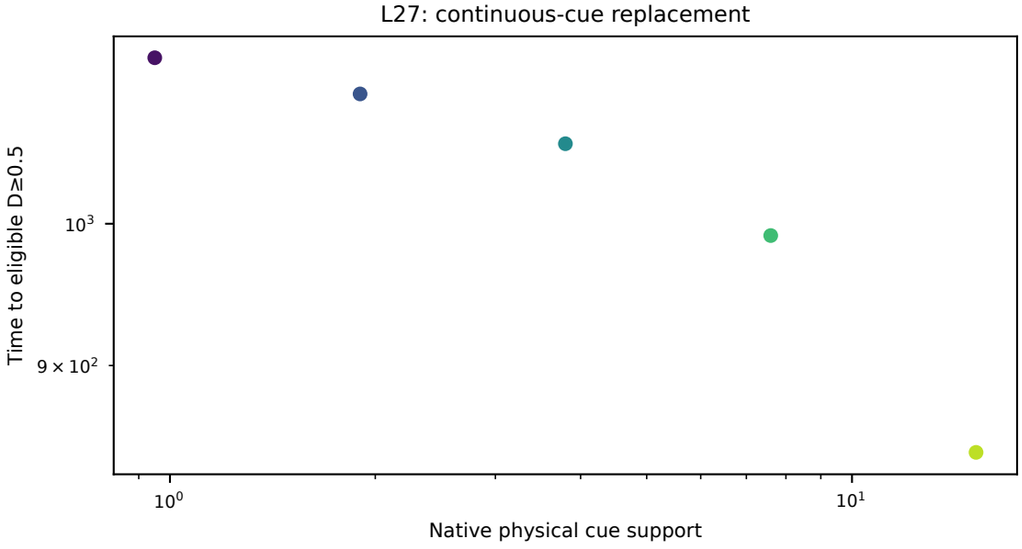


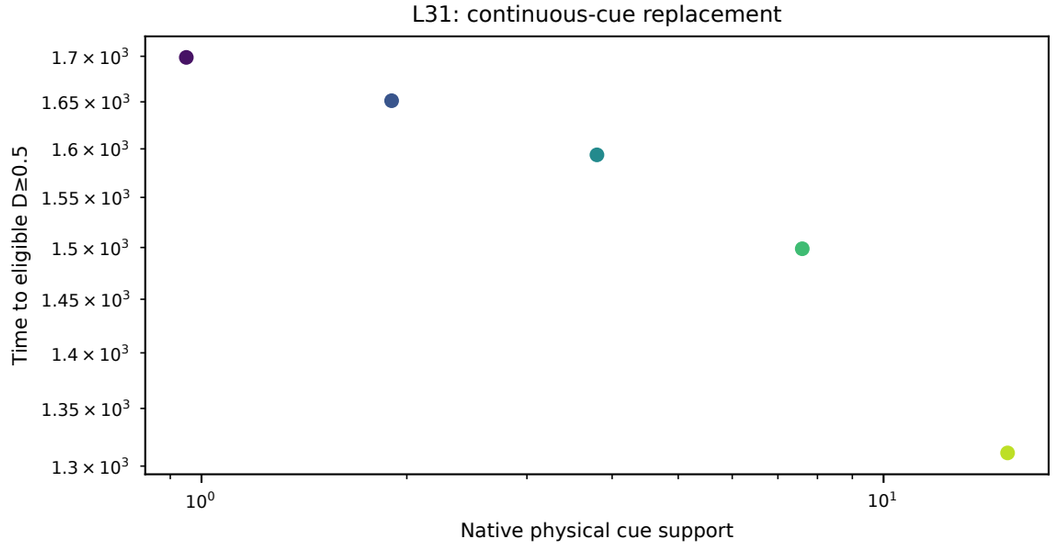


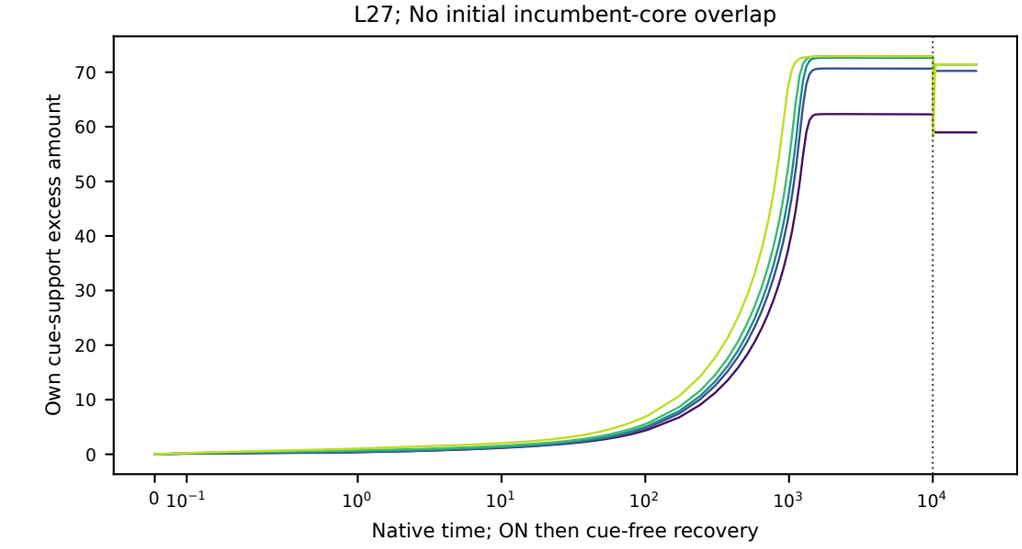


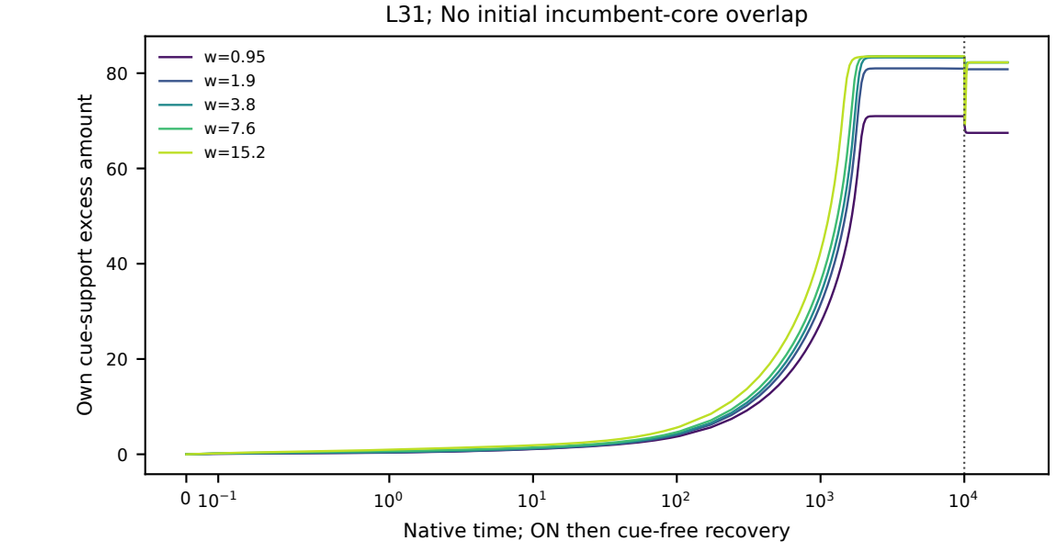
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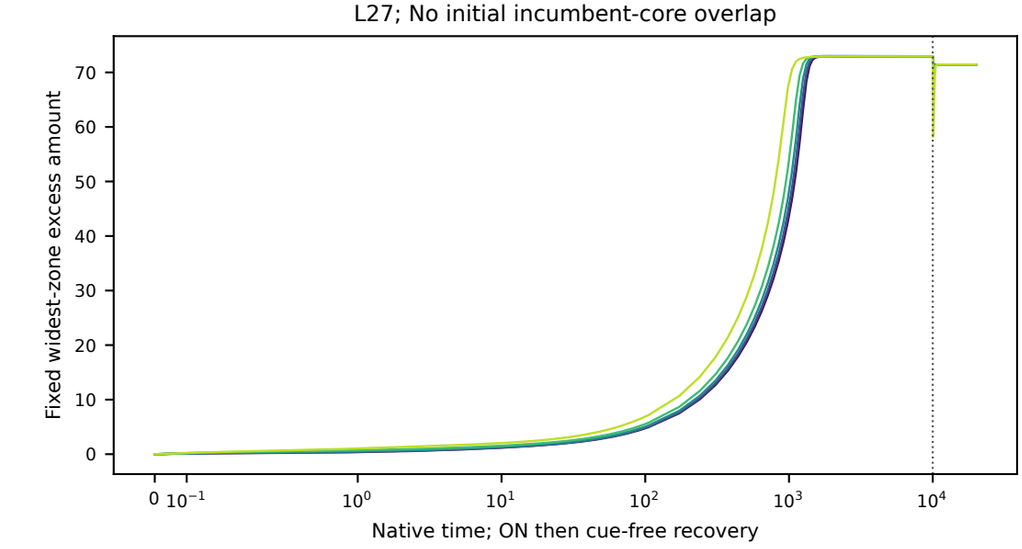
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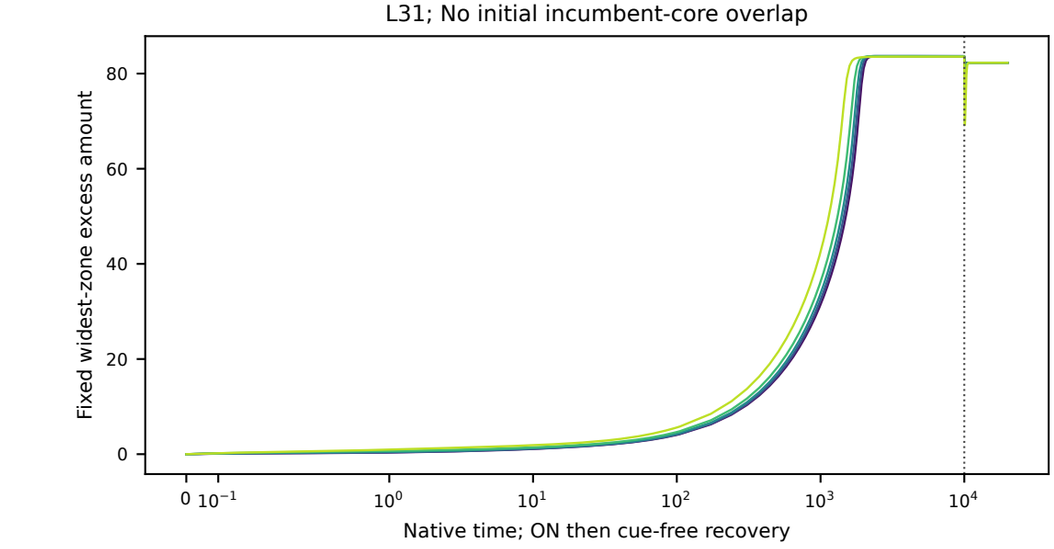






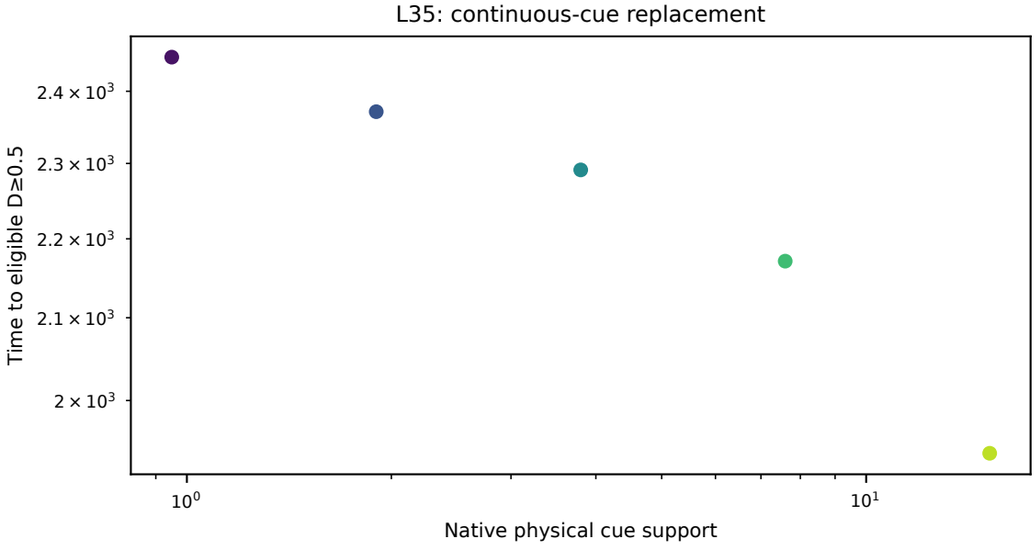


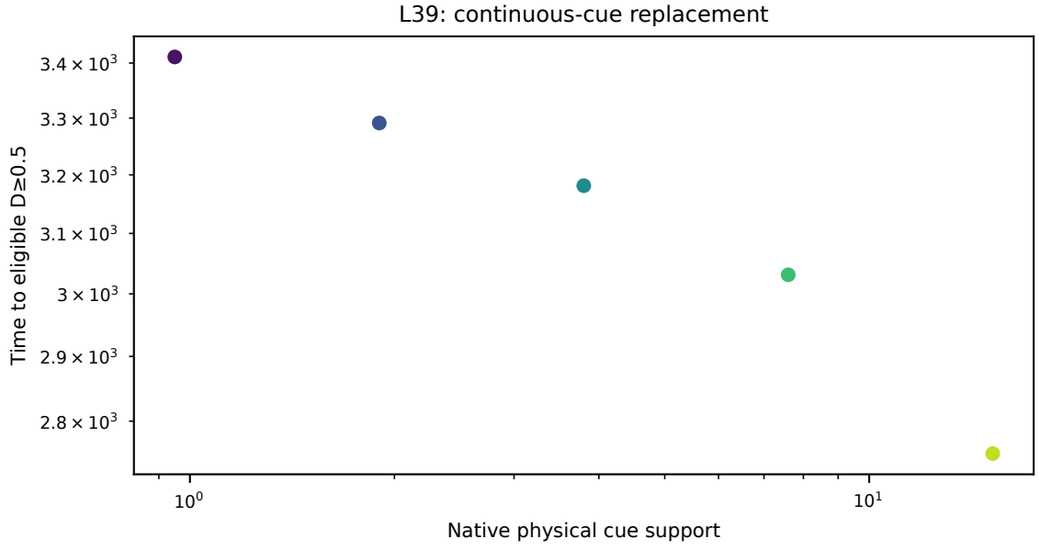


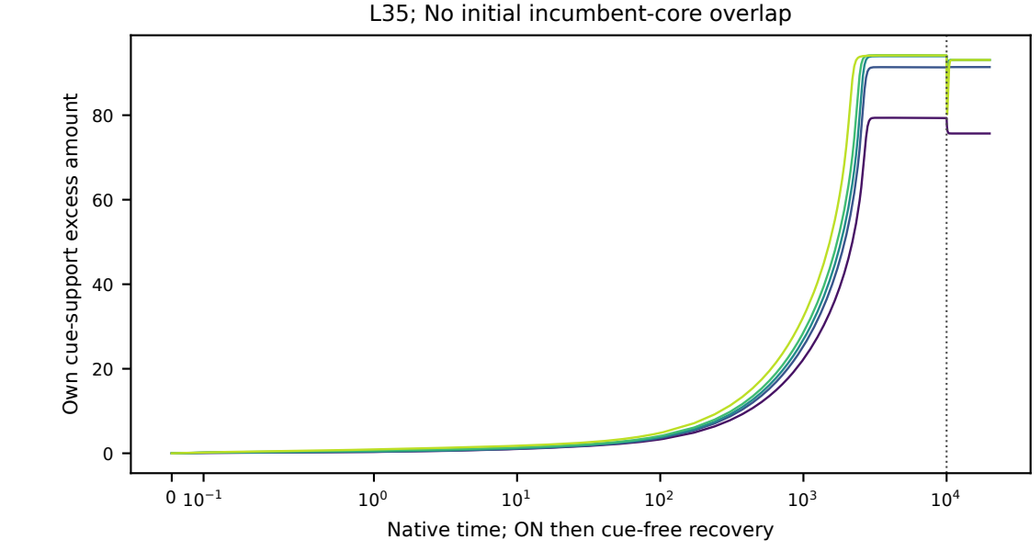


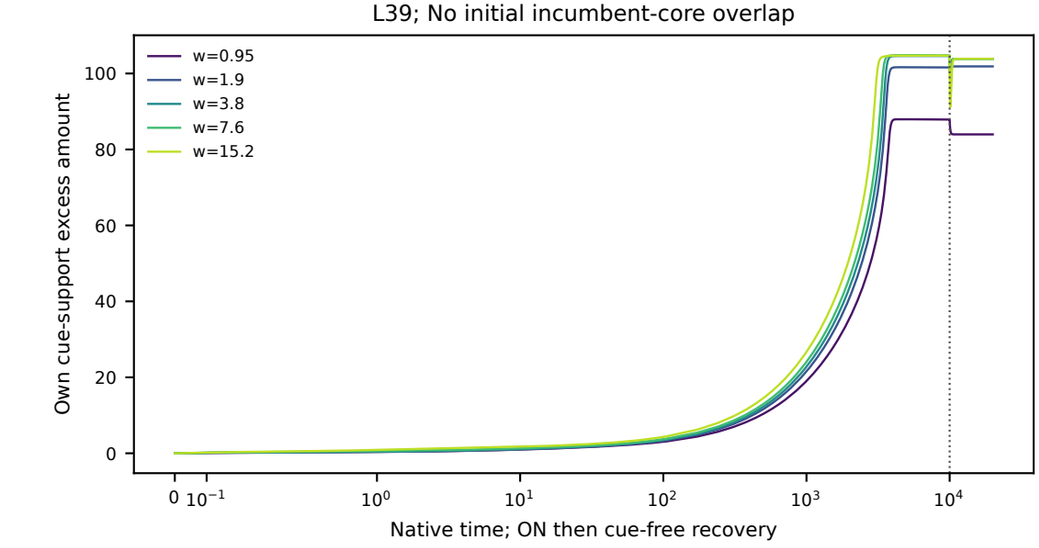
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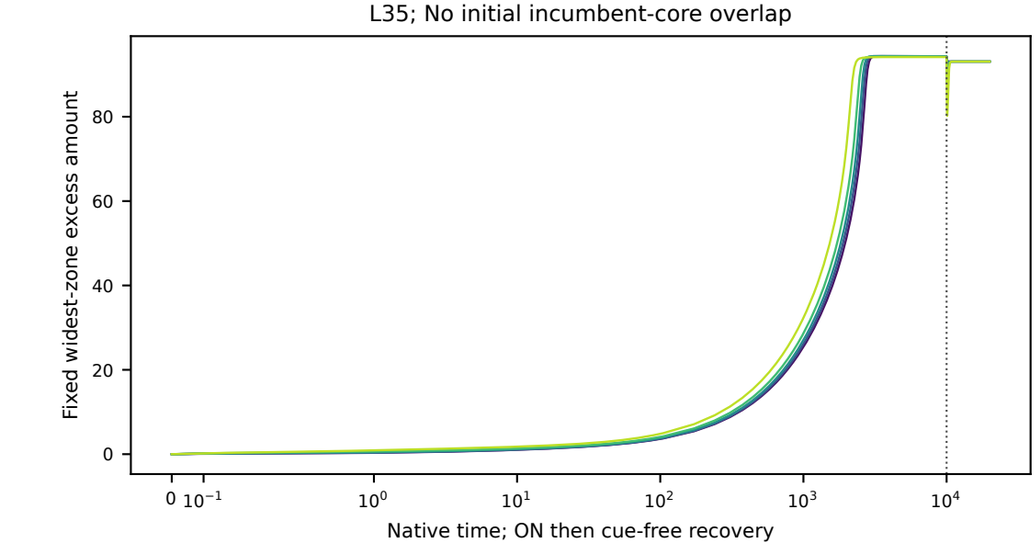
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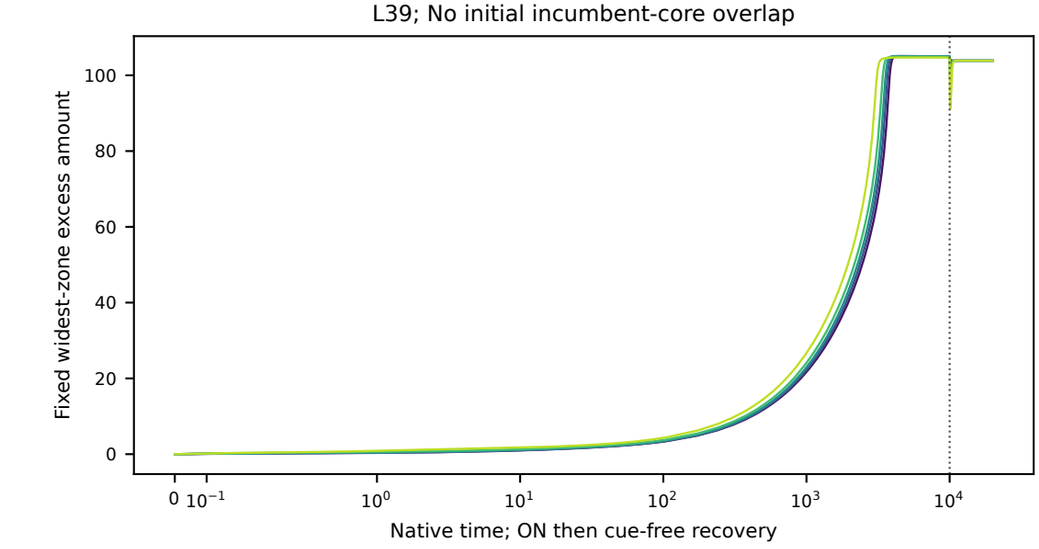






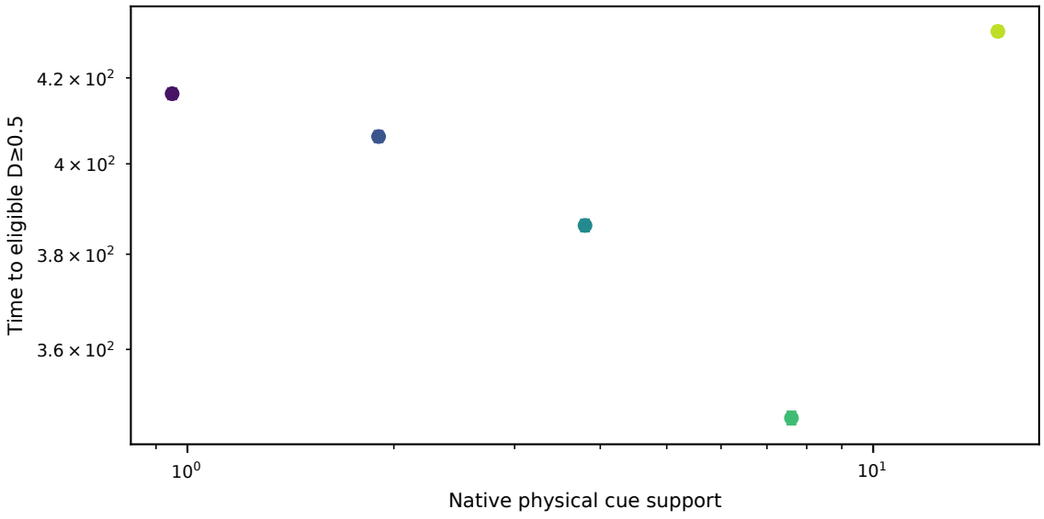


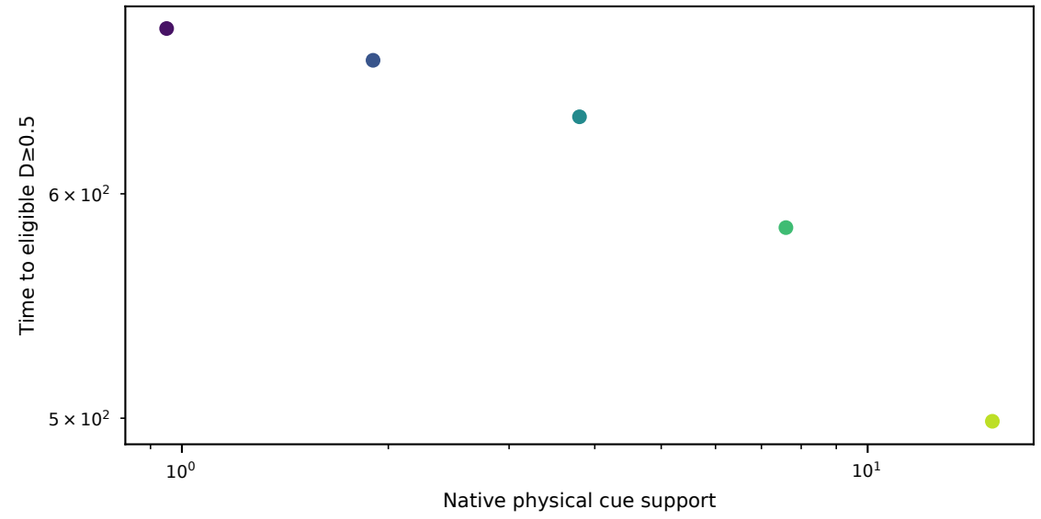


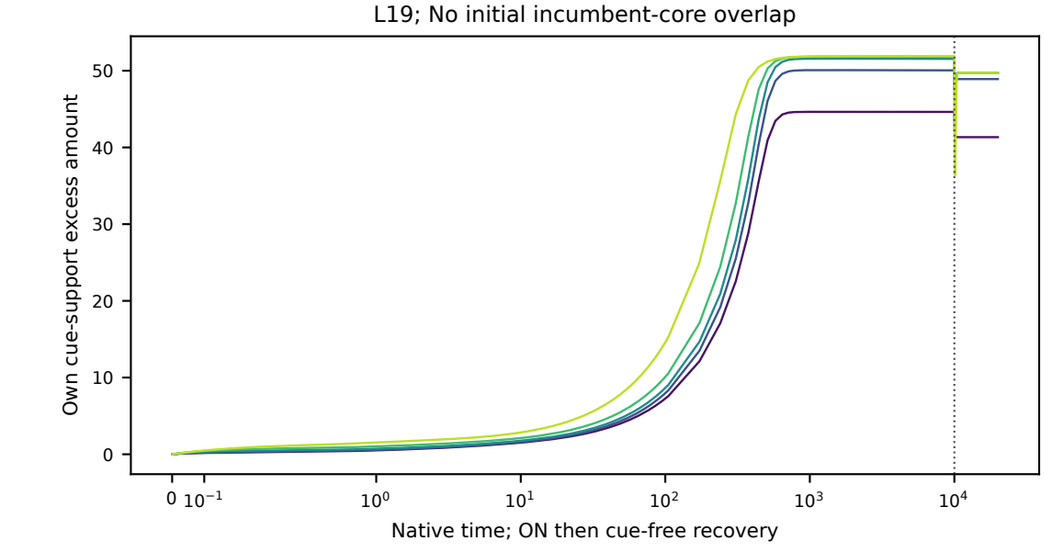


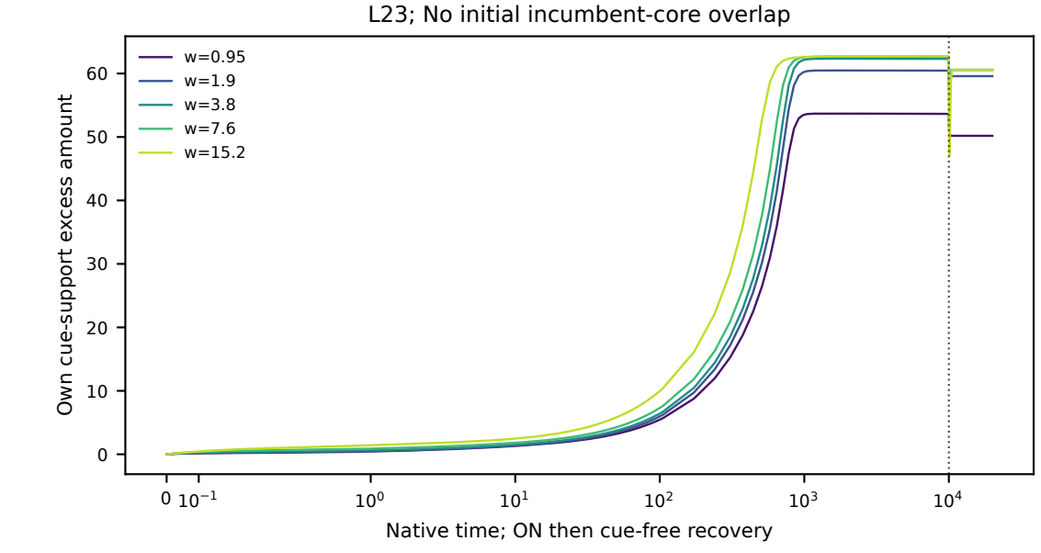
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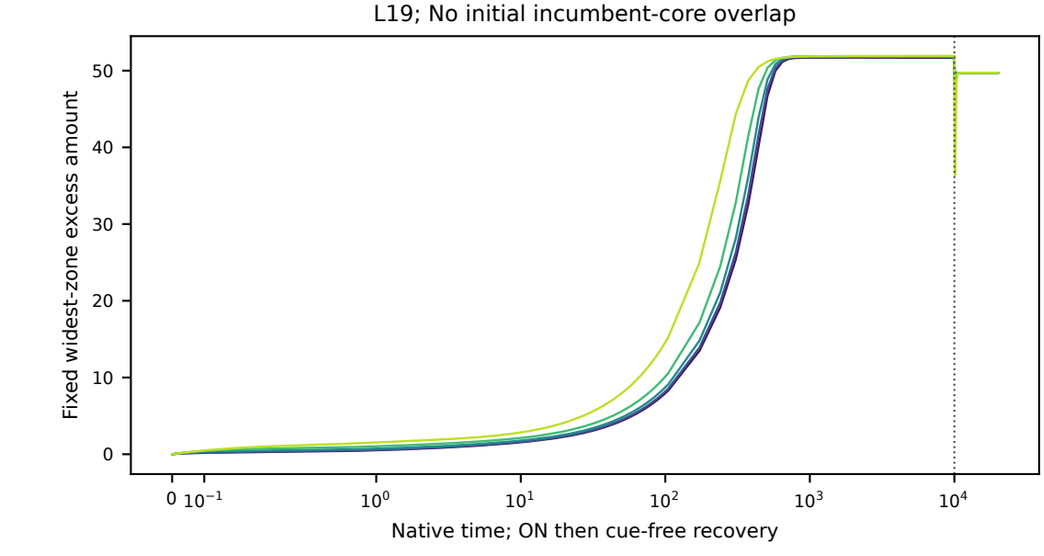
S1 Goryachev • commanded area 2× • native kinetics

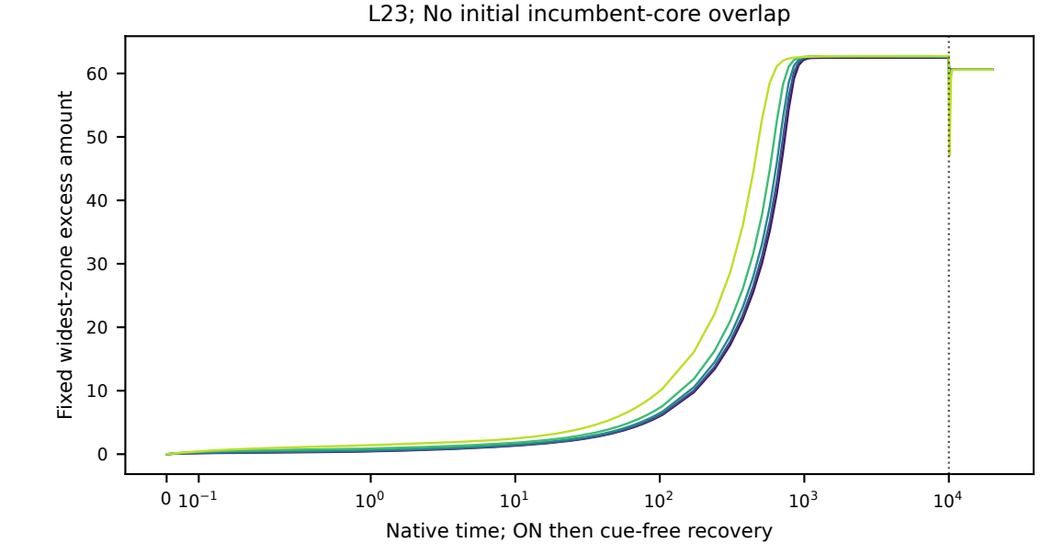






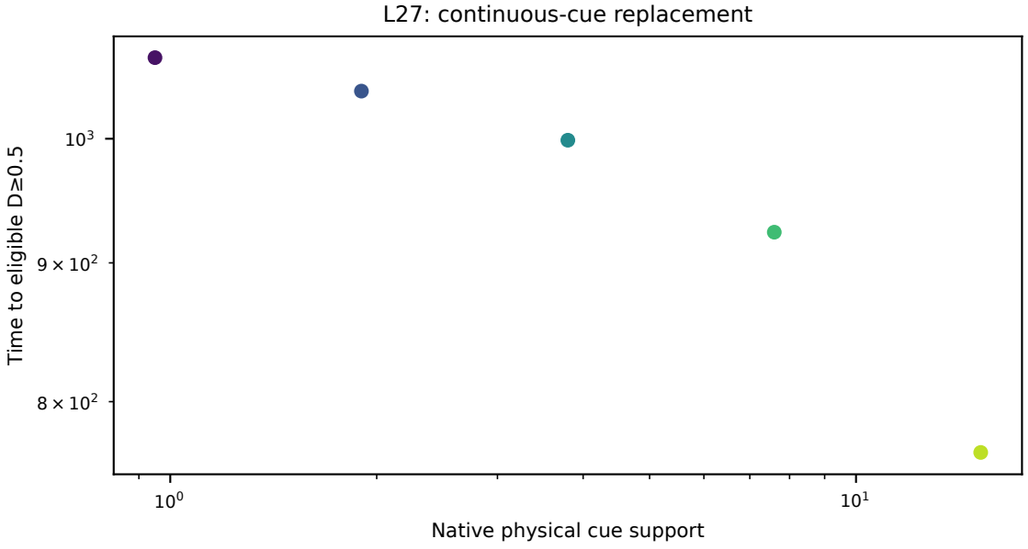


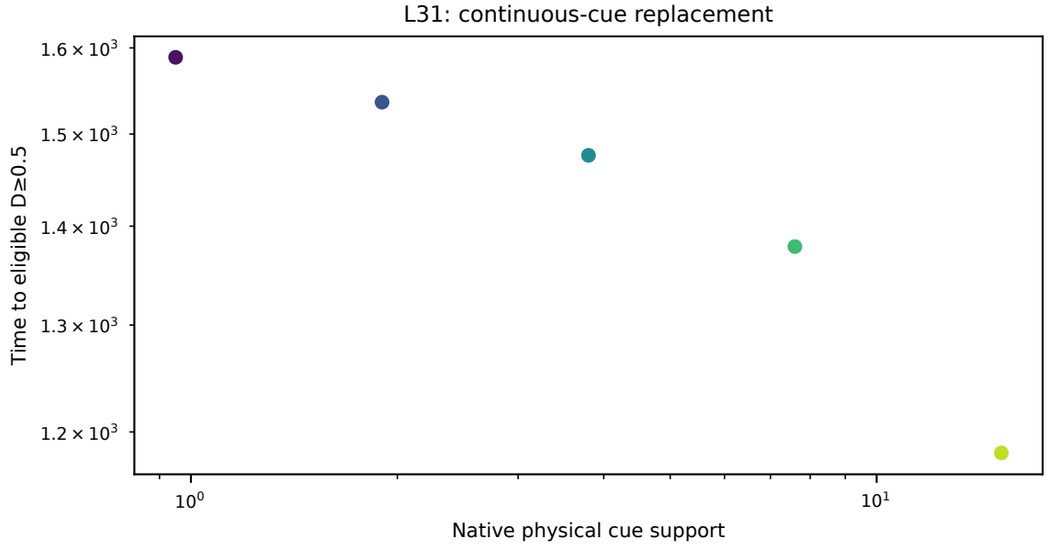


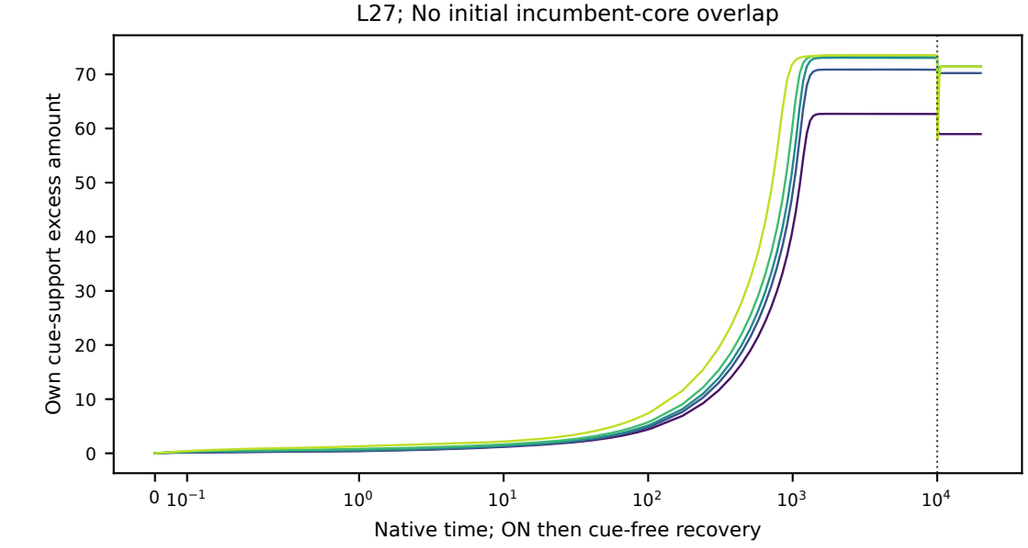


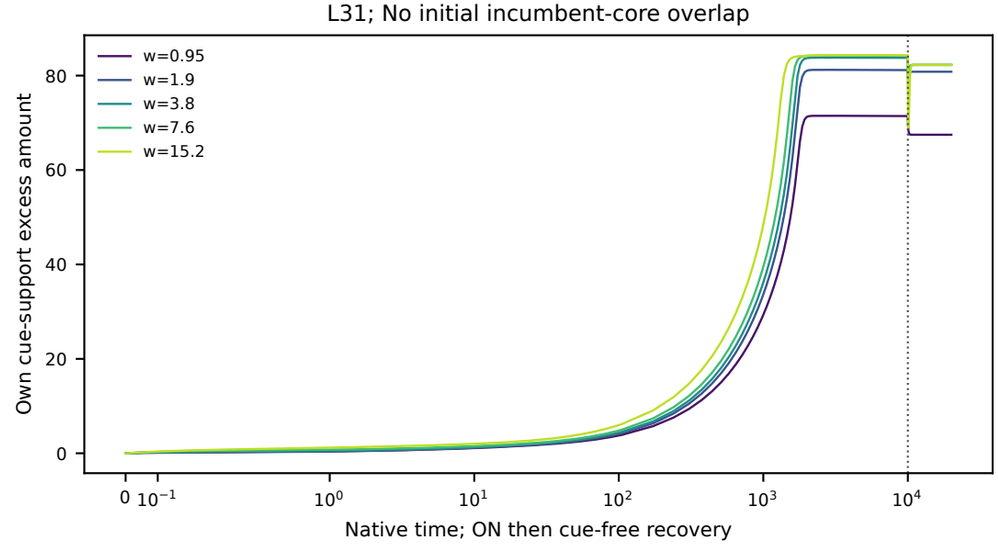
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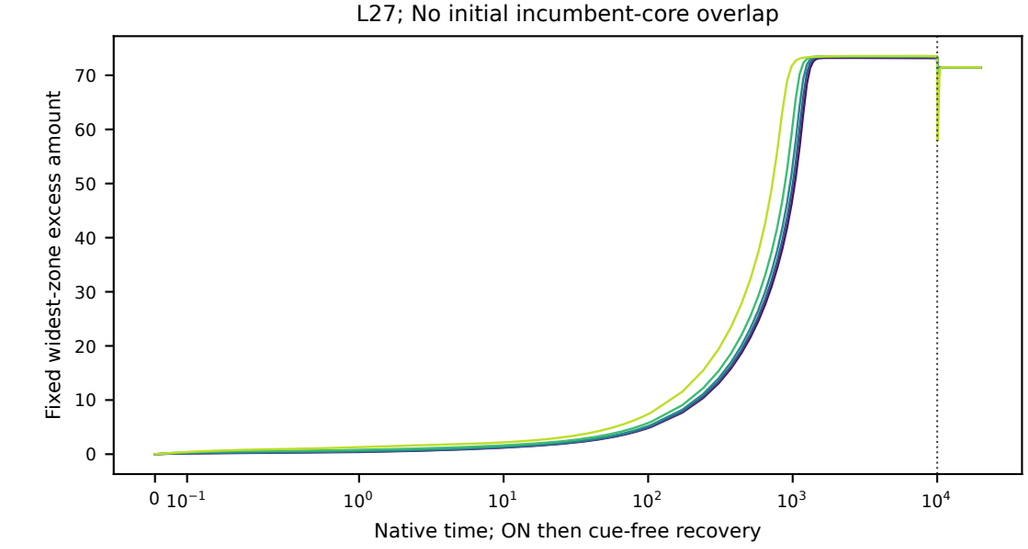
S1 Goryachev • commanded area 2× • native kinetics

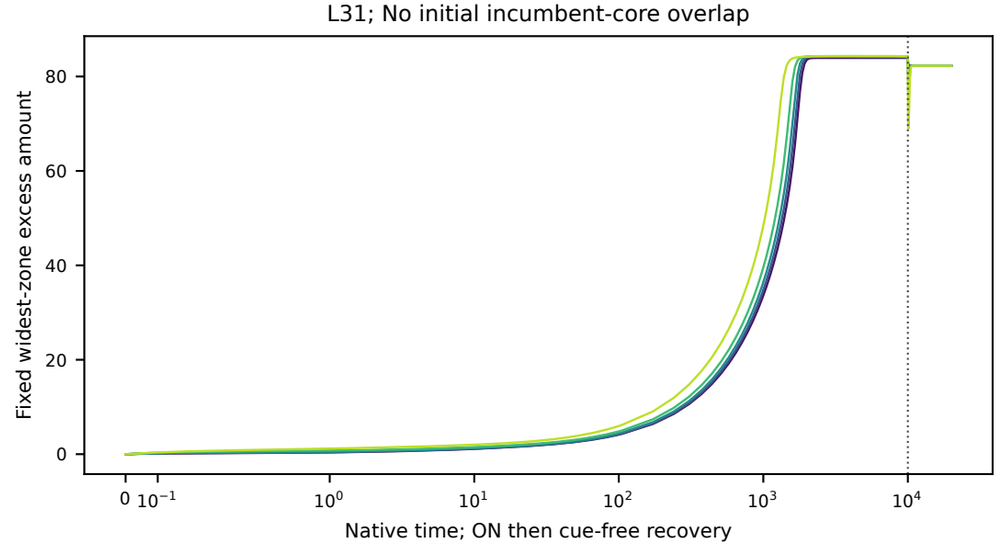






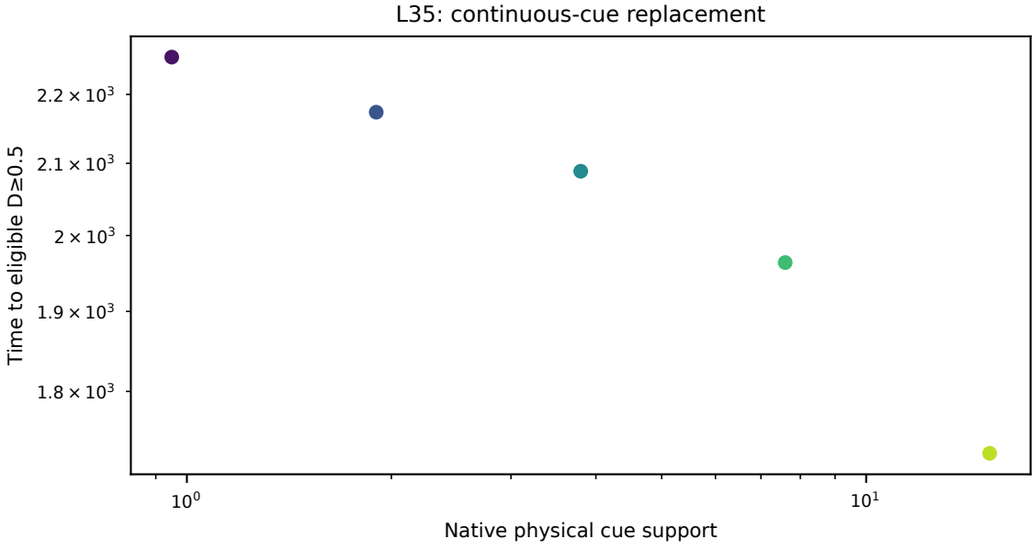


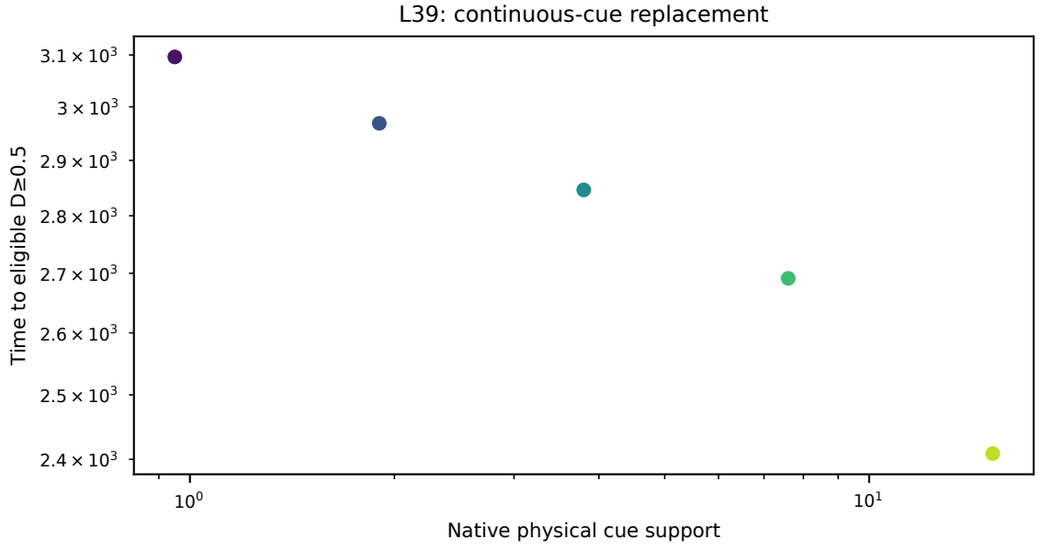


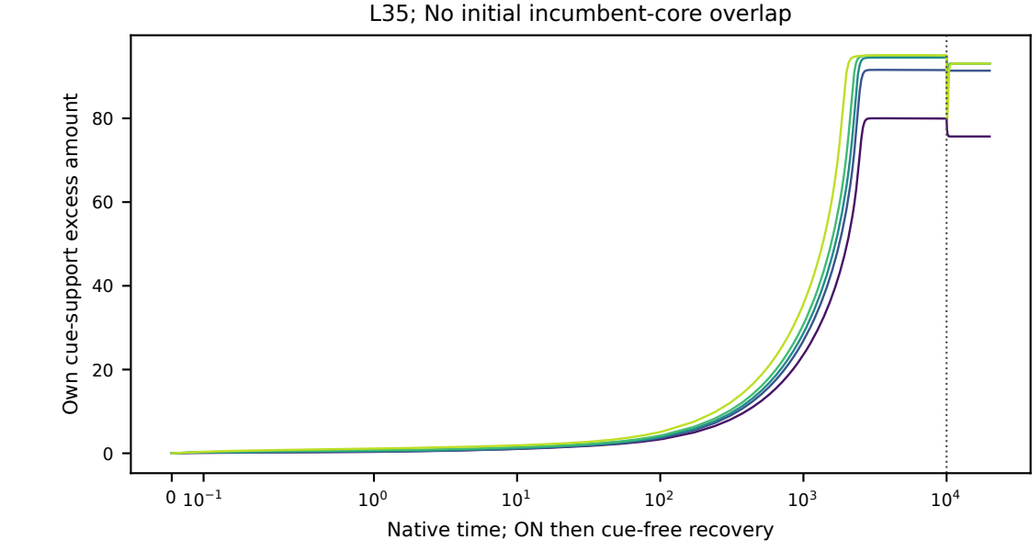


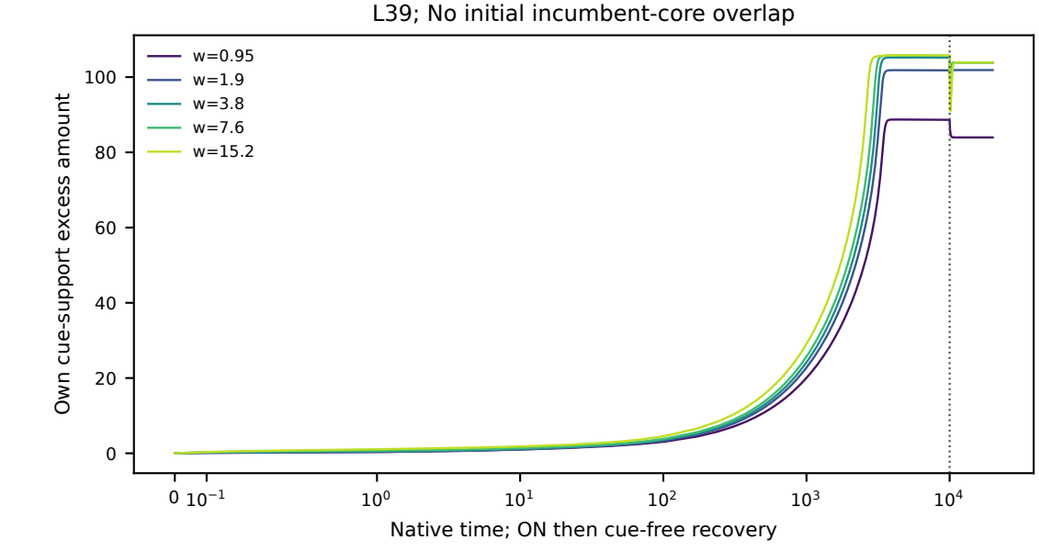
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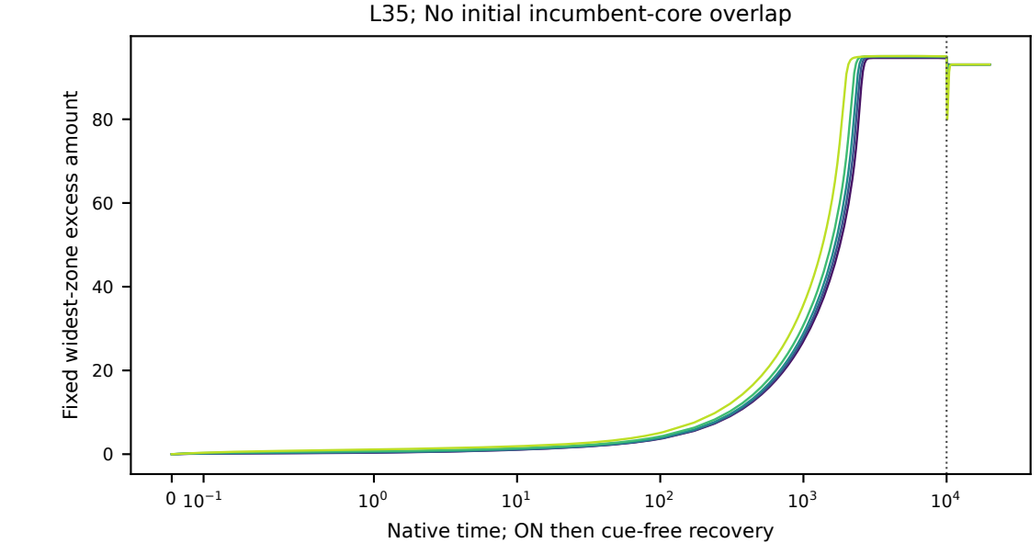
S1 Goryachev • commanded area 2× • native kinetics

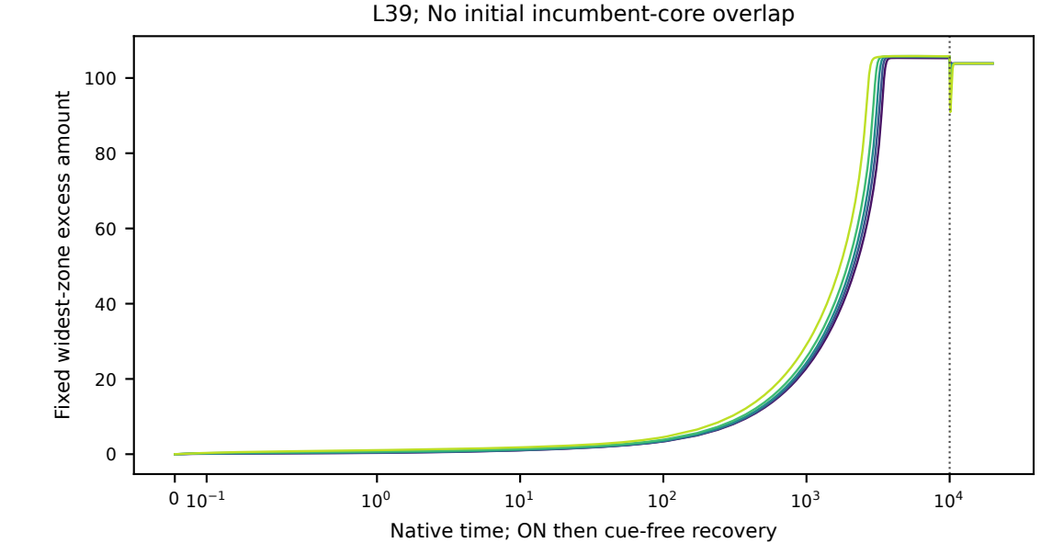






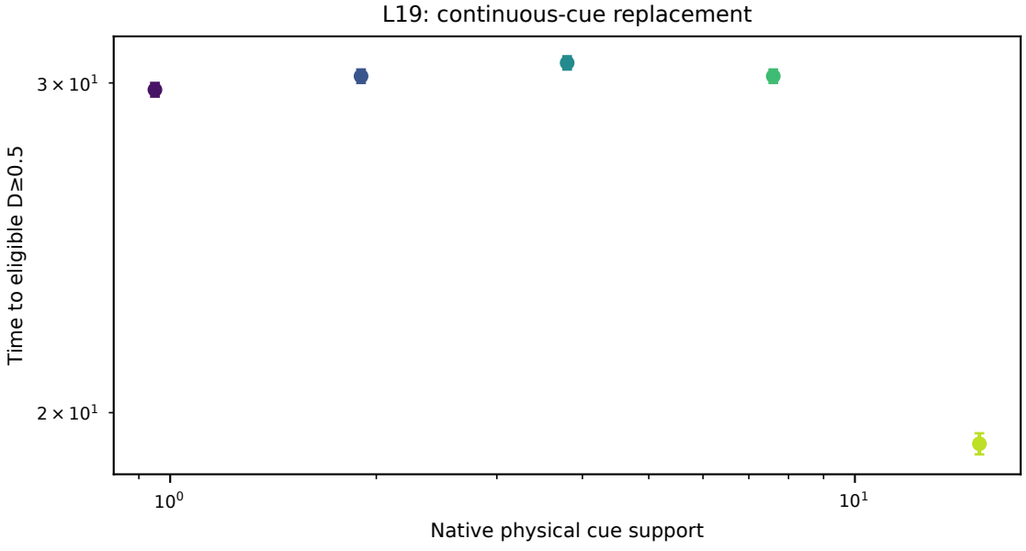


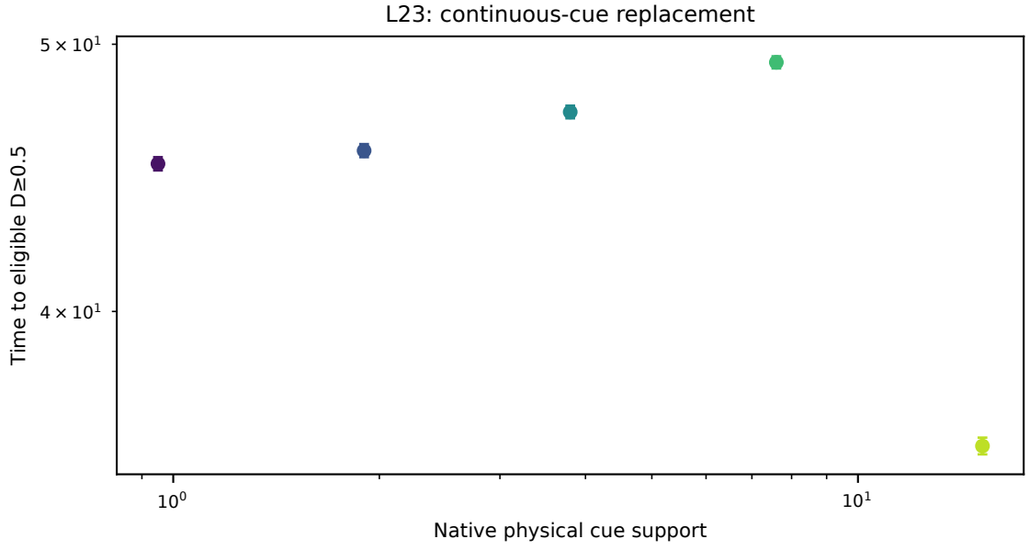


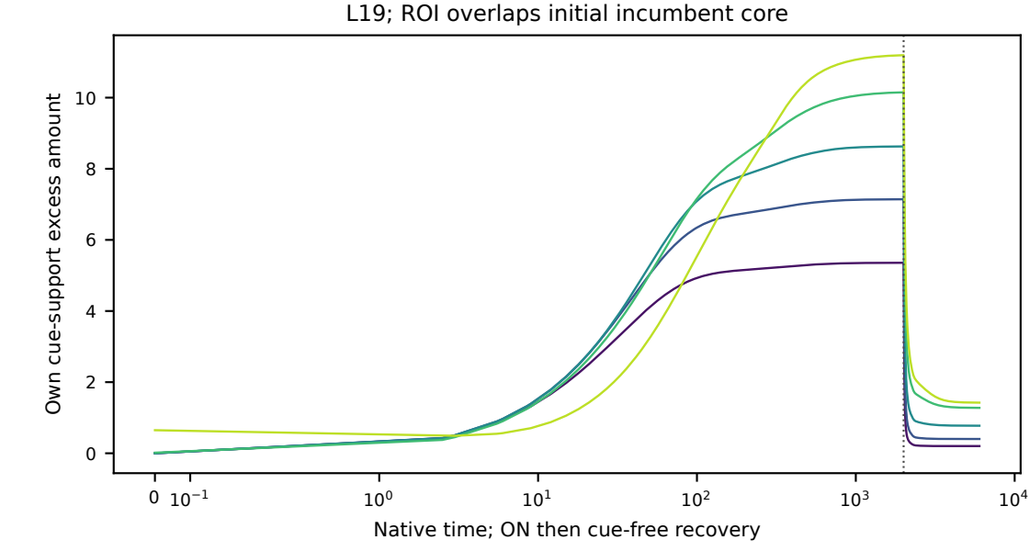


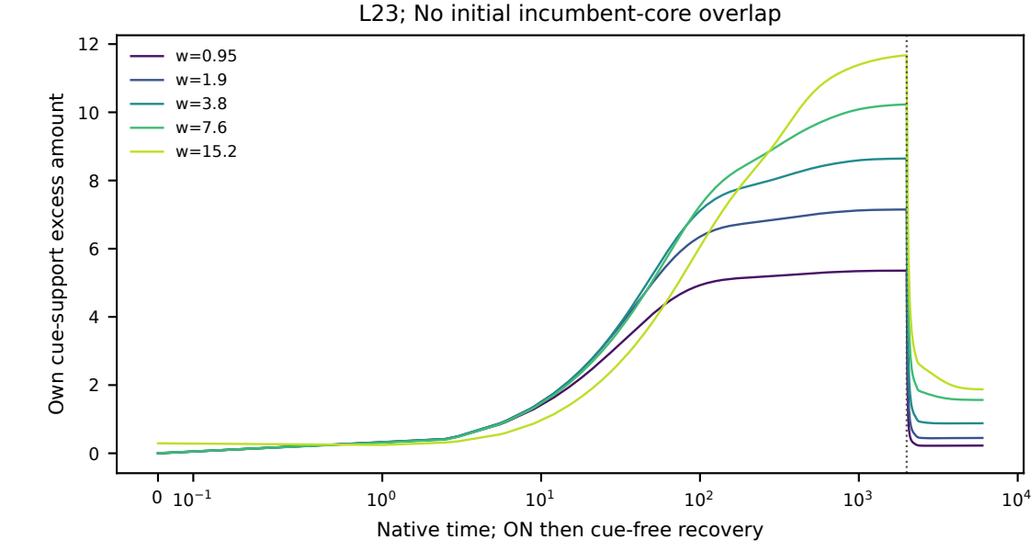
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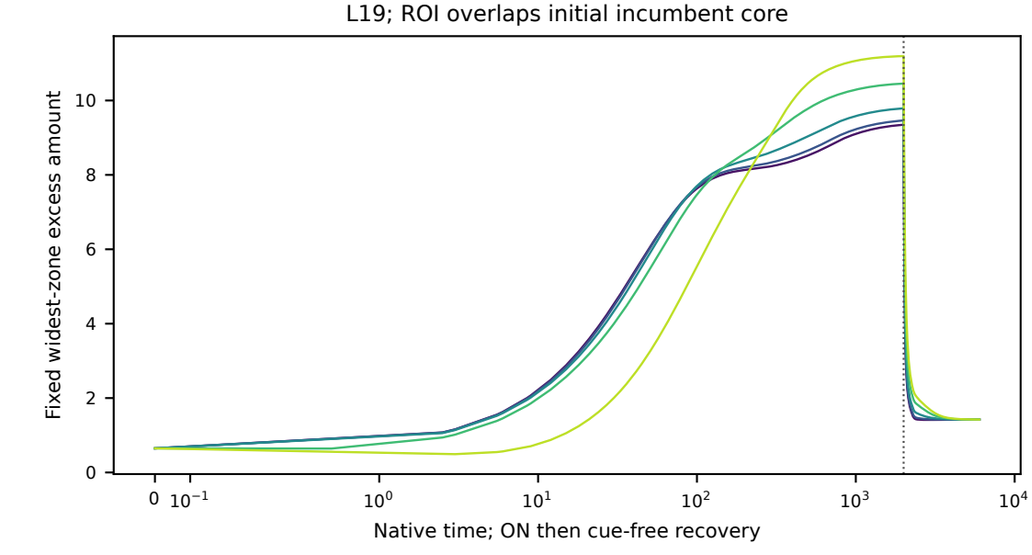
S1 deBelly • commanded area 0.5× • native kinetics

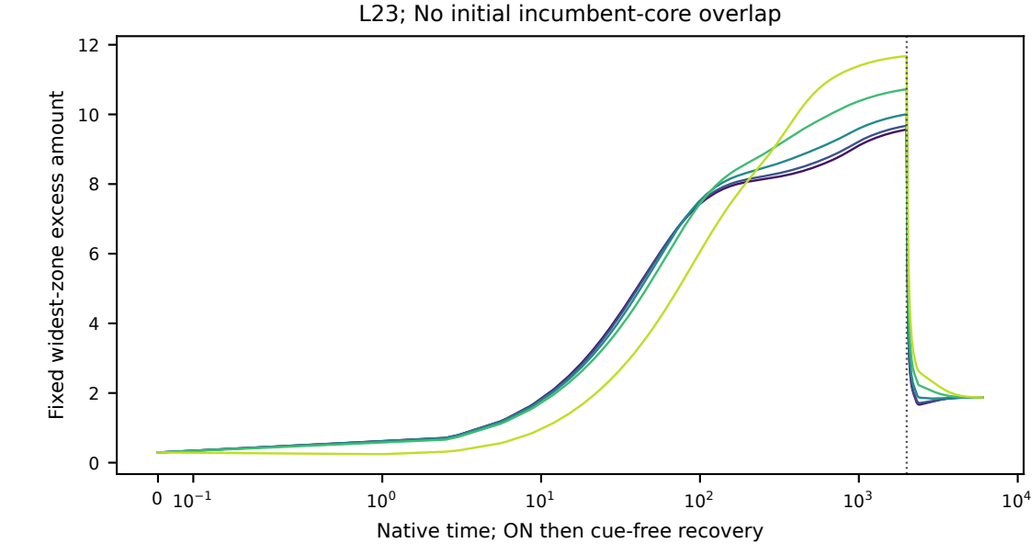






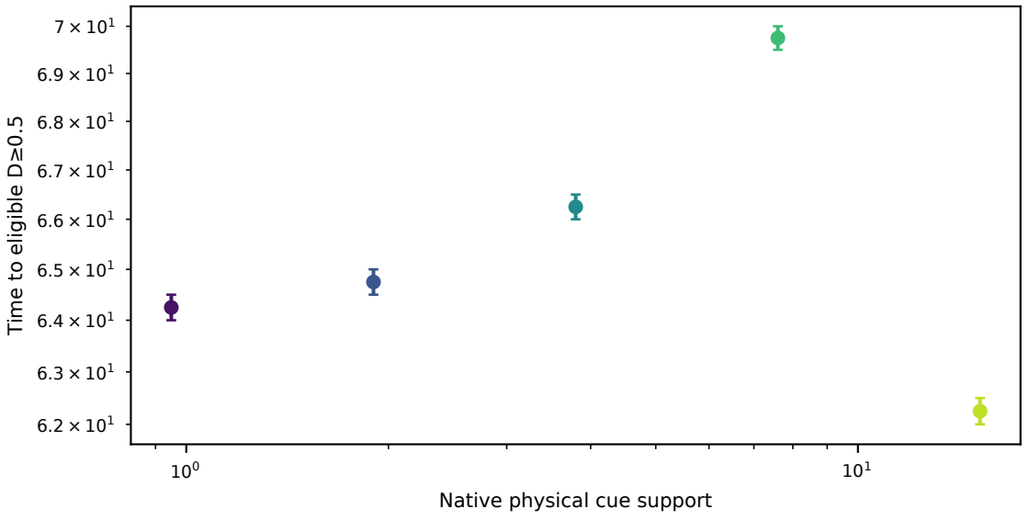


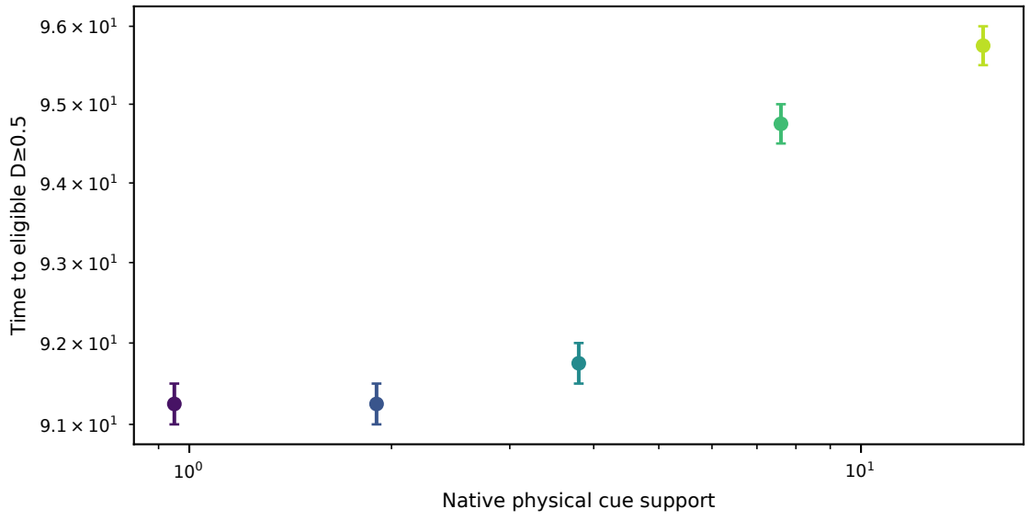


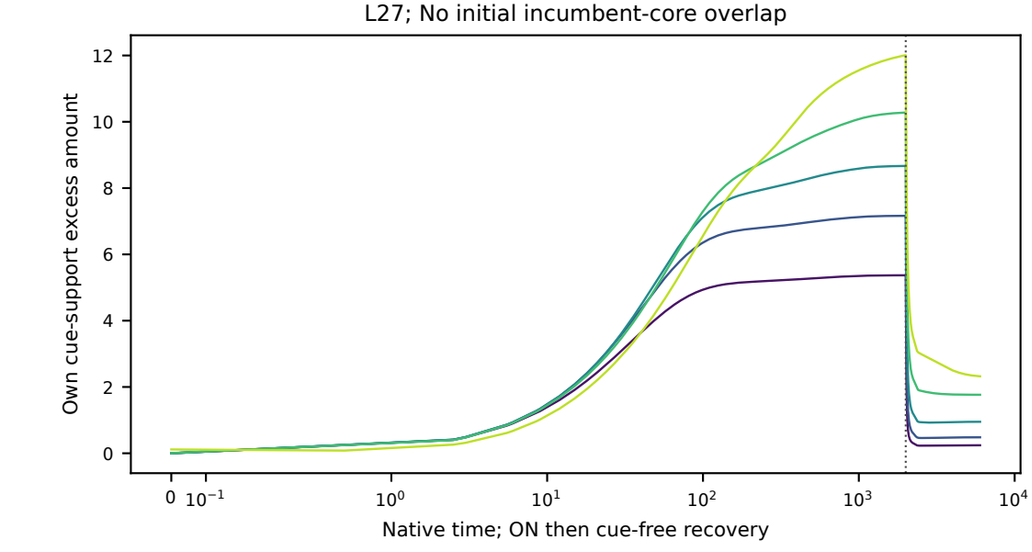


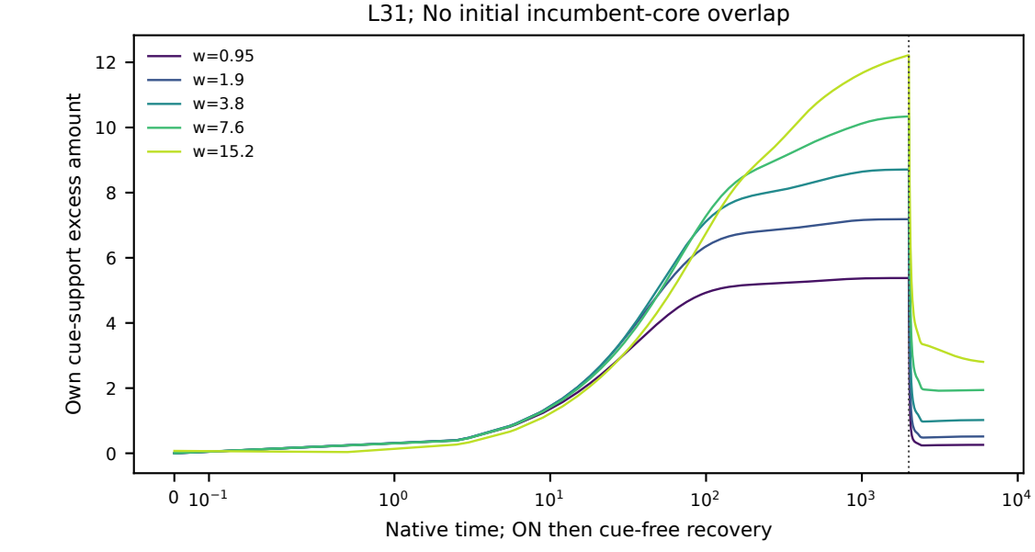
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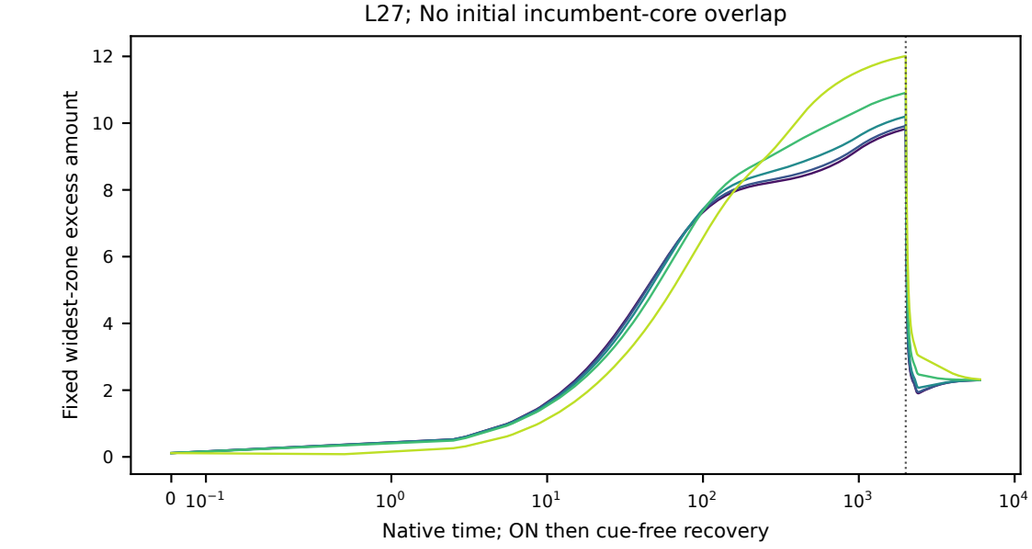
S1 deBelly • commanded area 0.5× • native kinetics

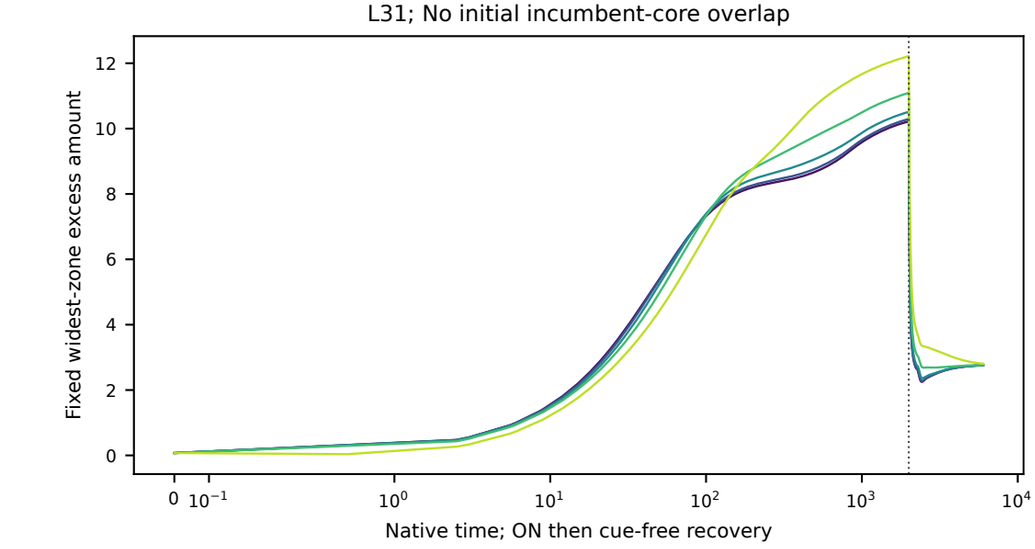






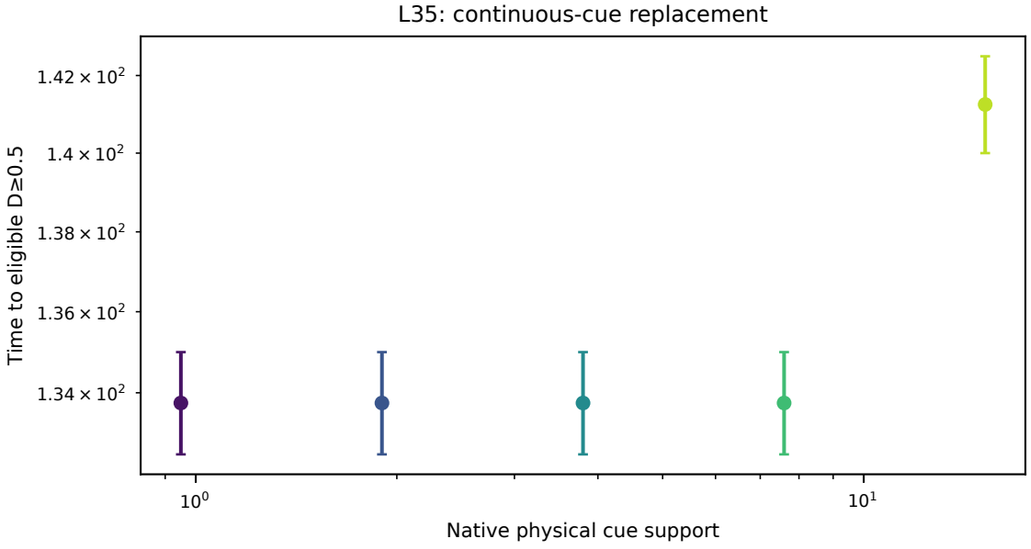


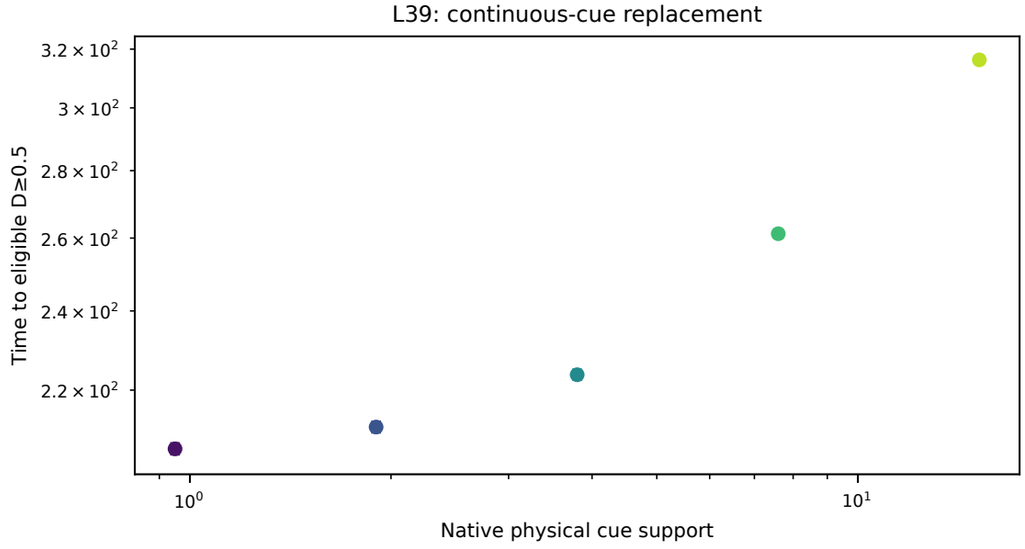


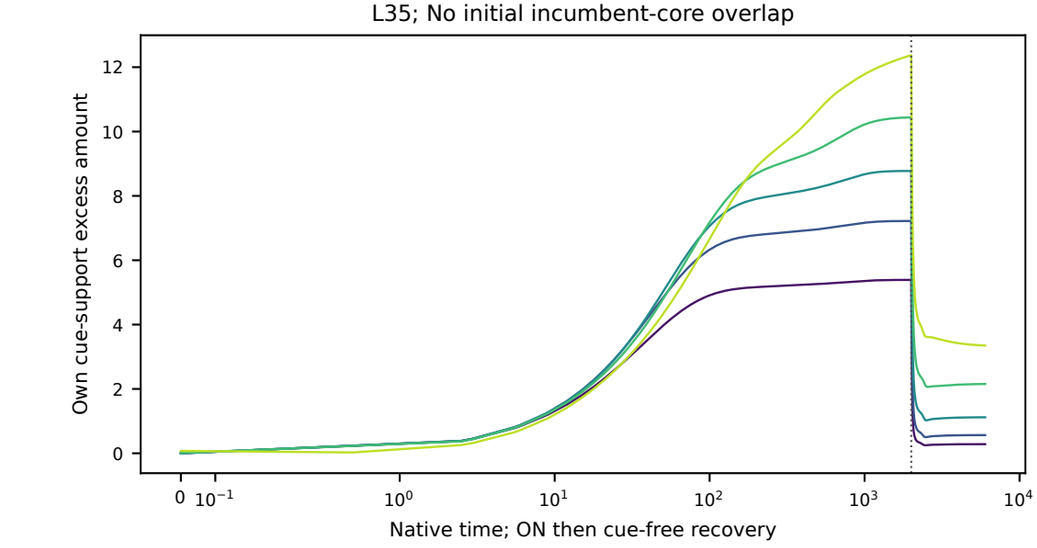


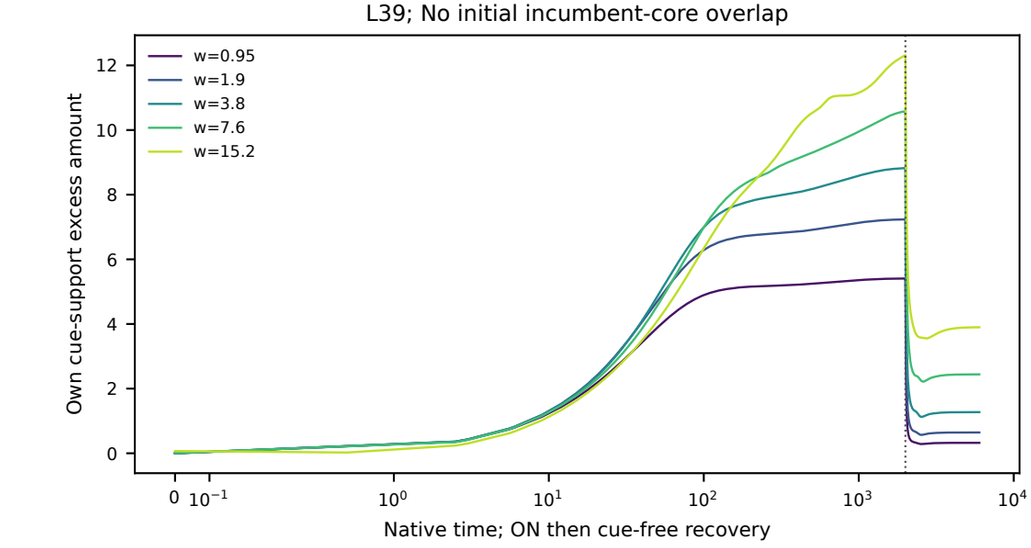
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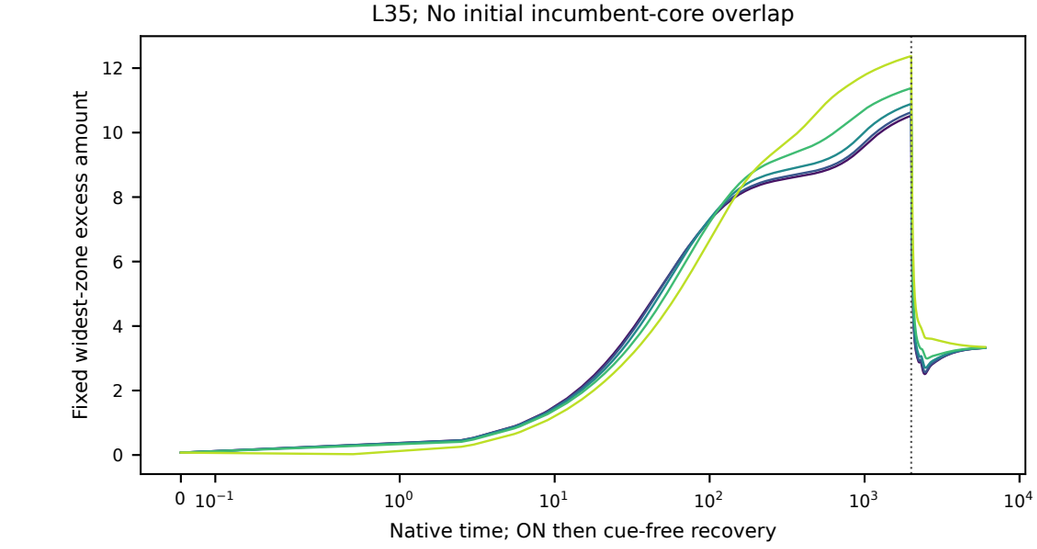
S1 deBelly • commanded area 0.5× • native kinetics

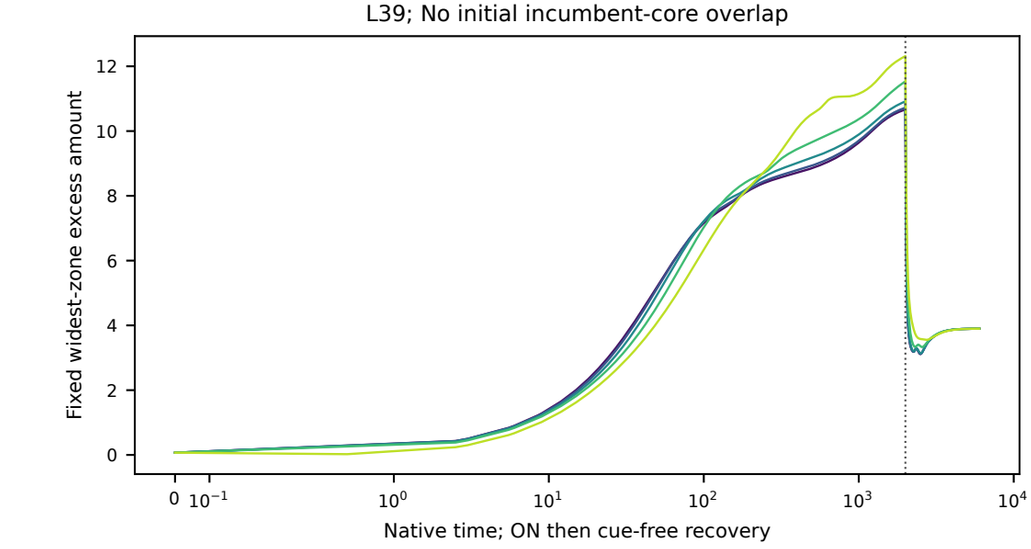






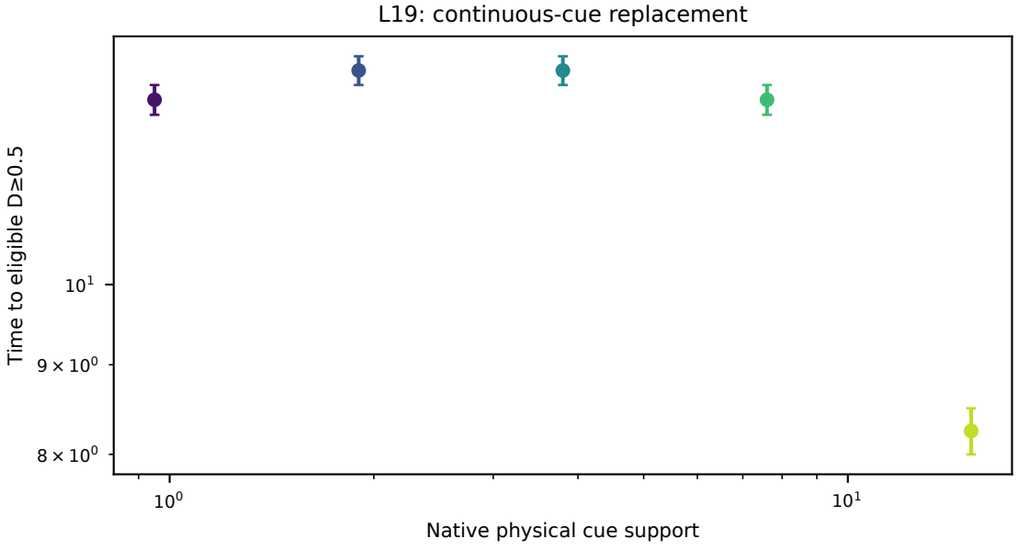


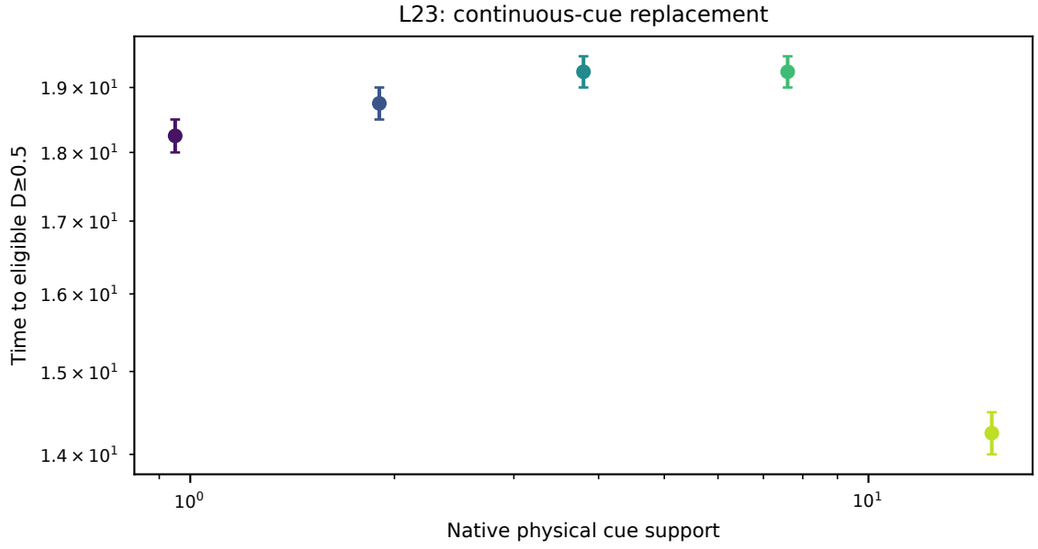


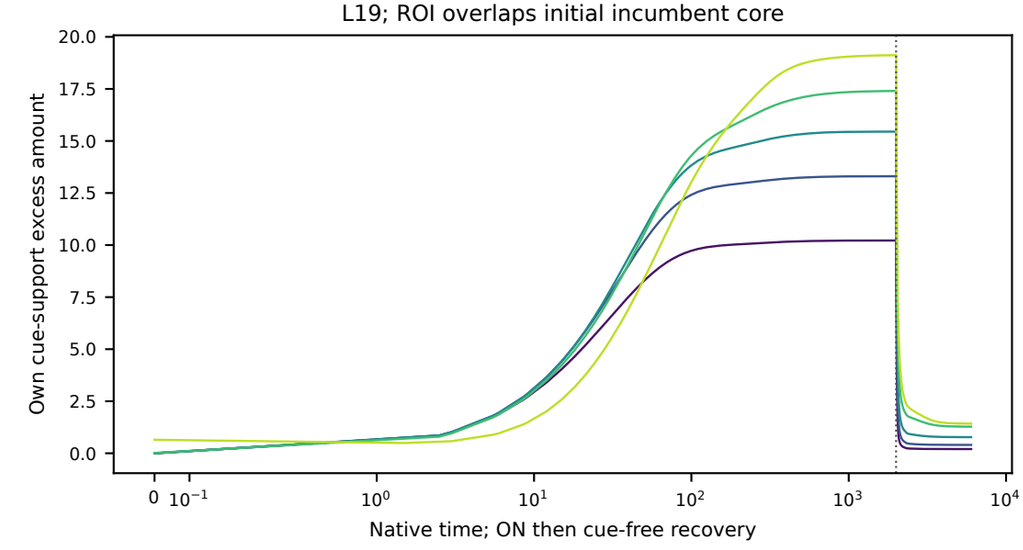


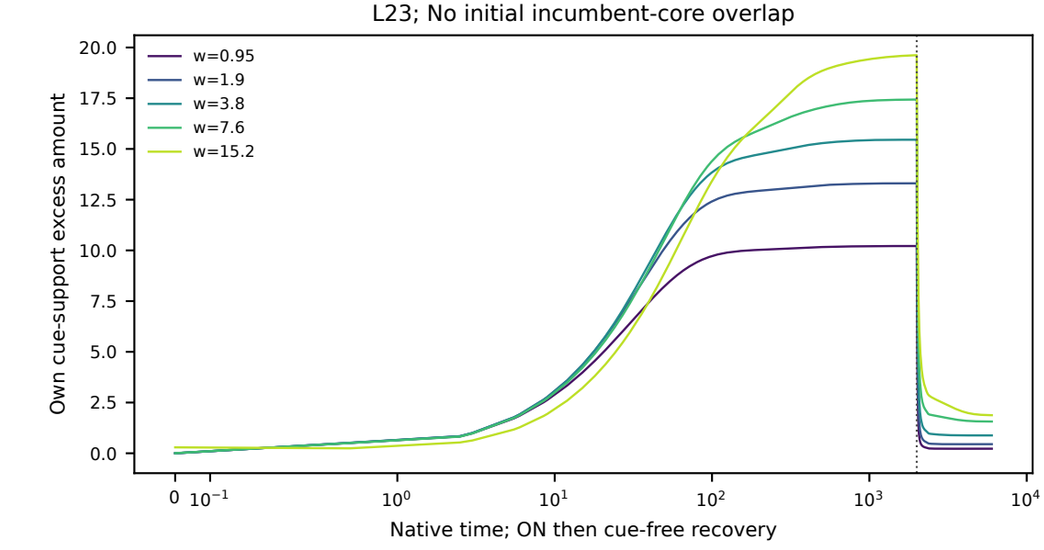
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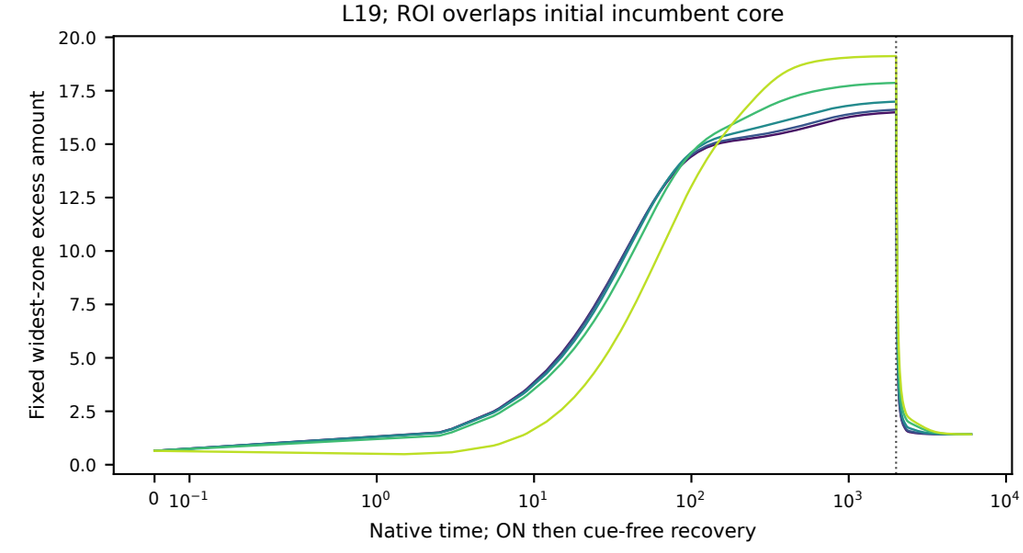
S1 deBelly • commanded area 1× • native kinetics

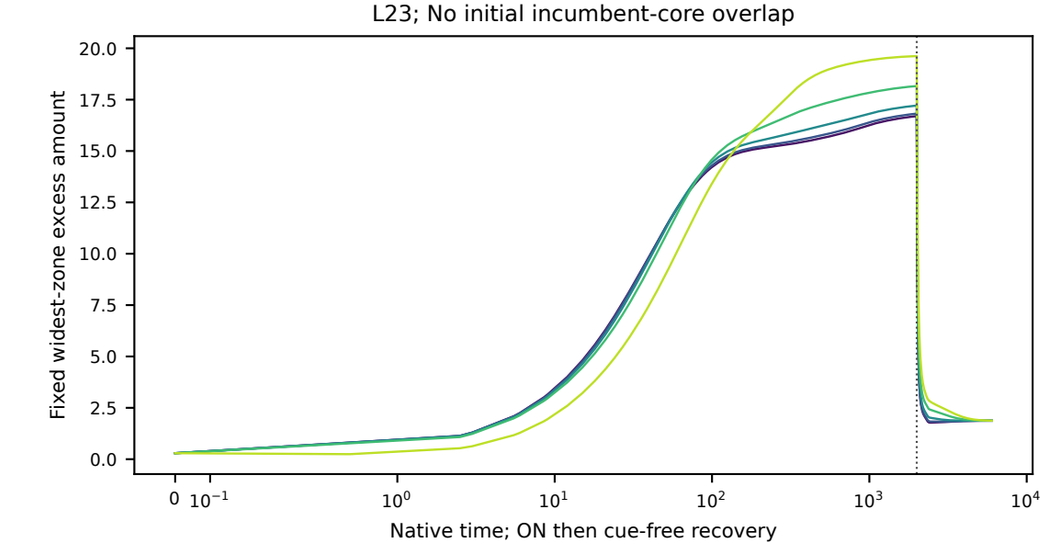






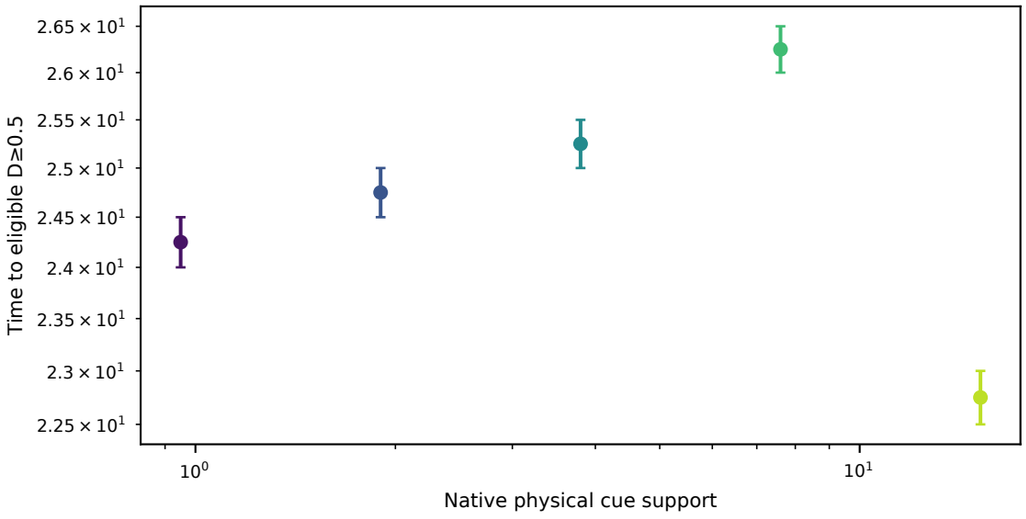


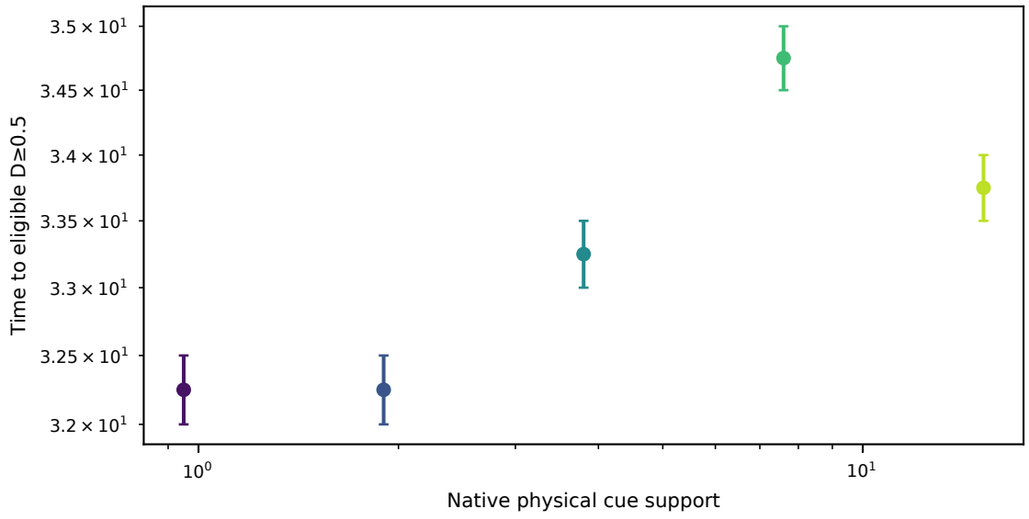


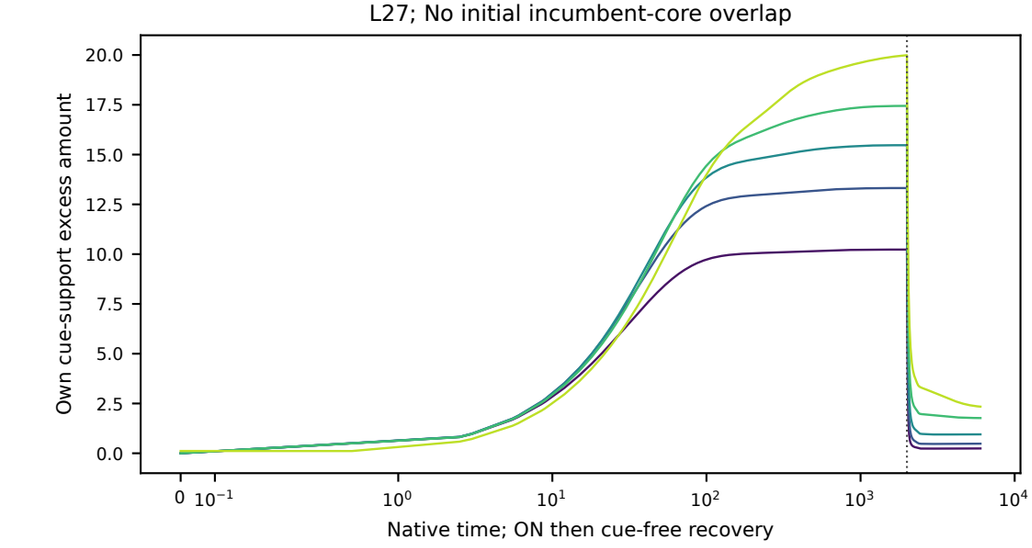


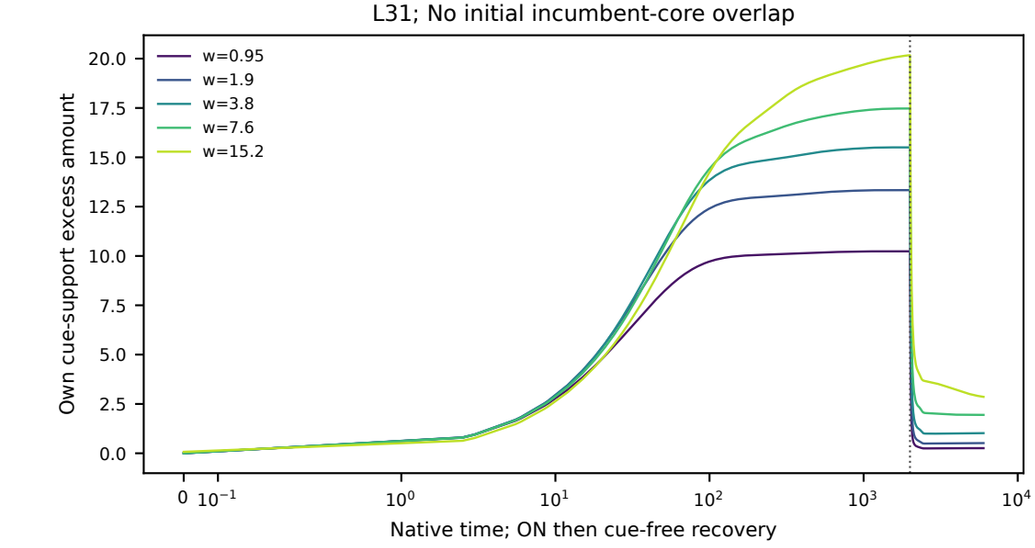
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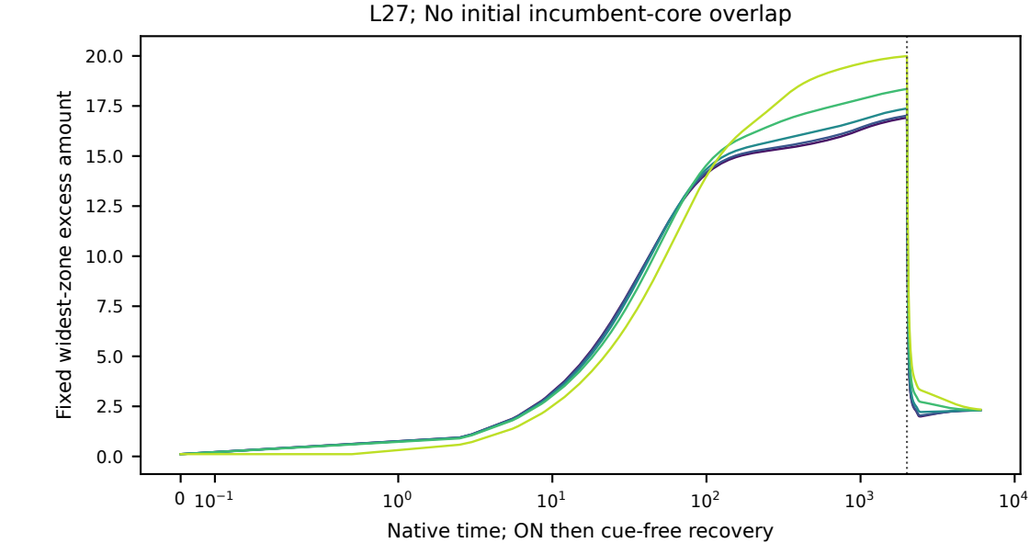
S1 deBelly • commanded area 1× • native kinetics

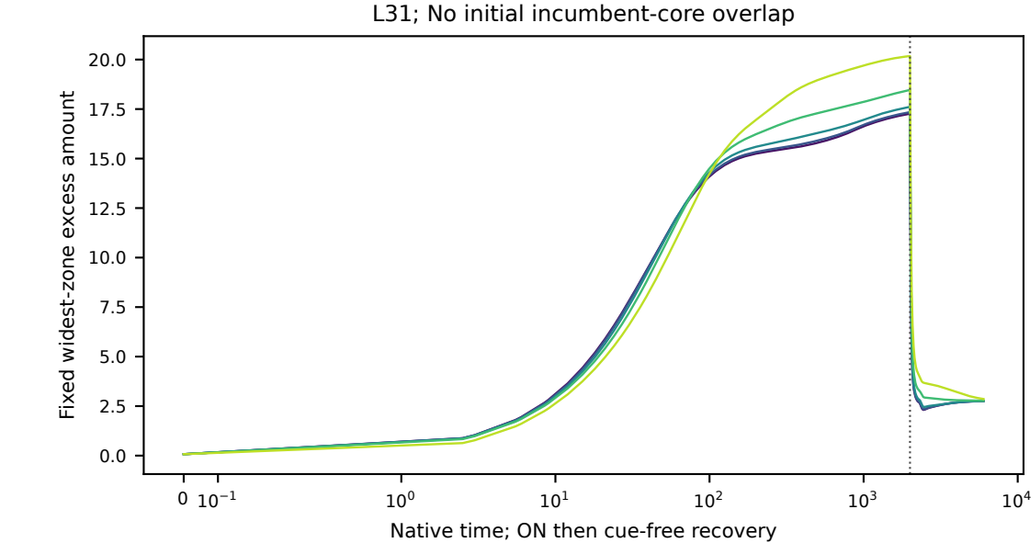






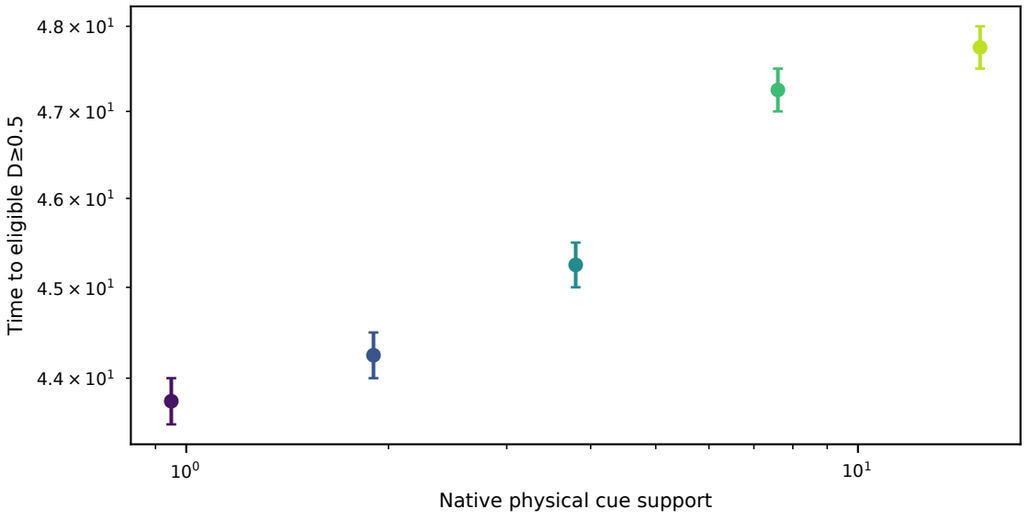


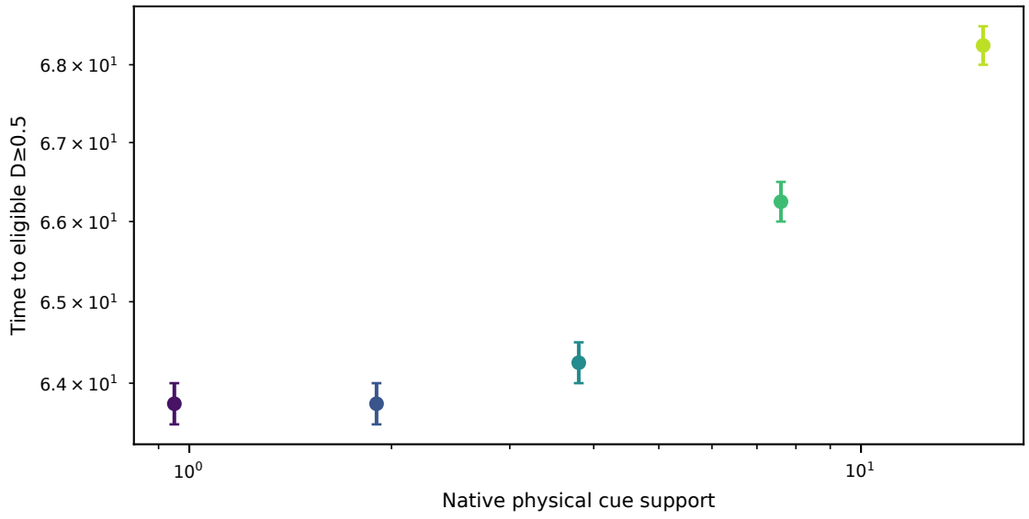


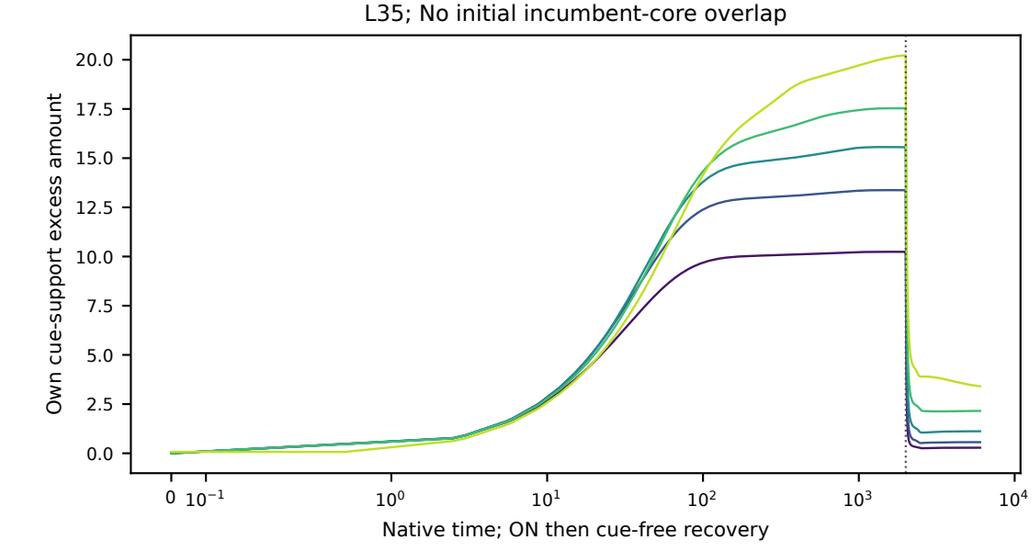


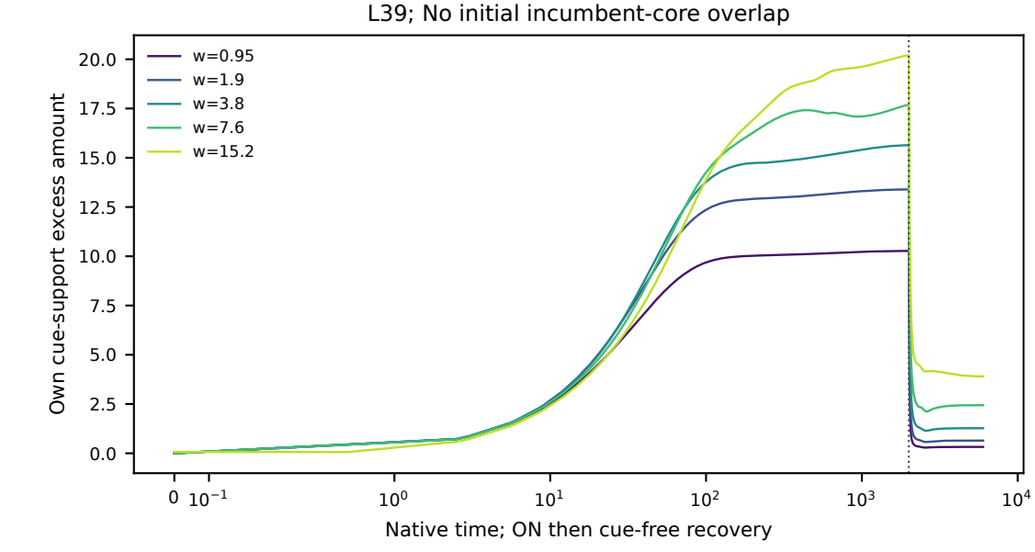
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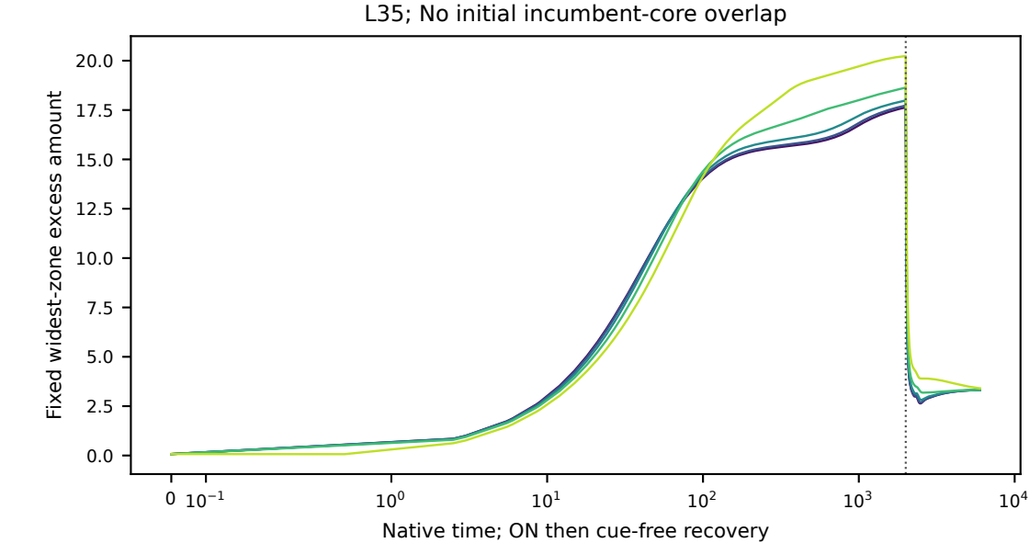
S1 deBelly • commanded area 1× • native kinetics

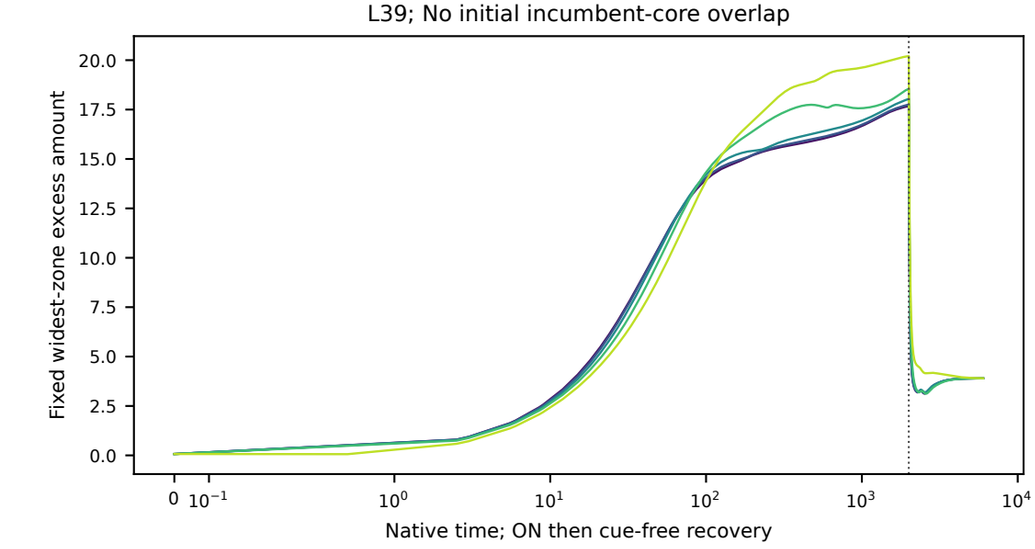






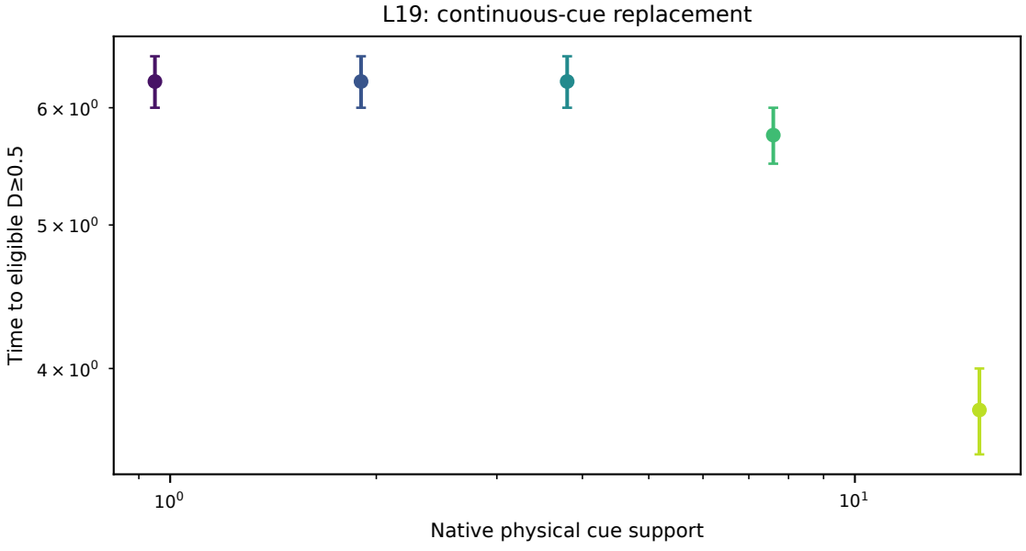


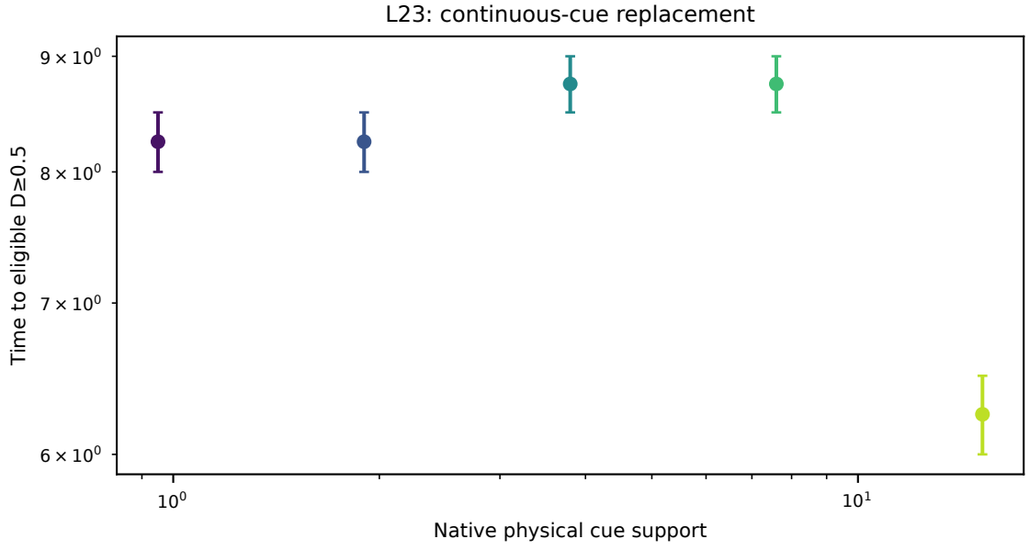


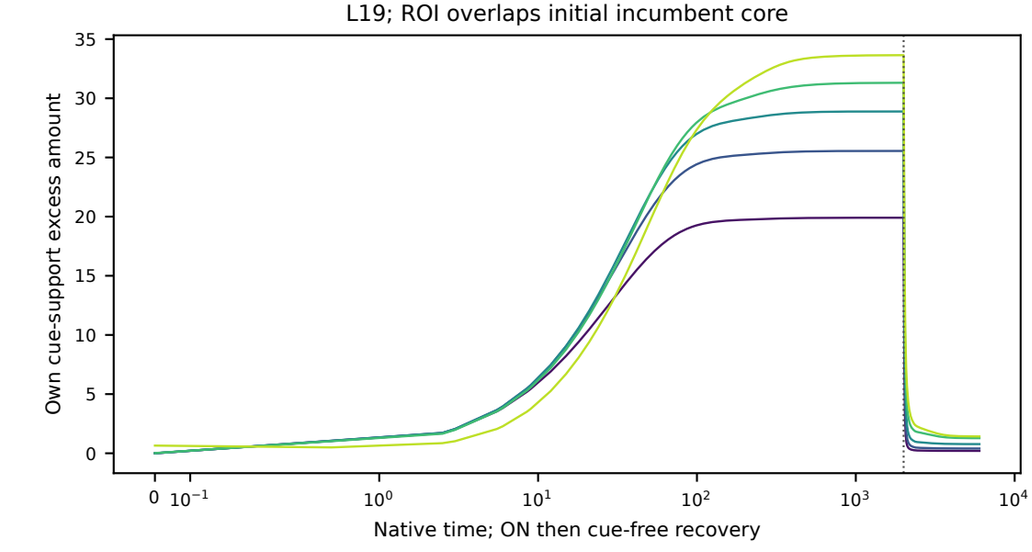


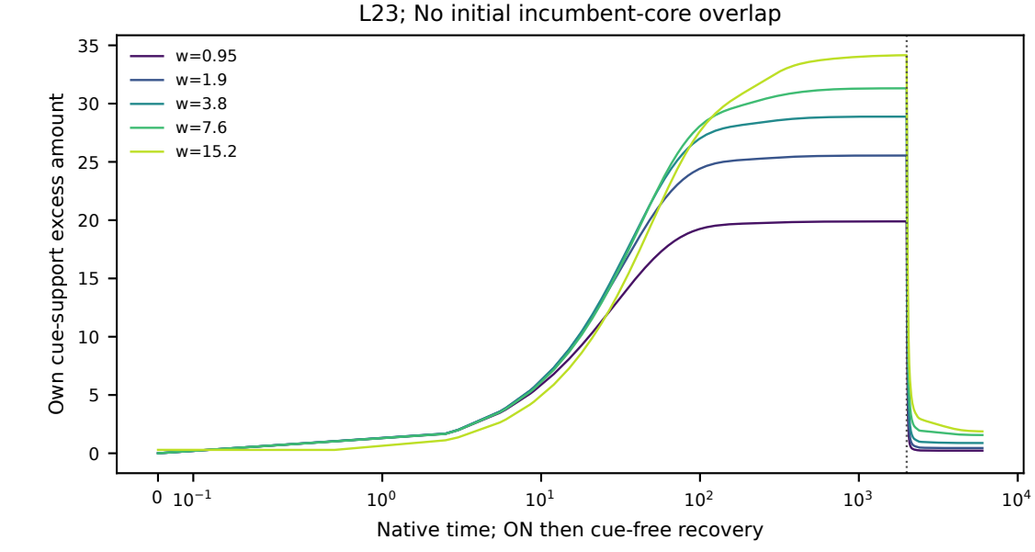
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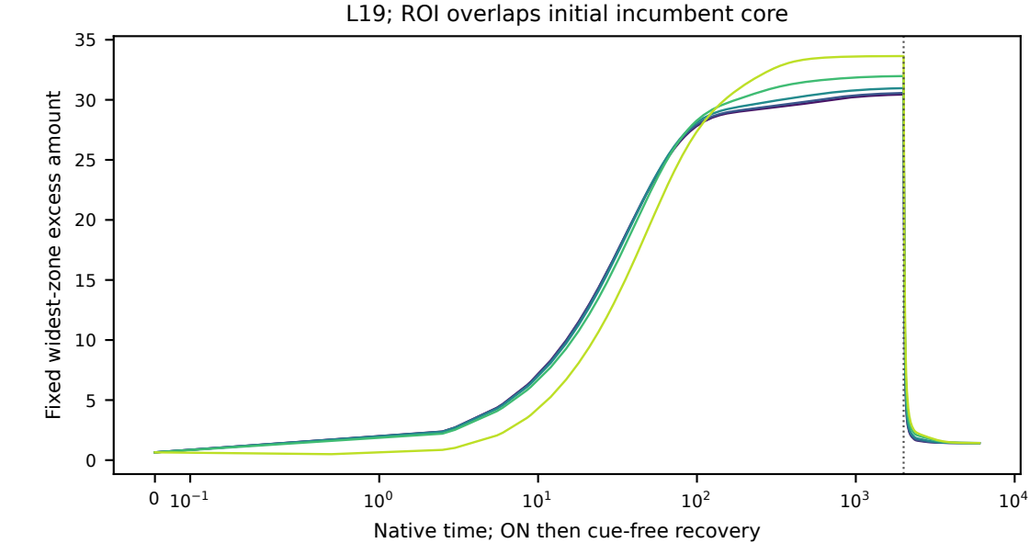
S1 deBelly • commanded area 2× • native kinetics

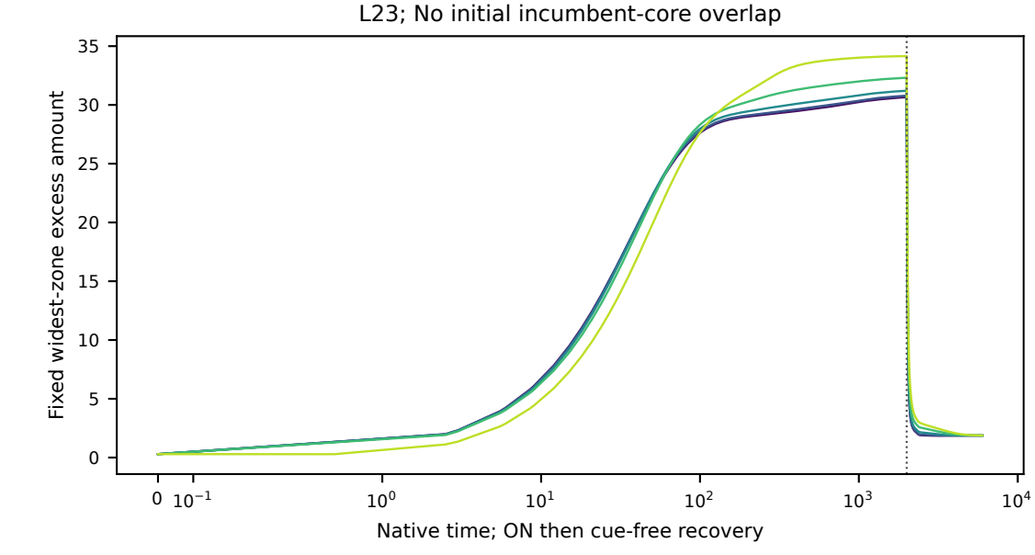






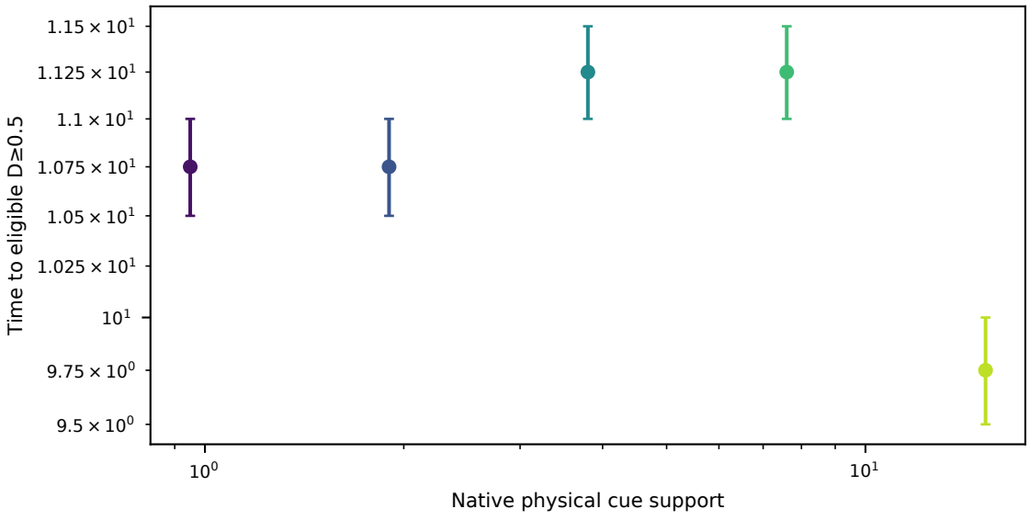


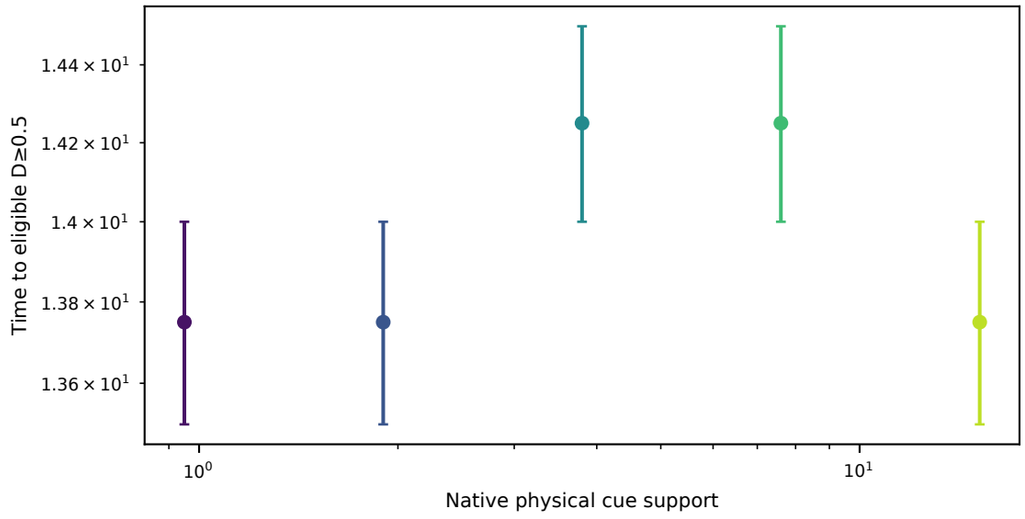


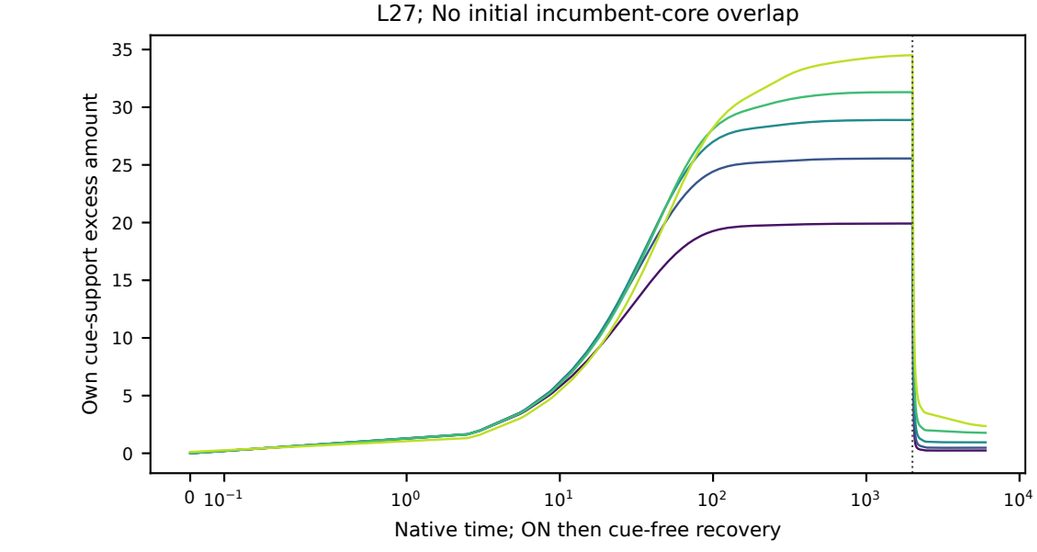


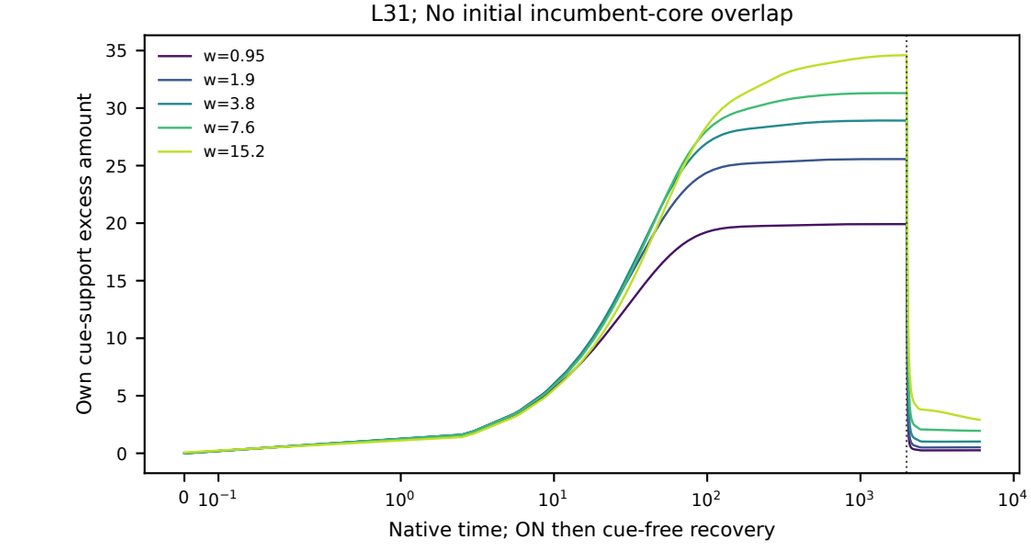
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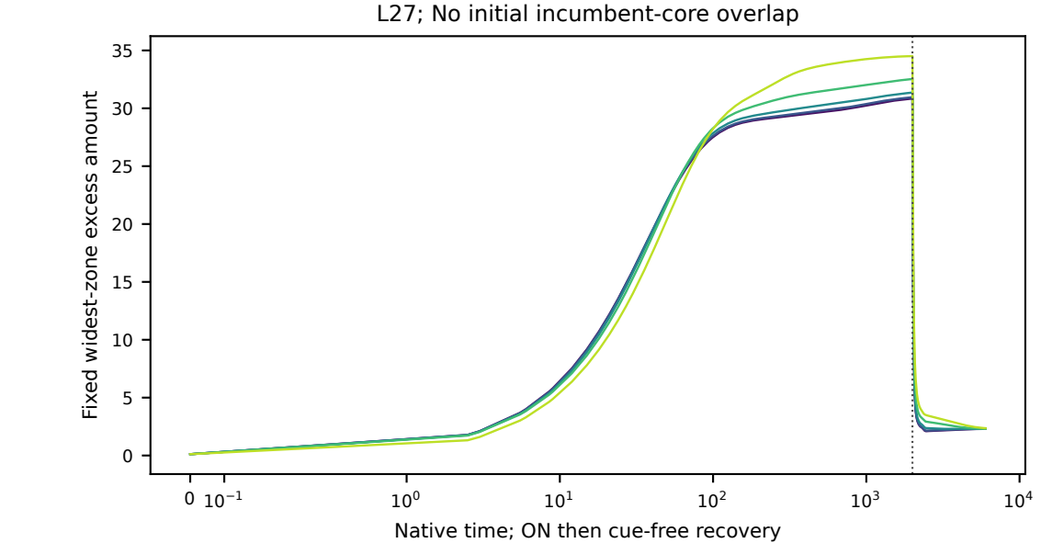
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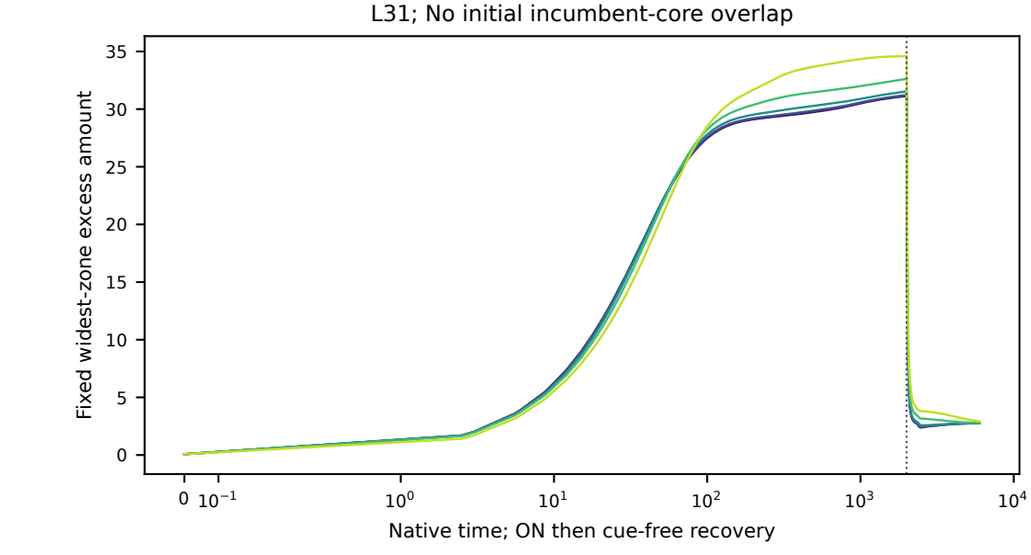






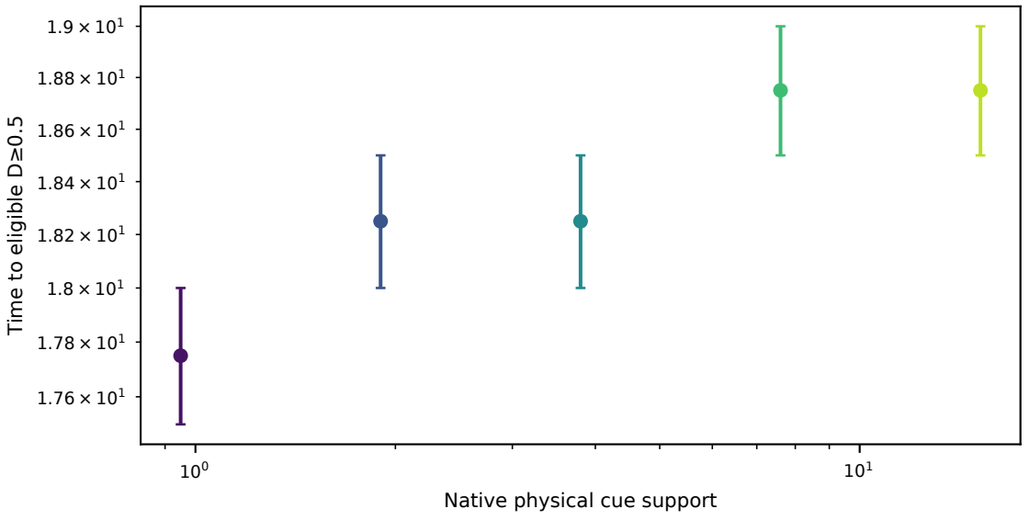


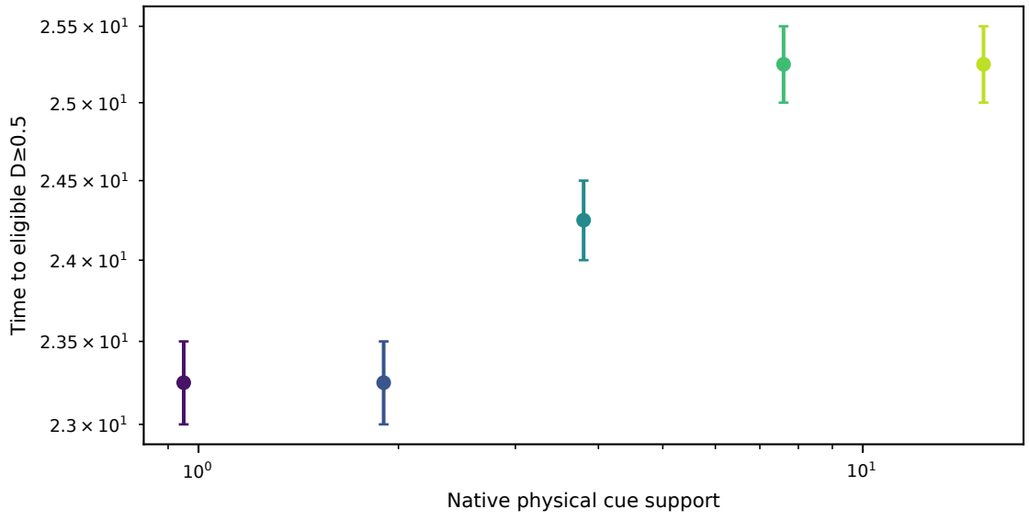


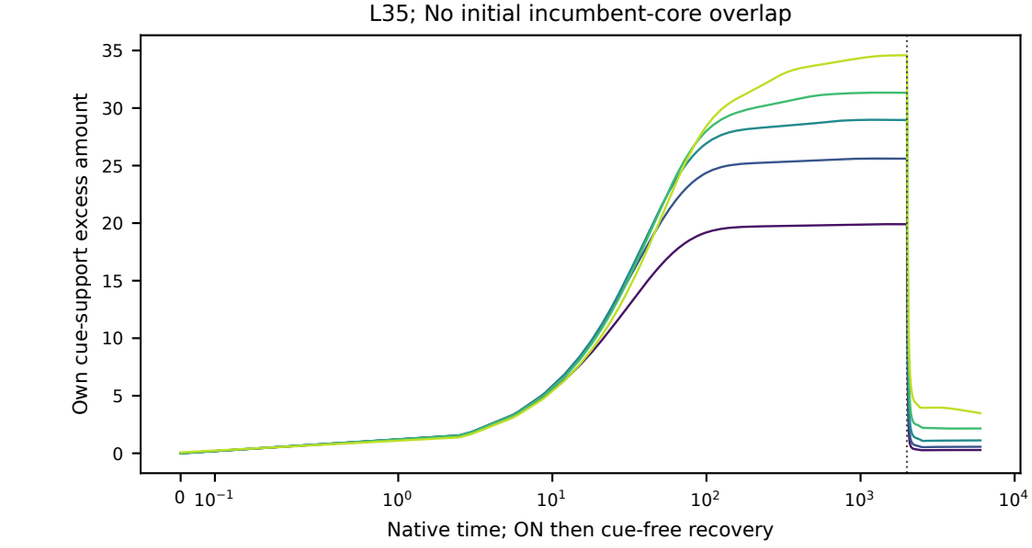


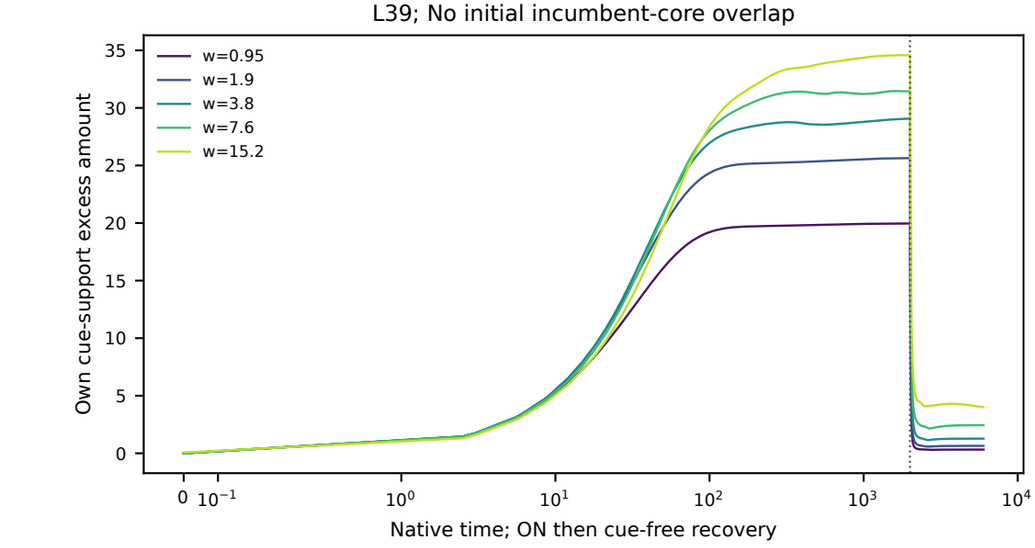
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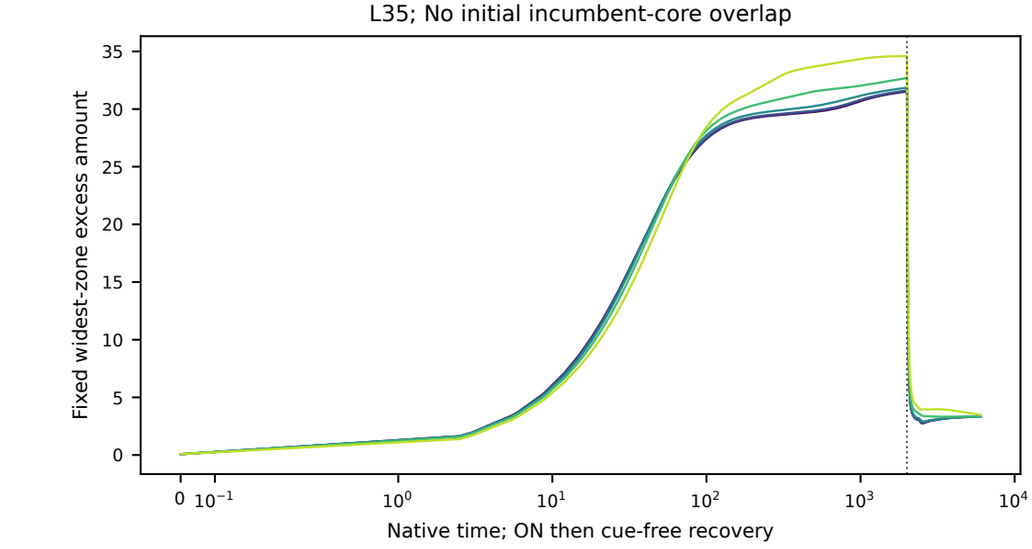
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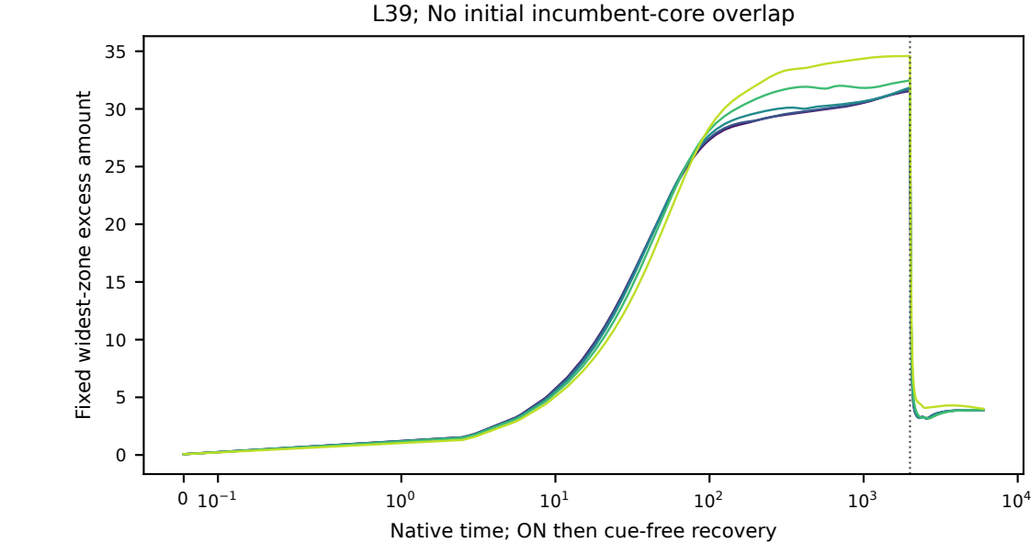






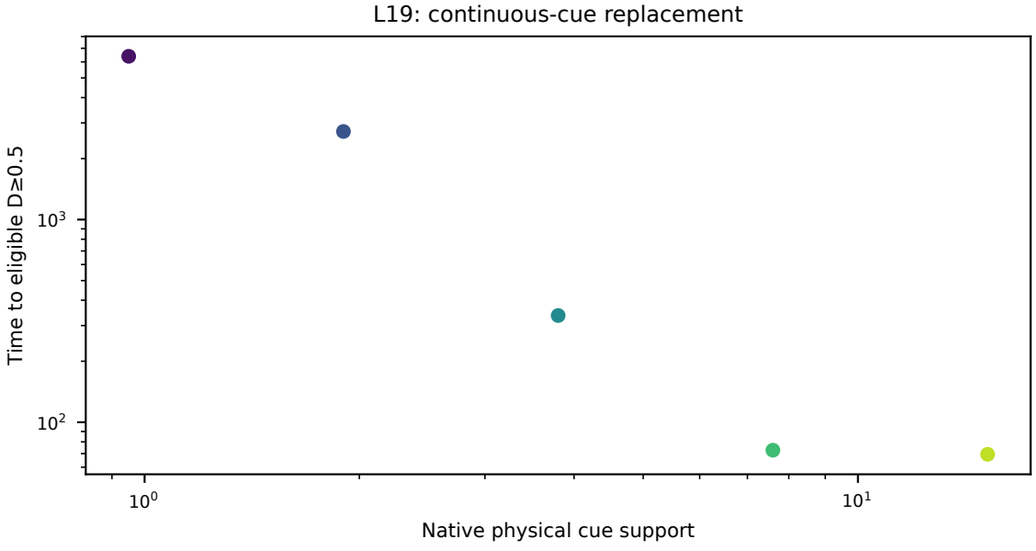


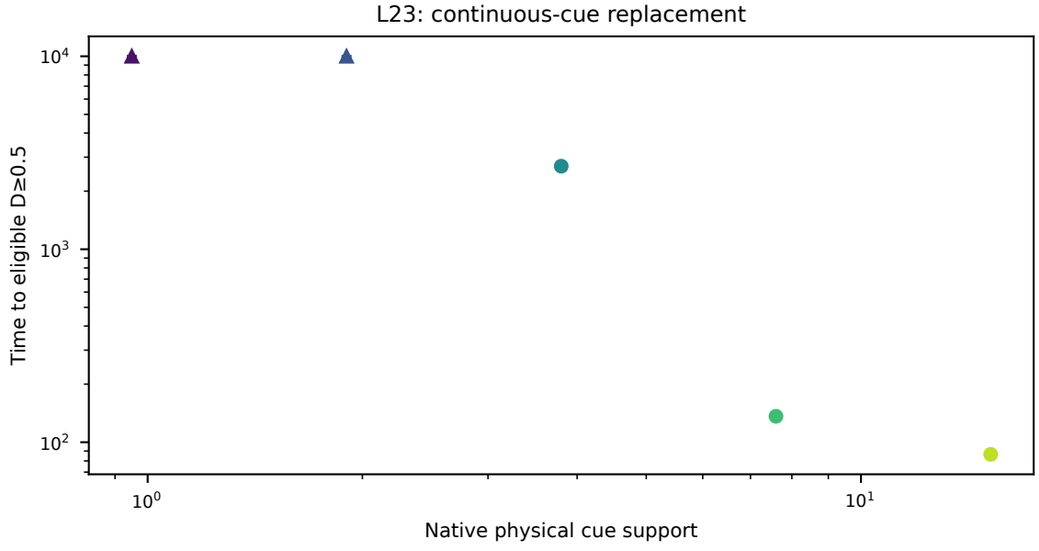


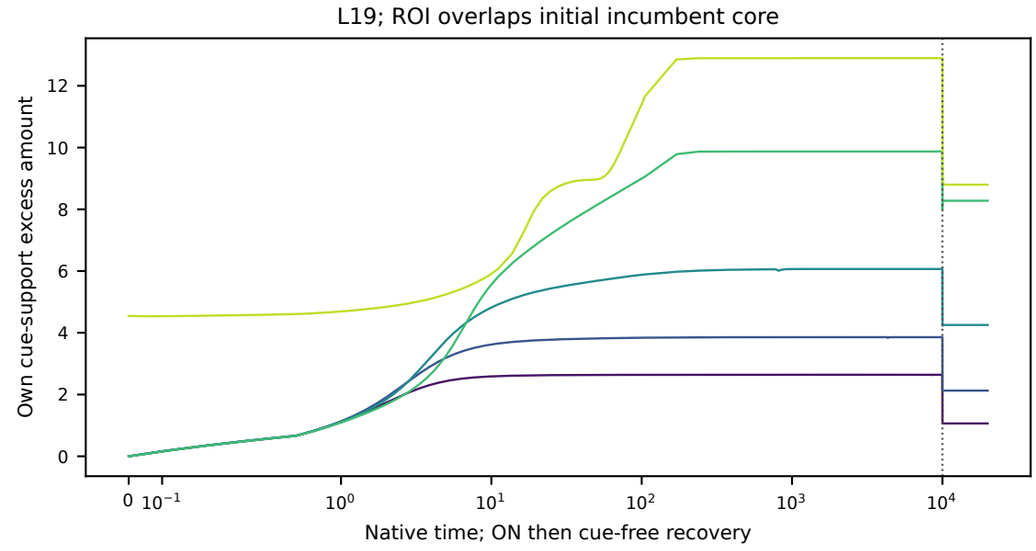


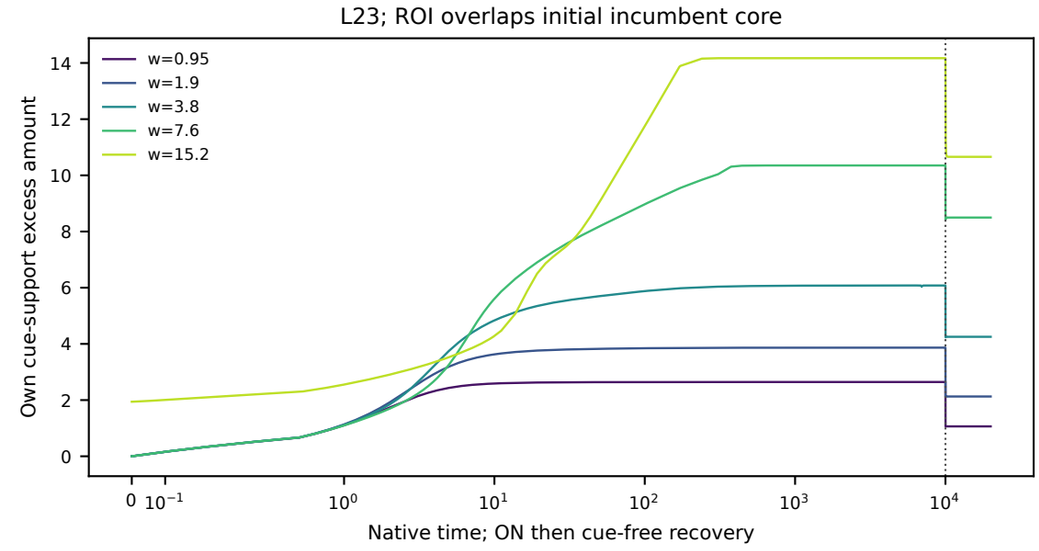
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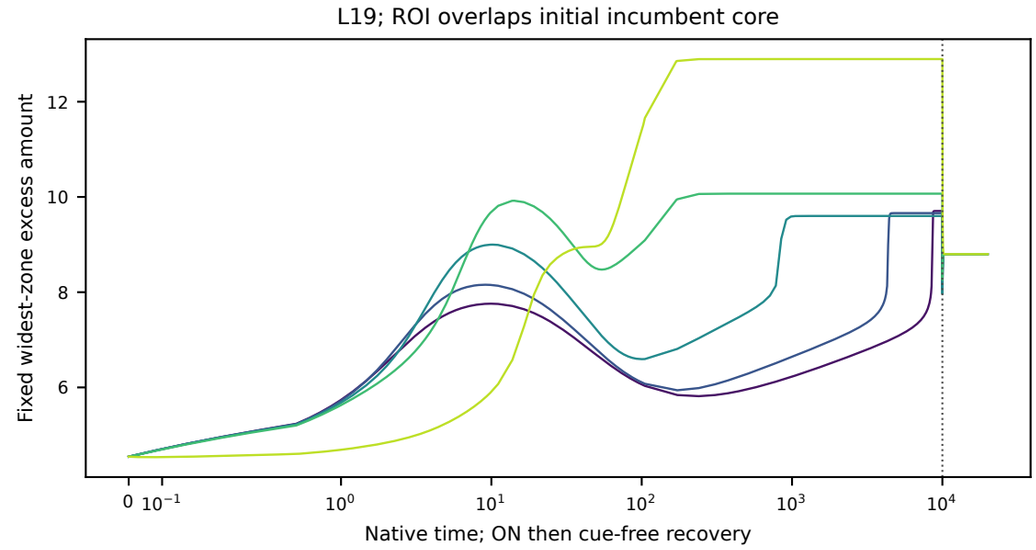
S1 Holmes 3 • commanded area 0.5× • native kinetics

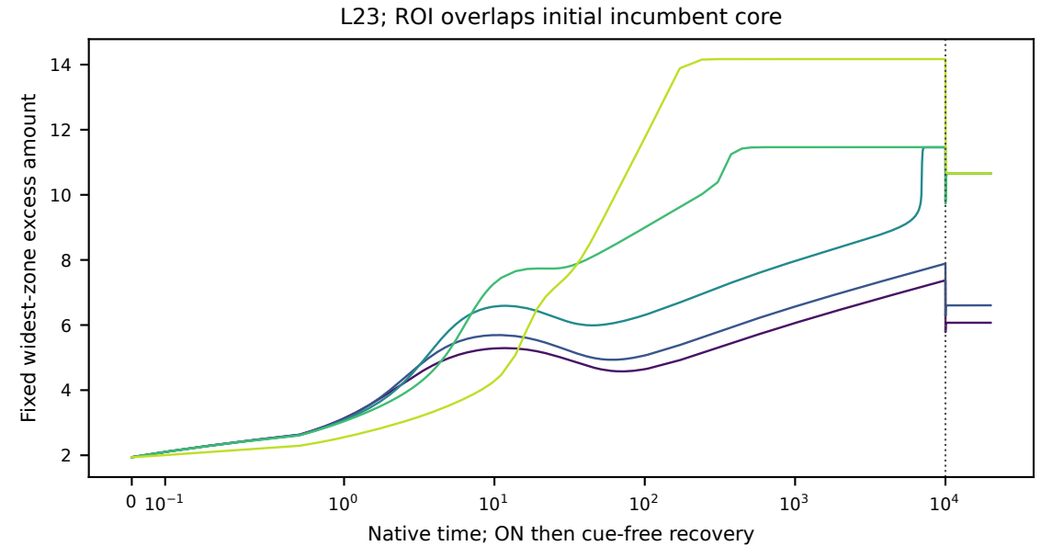






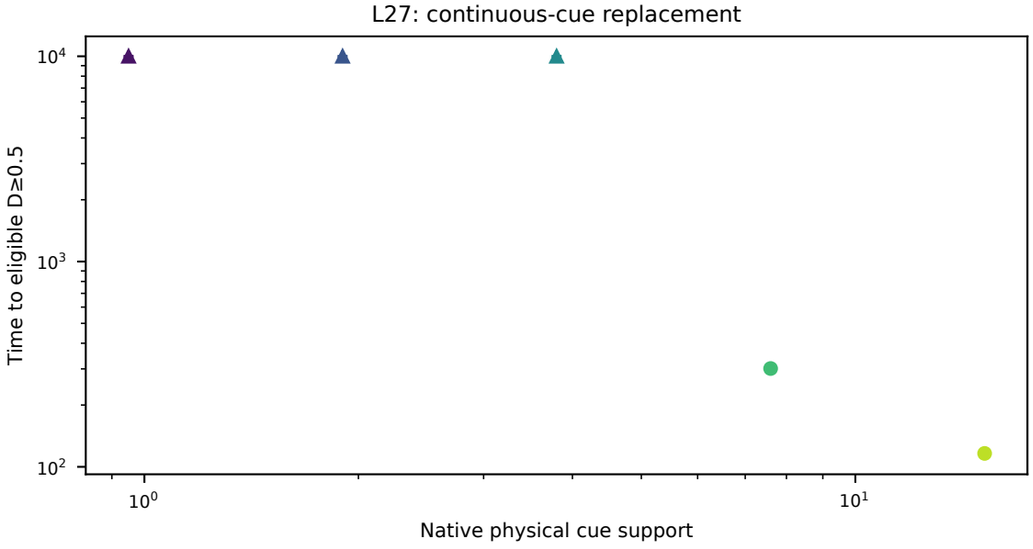


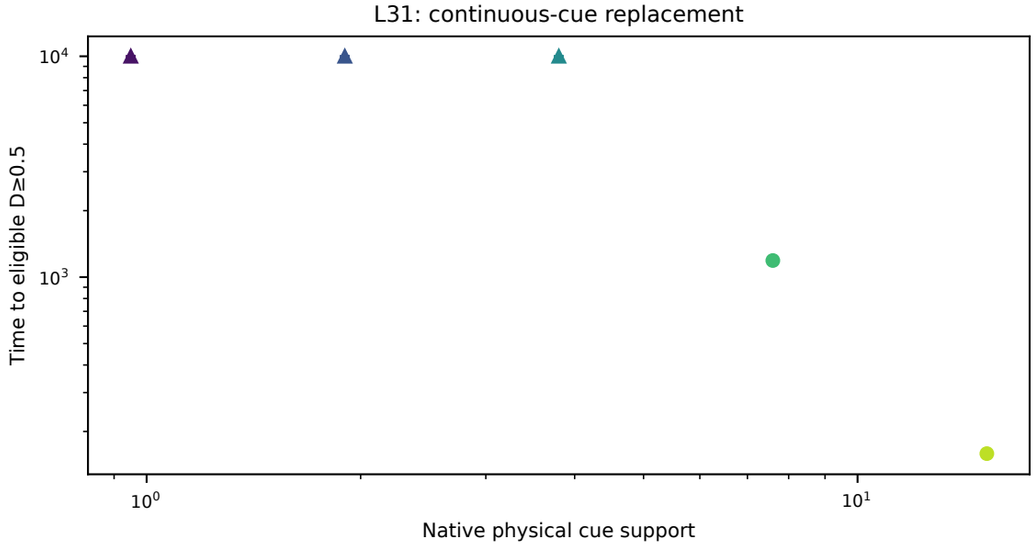


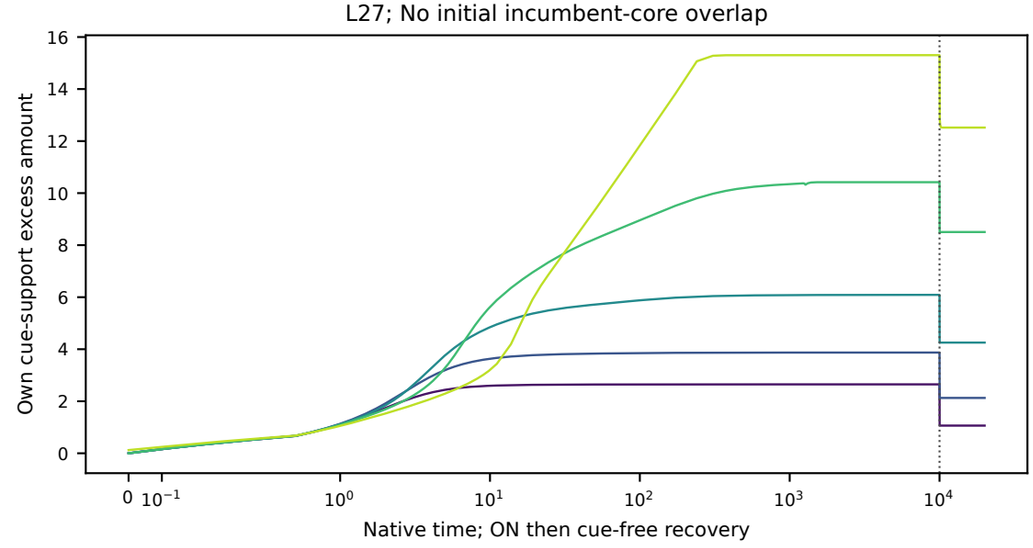


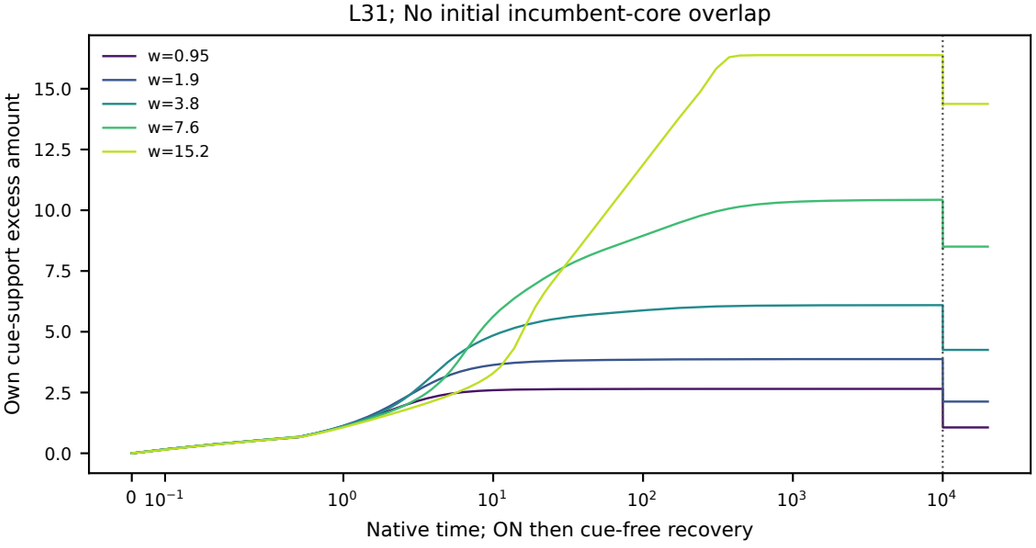
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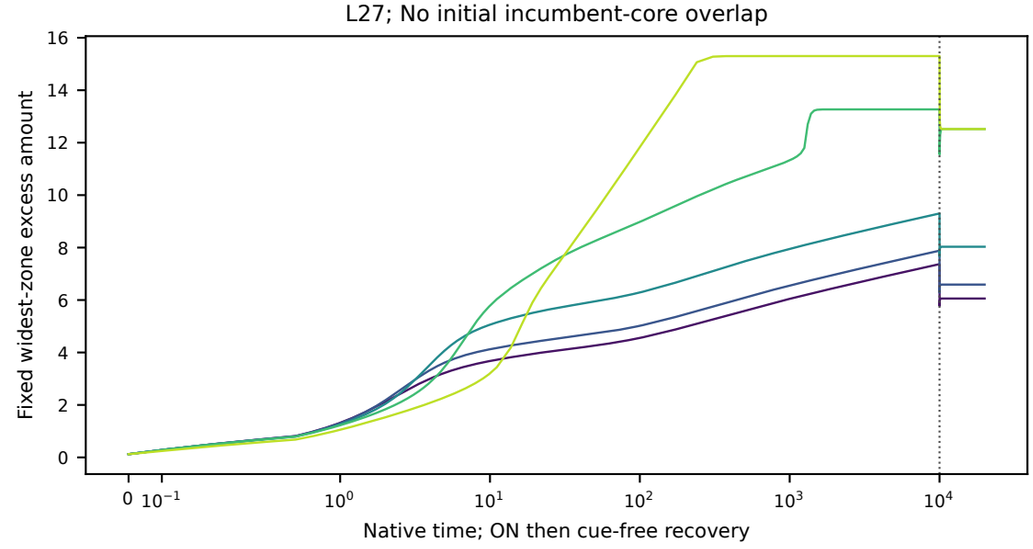
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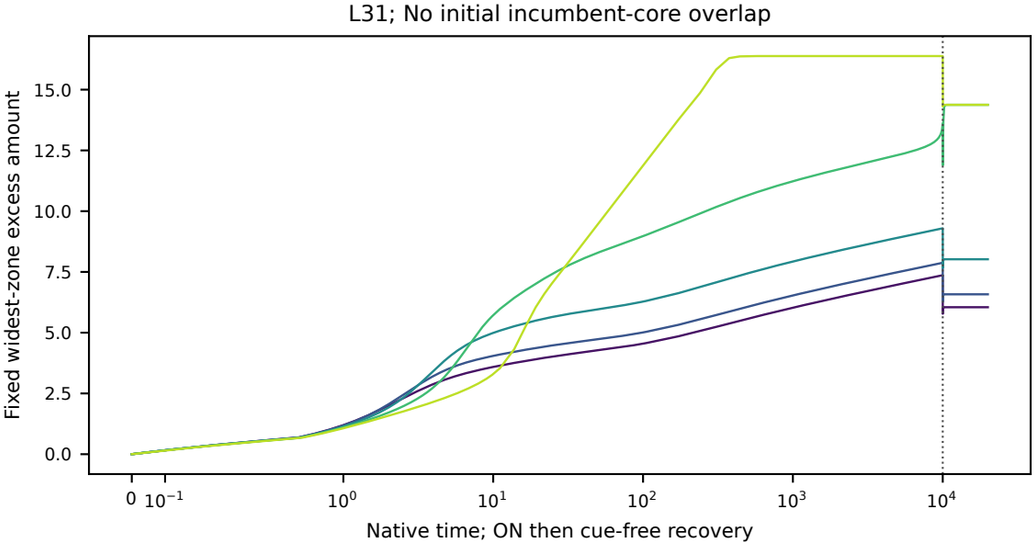






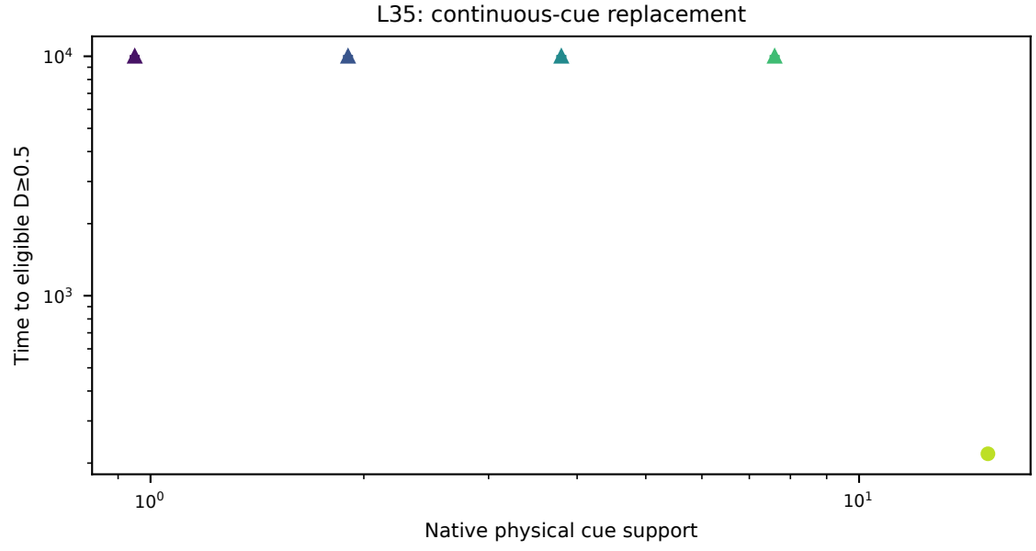


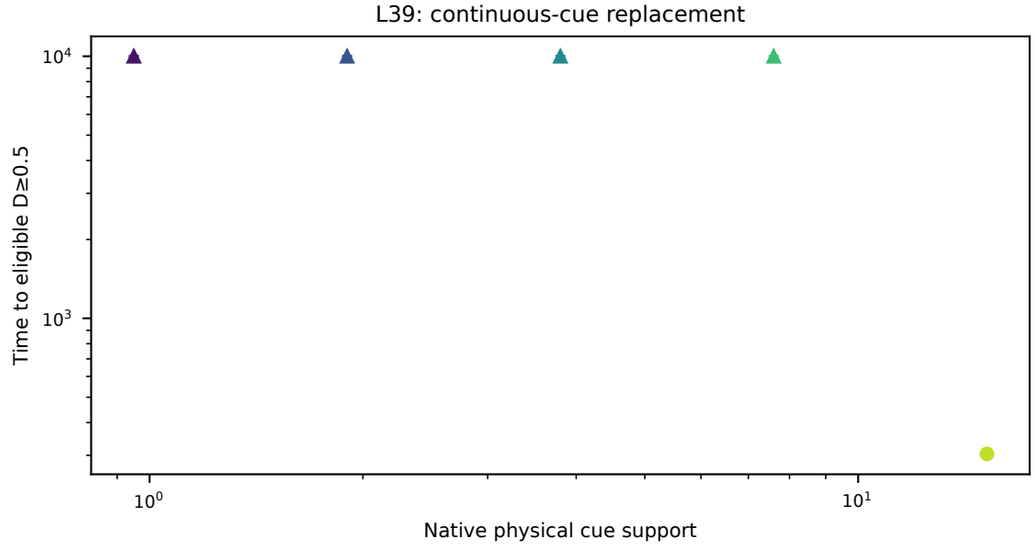


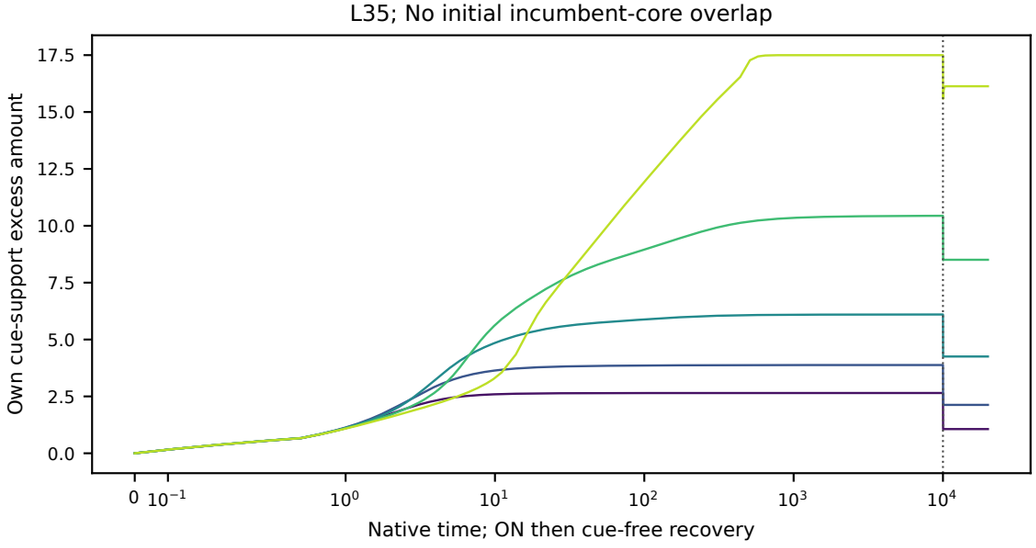


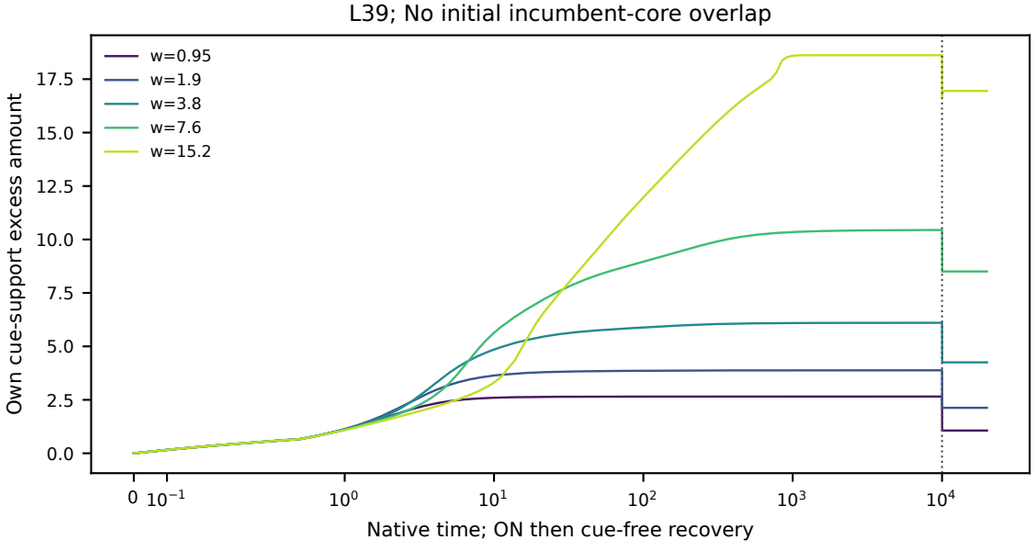
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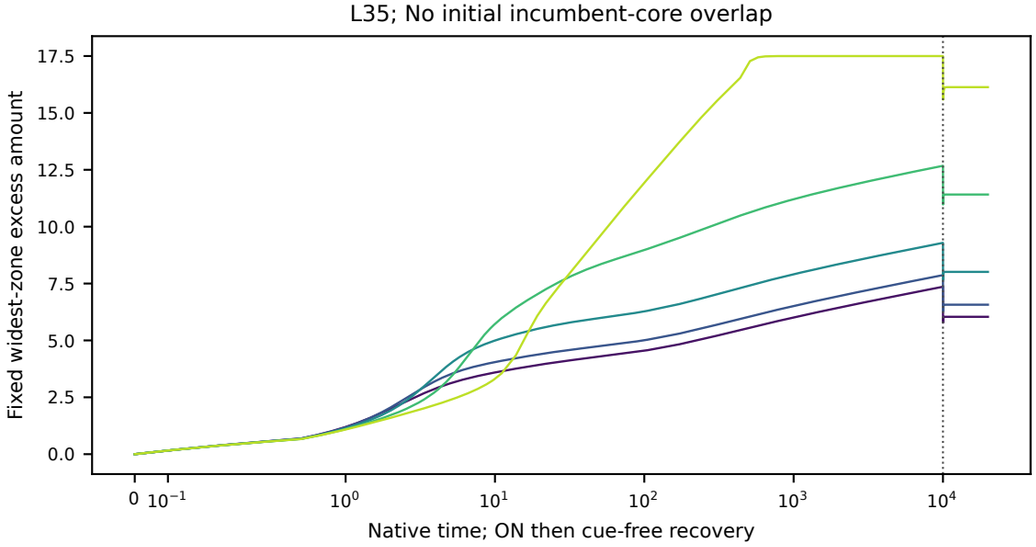
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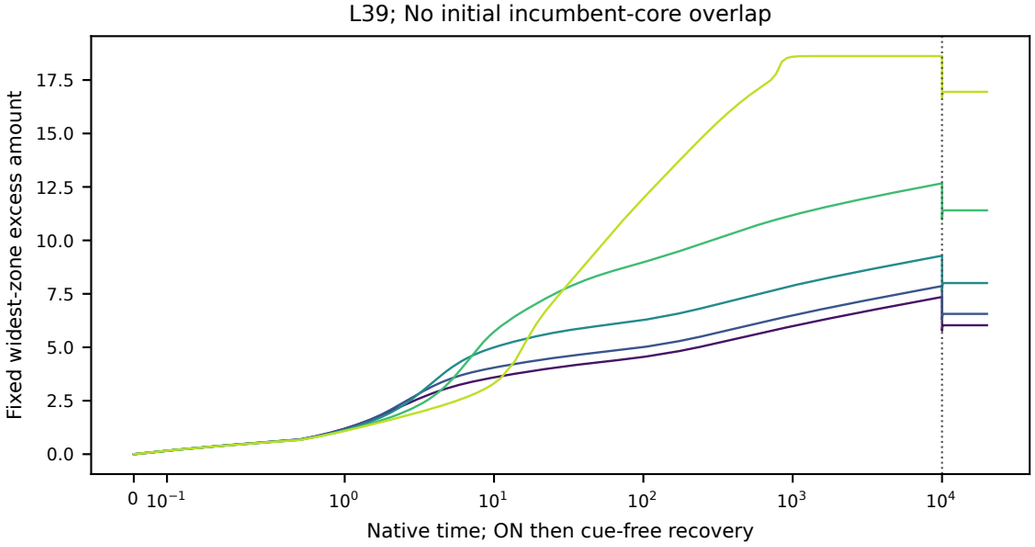






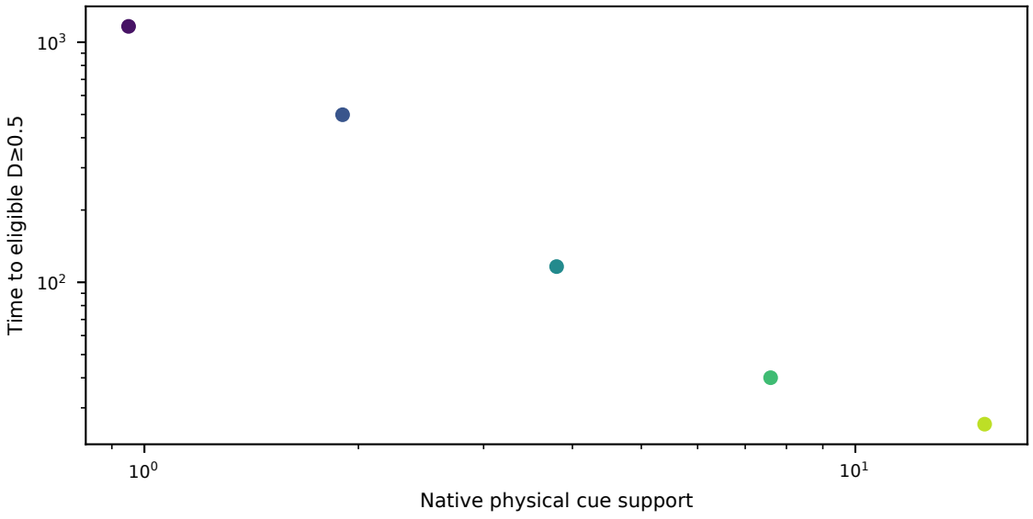


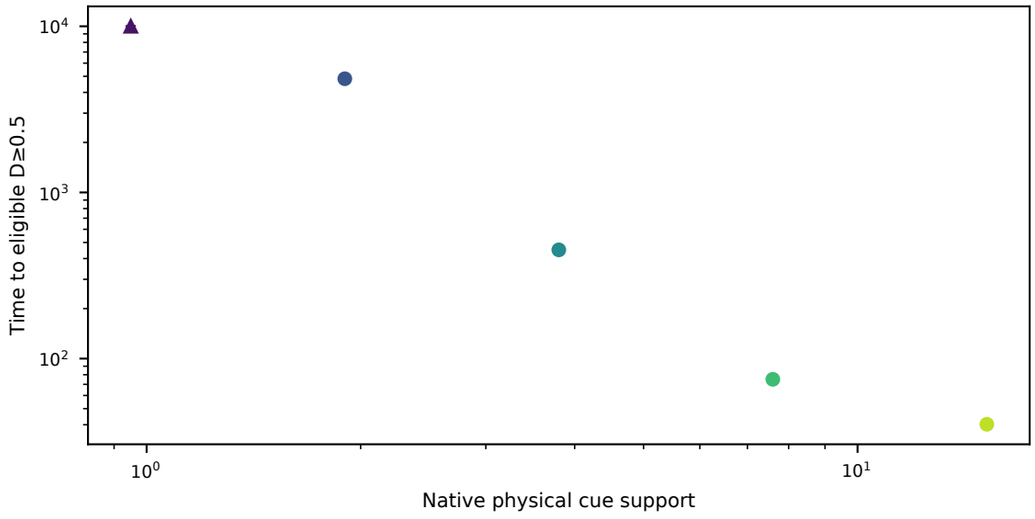


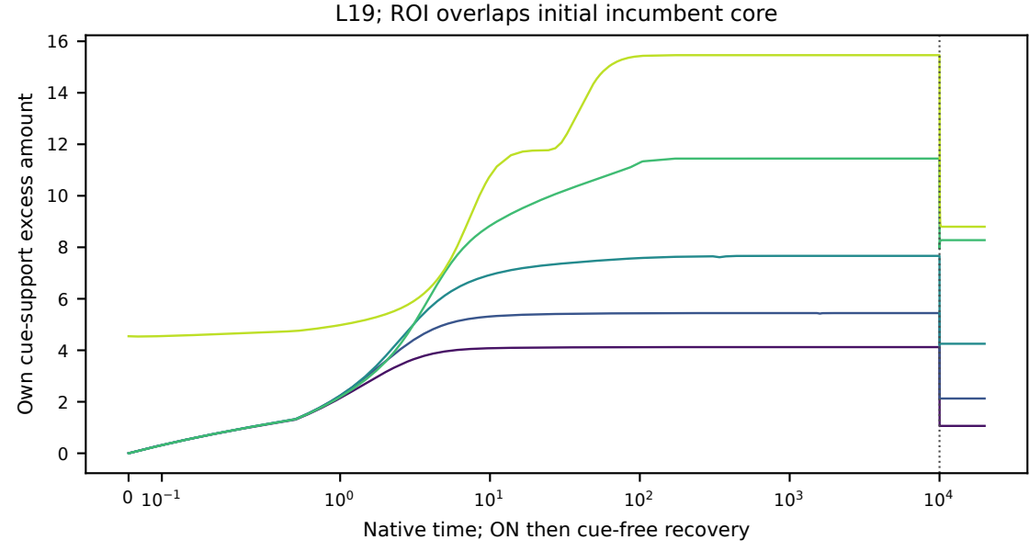


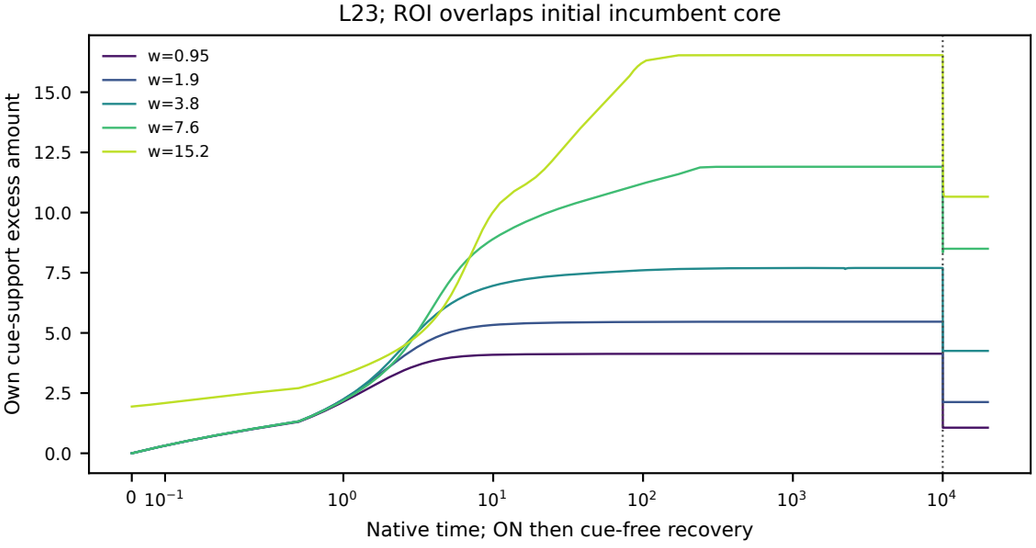
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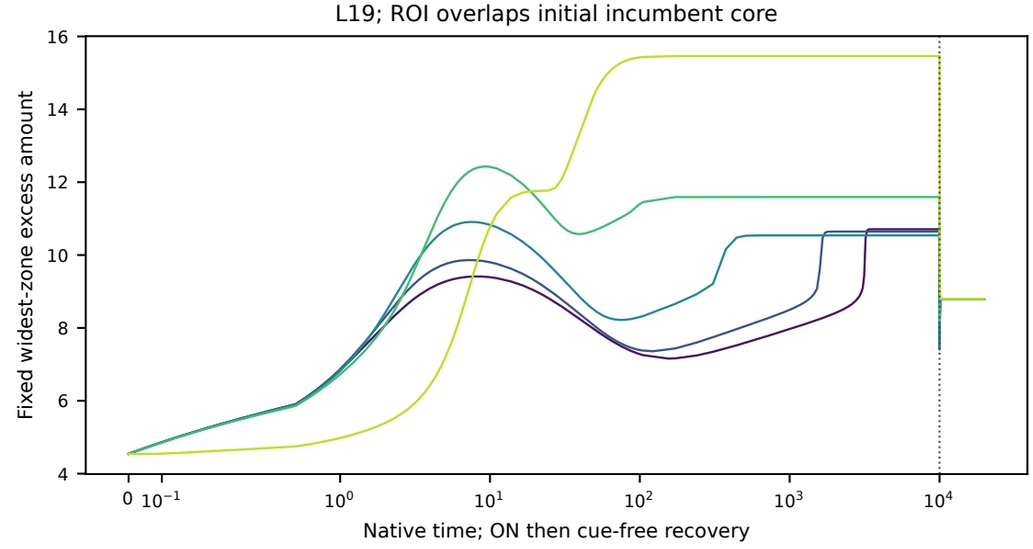
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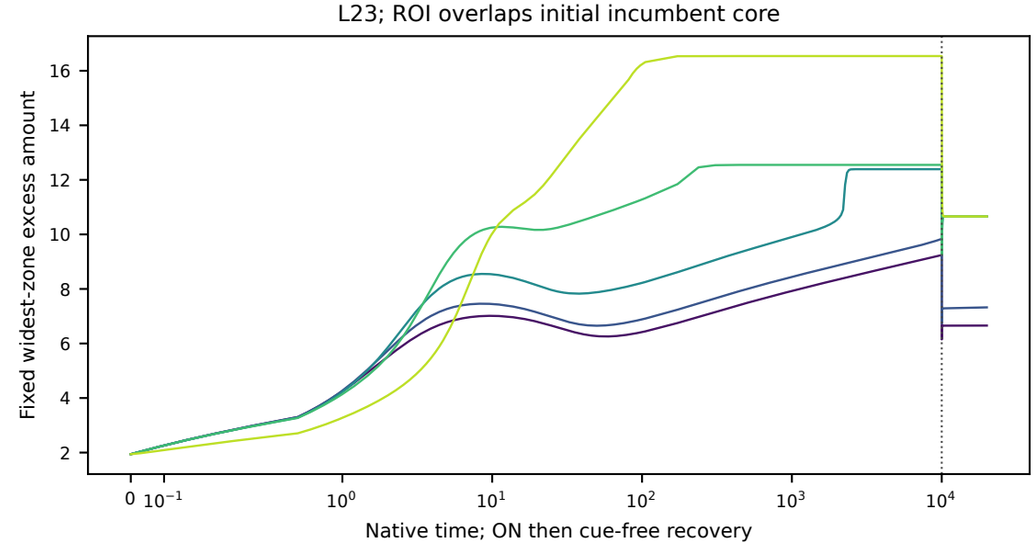






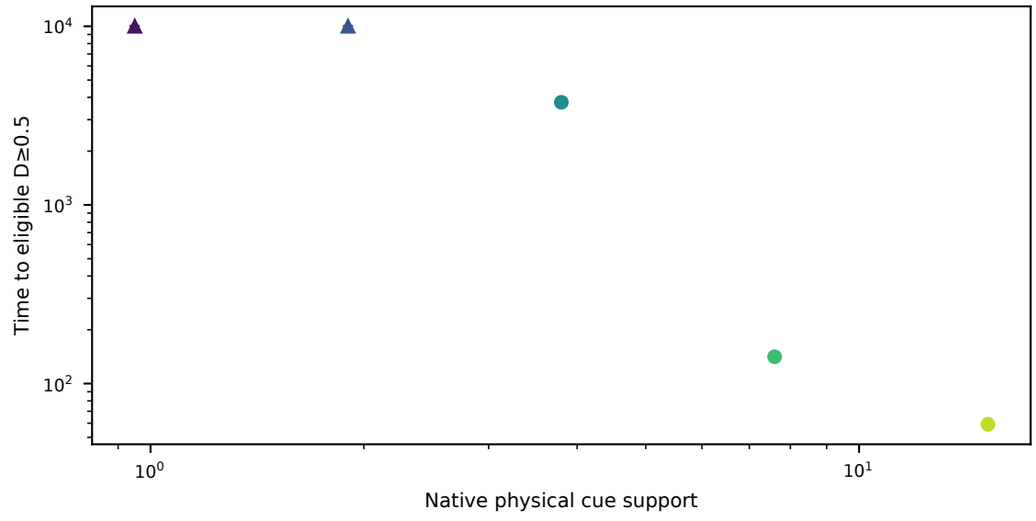


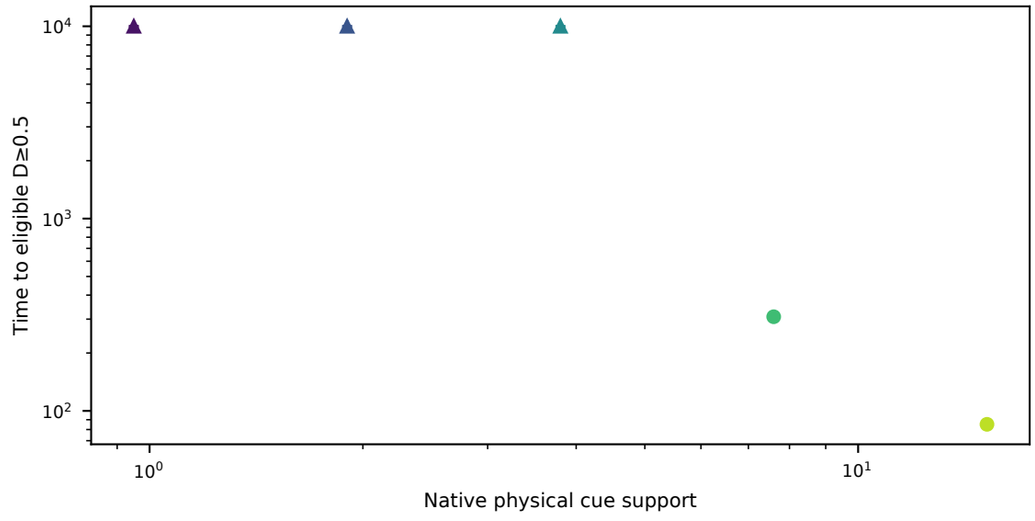


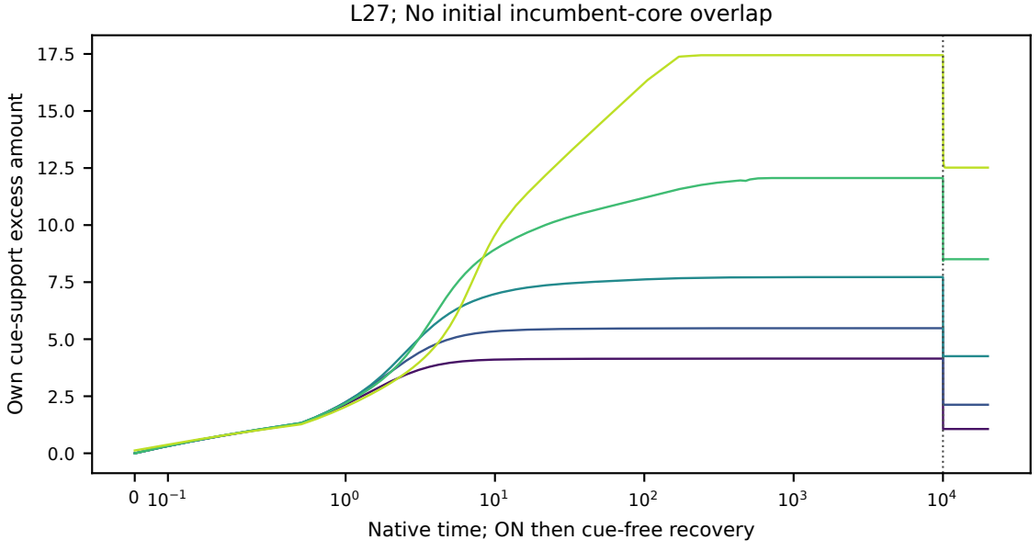


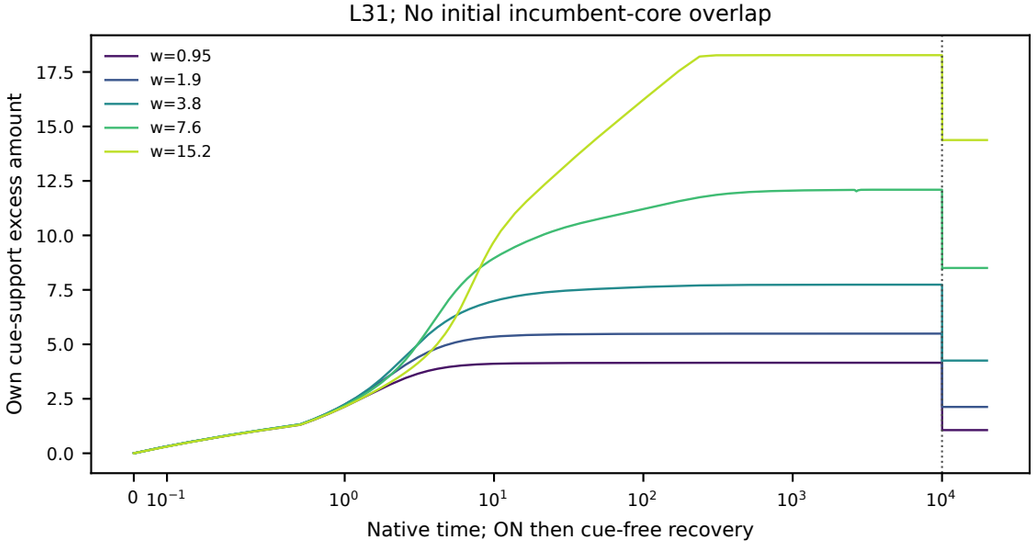
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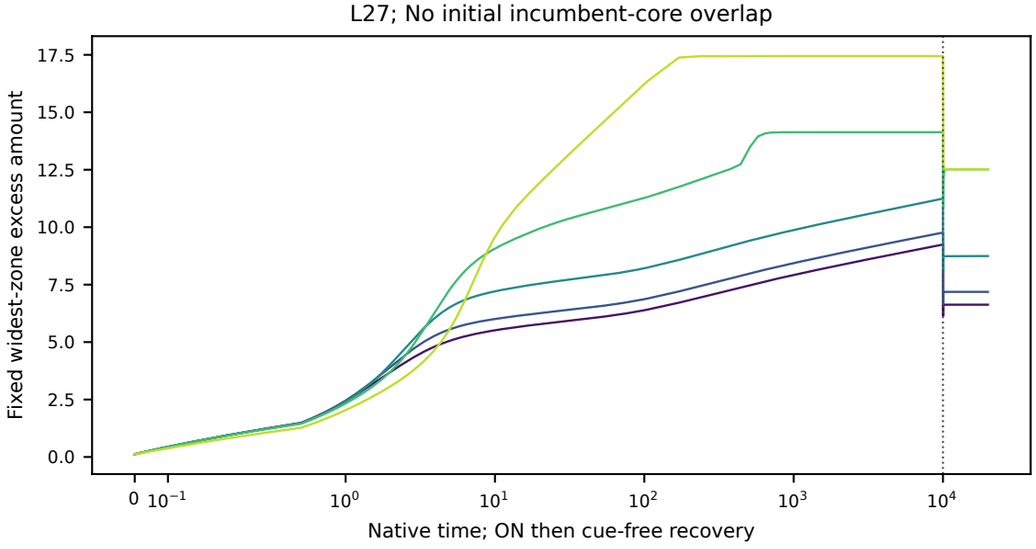
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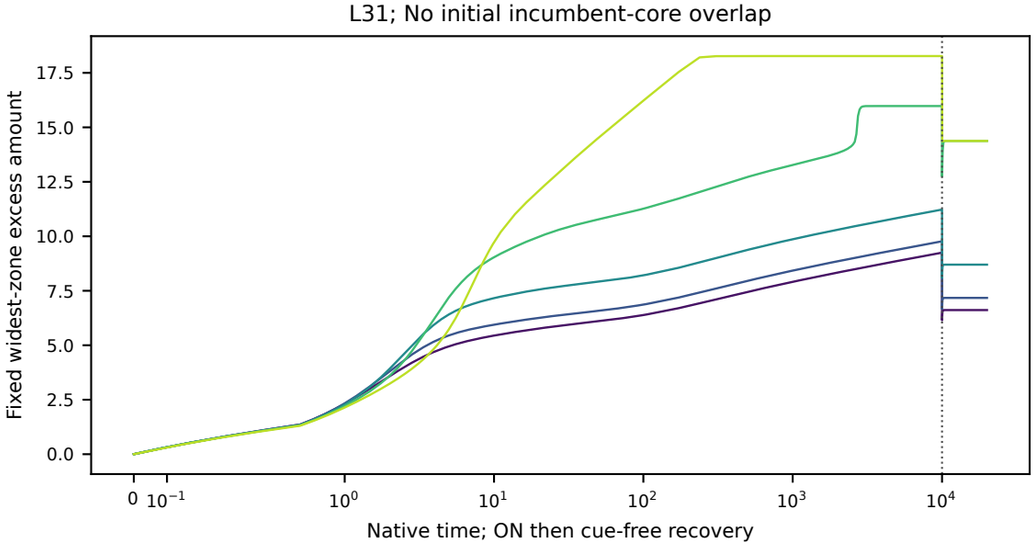






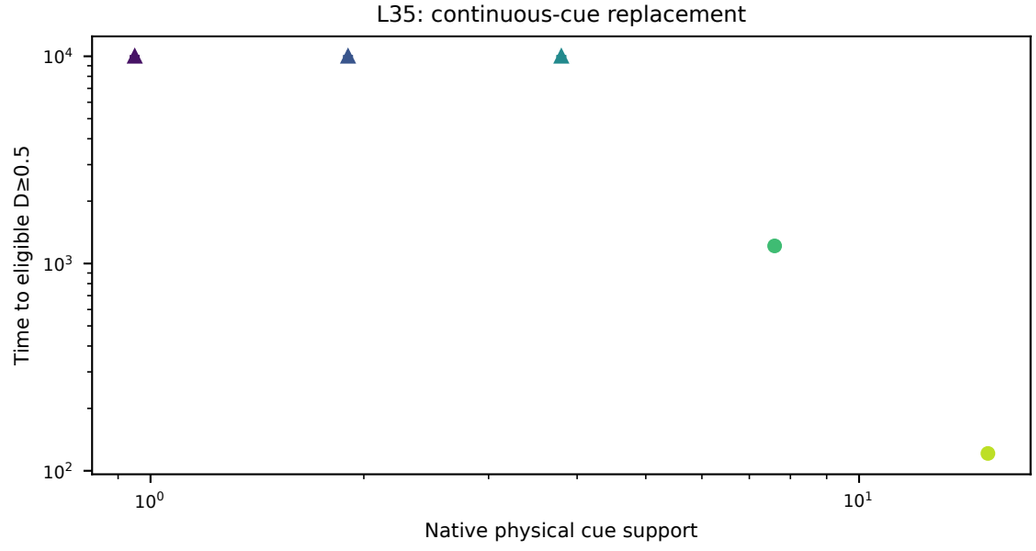


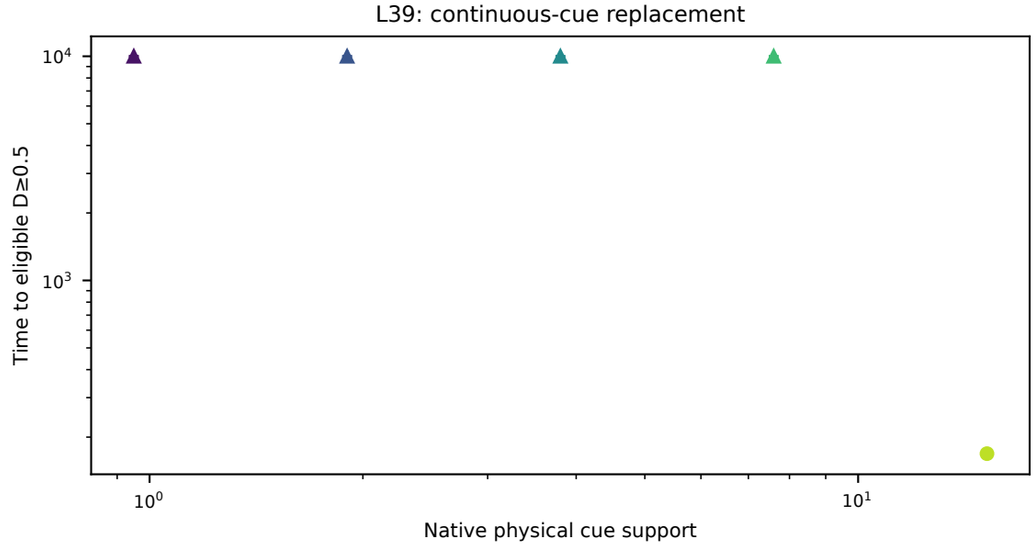


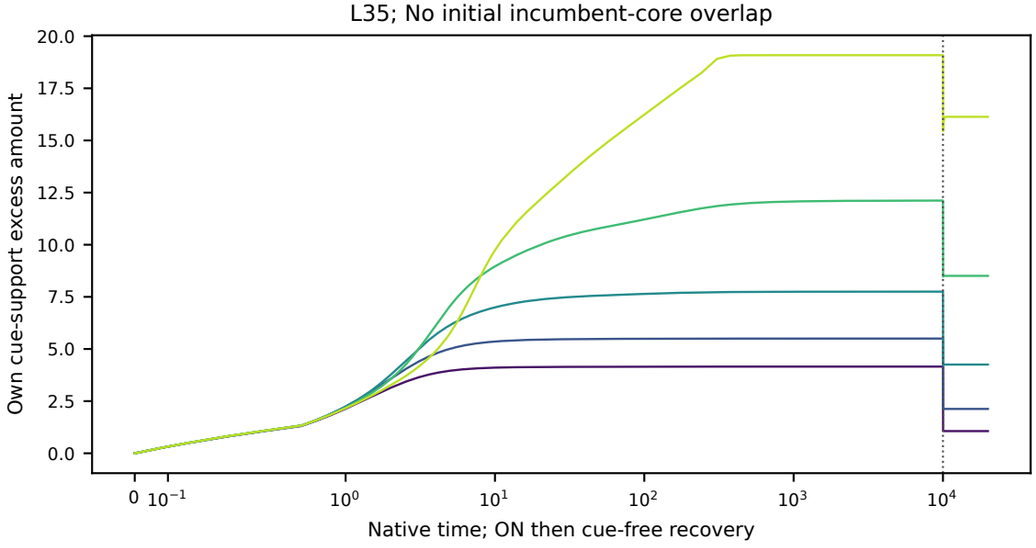


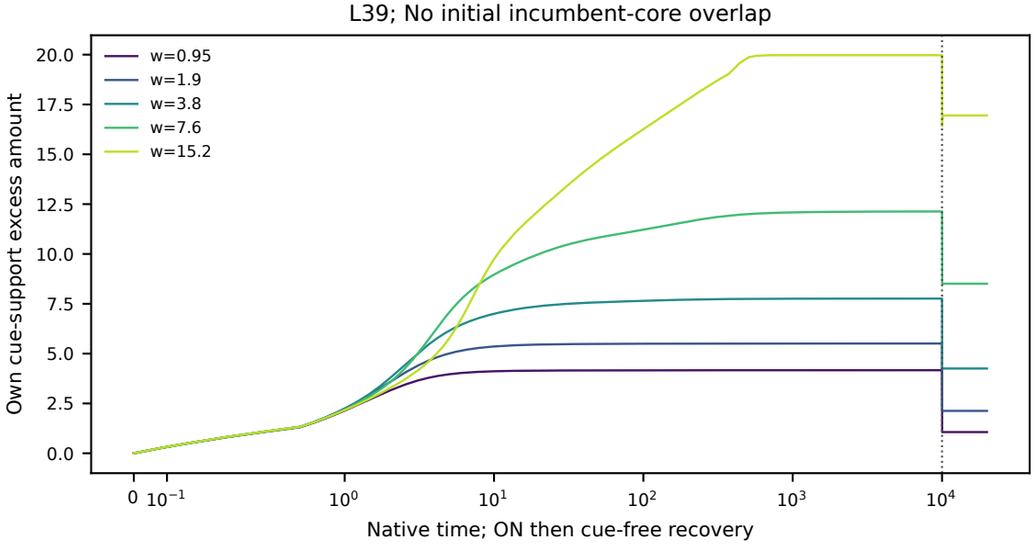
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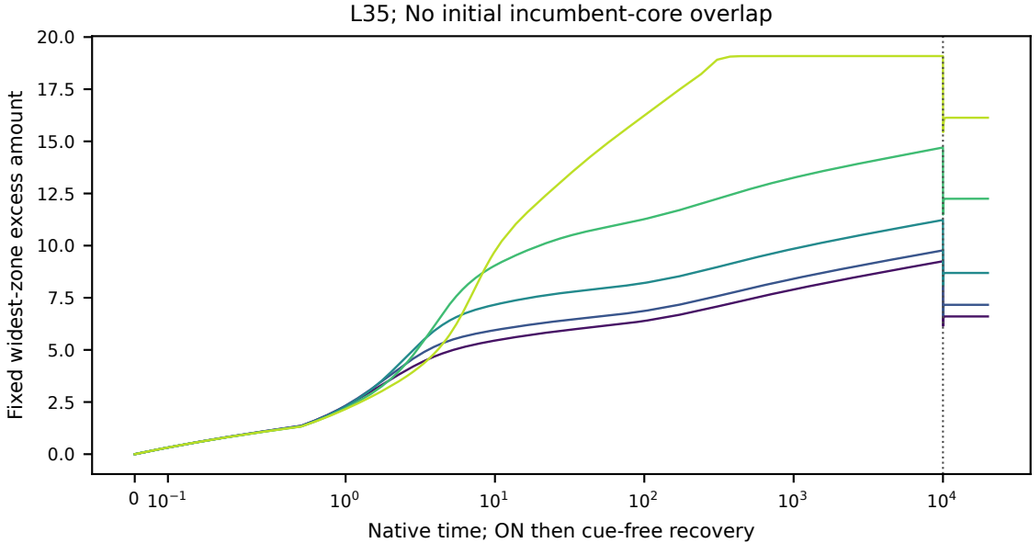
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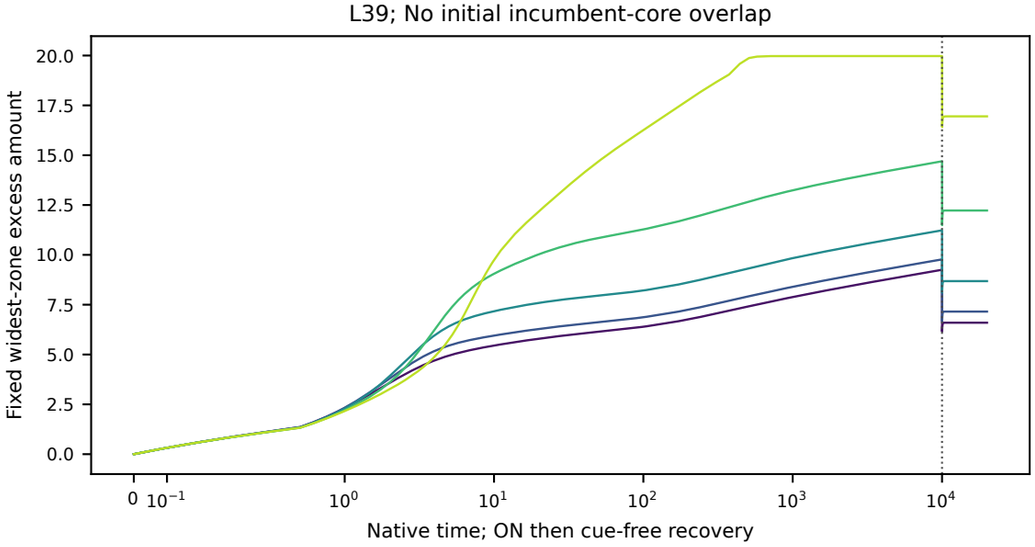






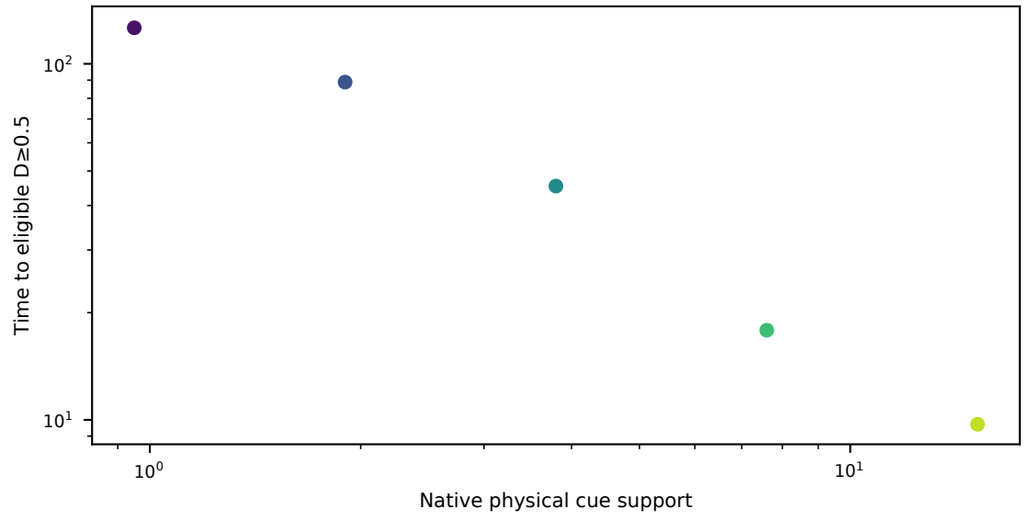


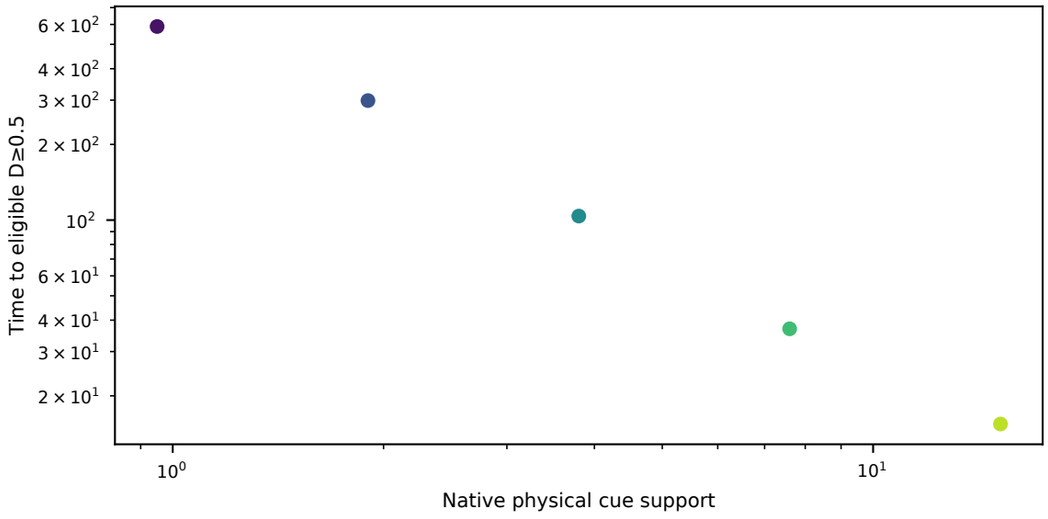


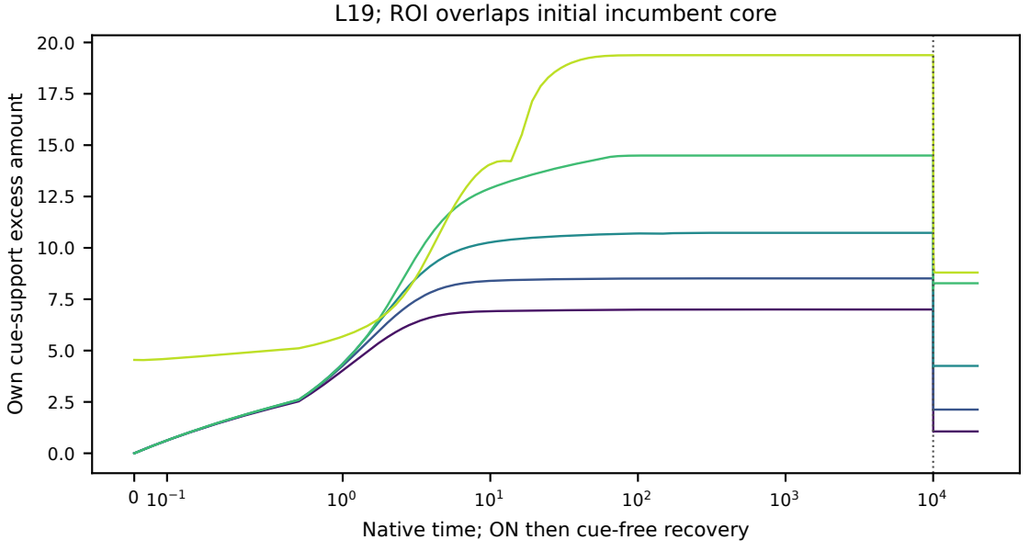


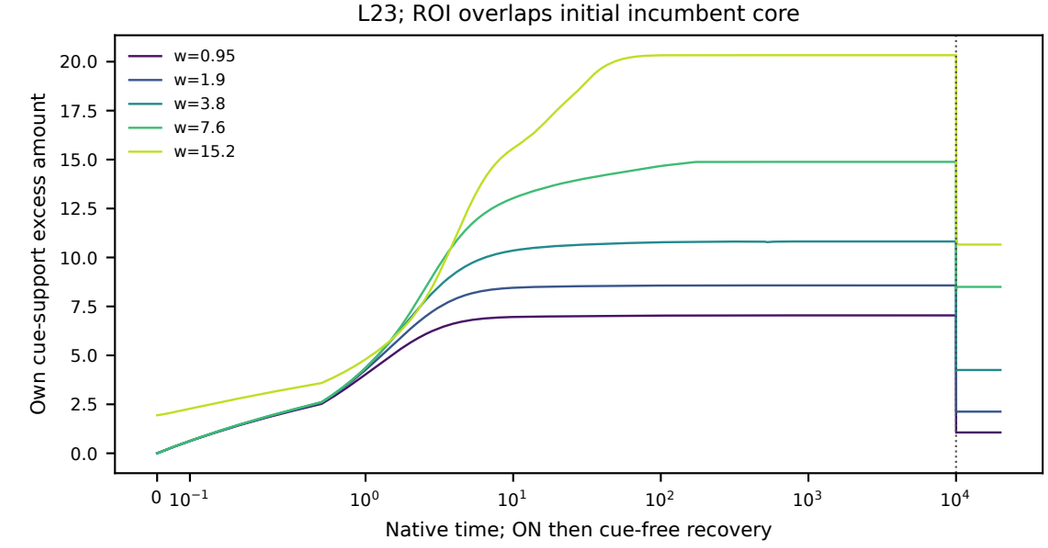
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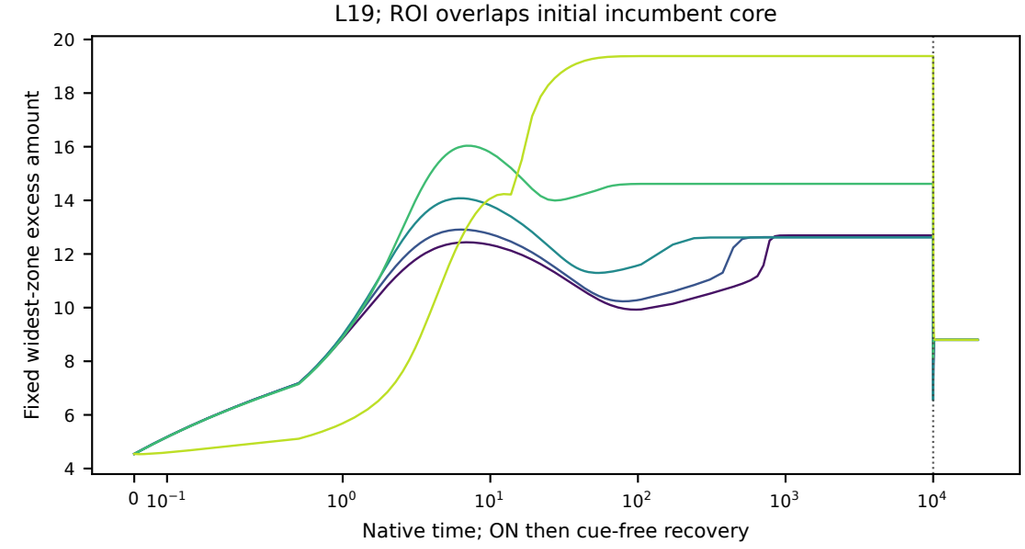
S1 Holmes 3 • commanded area 2× • native kinetics

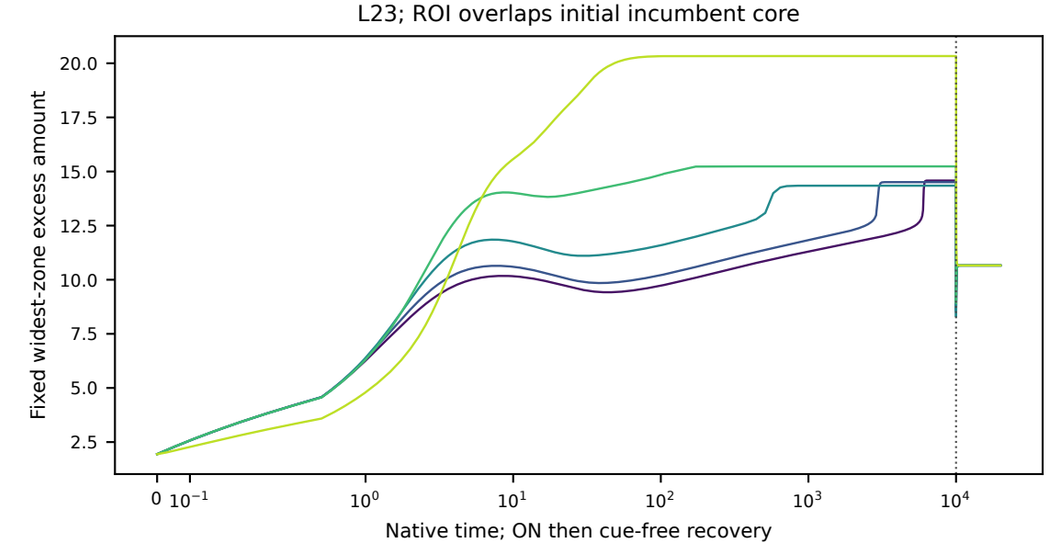






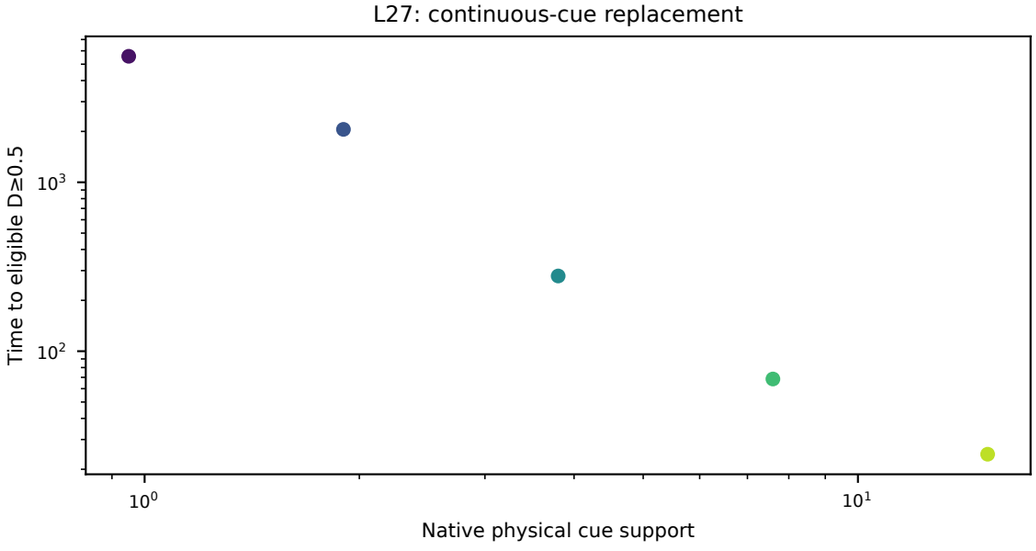


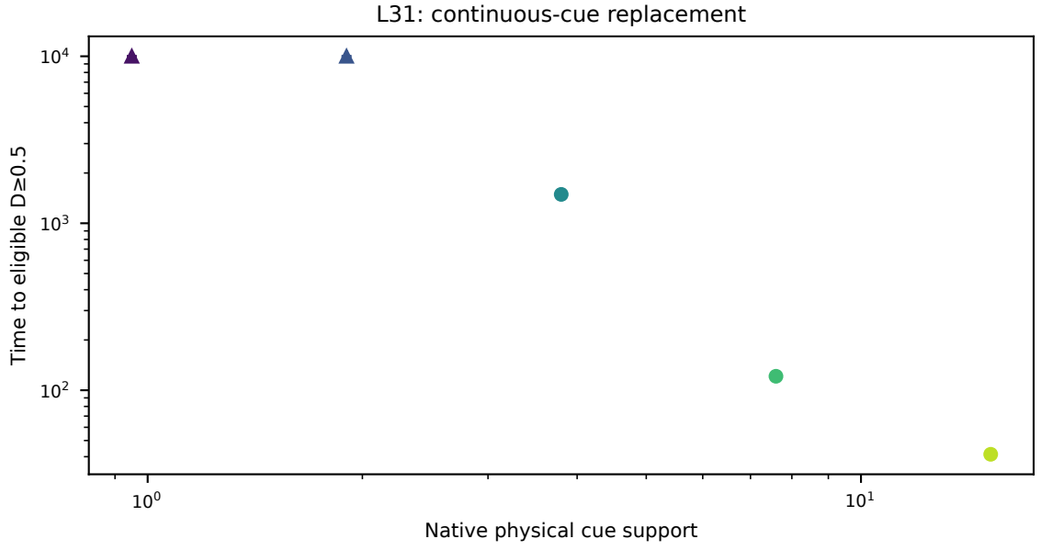


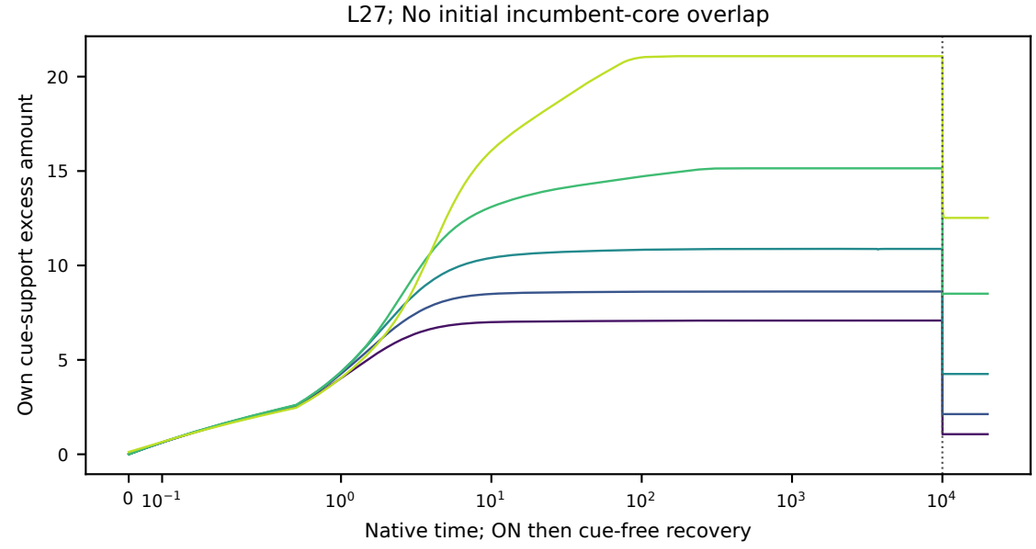


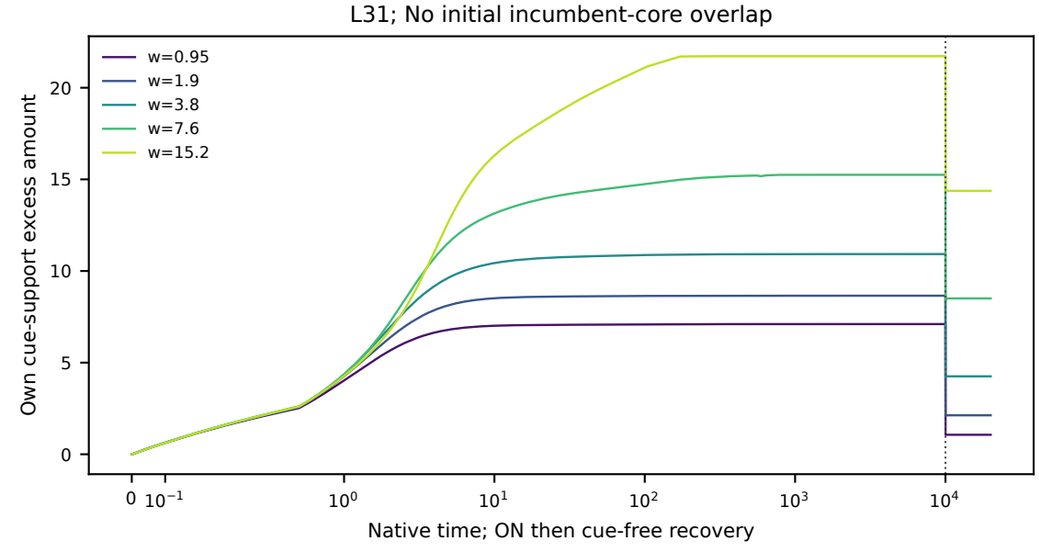
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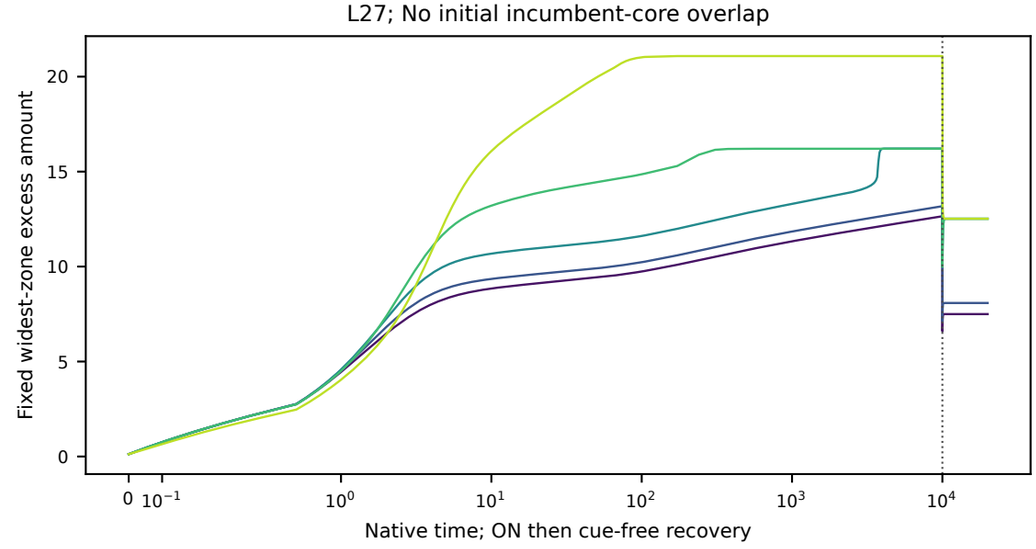
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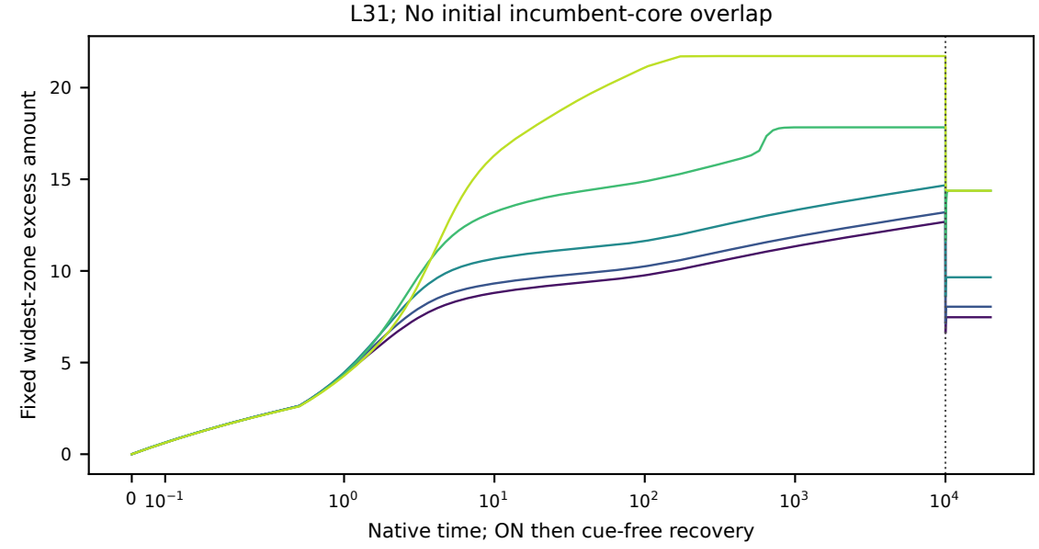






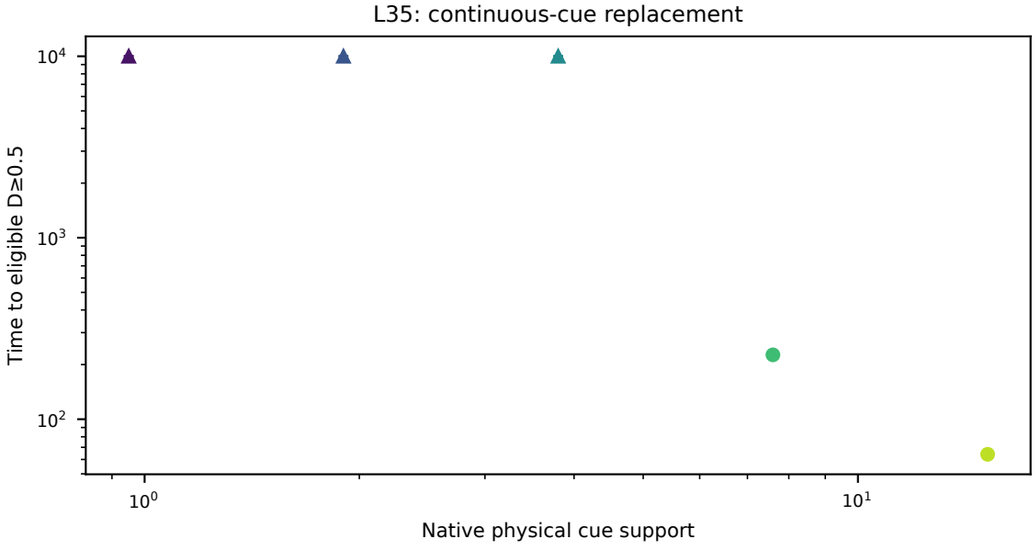


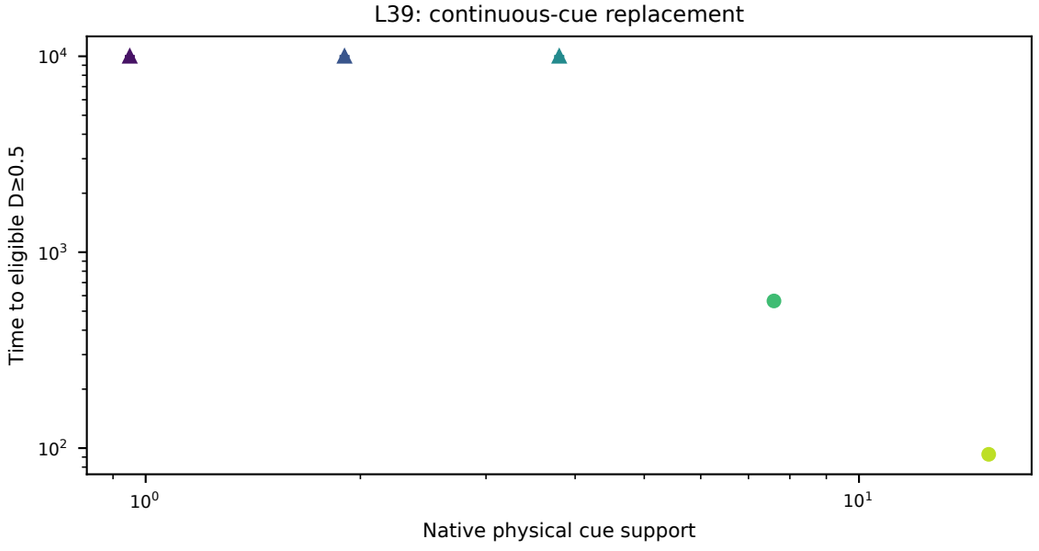


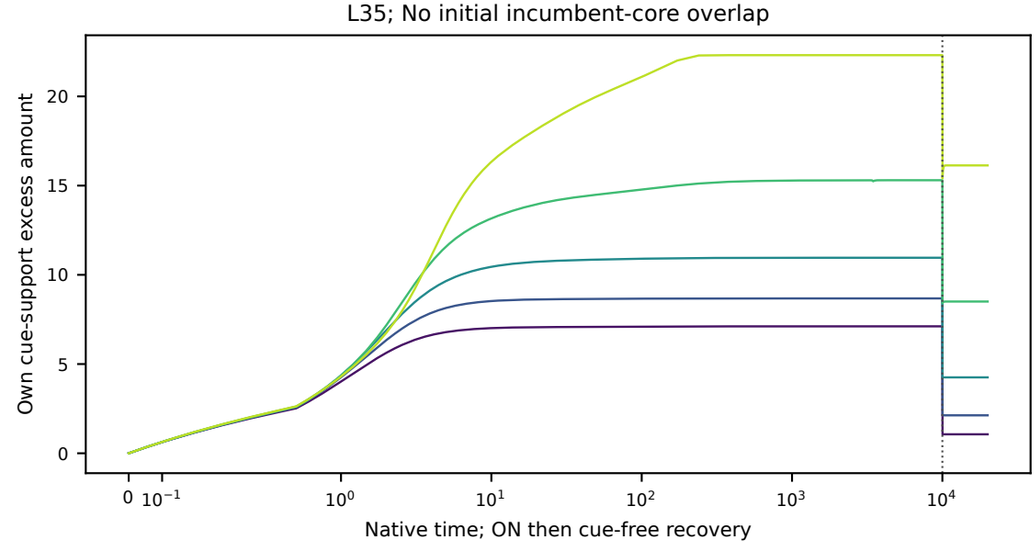


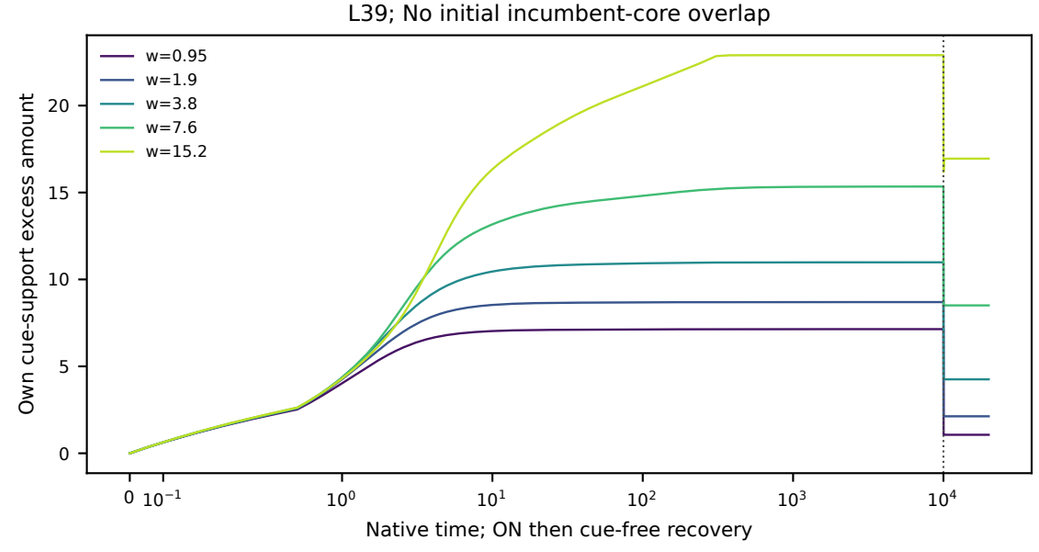
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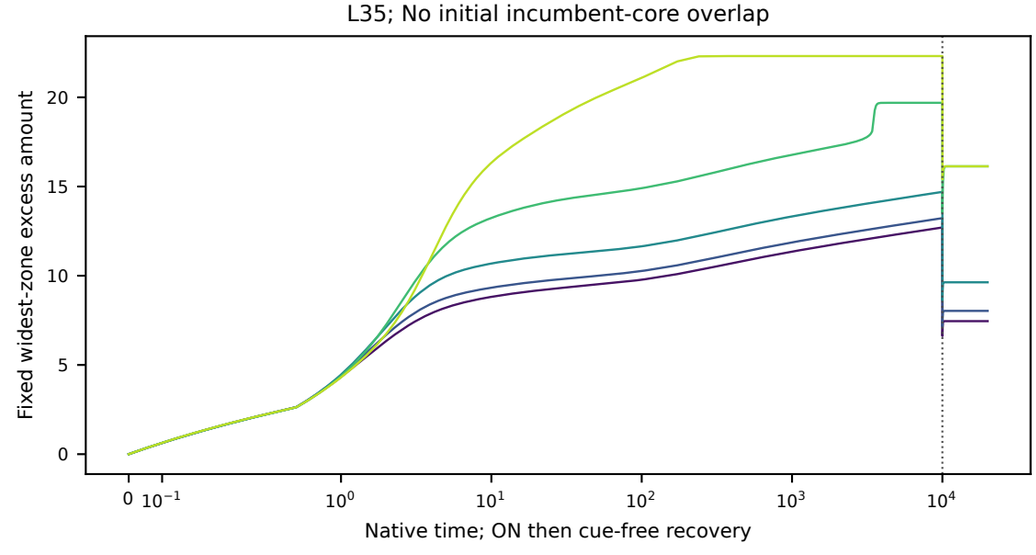
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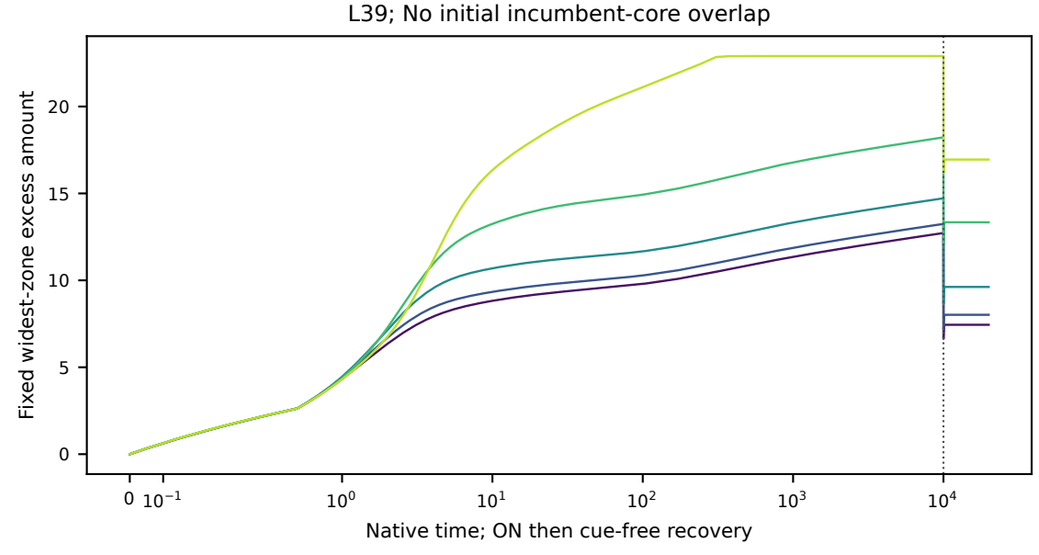






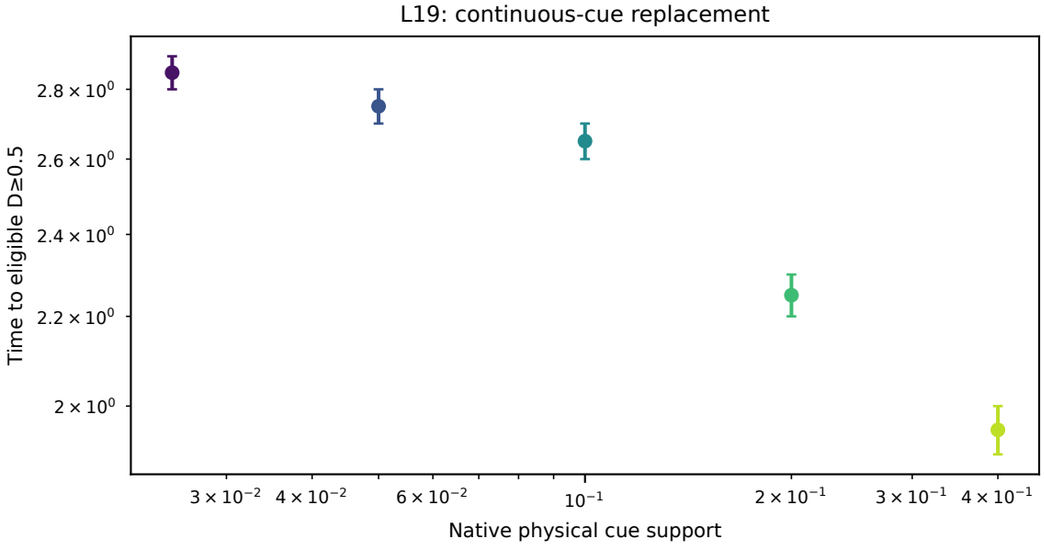


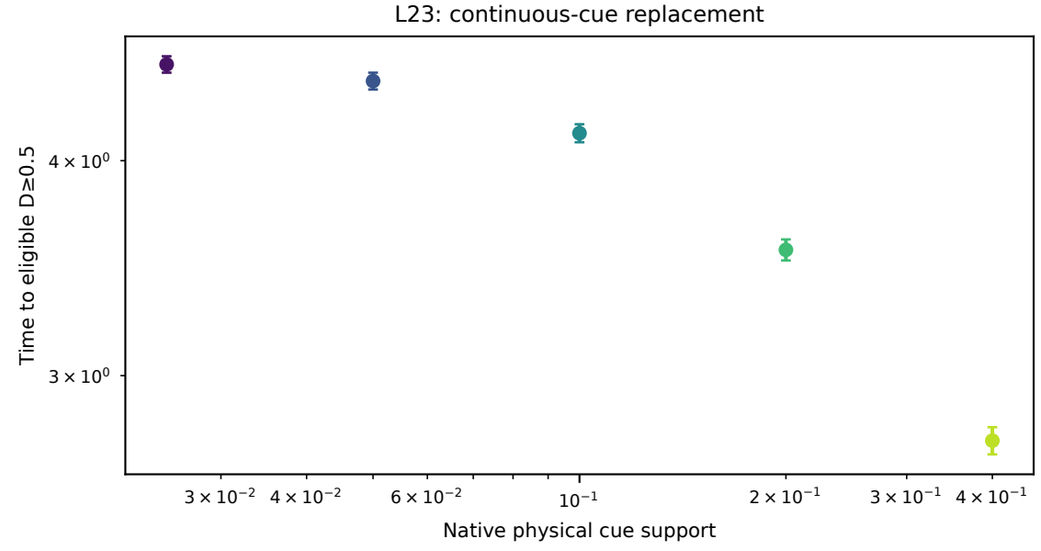


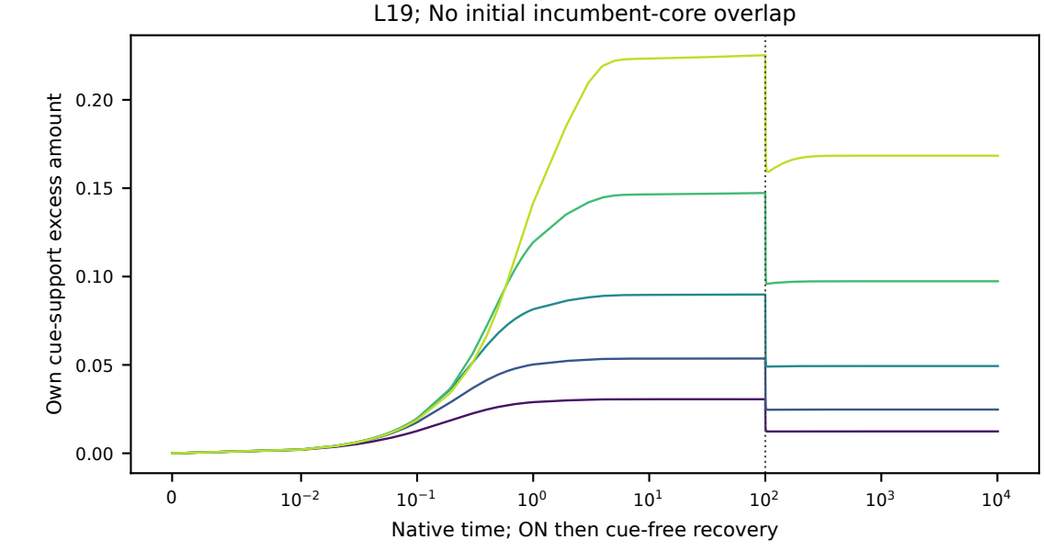


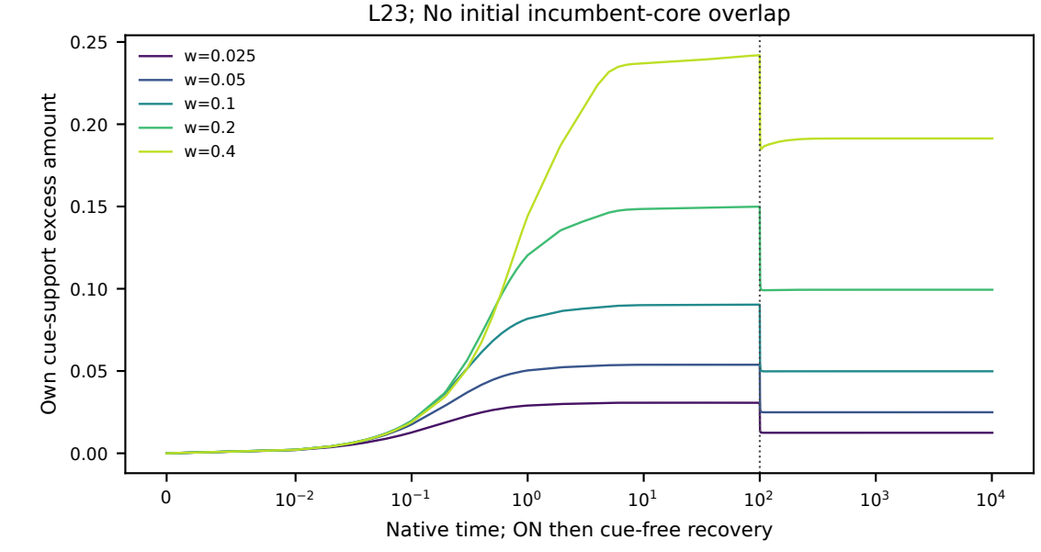
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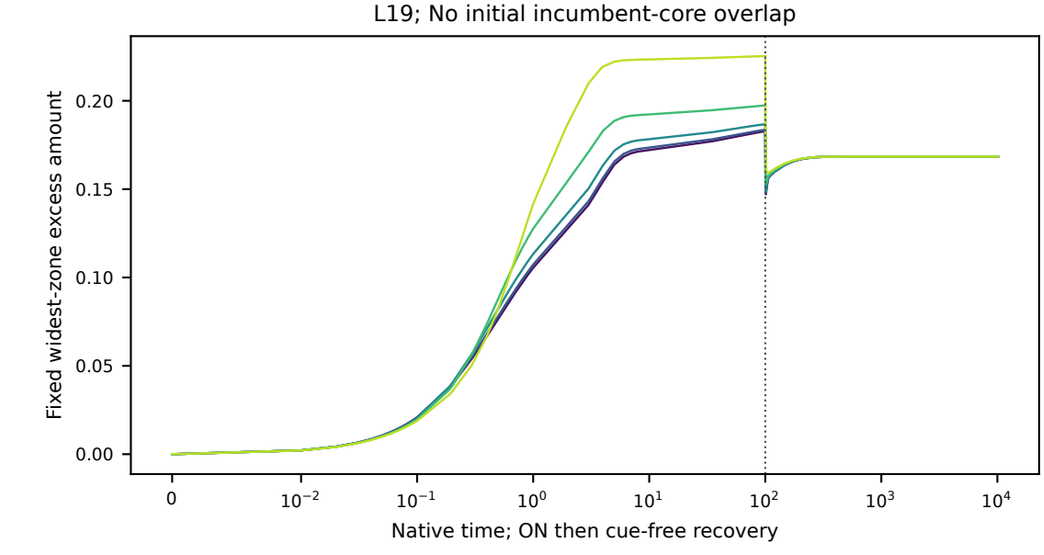
S1 Spring • commanded area 0.5× • native kinetics

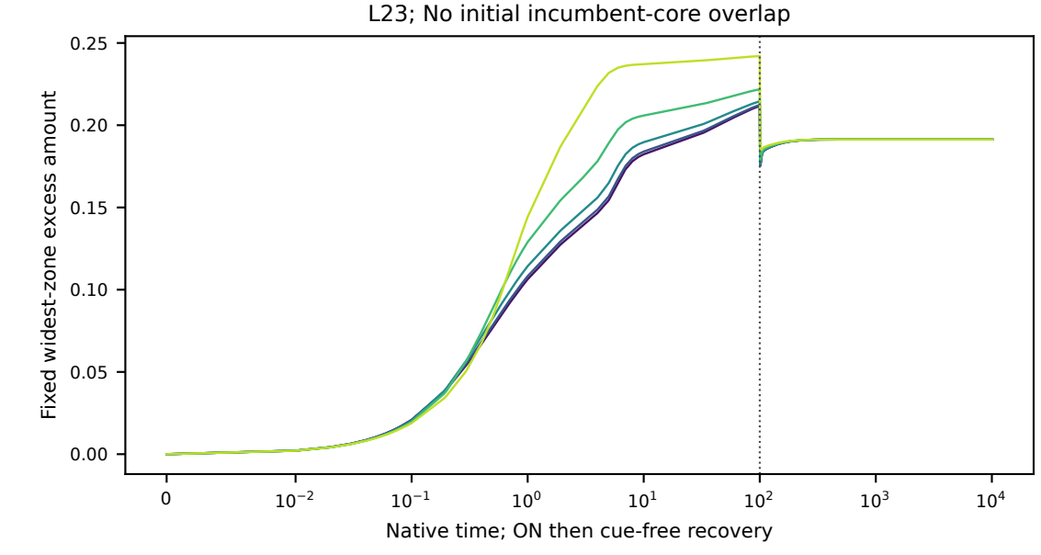






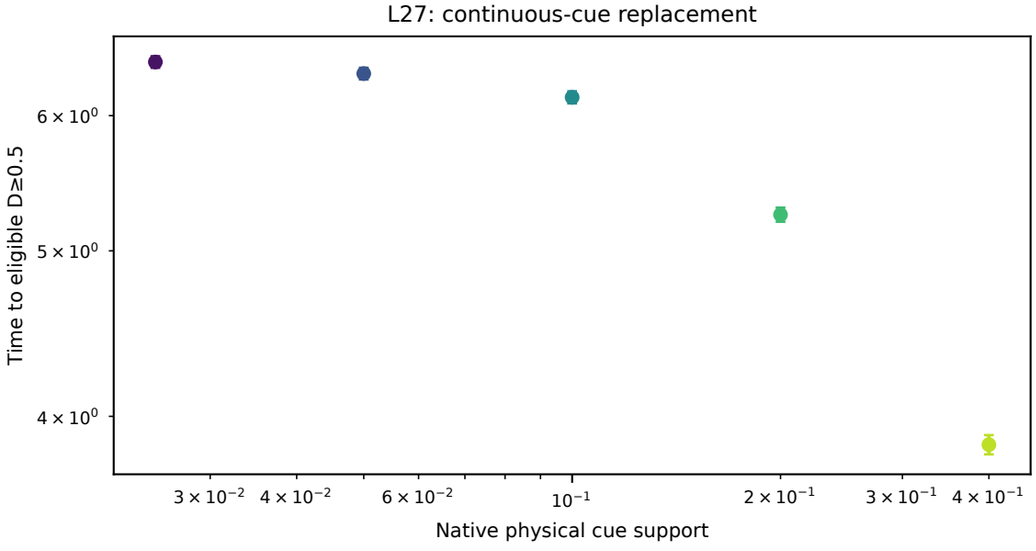


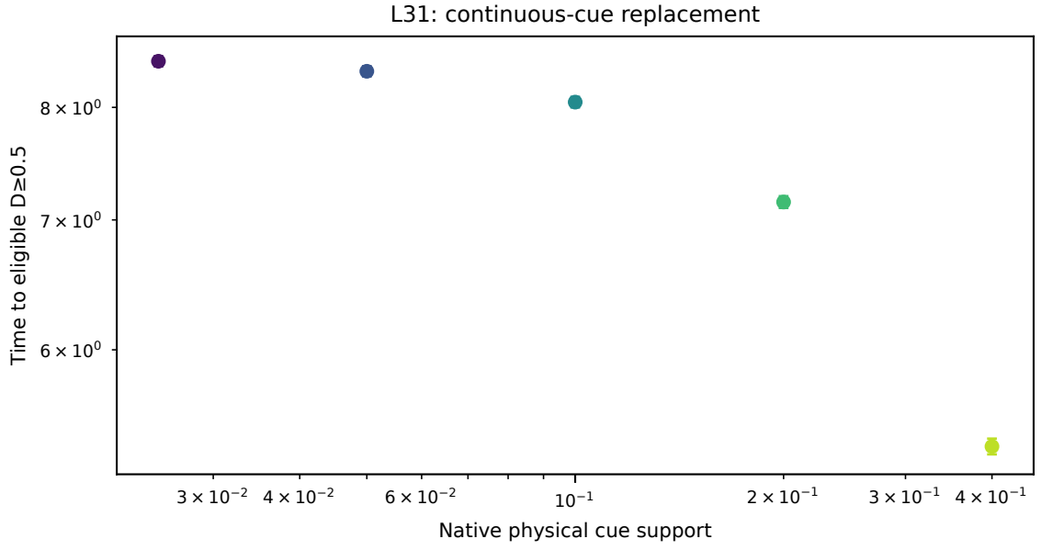


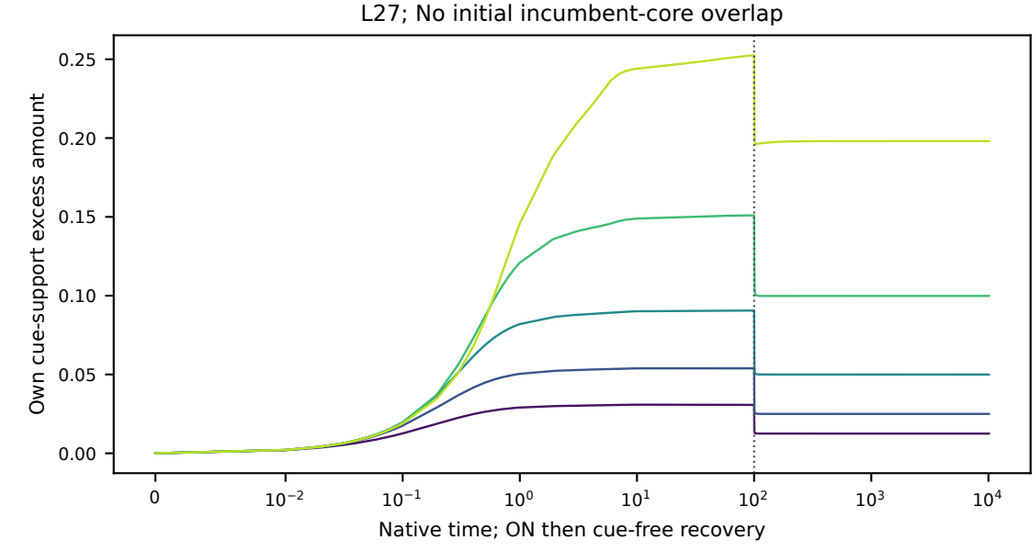


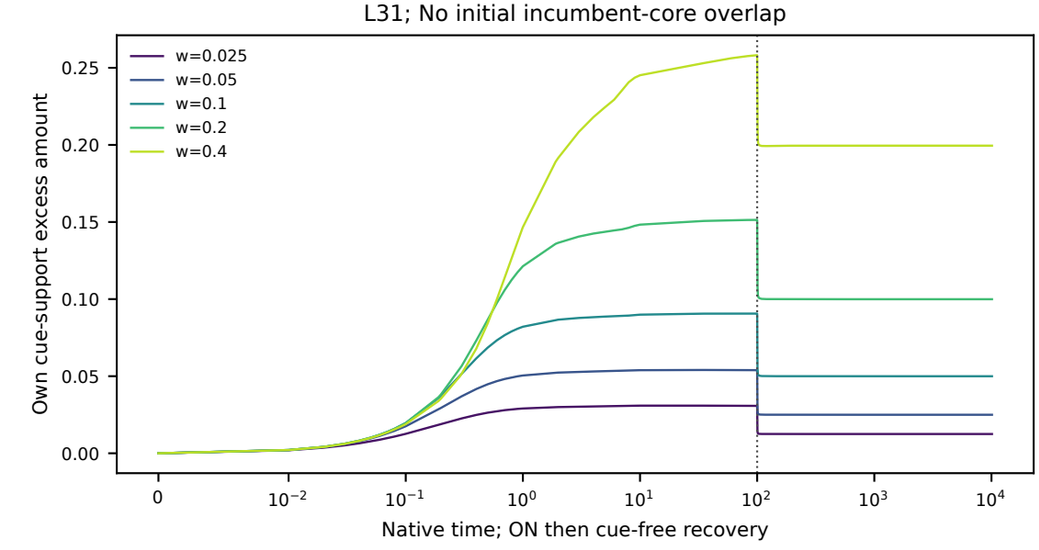
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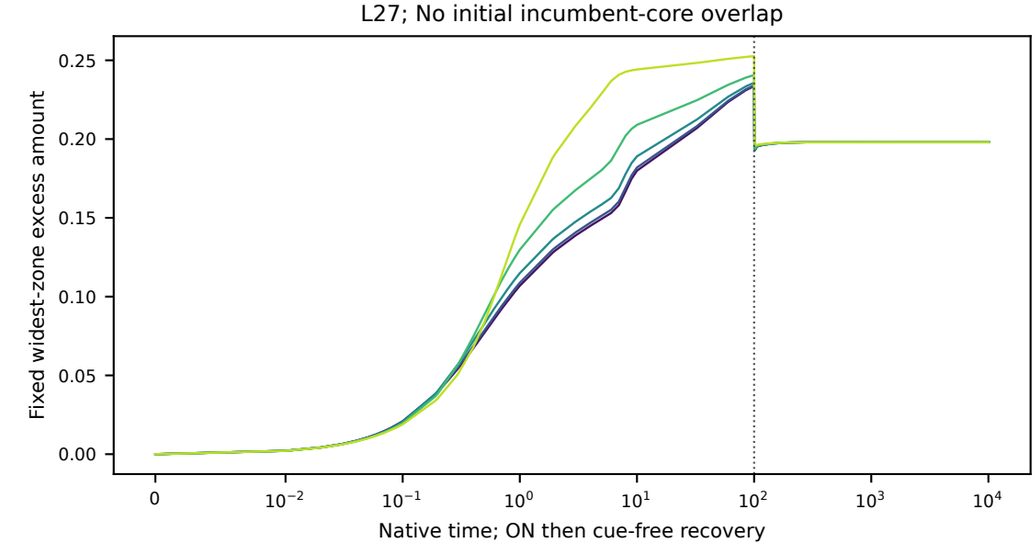
S1 Spring • commanded area 0.5× • native kinetics

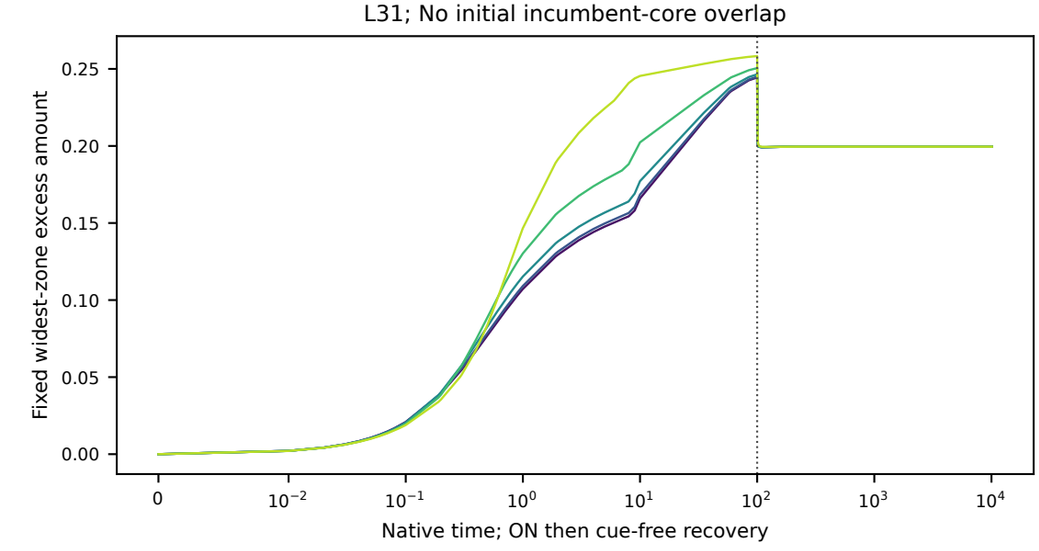






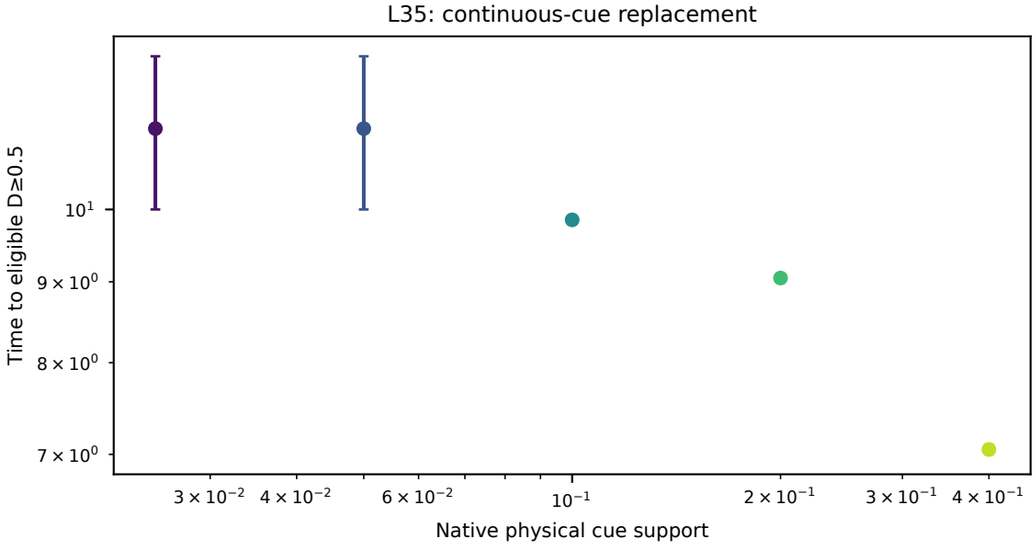


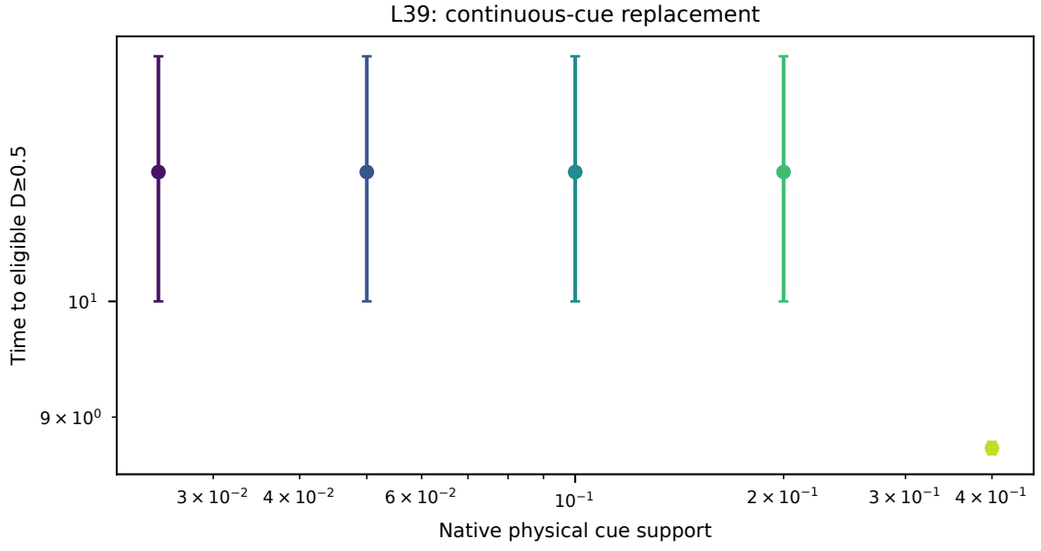


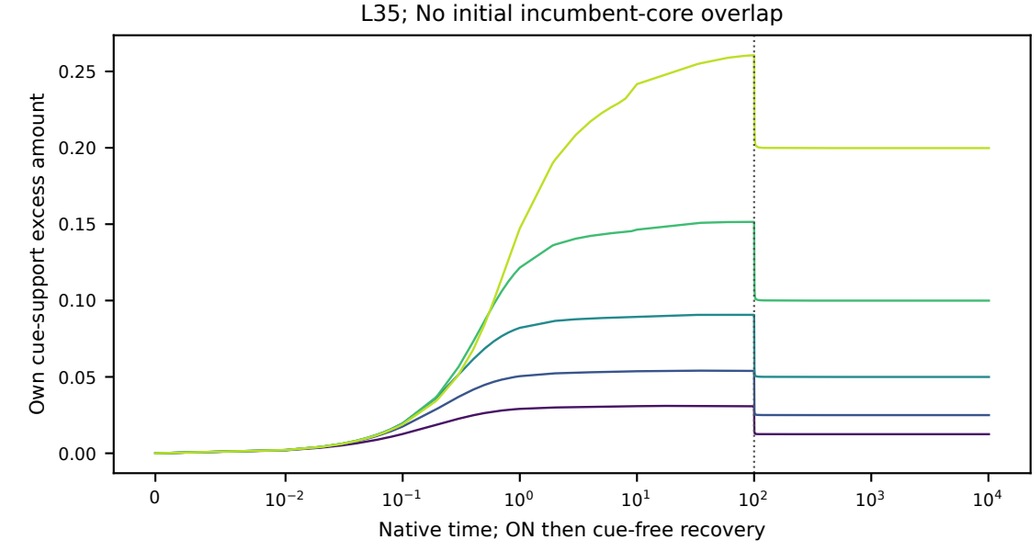


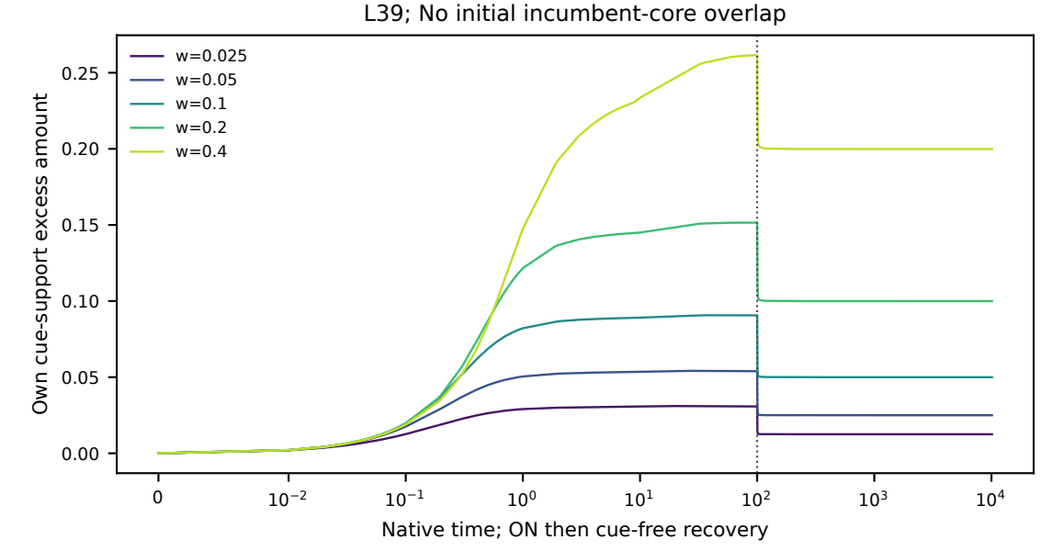
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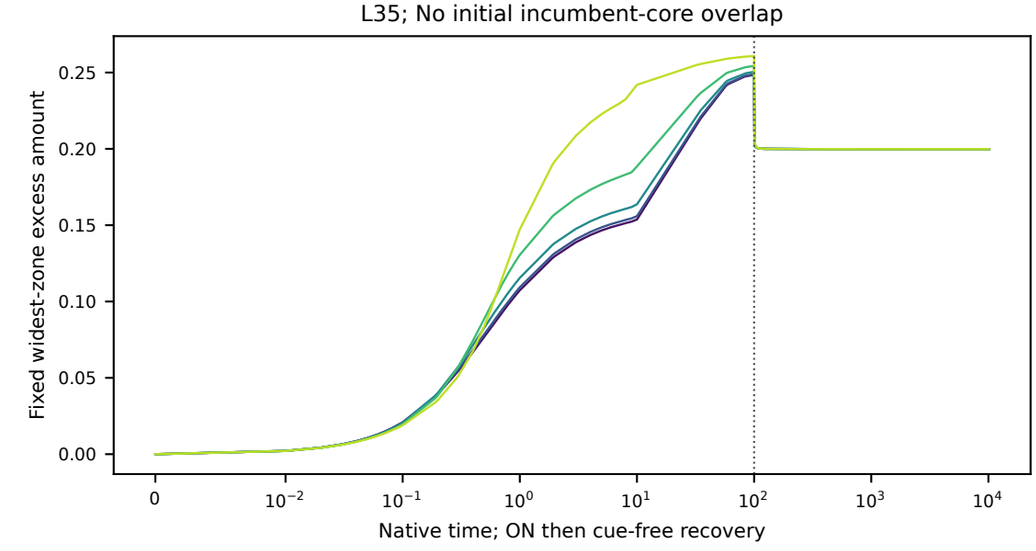
S1 Spring • commanded area 0.5× • native kinetics

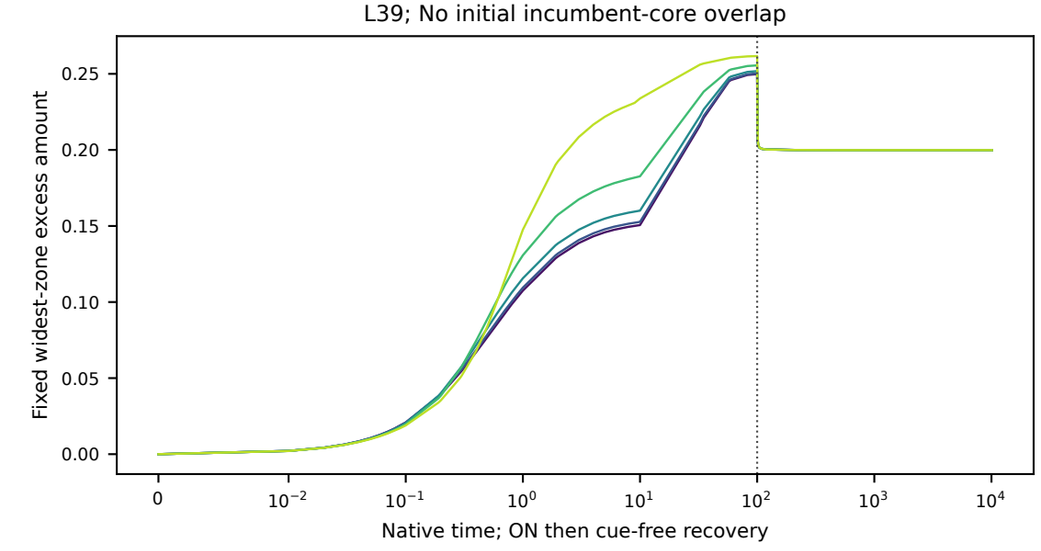






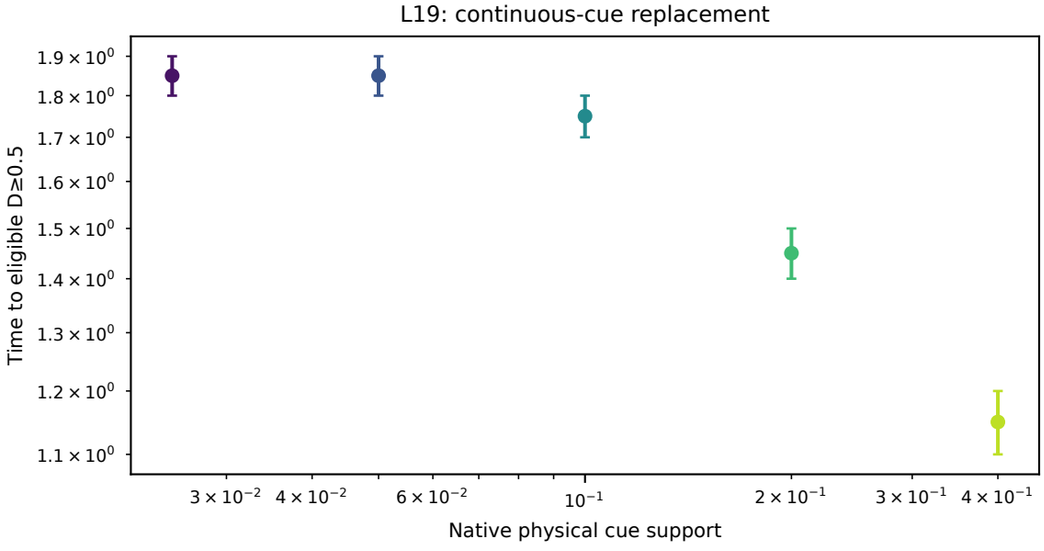


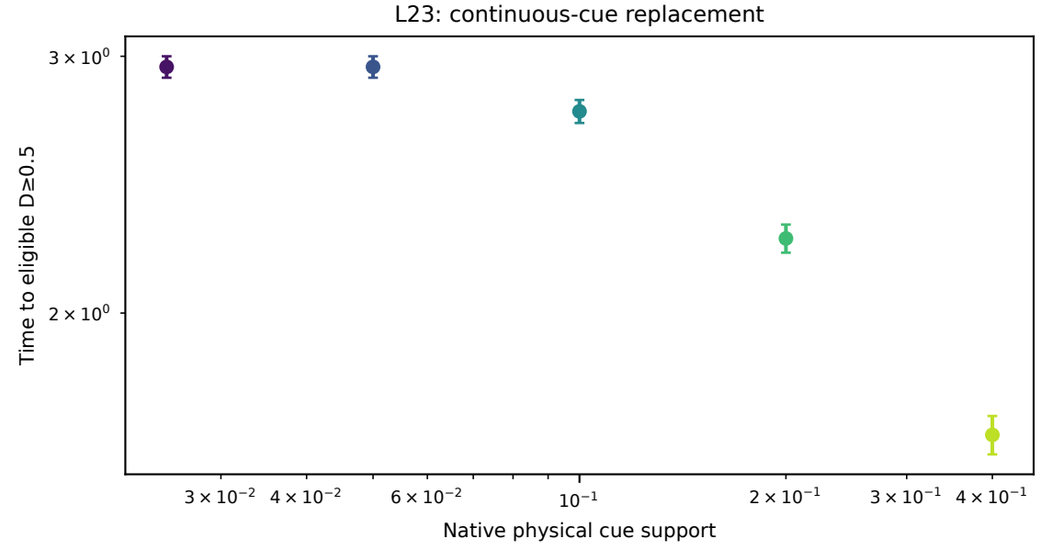


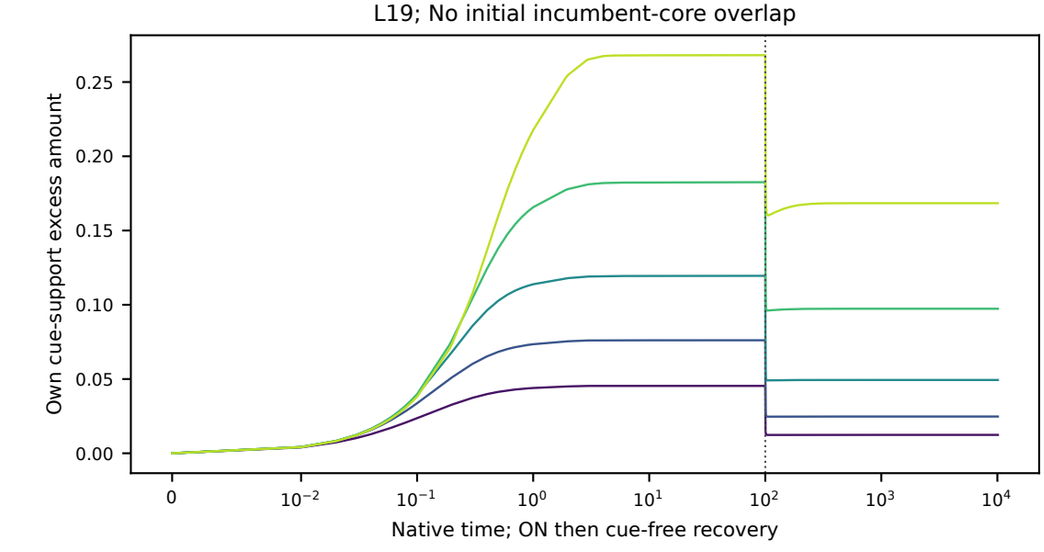


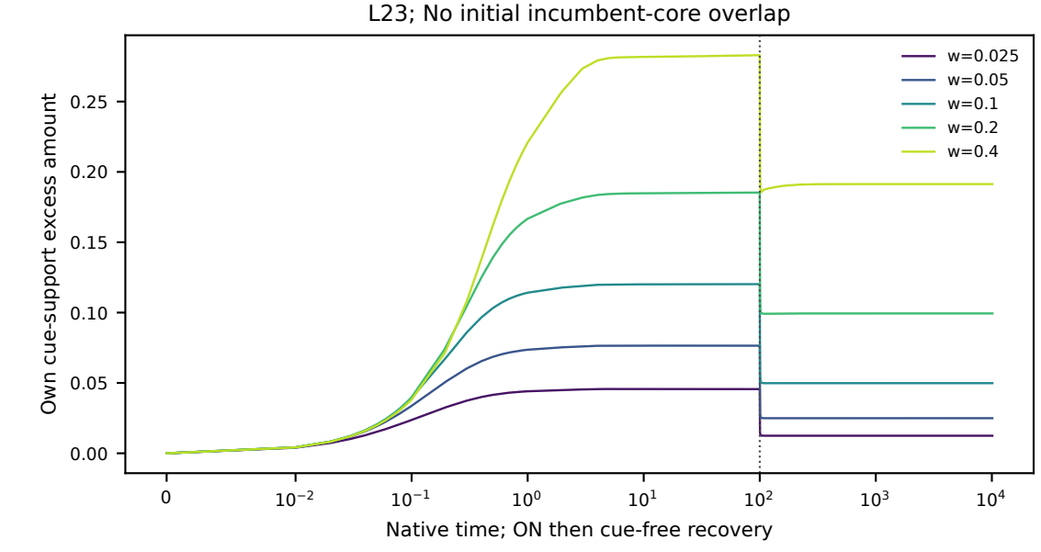
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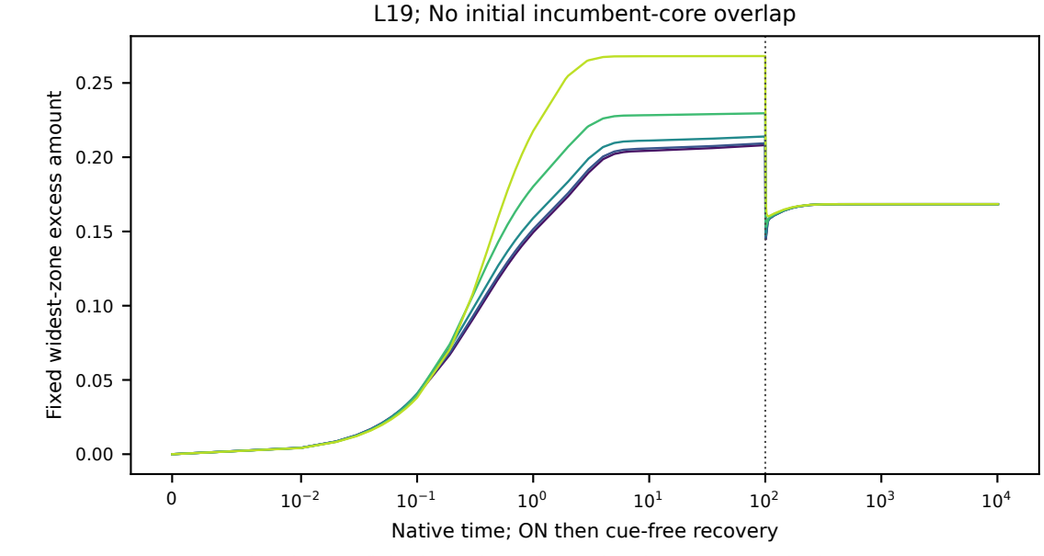
S1 Spring • commanded area 1× • native kinetics

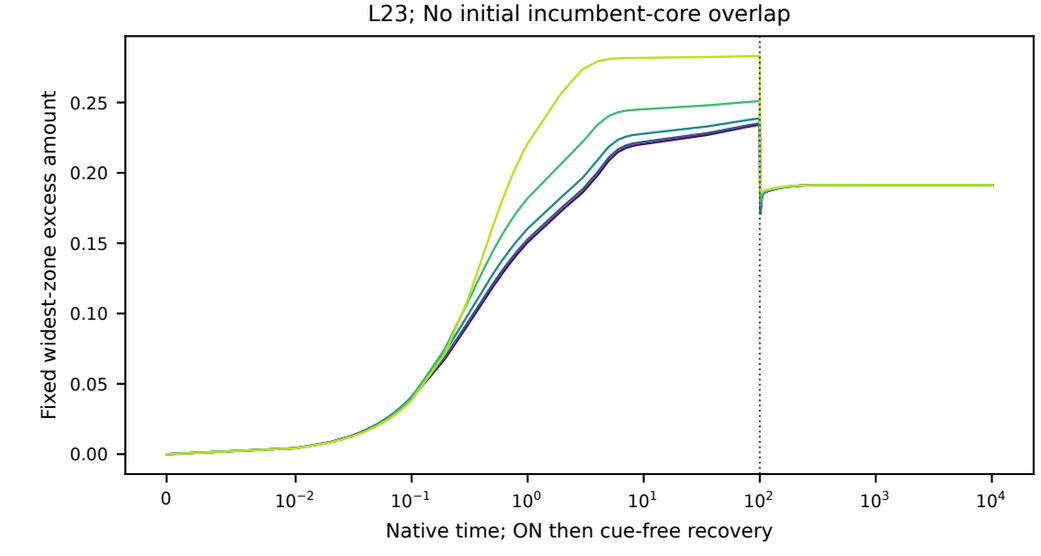






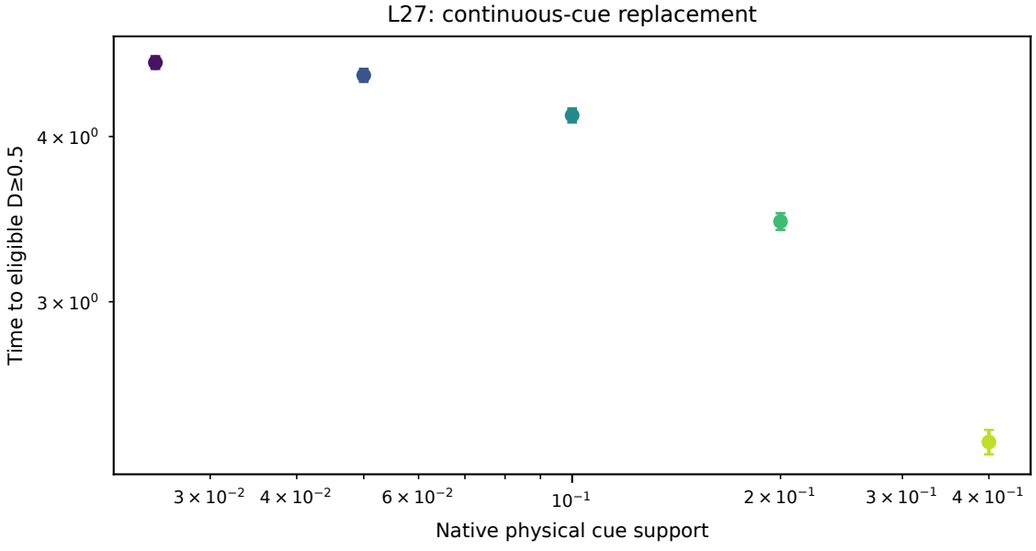


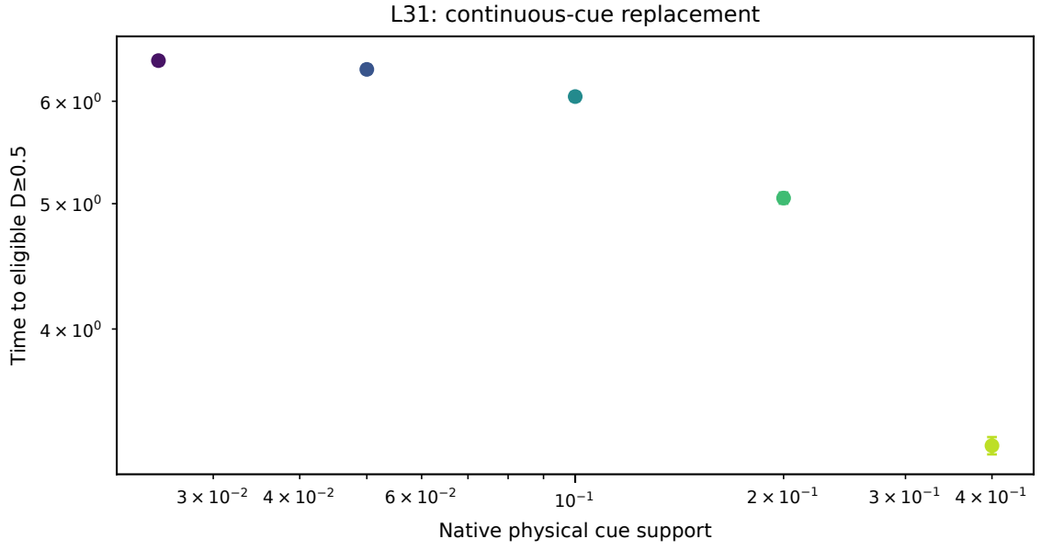


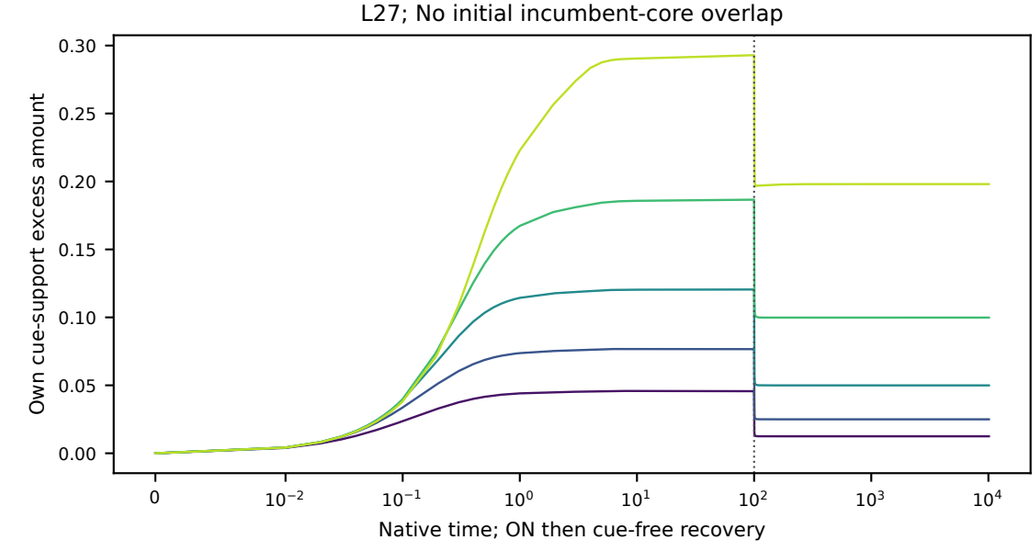


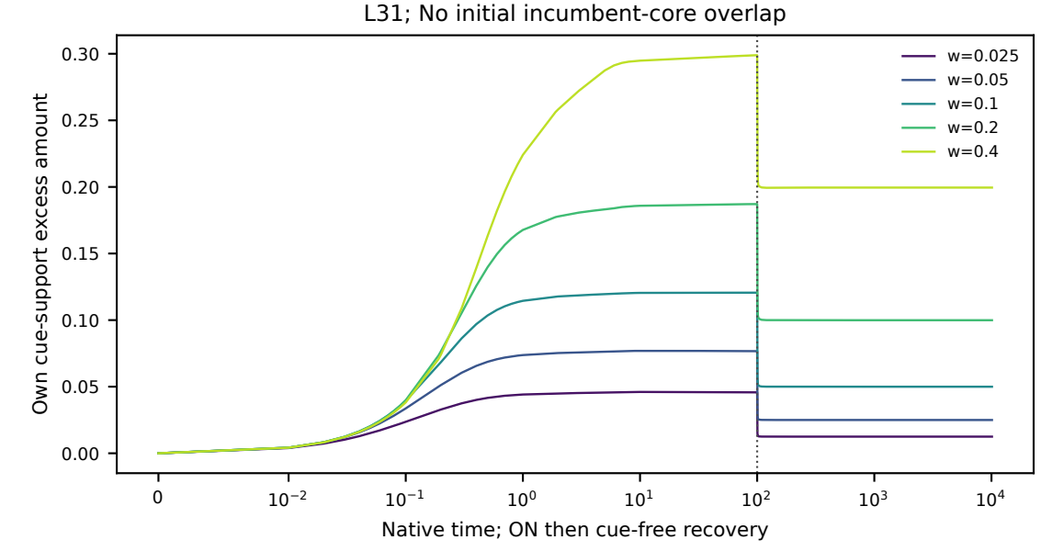
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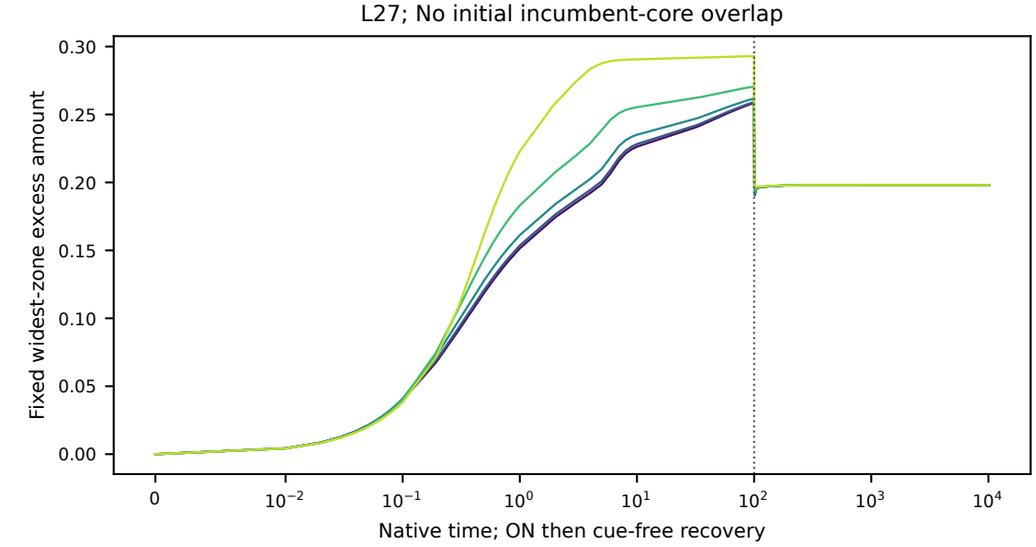
S1 Spring • commanded area 1× • native kinetics

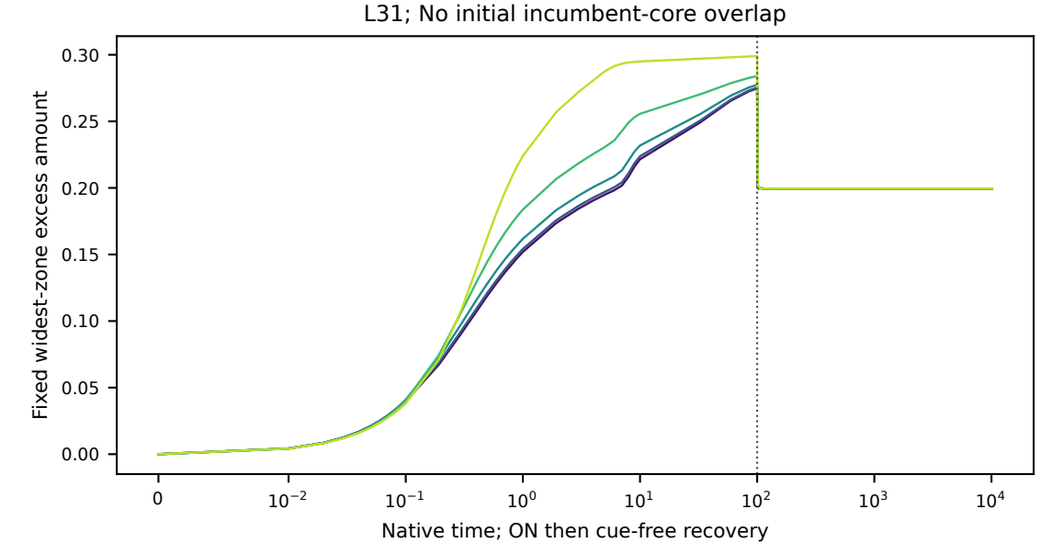






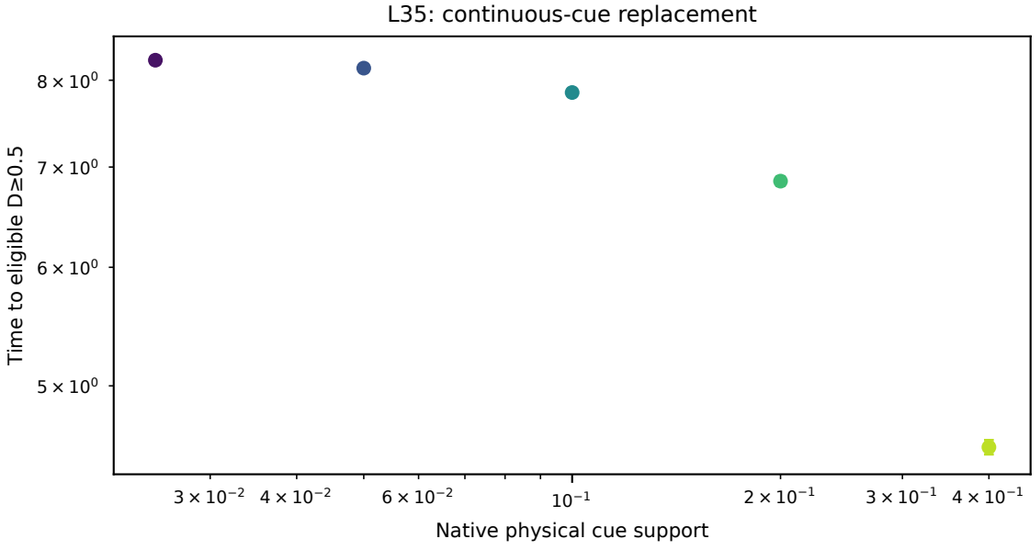


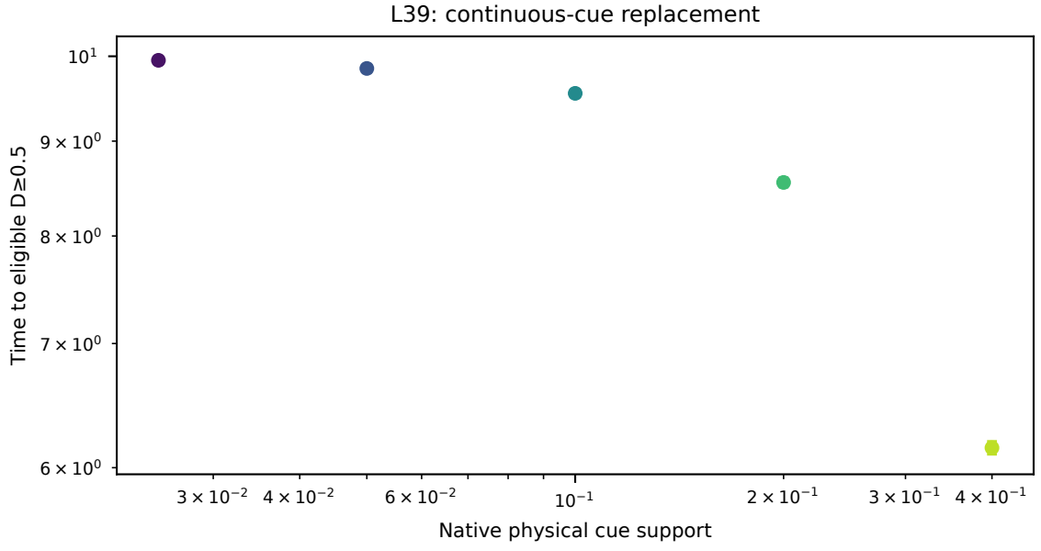


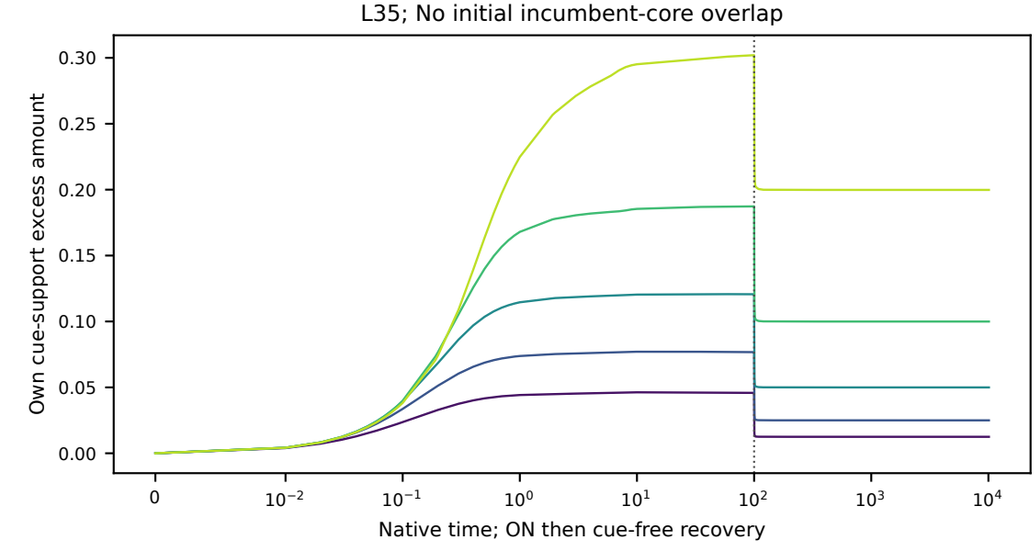


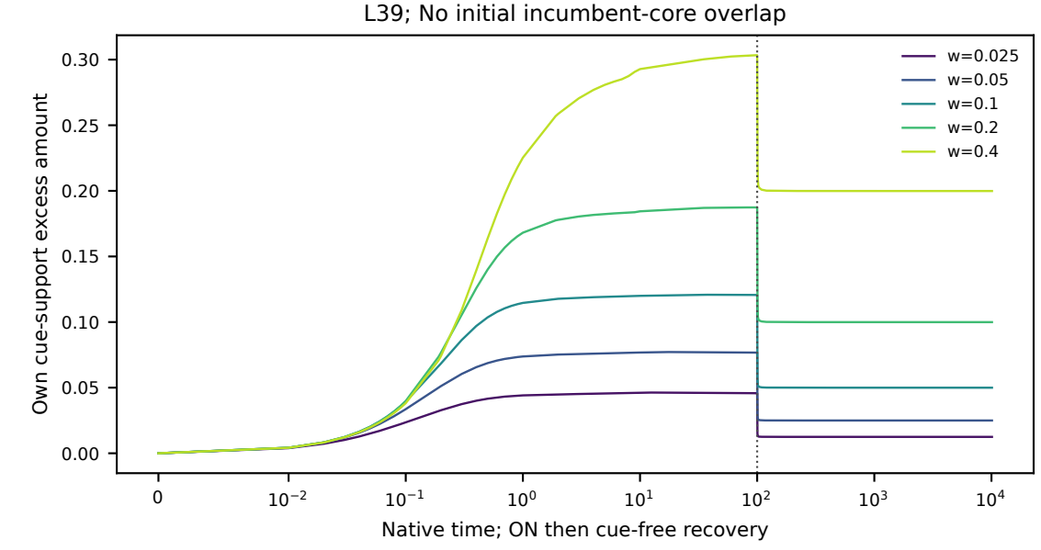
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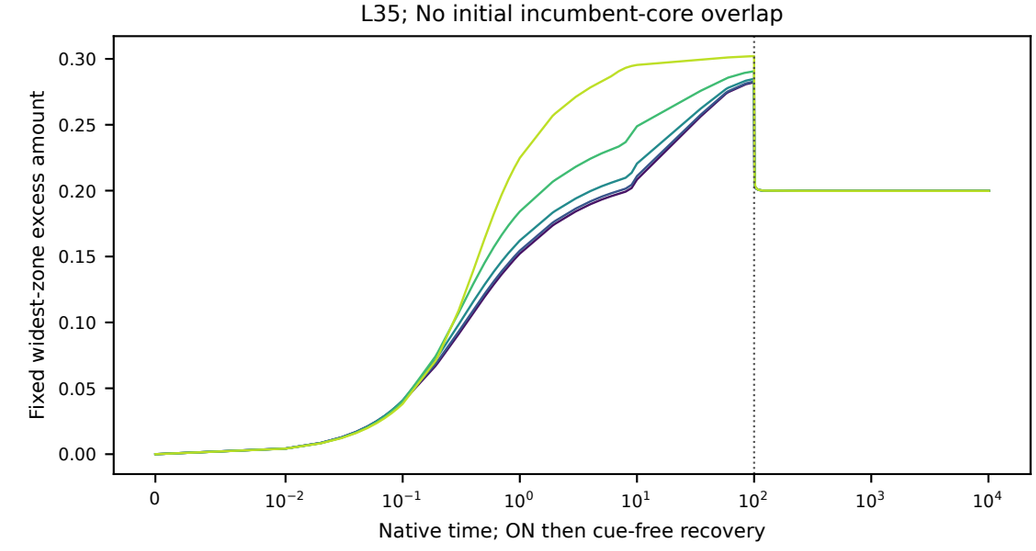
S1 Spring • commanded area 1× • native kinetics

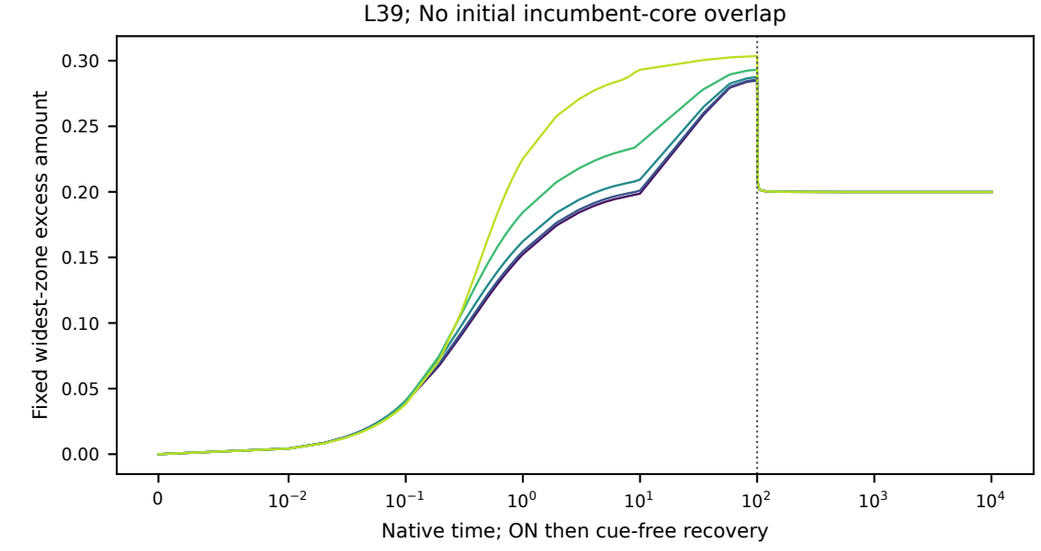






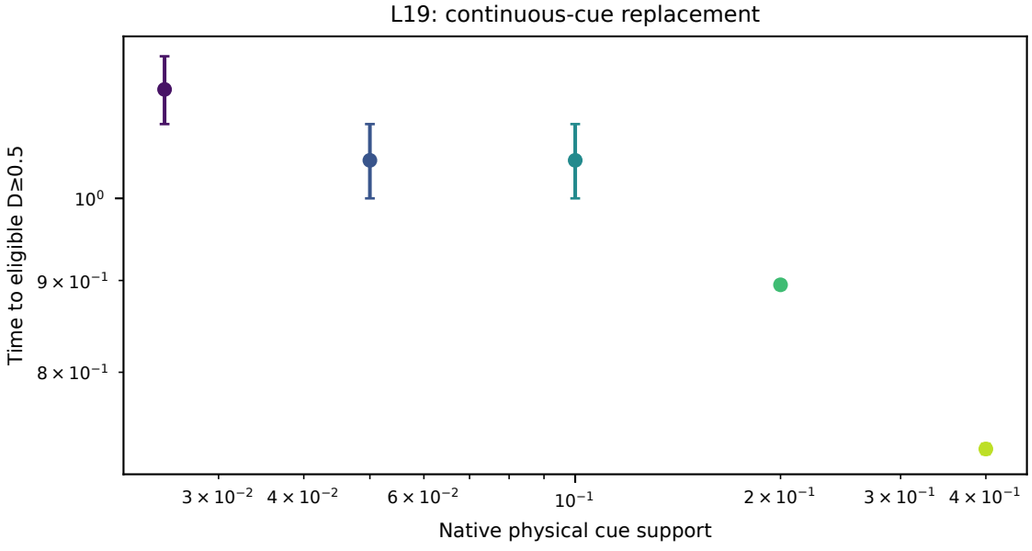


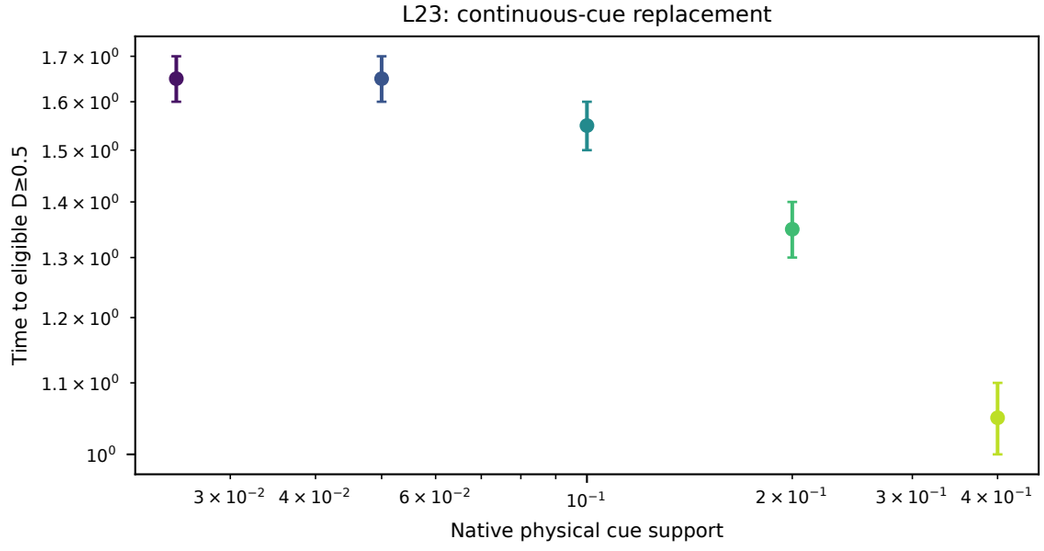


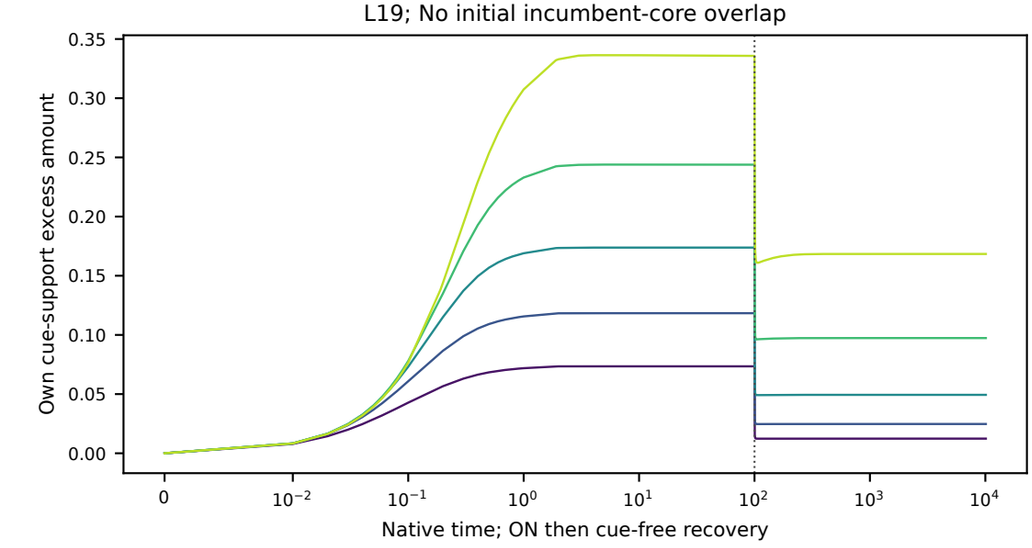


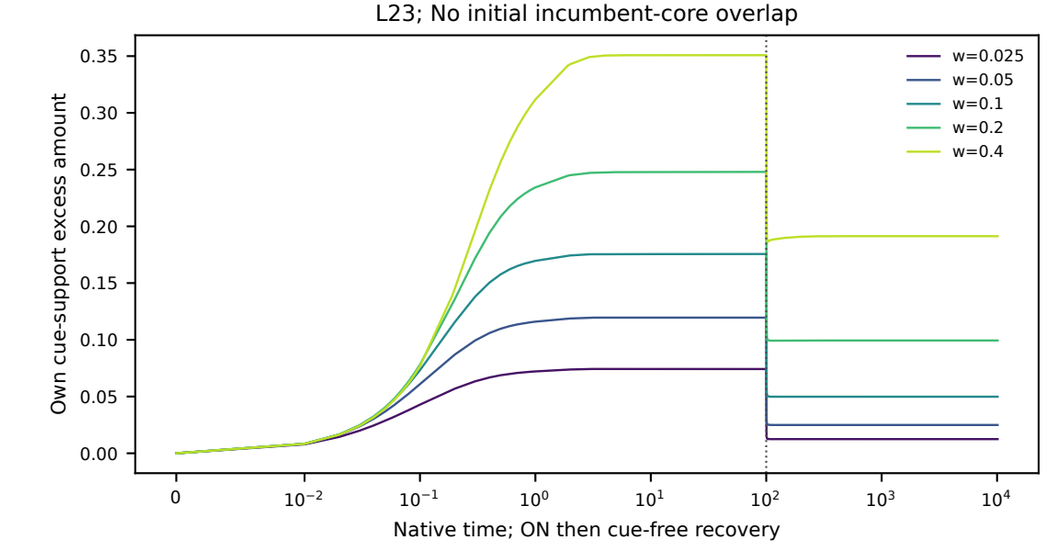
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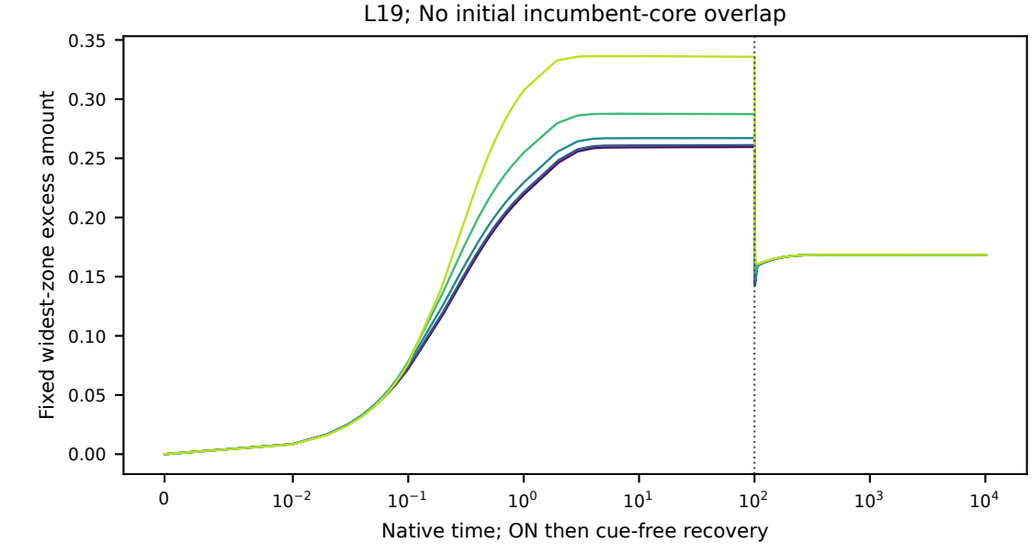
S1 Spring • commanded area 2× • native kinetics

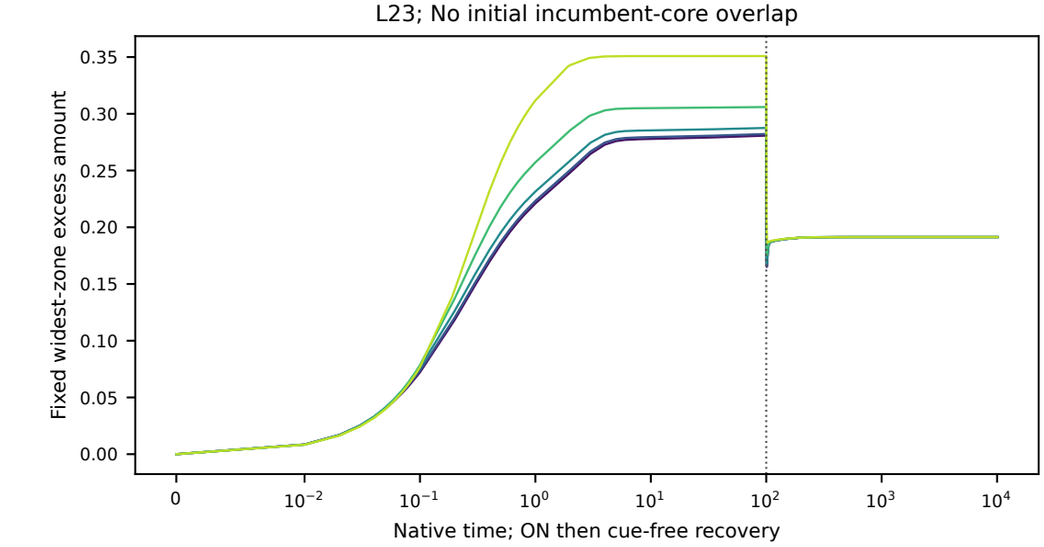






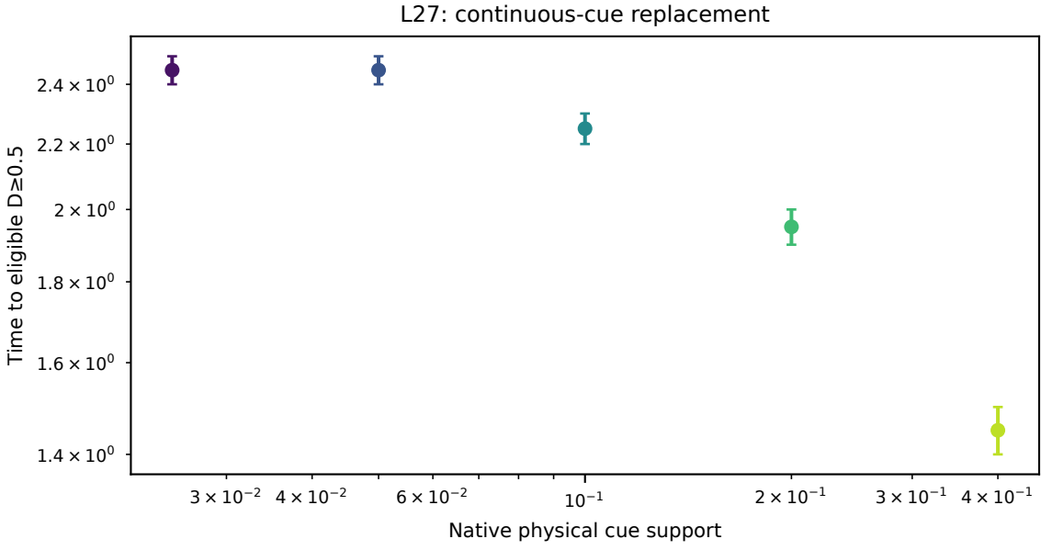


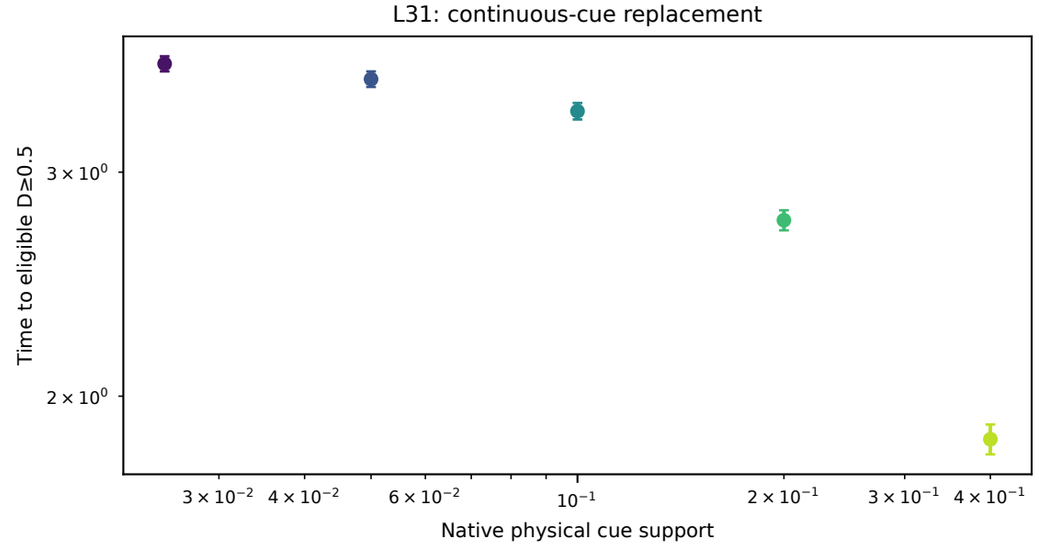


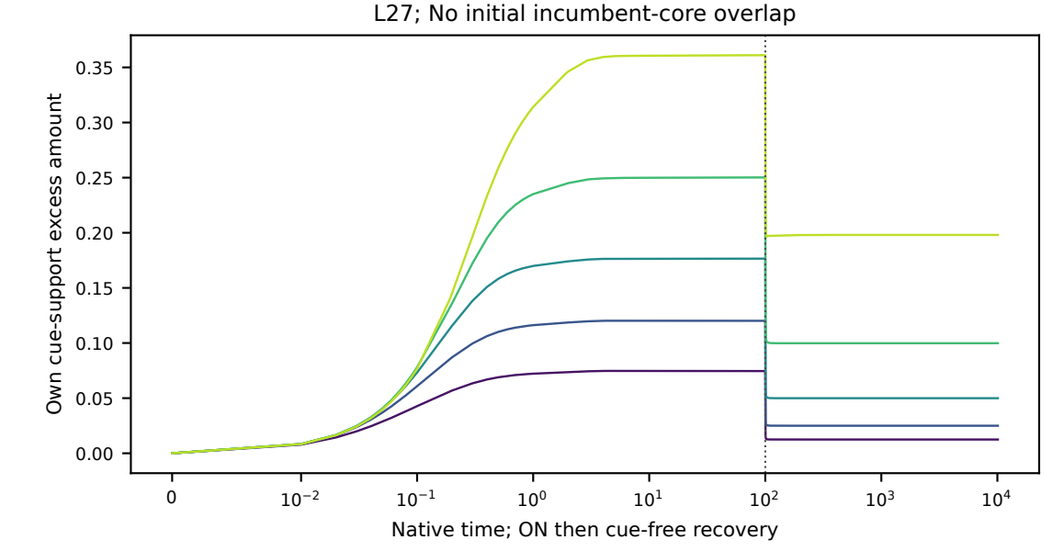


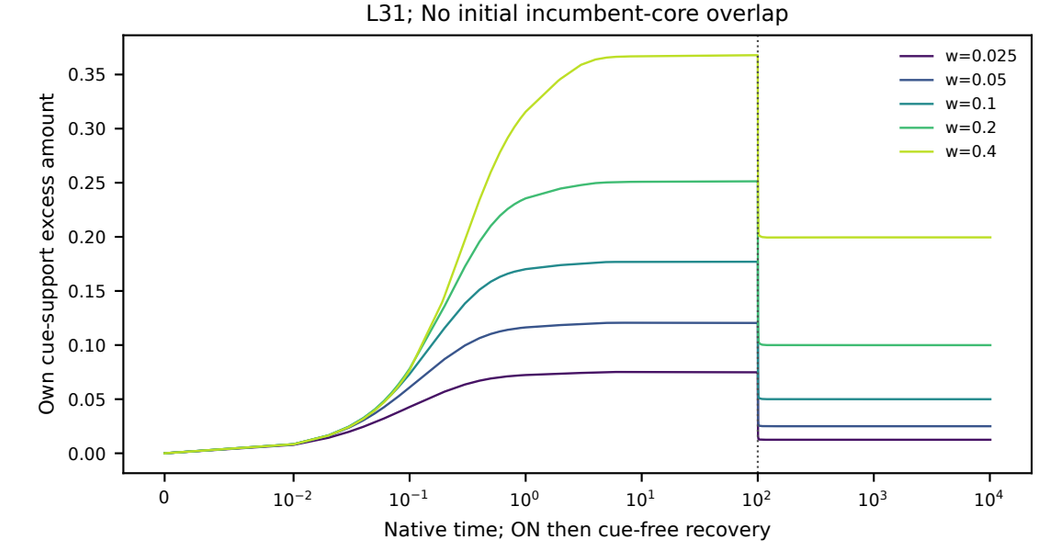
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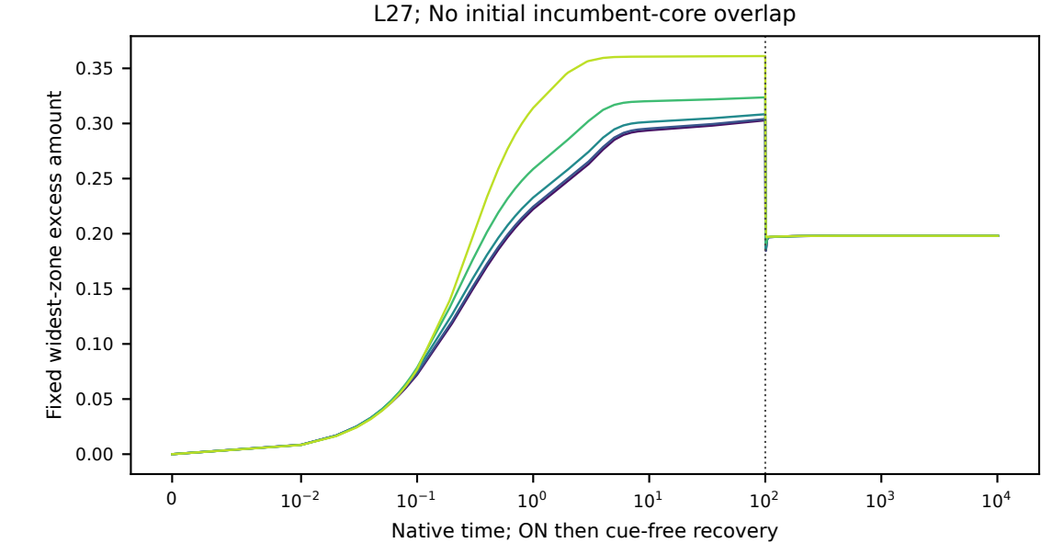
S1 Spring • commanded area 2× • native kinetics

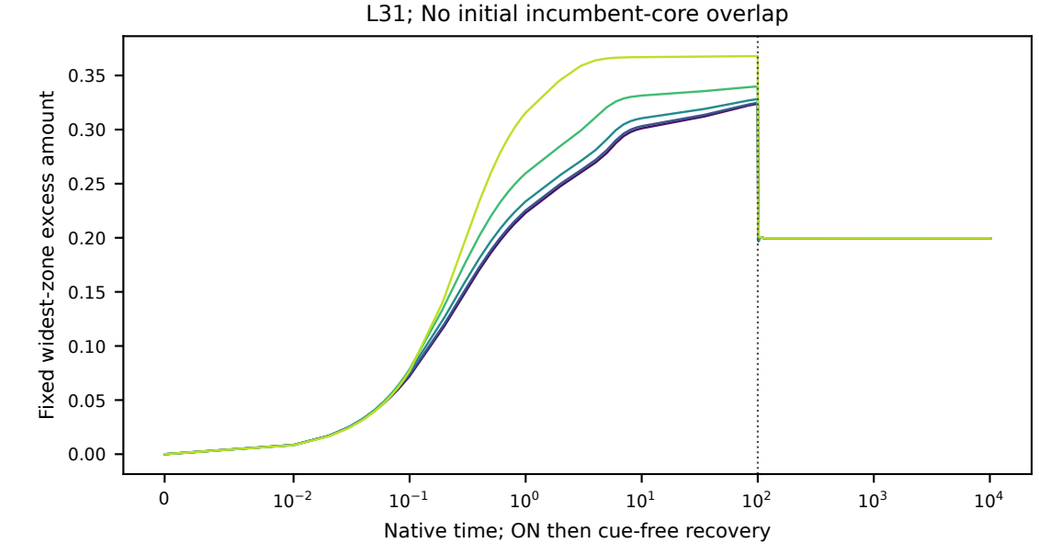






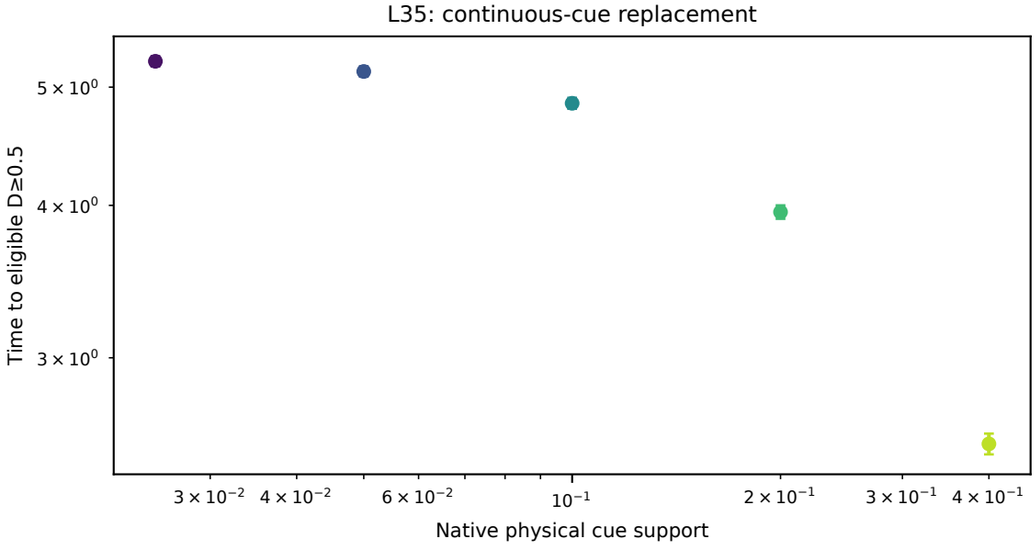


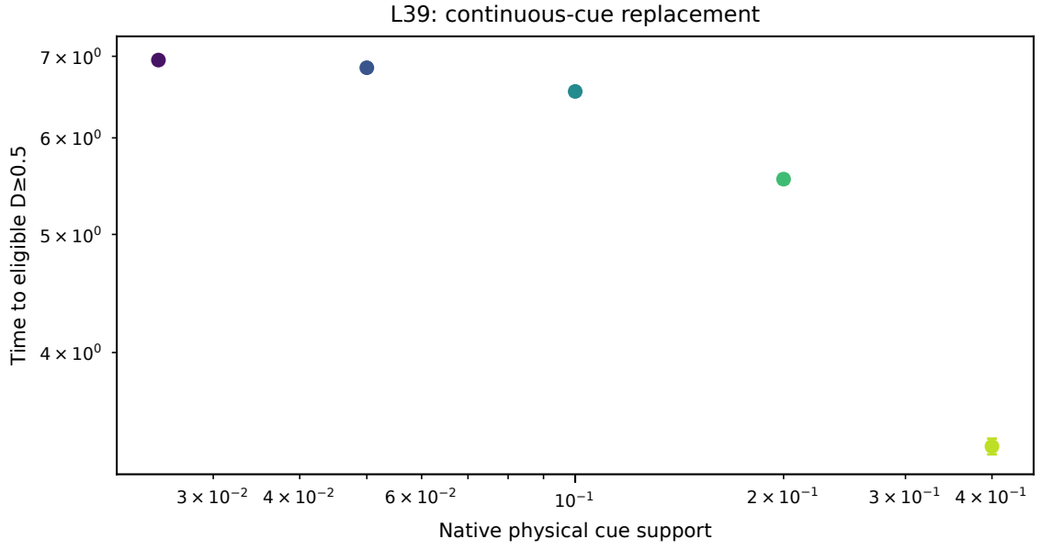


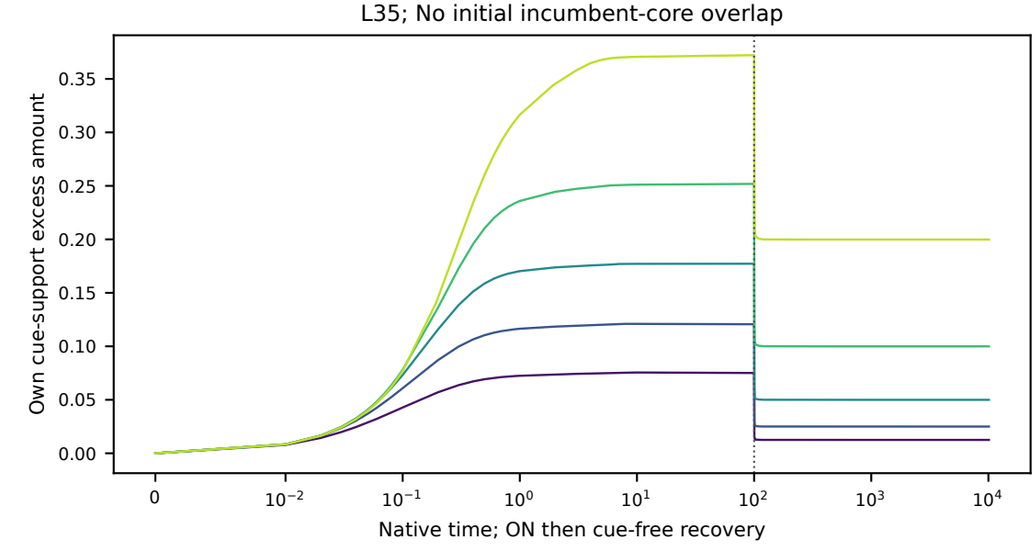


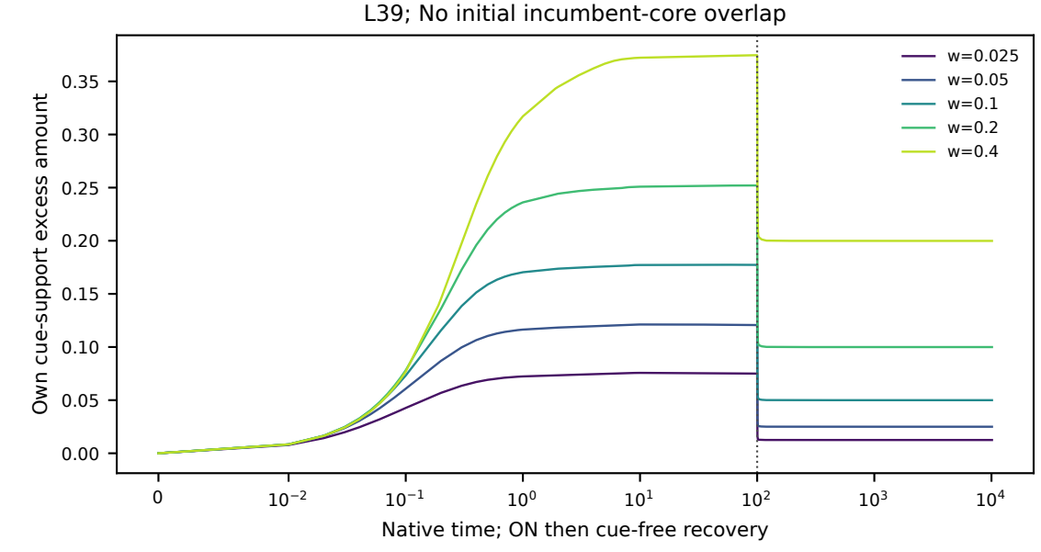
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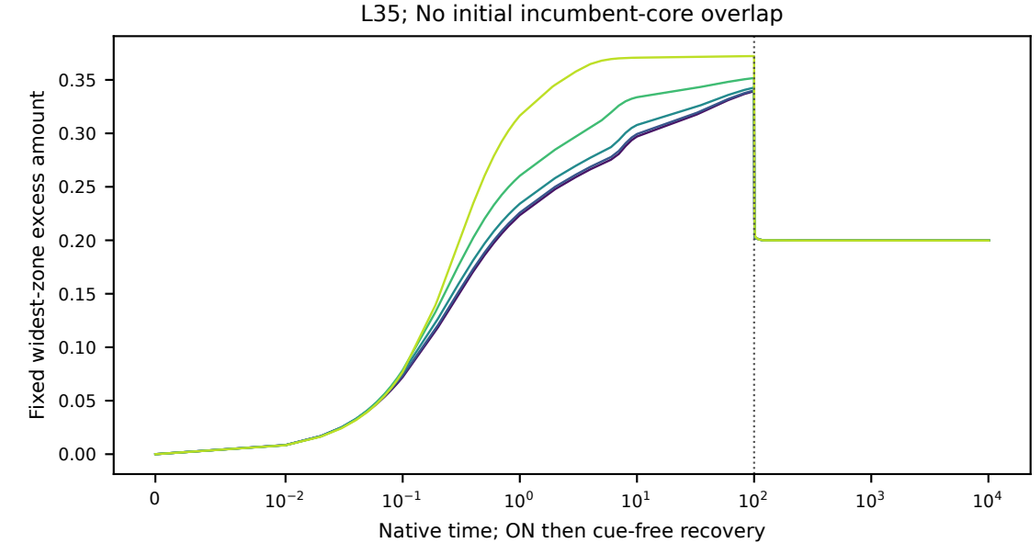
S1 Spring • commanded area 2× • native kinetics

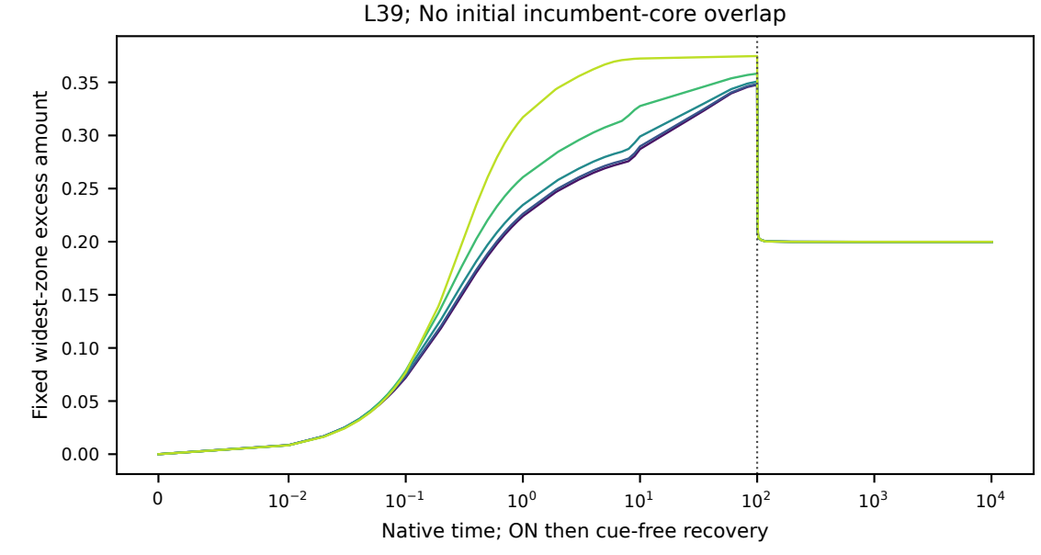




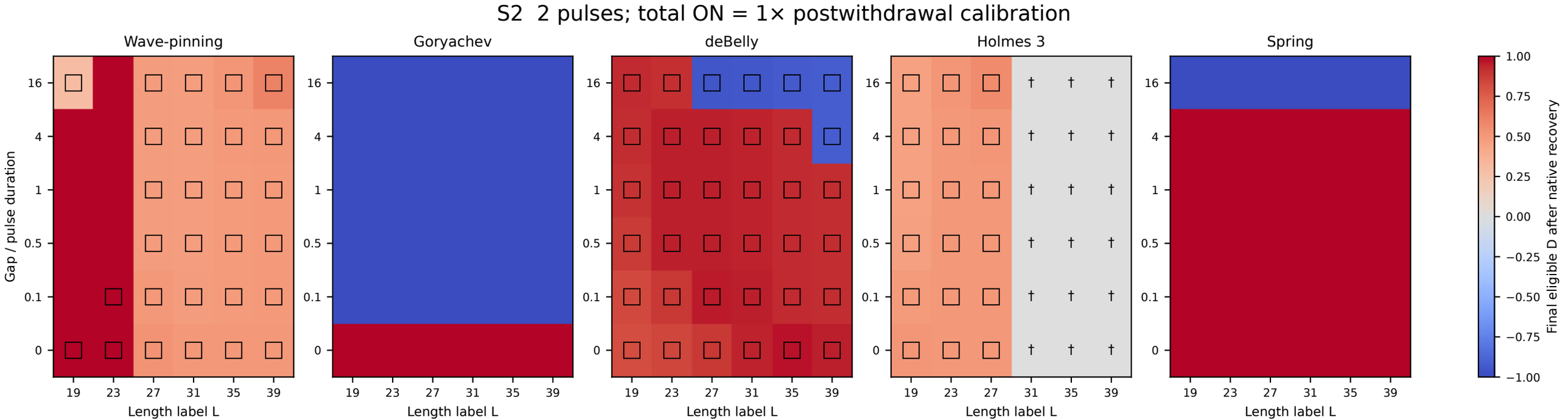




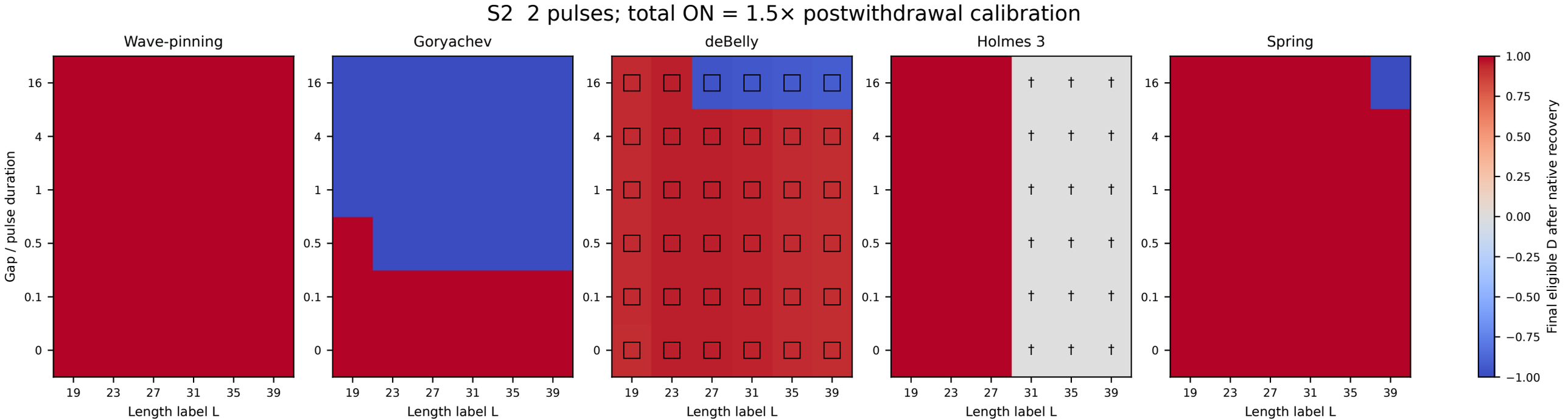


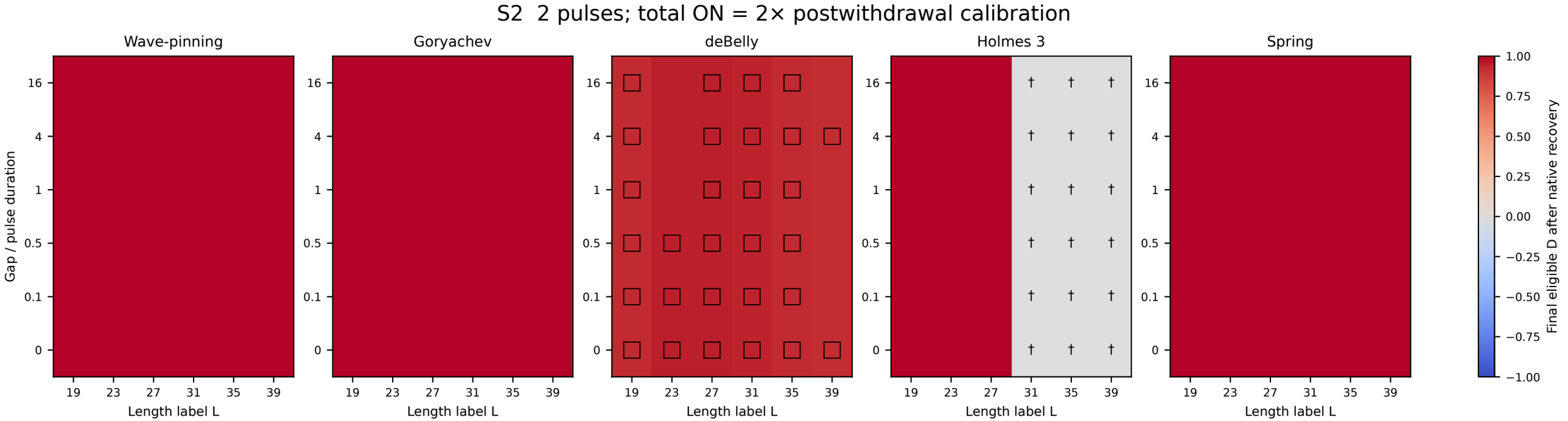


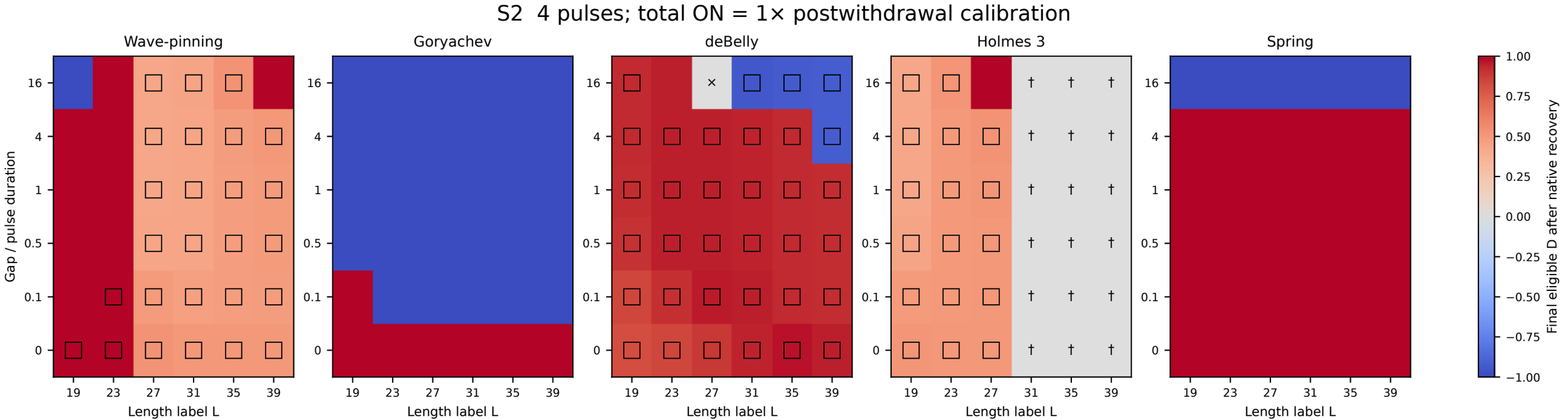
Triangles are finite lower bounds. Regional excess uses the instantaneous spatial minimum; ROI conventions can overlap the initial incumbent and do not define the primary disjoint-half D. Every field-tail flag remains in the source table.



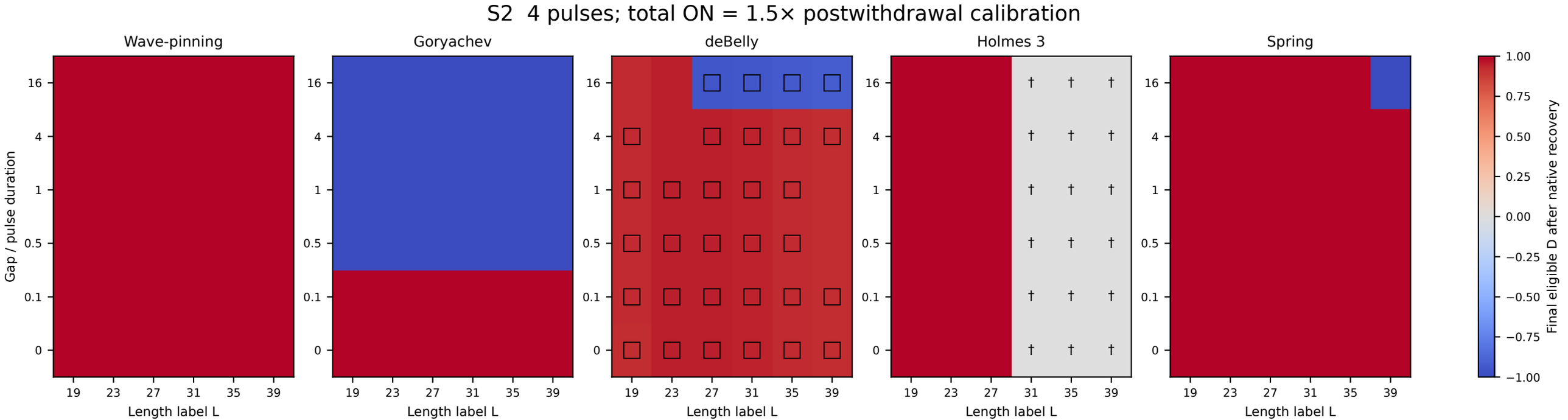
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).

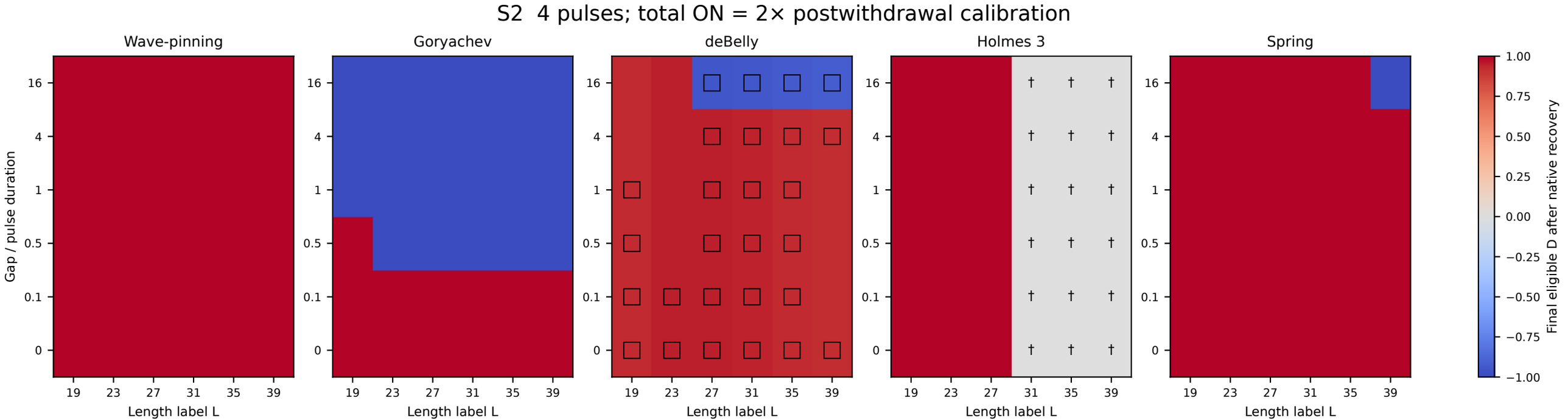

S2 2 pulses; total ON = 1.5× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
Length label L
Final eligible D after native recovery
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).

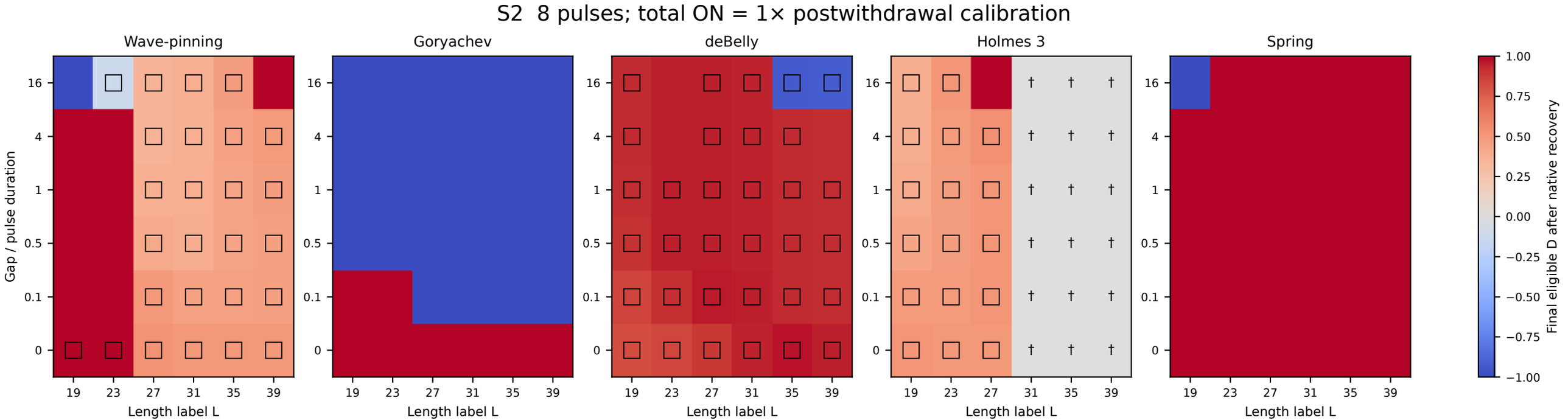

S2 2 pulses; total ON = 2× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
16
4
1
0.5
0.1
0
19
23
27
31
35
39
Length label L
Final eligible D after native recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes
are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).



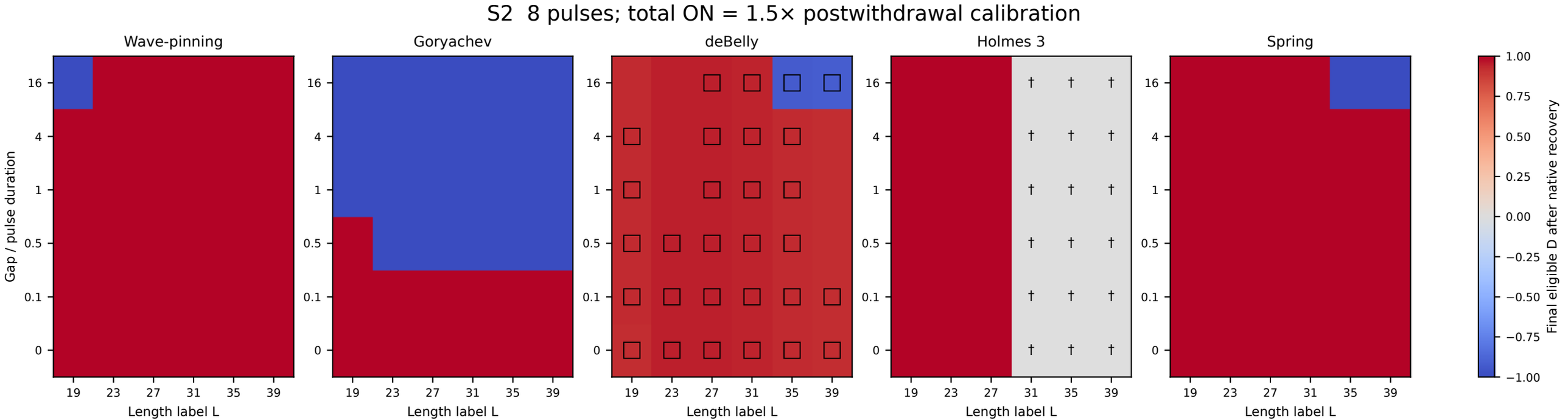
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).

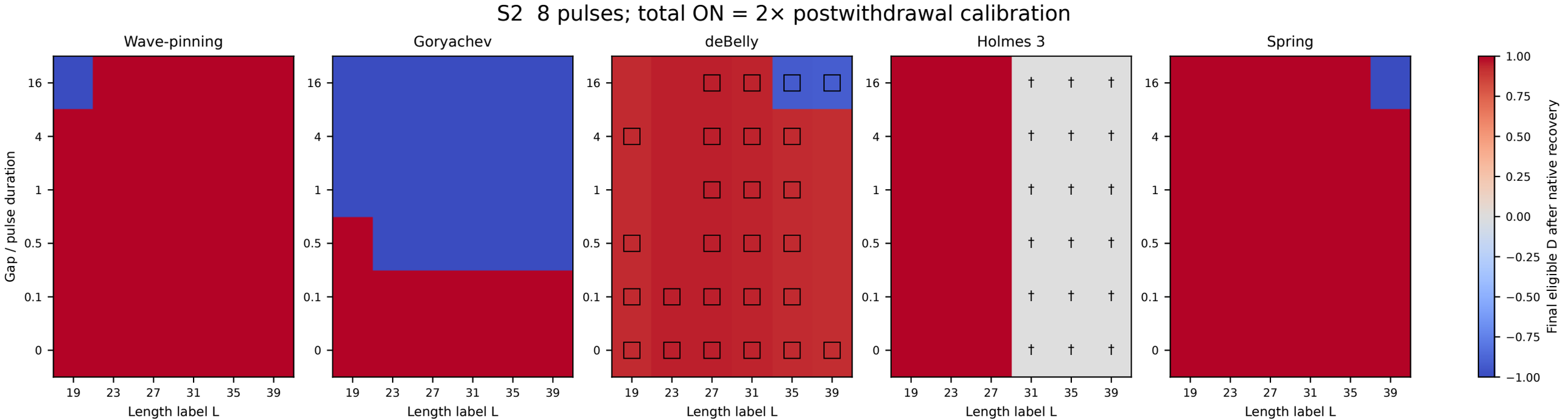

S2 4 pulses; total ON = 1.5× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
Length label L
Final eligible D after native recovery
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).

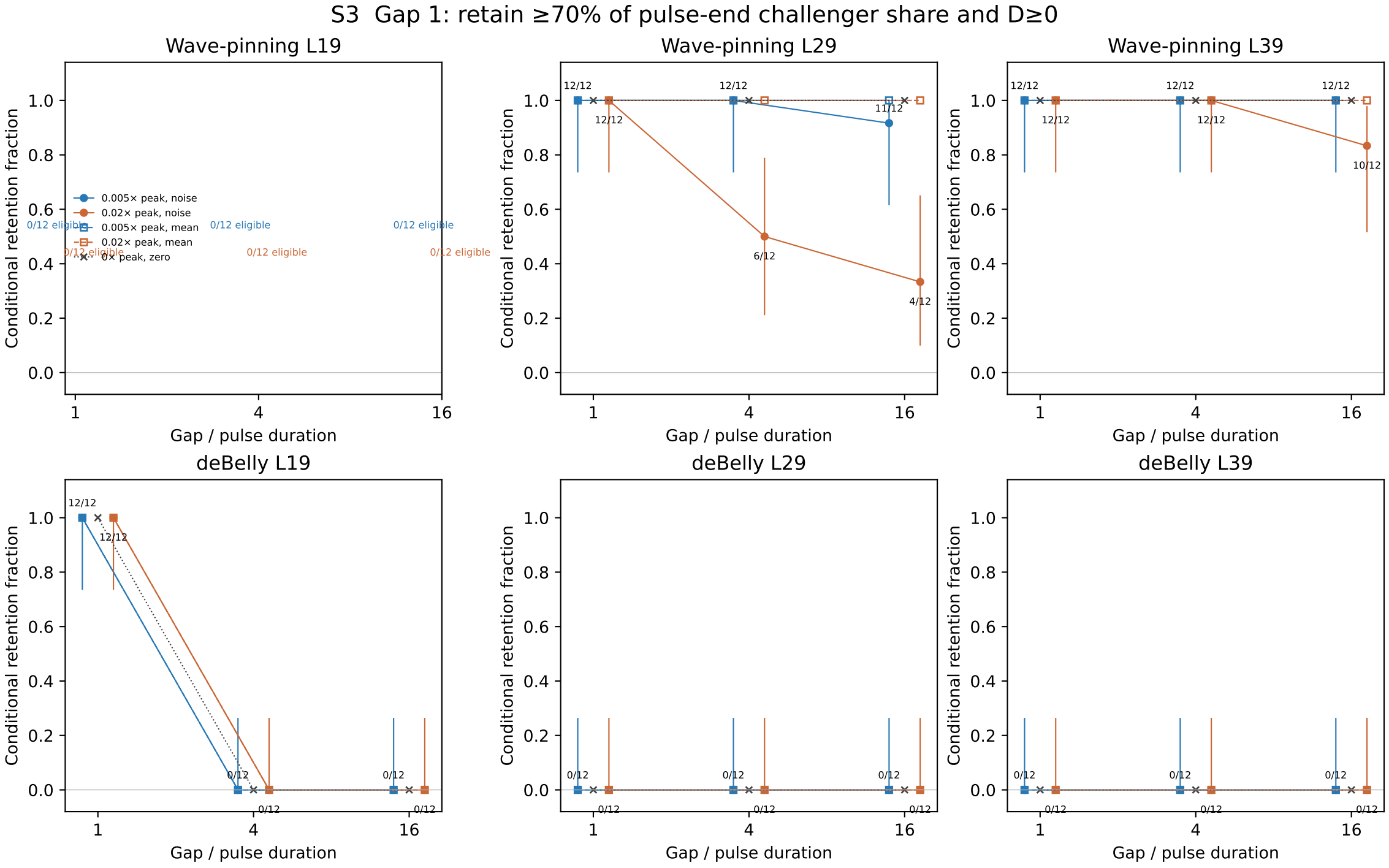

S2 4 pulses; total ON = 2× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
16
4
1
0.5
0.1
0
19
23
27
31
35
39
Length label L
Final eligible D after native recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes
are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).



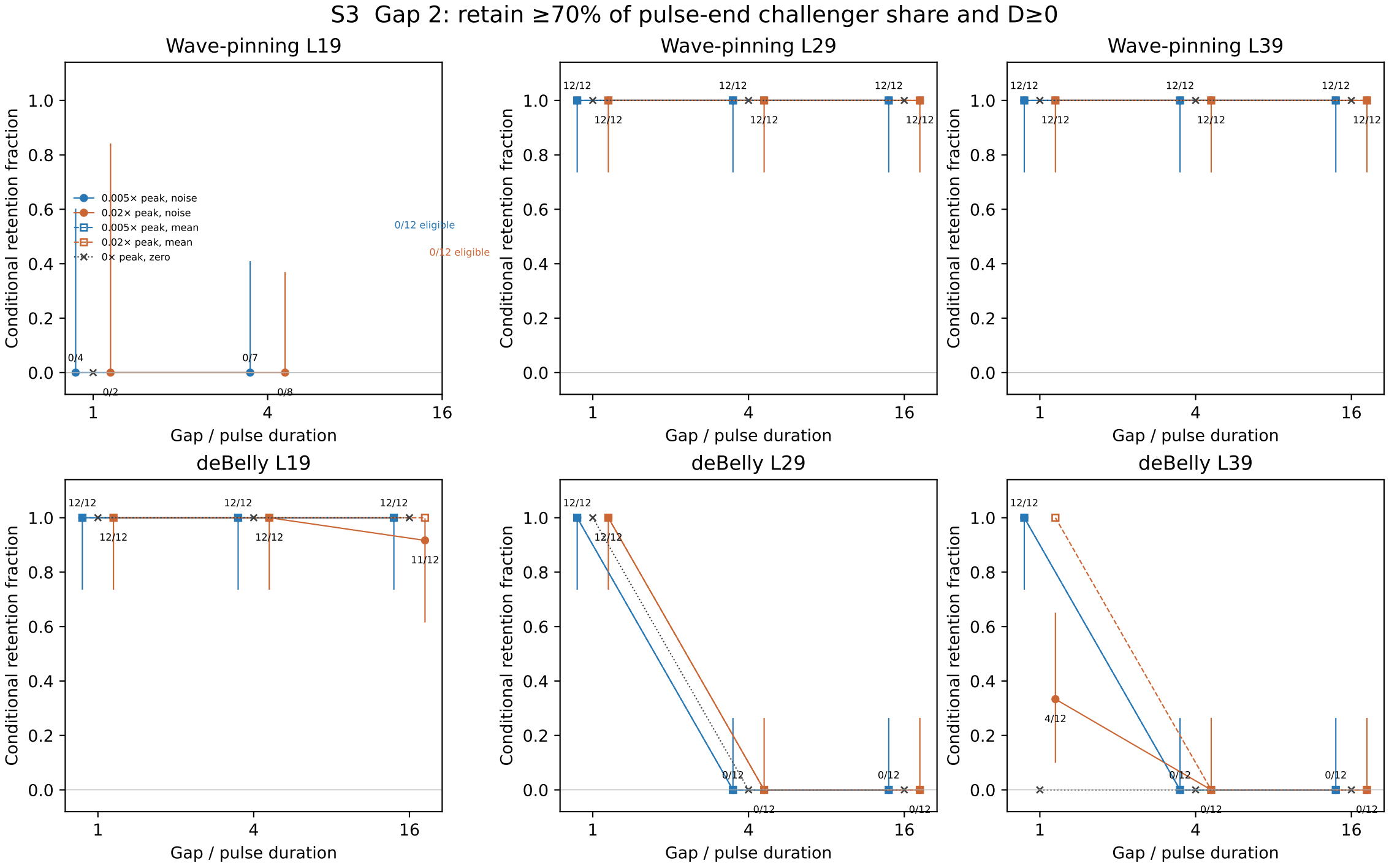
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).


S2 8 pulses; total ON = 1.5× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
16
4
1
0.5
0.1
0
19
23
27
31
35
39
Length label L
Final eligible D after native recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes
are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).

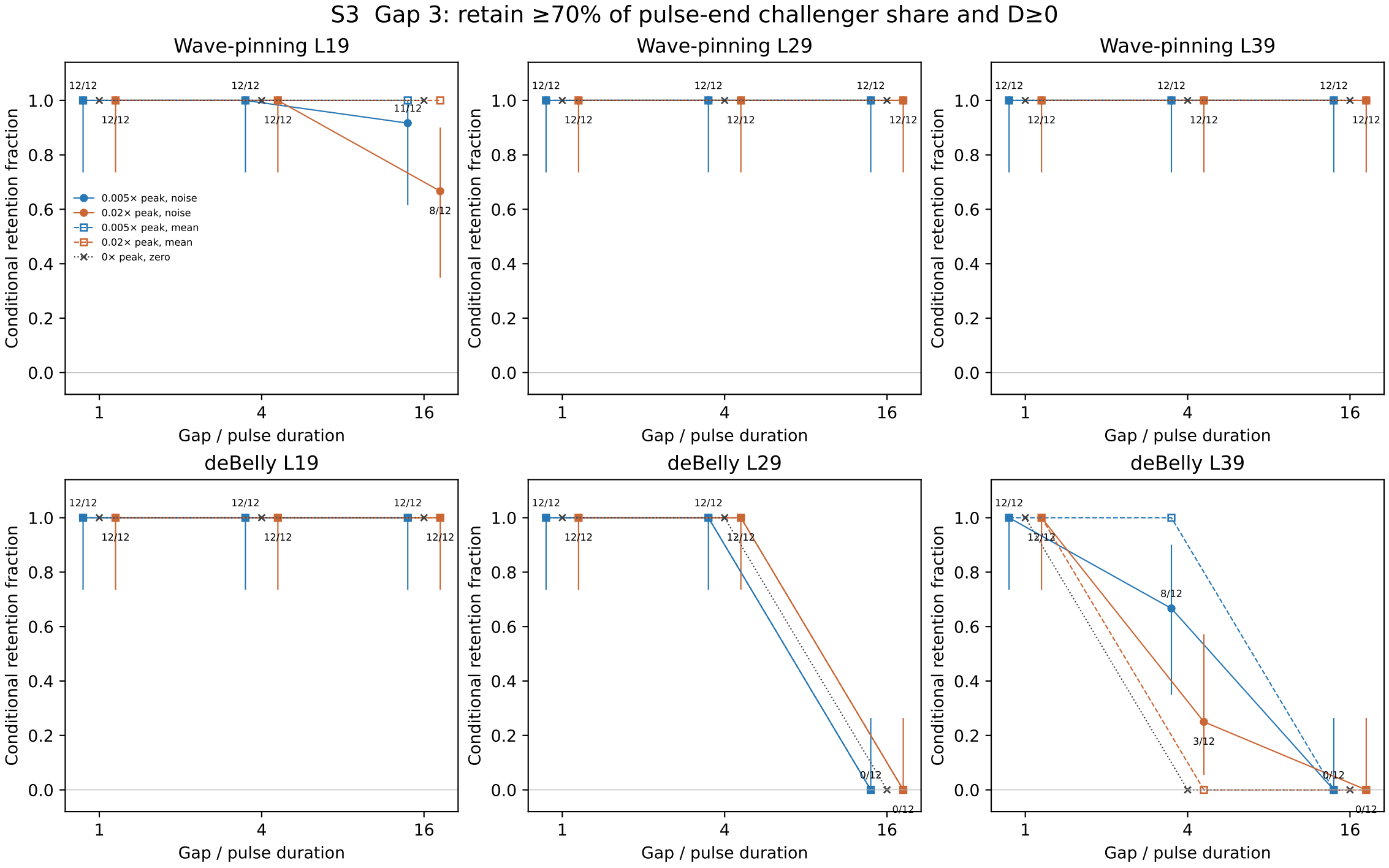

S2 8 pulses; total ON = 2× postwithdrawal calibration
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Gap / pulse duration
16
4
1
0.5
0.1
0
19
23
27
31
35
39
Length label L
Final eligible D after native recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
□: failed native-field tail, independent of fate; ×: known low activity; †: no sampled calibration at the finite cap; plain grey: missing independent native audit. Native numerical schemes
are retained; deBelly uses the 0.5-unit update and later-gap timing is sensitive to its qualified quarter-step companion (S5).



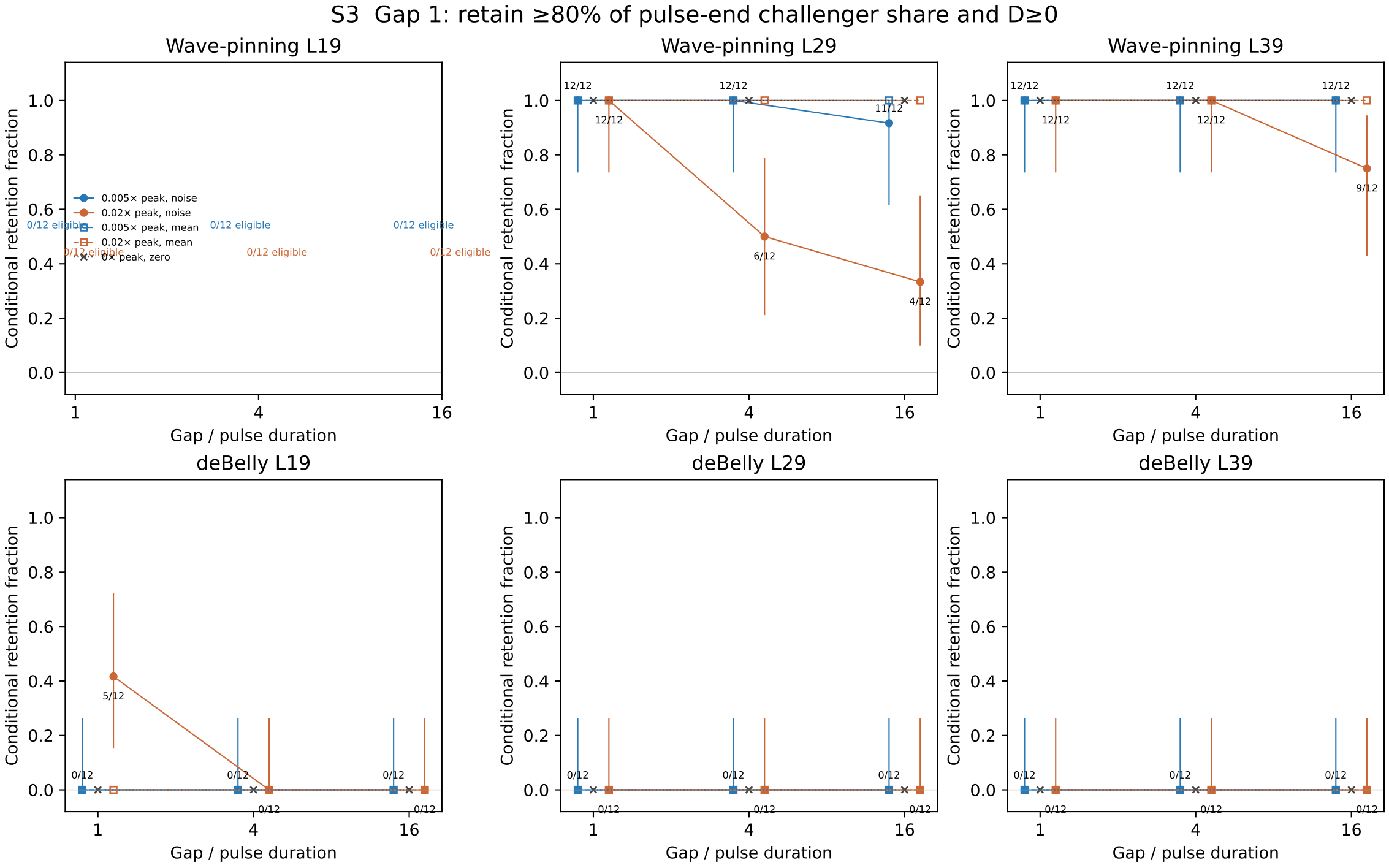
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.

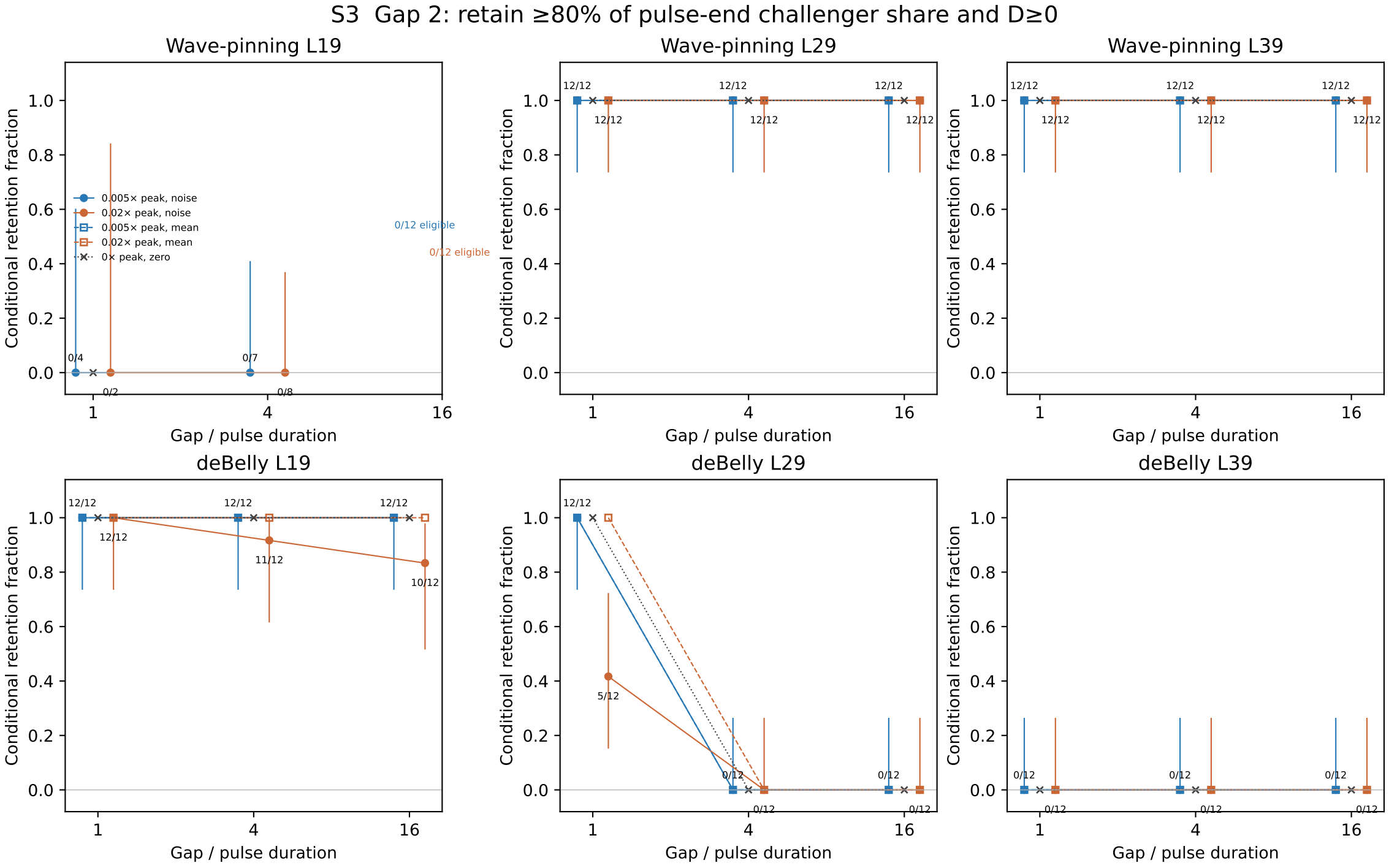


Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.

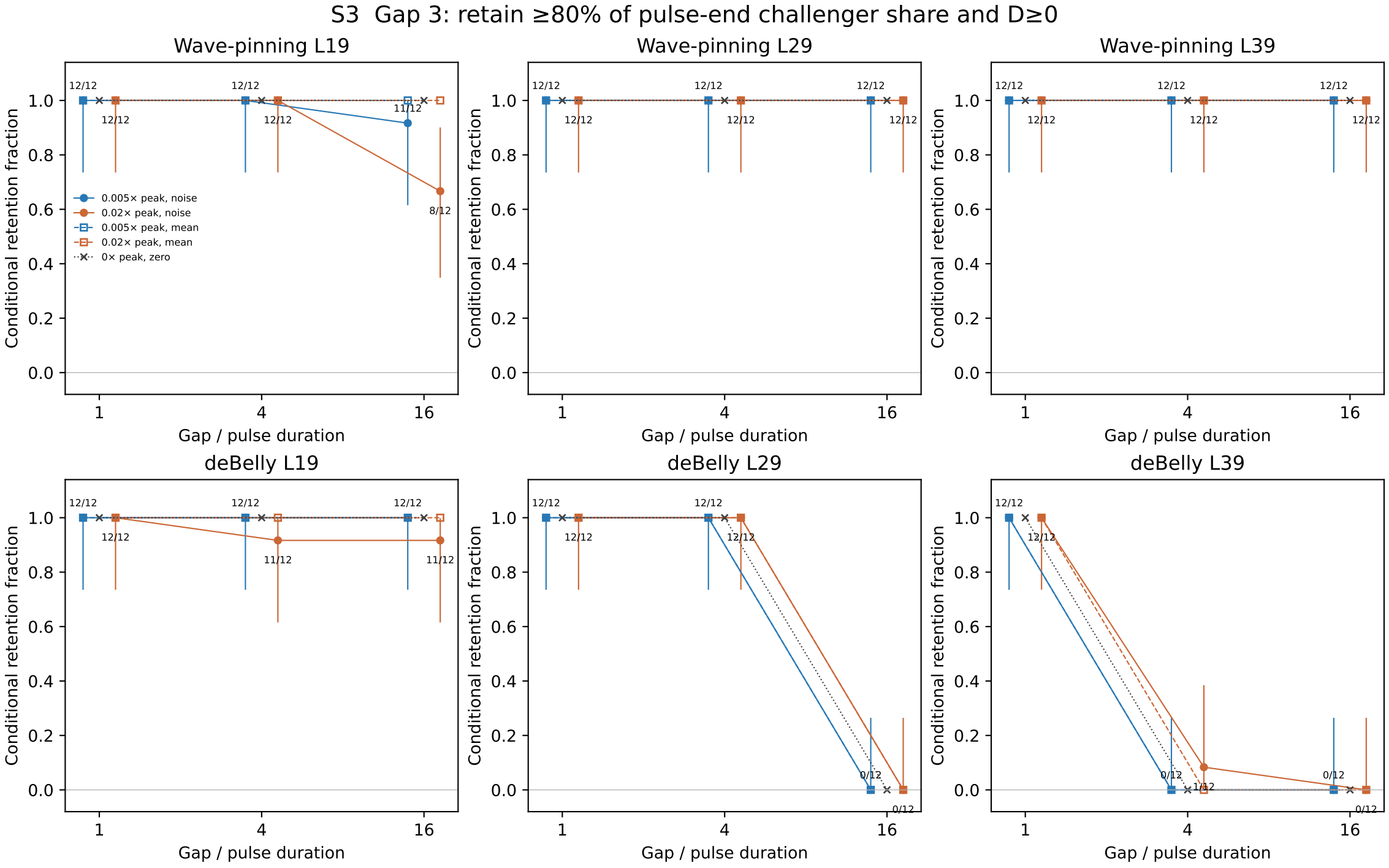


Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.

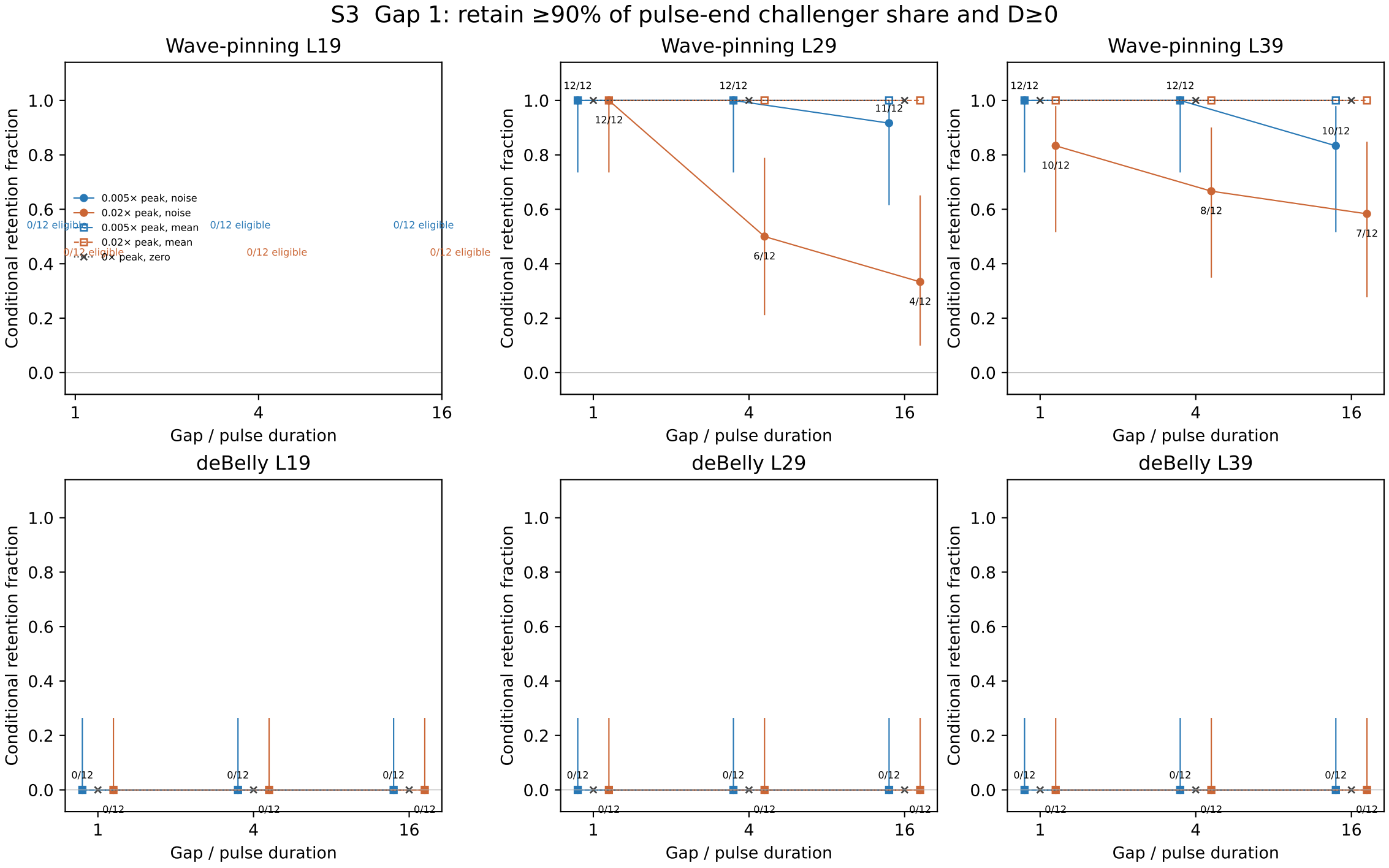

S3 Gap 1: retain ≥80% of pulse-end challenger share and D≥0
Wave-pinning L19
Wave-pinning L29
Wave-pinning L39
deBelly L19
deBelly L29
deBelly L39
Conditional retention fraction
Gap / pulse duration
0.005× peak, noise
0.02× peak, noise
0.005× peak, mean
0.02× peak, mean
0× peak, zero
0/12 eligible
12/12
11/12
9/12
6/12
4/12
5/12
0/12
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



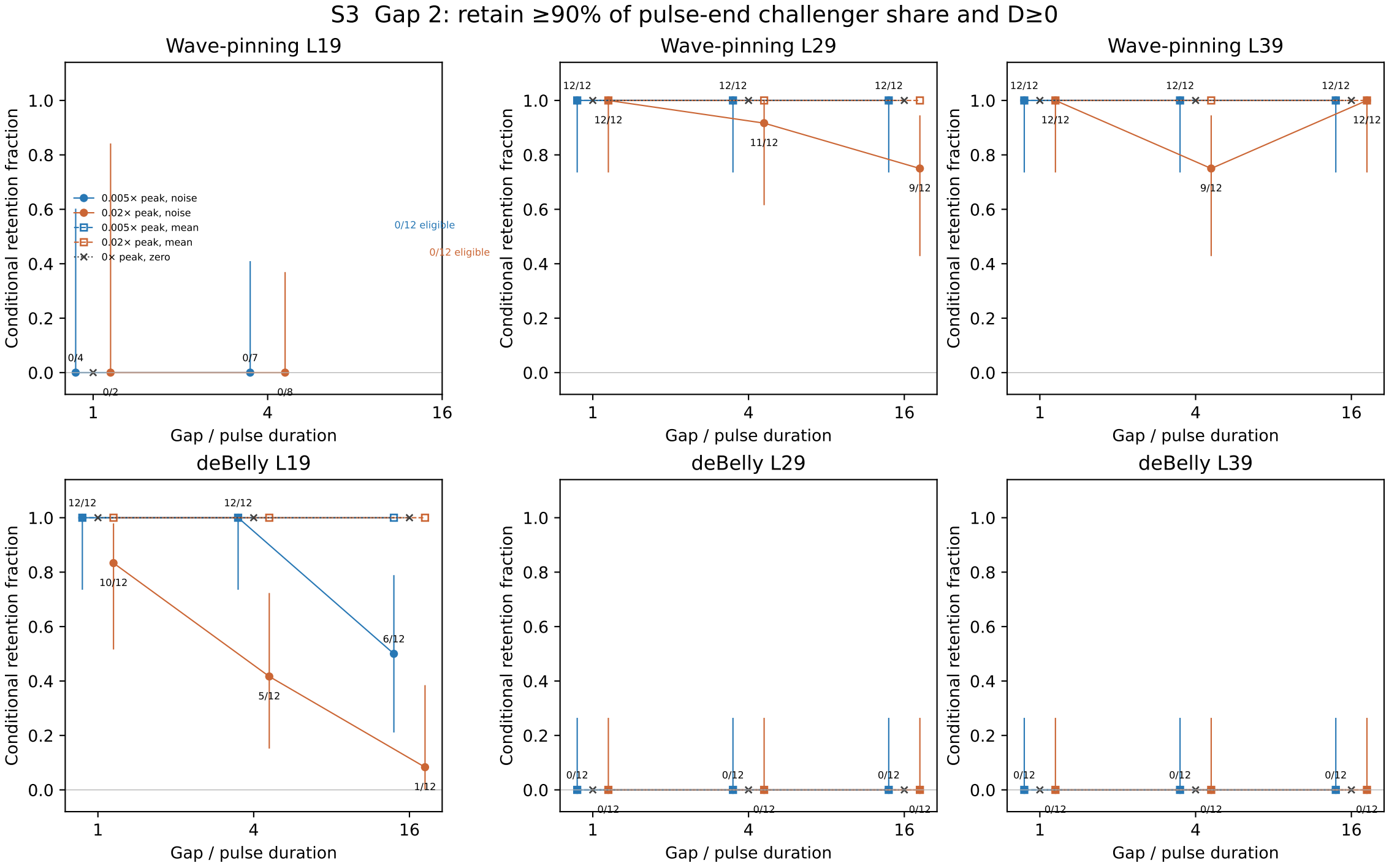
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



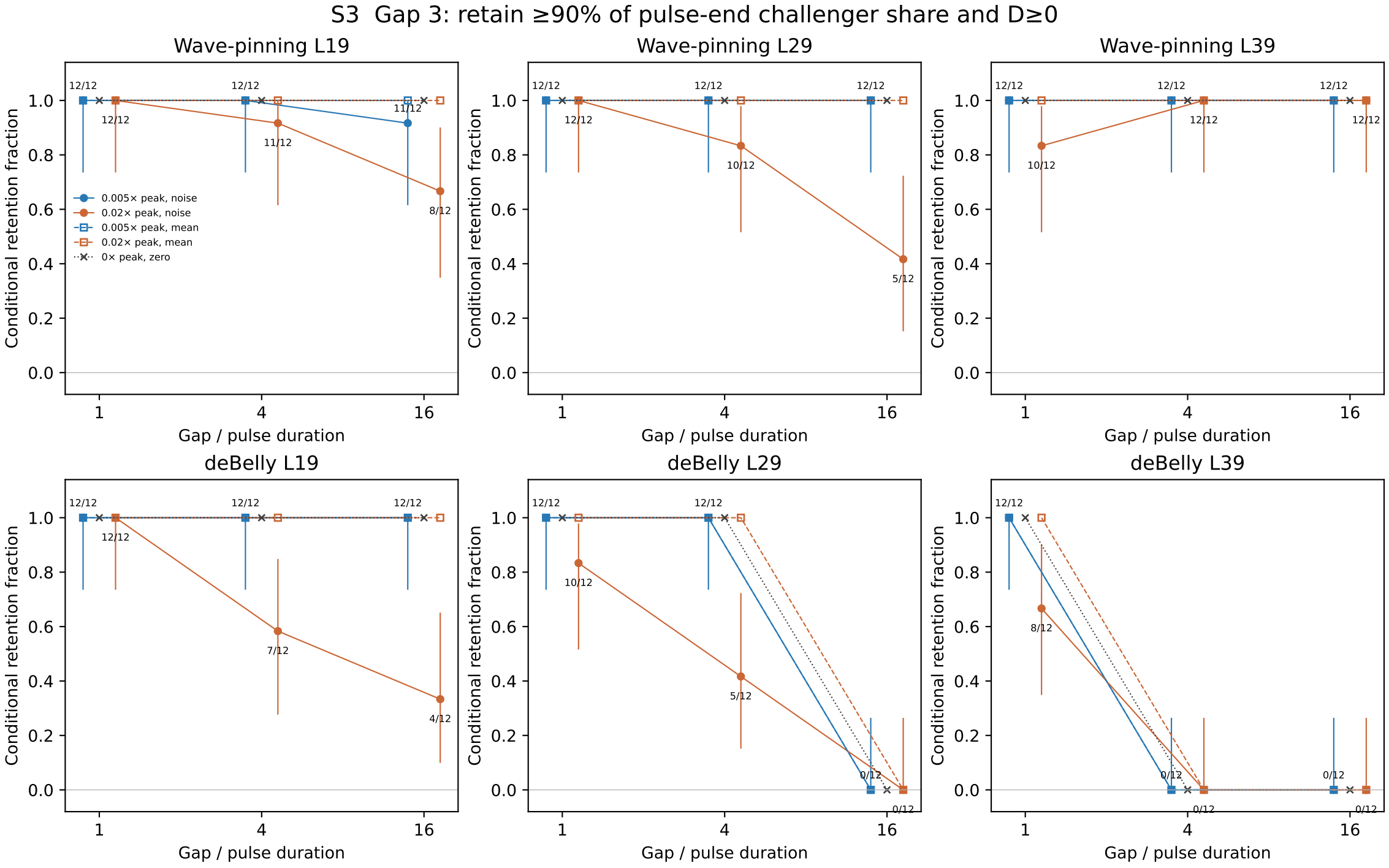
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



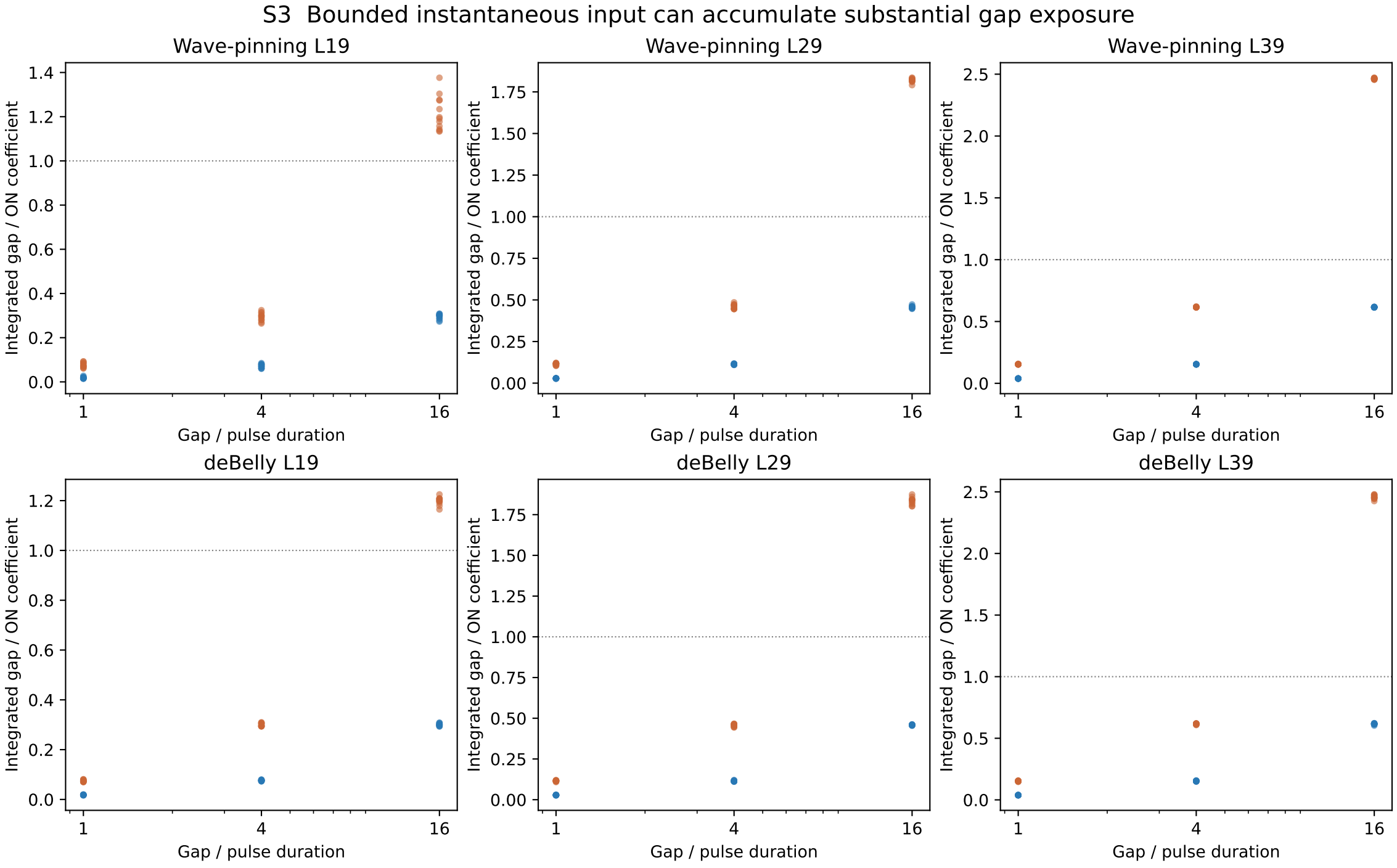
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



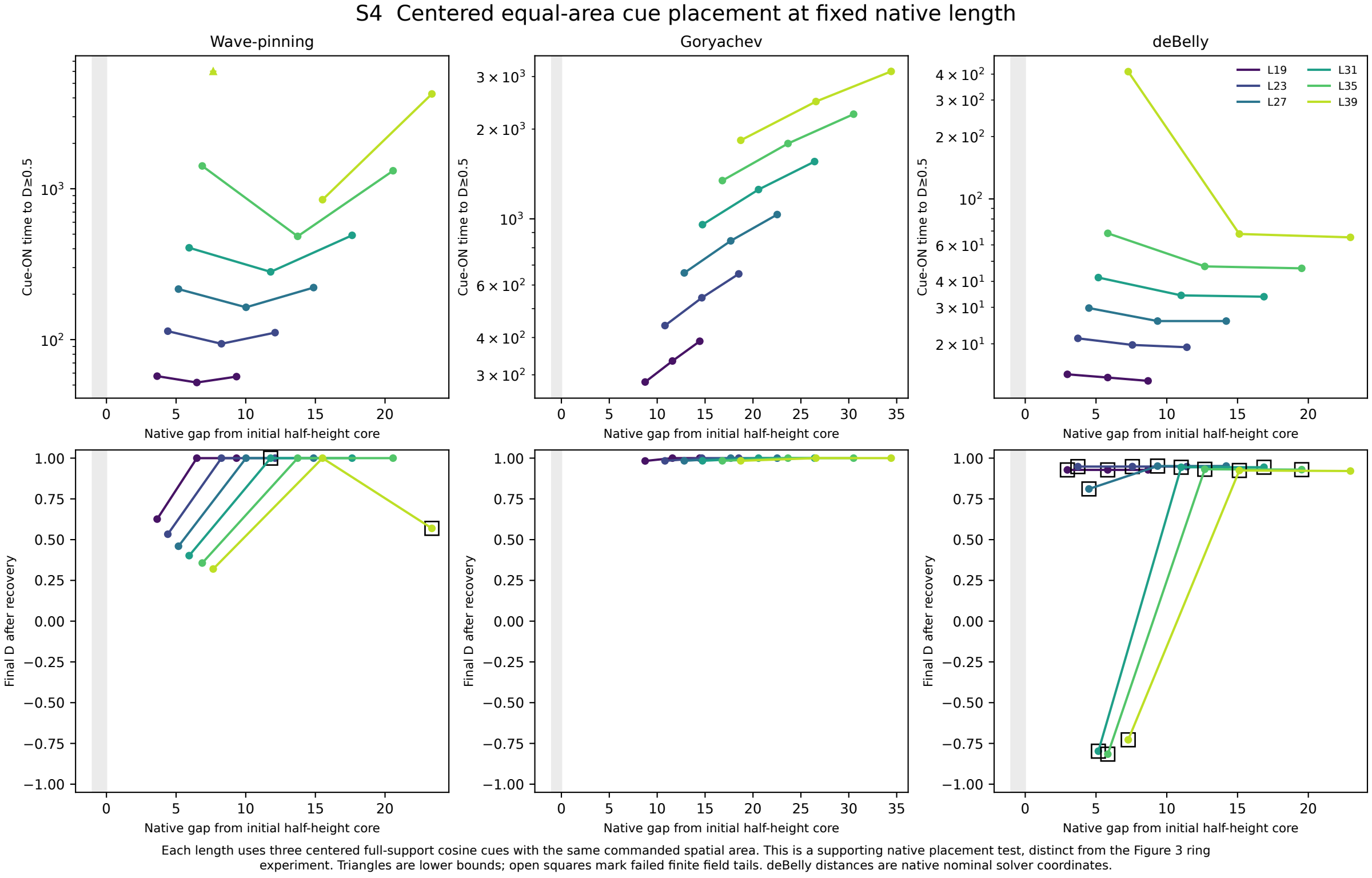
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



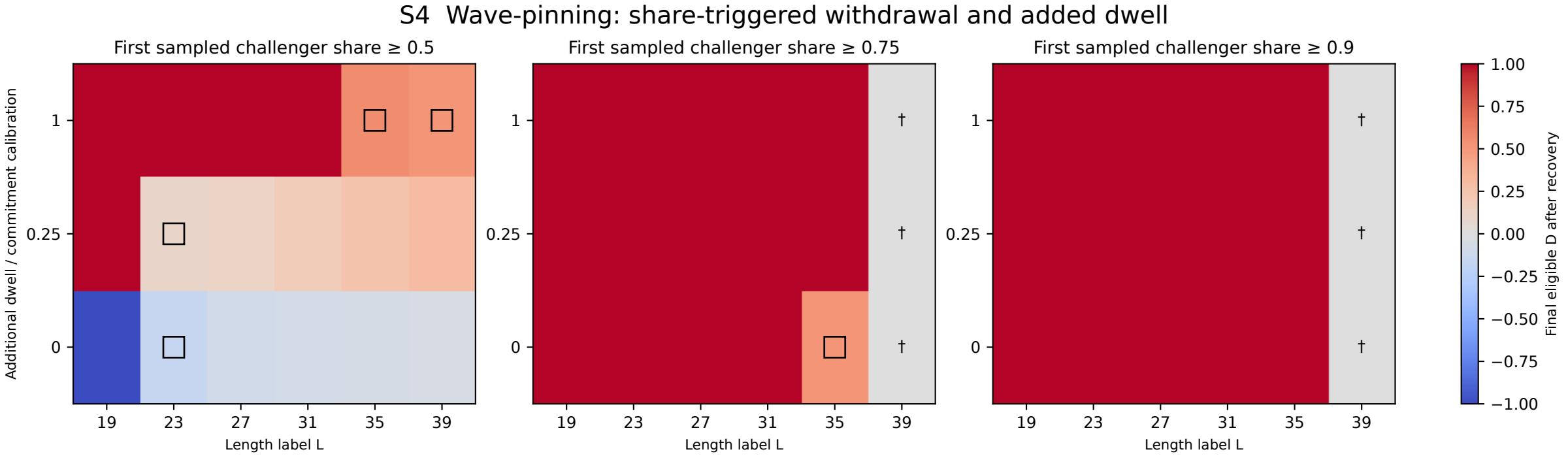
Each noisy denominator contains at most 12 independent trajectories; three gaps from one trajectory are not independent replicates. Eligibility requires pulse-end challenger bias, not eventual replacement. Known gap-end bias/activity loss counts as failure. Error bars: exact 95% finite-seed intervals. deBelly probabilities are conditional on its native0.5 update; later-gap timing is sensitive in S5.



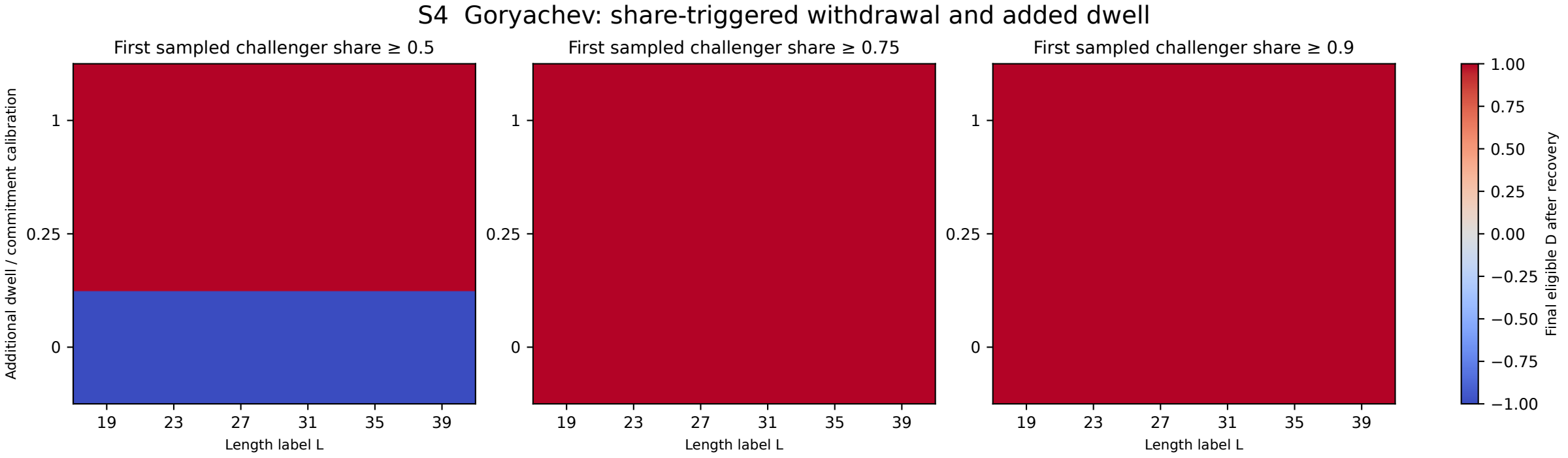
Weak describes the instantaneous bounded coefficient (maximum1% or4% of native peak), not cumulative exposure. Spatially distributed gap input is integrated over the whole native domain; at L39/gap16/mean2% its expected commanded exposure is about2.46× targeted ON exposure. This is not calibrated photon dose or biochemical activation. Native guards, all exposure summaries, final fate and field-tail flags remain separately in the source JSON.

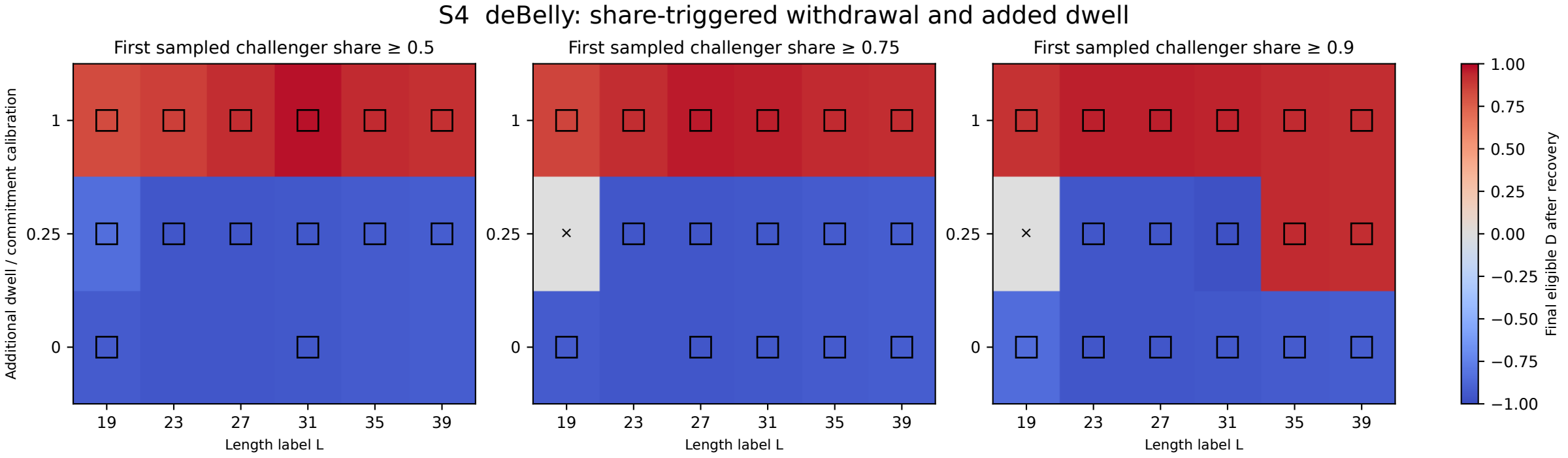


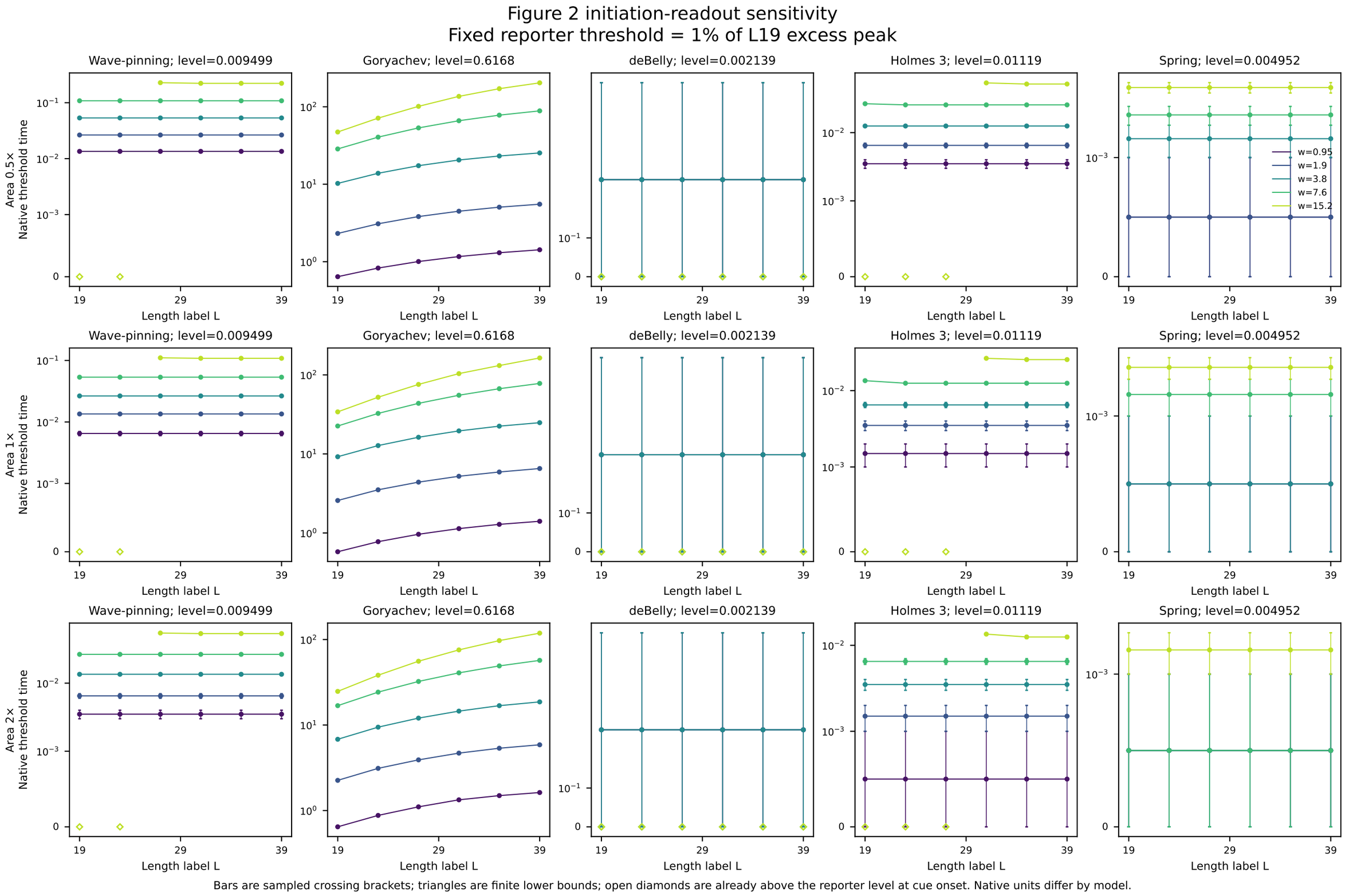
Each length uses three centered full-support cosine cues with the same commanded spatial area. This is a supporting native placement test, distinct from the Figure 3 ring experiment. Triangles are lower bounds; open squares mark failed finite field tails. deBelly distances are native nominal solver coordinates.

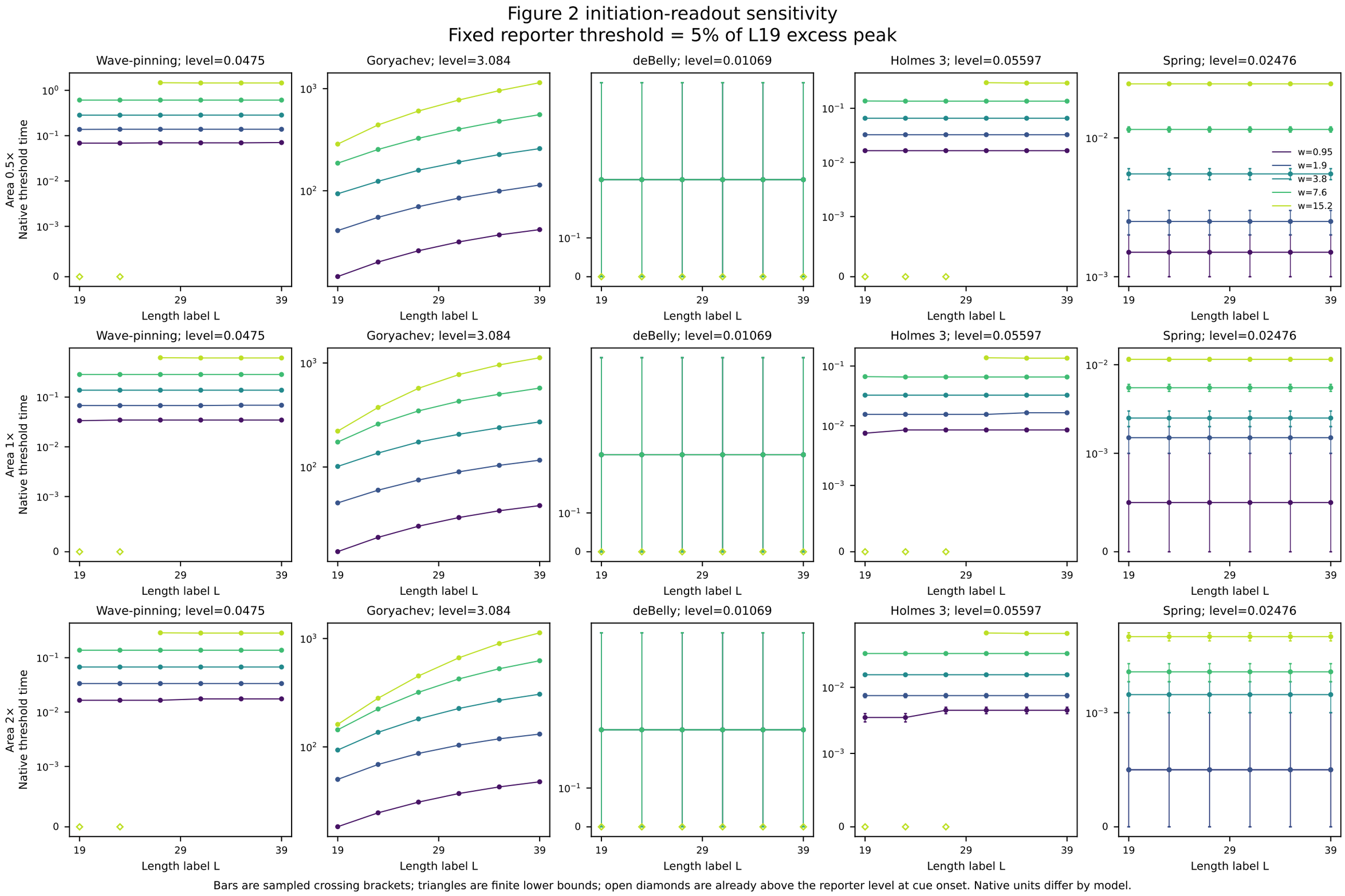


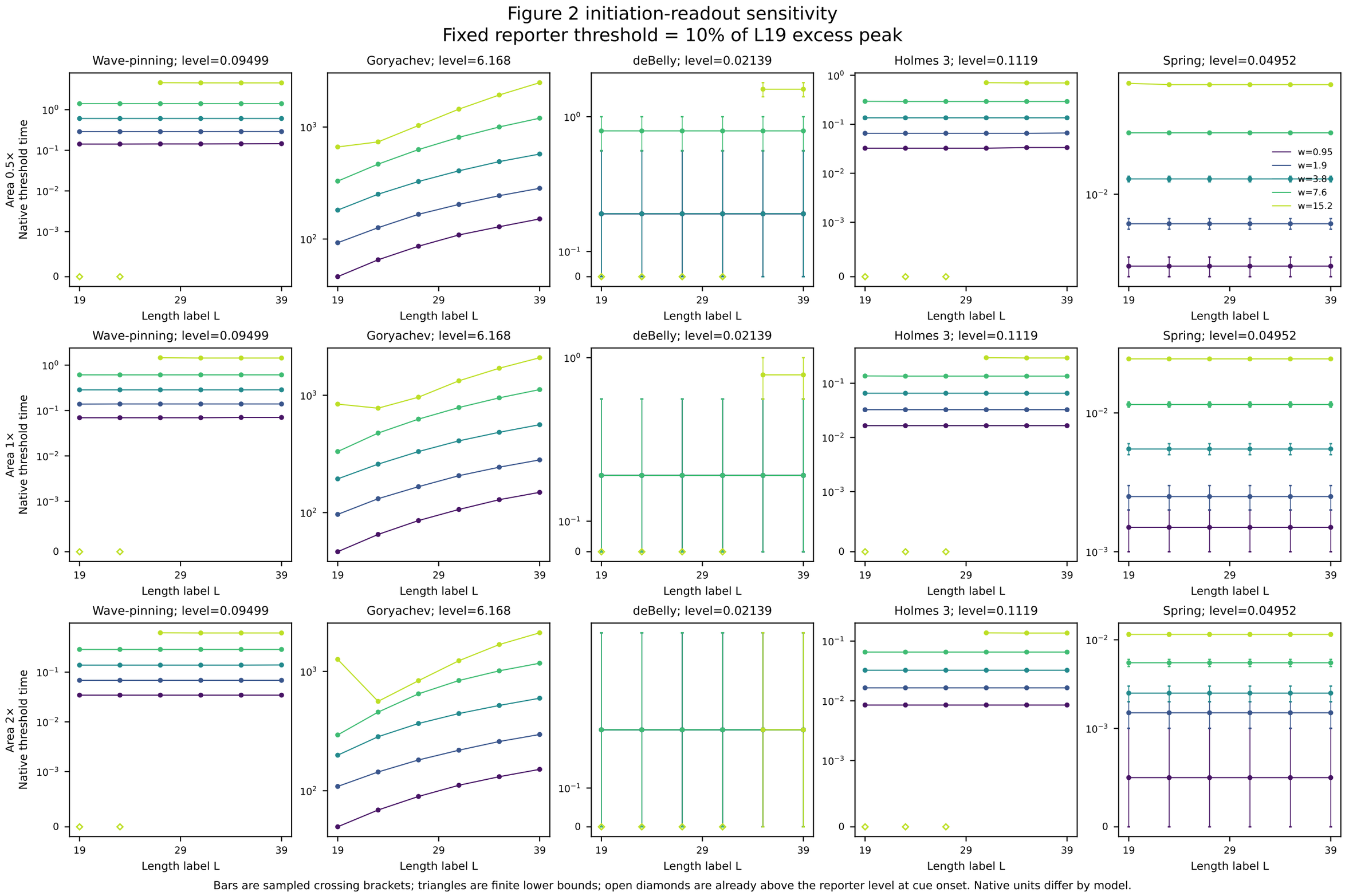
†: share target not reached in the finite shared native prefix; no withdrawal was executed. ×: observed low activity. □: failed finite native-field tail, independent of fate. Exact first-hit native states are reused without interpolation; threshold and added dwell remain separate. deBelly retains its native 0.5-unit clock.

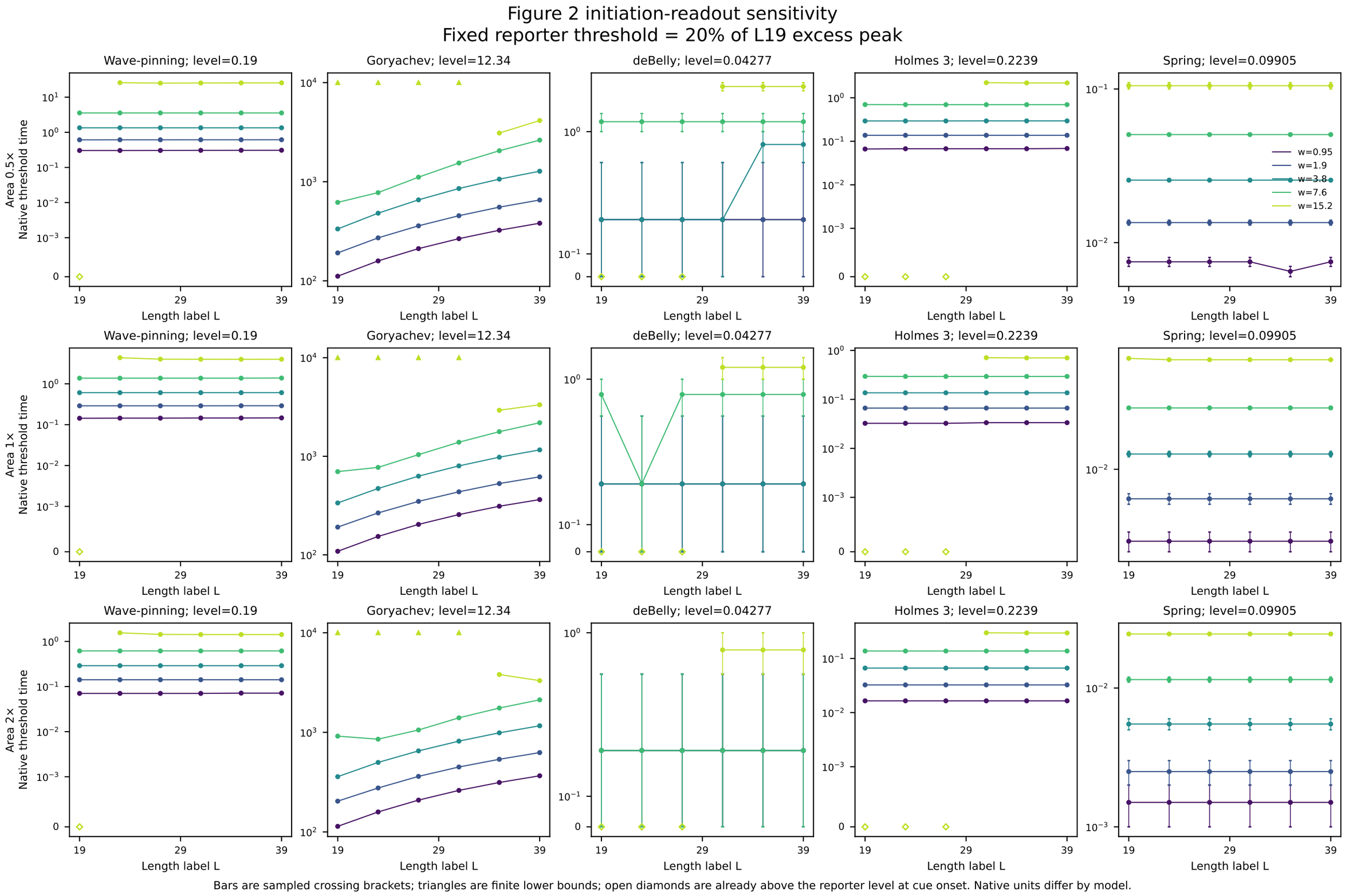

S4 Goryachev: share-triggered withdrawal and added dwell
First sampled challenger share ≥ 0.5
First sampled challenger share ≥ 0.75
First sampled challenger share ≥ 0.9
Additional dwell / commitment calibration
1
0.25
0
19
23
27
31
35
39
Length label L
Final eligible D after recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
†: share target not reached in the finite shared native prefix; no withdrawal was executed. ×: observed low activity. □: failed finite native-field tail, independent of fate. Exact first-hit native states are reused without interpolation; threshold and added dwell remain separate. deBelly retains its native 0.5-unit clock.

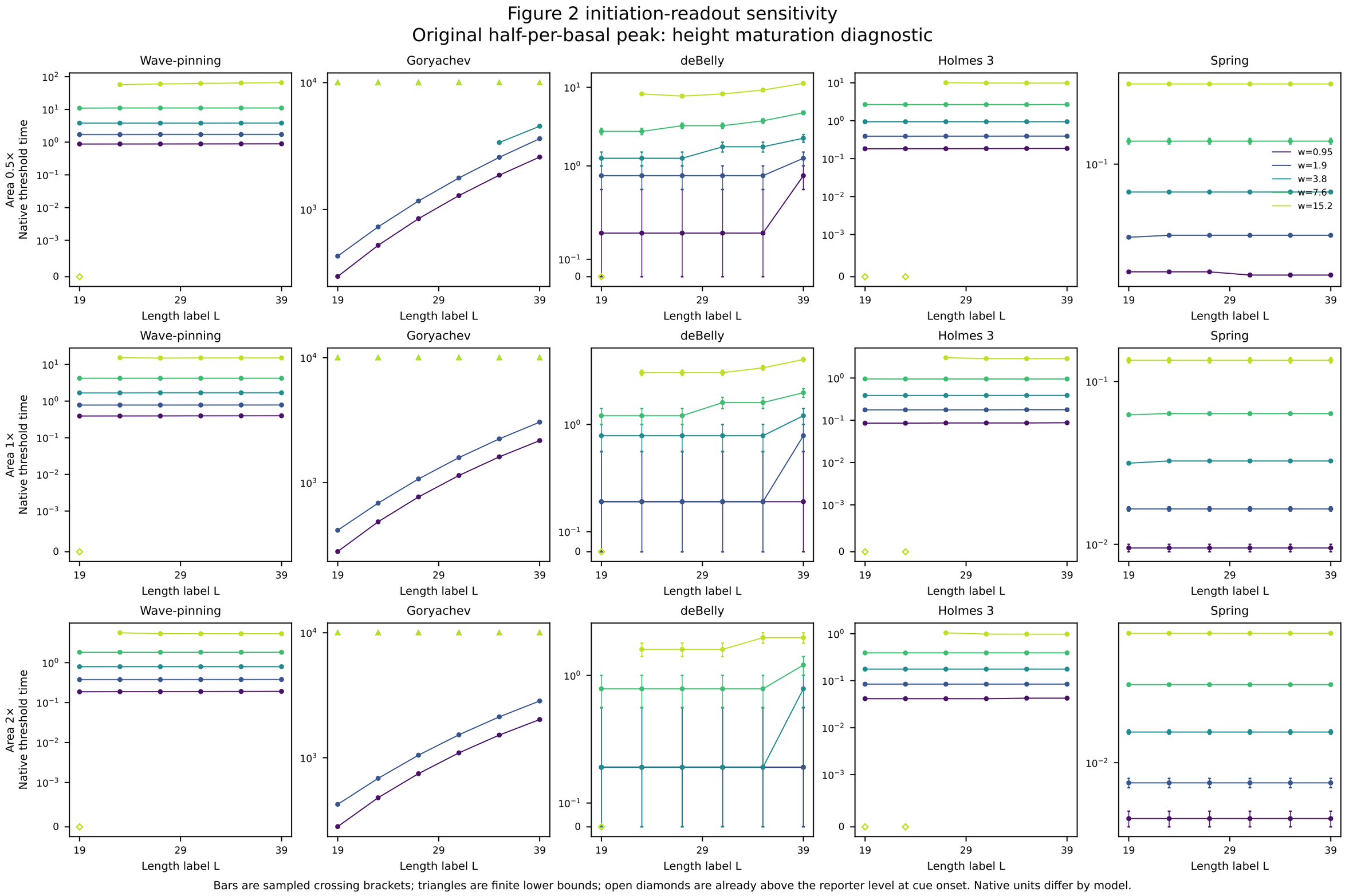

S4 deBelly: share-triggered withdrawal and added dwell
First sampled challenger share ≥ 0.5
First sampled challenger share ≥ 0.75
First sampled challenger share ≥ 0.9
Additional dwell / commitment calibration
1
0.25
0
19
23
27
31
35
39
Length label L
Final eligible D after recovery
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
†: share target not reached in the finite shared native prefix; no withdrawal was executed. ×: observed low activity. □: failed finite native-field tail, independent of fate. Exact first-hit native states are reused without interpolation; threshold and added dwell remain separate. deBelly retains its native 0.5-unit clock.


Figure 2 initiation-readout sensitivity
Fixed reporter threshold = 1% of L19 excess peak
Wave-pinning; level=0.009499
Goryachev; level=0.6168
deBelly; level=0.002139
Holmes 3; level=0.01119
Spring; level=0.004952
Area 0.5×
Area 1×
Area 2×
Native threshold time
Length label L
w=0.95
w=1.9
w=3.8
w=7.6
w=15.2
Bars are sampled crossing brackets; triangles are finite lower bounds; open diamonds are already above the reporter level at cue onset. Native units differ by model.


Figure 2 initiation-readout sensitivity
Fixed reporter threshold = 5% of L19 excess peak
Wave-pinning; level=0.0475
Goryachev; level=3.084
deBelly; level=0.01069
Holmes 3; level=0.05597
Spring; level=0.02476
Area 0.5×
Native threshold time
Area 1×
Native threshold time
Area 2×
Native threshold time
Length label L
w=0.95
w=1.9
w=3.8
w=7.6
w=15.2
Bars are sampled crossing brackets; triangles are finite lower bounds; open diamonds are already above the reporter level at cue onset. Native units differ by model.


Figure 2 initiation-readout sensitivity
Fixed reporter threshold = 10% of L19 excess peak
Wave-pinning; level=0.09499
Goryachev; level=6.168
deBelly; level=0.02139
Holmes 3; level=0.1119
Spring; level=0.04952
Area 0.5×
Area 1×
Area 2×
Native threshold time
Length label L
w=0.95
w=1.9
w=3.8
w=7.6
w=15.2
Bars are sampled crossing brackets; triangles are finite lower bounds; open diamonds are already above the reporter level at cue onset. Native units differ by model.


Figure 2 initiation-readout sensitivity
Fixed reporter threshold = 20% of L19 excess peak
Wave-pinning; level=0.19
Goryachev; level=12.34
deBelly; level=0.04277
Holmes 3; level=0.2239
Spring; level=0.09905
Area 0.5×
Area 1×
Area 2×
Native threshold time
Length label L
w=0.95
w=1.9
w=3.8
w=7.6
w=15.2
Bars are sampled crossing brackets; triangles are finite lower bounds; open diamonds are already above the reporter level at cue onset. Native units differ by model.


Figure 2 initiation-readout sensitivity
Original half-per-basal peak: height maturation diagnostic
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
Area 0.5×
Area 1×
Area 2×
Native threshold time
Length label L
w=0.95
w=1.9
w=3.8
w=7.6
w=15.2
Bars are sampled crossing brackets; triangles are finite lower bounds; open diamonds are already above the reporter level at cue onset. Native units differ by model.

S5 Wave-pinning: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	8.617e-07	0.0002667	0.9999 / 0.9999	True / True	none
19	width_7.6	mesh401	True	8.617e-07	0.0002667	0.9999 / 0.9999	True / True	none
39	width_0.95	mesh401	True	0.0009561	0.002449	0.09274 / 0.09184	True / True	none
39	width_7.6	mesh401	True	1.56e-09	0.00112	1 / 1	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0001612	0.0002667	0.9999 / 0.9999	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.001581	0.00112	1 / 1	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.

S5 Goryachev: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	4.417e-12	0.003297	1 / 1	True / True	none
19	width_7.6	mesh401	True	1.311e-07	0.003296	1 / 1	True / True	none
39	width_0.95	mesh401	True	0	0.008964	1 / 1	True / True	none
39	width_7.6	mesh401	True	0	0.008964	1 / 1	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0006152	0.003297	-1 / -1	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.00109	0.008964	-1 / -1	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.

S5 deBelly: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	5.627e-05	0.0005137	0.9277 / 0.9278	True / True	none
19	width_0.95	quarter step201	True	9.114e-05	0.0003046	0.9277 / 0.9277	True / True	none
19	width_7.6	mesh401	True	5.375e-05	0.0004833	0.9278 / 0.9279	False / False	none
19	width_7.6	quarter step201	True	0.0001901	0.001004	0.9278 / 0.9277	False / True	tail
39	width_0.95	mesh401	True	3.826e-05	0.008586	0.9209 / 0.9209	True / True	none
39	width_0.95	quarter step201	False	0.0002683	0.02642	0.9209 / 0.9209	True / True	NC
39	width_7.6	mesh401	True	9.478e-05	0.008585	0.9209 / 0.9209	True / True	none
39	width_7.6	quarter step201	False	0.000661	0.02648	0.9209 / 0.9209	True / True	NC
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0008136	0.0005952	0.9276 / 0.9276	False / False	none
19	4 pulses, gap4, ON1.5τ	quarter step201	True	0.05893	0.001933	0.9276 / 0.9277	False / True	tail, eligibility, D
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.02738	0.0554	0.9165 / 0.9133	False / False	fields
39	4 pulses, gap4, ON1.5τ	quarter step201	False	0.7739	0.08341	0.9165 / 0.9209	False / True	NC, tail, fields, D, bias
19	width_0.95	qualified quarter step201	True	9.114e-05	0.0003046	0.9277 / 0.9277	True / True	none
19	width_7.6	qualified quarter step201	True	0.0001901	0.001004	0.9278 / 0.9277	False / True	tail
39	width_0.95	qualified quarter step201	True	0.0002655	0.02642	0.9209 / 0.9209	True / True	none
39	width_7.6	qualified quarter step201	True	0.000646	0.02648	0.9209 / 0.9209	True / True	none
19	4 pulses, gap4, ON1.5τ	qualified quarter step201	True	0.05893	0.001933	0.9276 / 0.9277	False / True	tail, eligibility, D
39	4 pulses, gap4, ON1.5τ	qualified quarter step201	True	0.775	0.08341	0.9165 / 0.9209	False / True	tail, fields, D, bias

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules. Both exact-native-start quarter-step diagnostics and separately qualified quarter-step basals are retained. The L39 mesh401 train has native-field endpoint sensitivity; qualified quarter-step trains preserve early bias loss and final fate but later-gap timing, eligibility and tail status differ.

S5 Holmes 3: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	1.153e-05	0.000299	0.9963 / 0.9964	True / True	none
19	width_7.6	mesh401	True	1.153e-05	0.000299	0.9963 / 0.9964	True / True	none
39	width_0.95	mesh401	True	0.001687	0.00499	-0.2697 / -0.2714	True / True	none
39	width_7.6	mesh401	True	0.0003869	0.001925	0.3536 / 0.3533	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0003293	0.000299	0.9963 / 0.9964	True / True	none

Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules. The requested L39 train has no sampled commitment calibration through100,000 ON; its numerical companion is explicitly unavailable. It is not replaced with a forced-cap schedule.

S5 Spring: all mesh/time-step comparison gates

L	Assay	Companion	No-cue gate	Max phase |ΔD|	Max native field error	Final D: main / companion	Tail: main / companion	Qualification codes
19	width_0.95	mesh401	True	4.37e-06	1.62e-05	0.9964 / 0.9964	True / True	none
19	width_7.6	mesh401	True	4.37e-06	1.62e-05	0.9964 / 0.9964	True / True	none
39	width_0.95	mesh401	True	2.082e-06	6.576e-05	0.9997 / 0.9997	True / True	none
39	width_7.6	mesh401	True	2.082e-06	6.576e-05	0.9997 / 0.9997	True / True	none
19	4 pulses, gap4, ON1.5τ	mesh401	True	0.0006407	1.62e-05	0.9964 / 0.9964	True / True	none
39	4 pulses, gap4, ON1.5τ	mesh401	True	0.02262	6.576e-05	0.9997 / 0.9997	True / True	none

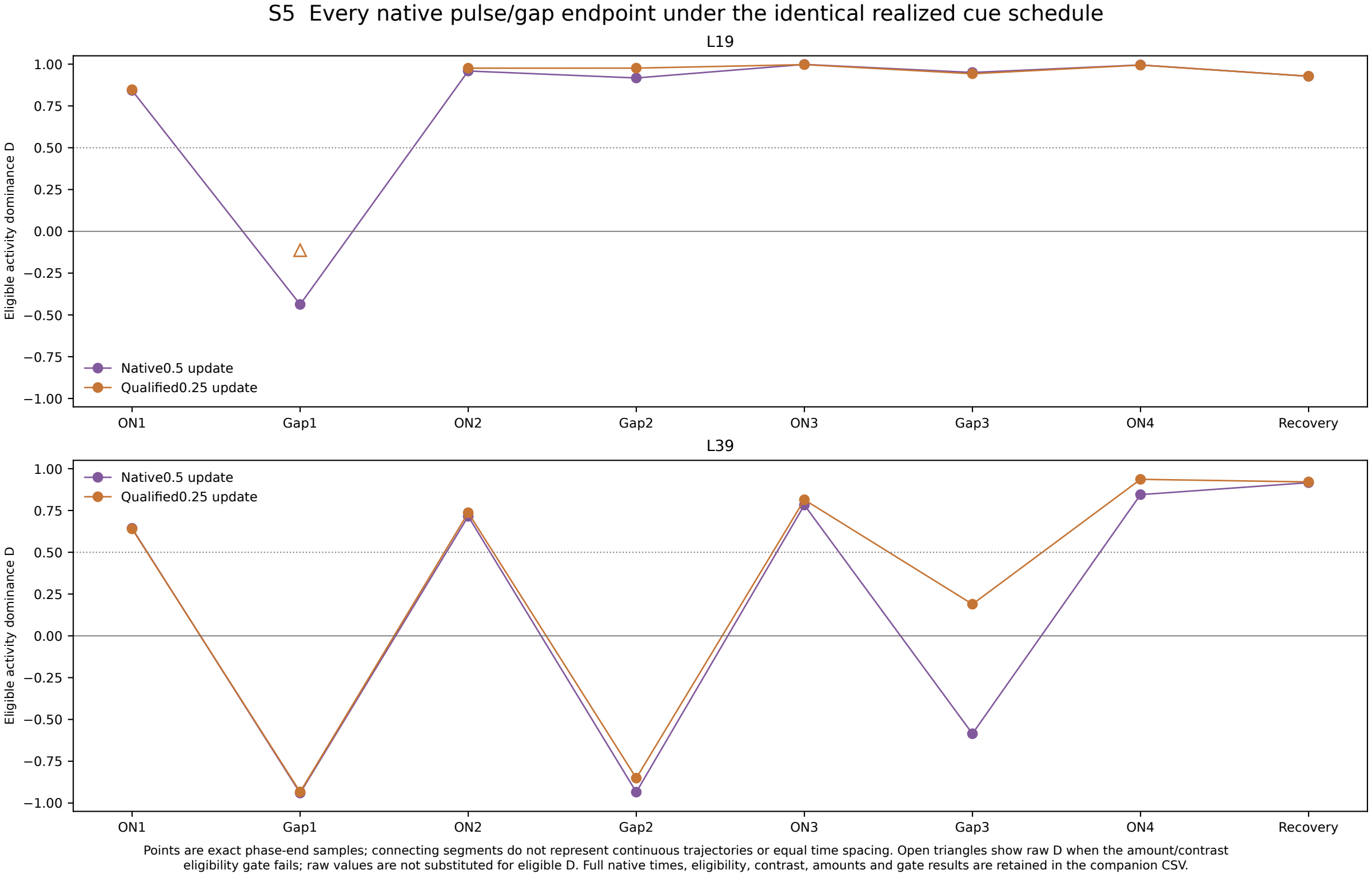
Codes: NC, companion no-cue gate failed; tail, finite-tail flag changed; eligibility, phase eligibility changed; D, phase |ΔD| exceeds tolerance; fields, native-field error exceeds tolerance; bias, challenger-bias sign changed. Full warning text is retained in CSV. Independent native identity/readout audits pass for every actual comparison. Numerical agreement is assessed separately for every requested readout; final fate agreement does not establish pulse/gap timing or full-field agreement. The mesh comparison is201 versus401 cells, using the same frozen native kinetics and realized schedules.

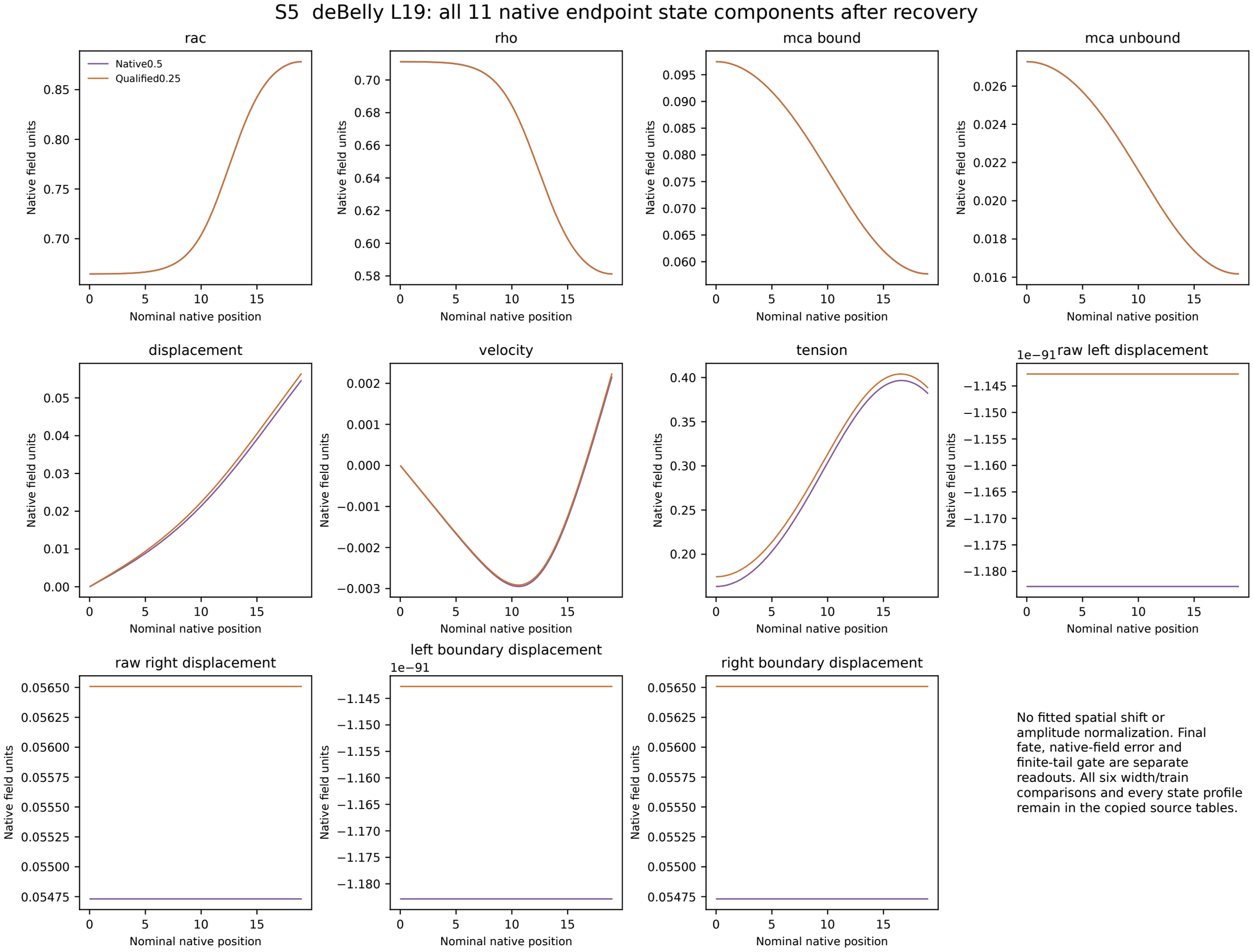
S5 deBelly: native identity audits pass; numerical agreement is incomplete

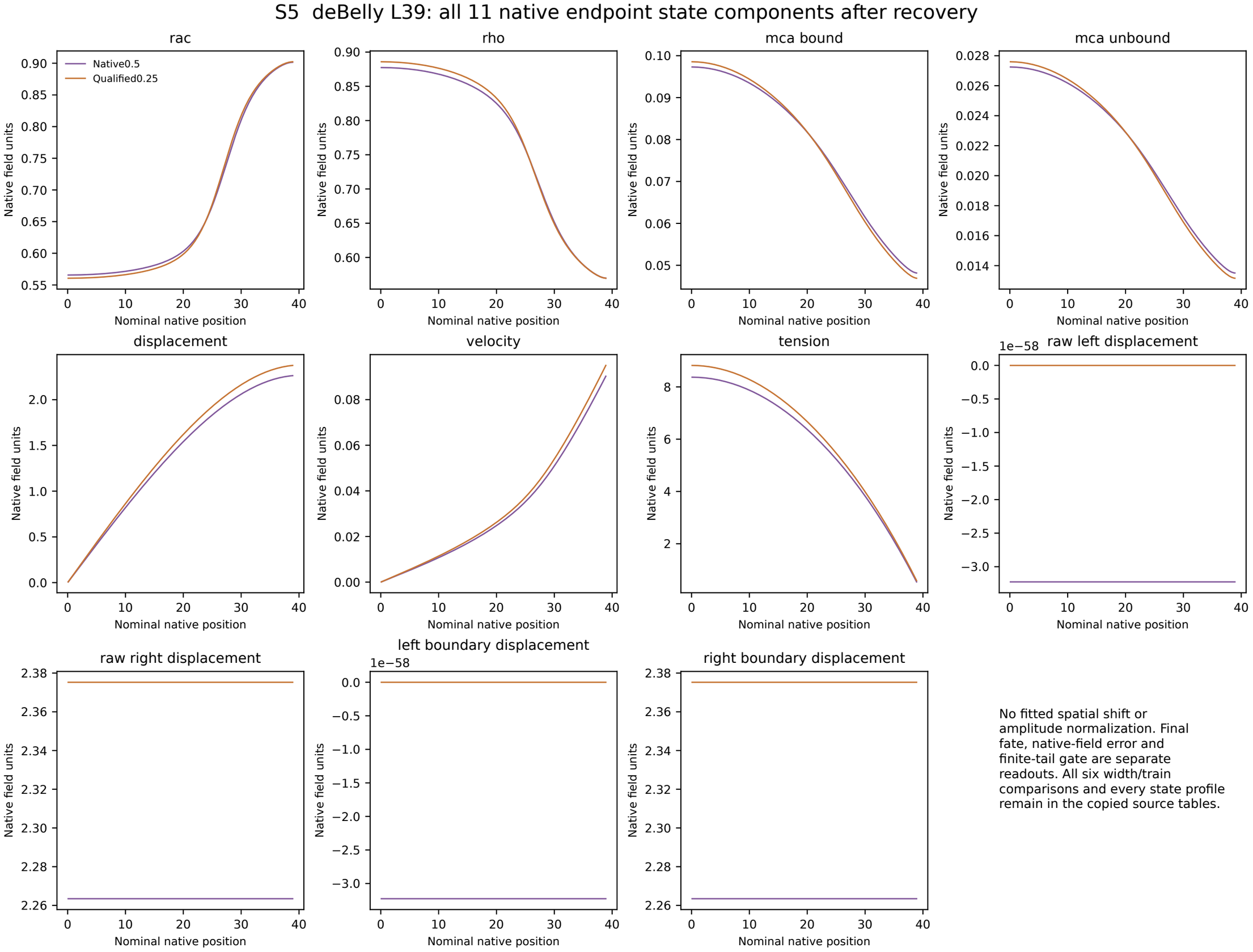
L	Assay	Basal field shift	Max endpoint |ΔD|	Max endpoint field error	Final D:0.5 /0.25	Tail:0.5 /0.25	Nonexclusive qualifications
19	width_0.95	0.0245%	9.11e-05	0.0305%	0.928 / 0.928	True / True	none
19	width_7.6	0.0245%	0.00019	0.1%	0.928 / 0.928	False / True	finite_field_tail_status_changed
39	width_0.95	2.62%	0.000266	2.64%	0.921 / 0.921	True / True	none
39	width_7.6	2.62%	0.000646	2.65%	0.921 / 0.921	True / True	none
19	train_4_gap4_on1p5	0.0245%	0.0589	0.193%	0.928 / 0.928	False / True	finite_field_tail_status_changed phase_endpoint_eligibility_changed phase_endpoint_D_difference_above_tolerance
39	train_4_gap4_on1p5	2.62%	0.775	8.34%	0.916 / 0.921	False / True	finite_field_tail_status_changed native_endpoint_field_difference_above_tolerance phase_endpoint_D_difference_above_tolerance phase_endpoint_challenger_bias_changed

The 0.25-unit companions use separately qualified basal states and a controller relaxation rate adjusted to preserve its physical rate. The native 0.5-unit campaign remains unchanged. Final fate agreement does not establish agreement in phase timing, eligibility, coupled native fields or finite-tail status.

At L39 the first two gaps still lose challenger bias, but third-gap D changes from −0.58556 to +0.18947. Final D changes from 0.91646 to 0.92085. Tension and velocity endpoint differences reach 8.341% and 6.802% of native scales. The first L19 gap also loses bias in both schemes while its eligibility flag changes. These are explicit numerical limits, not omitted failed simulations.


S5 Every native pulse/gap endpoint under the identical realized cue schedule
L19
L39
Eligible activity dominance D
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Native0.5 update
Qualified0.25 update
ON1
Gap1
ON2
Gap2
ON3
Gap3
ON4
Recovery
Points are exact phase-end samples; connecting segments do not represent continuous trajectories or equal time spacing. Open triangles show raw D when the amount/contrast eligibility gate fails; raw values are not substituted for eligible D. Full native times, eligibility, contrast, amounts and gate results are retained in the companion CSV.


S5 deBelly L19: all 11 native endpoint state components after recovery
rac
rho
mca bound
mca unbound
displacement
velocity
tension
raw left displacement
raw right displacement
left boundary displacement
right boundary displacement
Native0.5
Qualified0.25
Native field units
Nominal native position
1e−91
No fitted spatial shift or amplitude normalization. Final fate, native-field error and finite-tail gate are separate readouts. All six width/train comparisons and every state profile remain in the copied source tables.


S5 deBelly L39: all 11 native endpoint state components after recovery
rac
rho
mca bound
mca unbound
displacement
velocity
tension
raw left displacement
raw right displacement
left boundary displacement
right boundary displacement
Native0.5
Qualified0.25
Native field units
Nominal native position
1e−58
No fitted spatial shift or amplitude normalization. Final fate, native-field error and finite-tail gate are separate readouts. All six width/train comparisons and every state profile remain in the copied source tables.